MODULAR KOSZUL DUALITY AND THE SHADOW OBSTRUCTION TOWER: A SURVEY

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Abstract. We study holomorphic chiral factorisation (co)homology via bar and cobar chain constructions at various geometric locations on algebraic curves. E_n-chiral algebras at all operadic levels ($n = 1, 2, 3, \infty$) are factorisation algebras on configuration spaces, inherently geometric; the ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct, R-matrix, and Yangian structure is the categorical logarithm. The bar differential squares to zero by Arnold plus Borcherds at genus zero and acquires curvature $\kappa(\mathcal{A}) \cdot \omega_g$ from the Hodge bundle at higher genus. A single universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ controls the genus tower. Five theorems (A–D, H) organise the invariants: bar–cobar adjunction with Verdier intertwining, inversion on the Koszul locus, complementary Lagrangian decomposition, the modular characteristic, and polynomial chiral Hochschild cohomology concentrated in degrees $\{0, 1, 2\}$. The shadow obstruction tower partitions the standard landscape into four depth classes (Gaussian/Lie/contact/mixed). The derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ carries E_2/E_3 structure; its physical realisation as the bulk algebra of a 3d holomorphic–topological gauge theory gives rise to line operators, spectral braiding, and the bulk-boundary-line triangle, with 3d quantum gravity as the climax. Calabi–Yau quantum groups (CoHA, quantum toroidal algebras) are the concrete E_1-chiral quantum groups constructed from CY geometry via 6d holomorphic Chern–Simons and related theories. We survey the E_n hierarchy and operadic circle, the seven faces of the collision residue, the arithmetic of the shadow L-function, the open/closed bridge to 3d HT-QFT, modular PVA quantization, holographic reconstruction, and the outstanding frontier conjectures across three volumes totalling ~4,800 pages.

Contents

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1. The ordered bar complex

The primitive object of this programme is the *ordered bar complex* $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, an E_1-coalgebra with deconcatenation coproduct, built on ordered configurations of points on an algebraic curve. The symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$, and with it every statement about the modular characteristic κ and the shadow obstruction tower, is the Σ_n-coinvariant shadow obtained by averaging. The ordered bar carries the full R-matrix, the Yangian, the braiding; averaging collapses this profile to a scalar. What survives averaging is the subject of the five theorems; what is lost is the subject of the E_1 frontier.

The exposition constructs the ordered bar first, computes its chain-level structure on explicit examples, extracts the R-matrix as the degree-2 collision residue, and only then derives the symmetric bar by projection. No symmetric-bar result is stated without first exhibiting the E_1 object it descends from.

1.1. The 1-form. Let X be a smooth algebraic curve over $\mathbf{C}$. On the configuration space $C_2(X) = X \times X \setminus \Delta$, define the logarithmic 1-form

$$\eta_{12} \;=\; d\log(z_1 - z_2) \;=\; \frac{dz_1 - dz_2}{z_1 - z_2}.$$

This form has a simple pole along Δ with residue 1. The residue depends on no coordinate, no metric, no level. A double pole $dz/(z-w)^2$ transforms under coordinate change and has no coordinate-independent residue; a regular 1-form extracts nothing from the collision. The logarithmic derivative is the unique 1-form on $C_2(X)$ with a first-order pole along Δ and conformally invariant residue.

On $C_3(X)$, the three pullbacks $\eta_{12}, \eta_{23}, \eta_{31}$ satisfy the *Arnold relation*

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} \;=\; 0, \tag{0.1}$$

the differential-form image of the partial-fractions identity $\frac{1}{(z_1-z_2)(z_2-z_3)} + \frac{1}{(z_2-z_3)(z_3-z_1)} + \frac{1}{(z_3-z_1)(z_1-z_2)} = 0$. Arnold proved that (0.1), replicated across all triples $\{i, j, k\} \subset \{1, \ldots, n\}$, generates the complete ideal of relations in $H^*(\mathrm{Conf}_n(\mathbf{C}), \mathbf{C})$.

Under $z \mapsto w(z)$ the propagator transforms as $\eta_{ij} \mapsto \eta_{ij} + d\log\frac{w(z_i)-w(z_j)}{z_i - z_j}$; near the diagonal this reduces to $d\log w'(z_i)$, the standard $\mathrm{Diff}(X)$ cocycle. The propagator is a BRST ghost for diffeomorphisms.

1.2. The ordered bar complex on X. A *chiral algebra* $\mathcal{A}$ on a smooth algebraic curve X is a $\mathcal{D}_X$-module equipped with a chiral bracket $\mu^{\mathrm{ch}} \colon j_*j^*(\mathcal{A} \boxtimes \mathcal{A}) \to \Delta_!(\mathcal{A})$ extracting the singular part of sections with poles along the diagonal. In local coordinates this encodes the operator product expansion $a(z)\,b(w) \sim \sum_{n \geq 0} a_{(n)}b(w)/(z-w)^{n+1}$, governed by the Borcherds identity coupling all pole orders.

Place sections $a_1, \ldots, a_n \in \mathcal{A}$ at distinct ordered points $z_1 < \cdots < z_n$ on X and assign to each consecutive pair the form $\eta_{i,i+1} = d\log(z_i - z_{i+1})$. The Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n(X)$ resolves $\mathrm{Conf}_n(X)$ by iterated real oriented blowup along all diagonal strata (a blowup, not a complement); boundary codimension-one faces are indexed by subsets $S \subseteq \{1, \ldots, n\}$ with $|S| \geq 2$.

The *ordered chiral bar complex*:

$$\bar{B}^{\mathrm{ord}}_X(\mathcal{A}) \;=\; \bigoplus_{n \geq 1} (s^{-1}\bar{\mathcal{A}})^{\otimes n} \otimes \Gamma\big(\overline{\mathrm{FM}}_n(X), \Omega^{n-1}_{\log}\big).$$

Here $\bar{\mathcal{A}} = \ker(\varepsilon\colon \mathcal{A} \to \omega_X)$ is the augmentation ideal, $T^c(V) = \bigoplus_{n\geq 0} V^{\otimes n}$ is the cofree tensor coalgebra, and s^{-1} is desuspension with $|s^{-1}a| = |a| - 1$ in cohomological convention ($|d| = +1$).

This is the primitive object. It carries the *deconcatenation coproduct*

$$\Delta[s^{-1}a_1|\cdots|s^{-1}a_n] = \sum_{i=0}^{n}[s^{-1}a_1|\cdots|s^{-1}a_i] \otimes [s^{-1}a_{i+1}|\cdots|s^{-1}a_n],$$

splitting an ordered sequence at every position: $n+1$ terms for an ordered n-tuple. This makes $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ an E_1-coalgebra: coassociative, not cocommutative. The linear ordering of the points is retained; there is no Σ_n-quotient. Its E_1-Koszul dual $\mathcal{A}^!_{\mathrm{line}}$ governs the category of line operators: for affine Kac–Moody, $\mathcal{A}^!_{\mathrm{line}}$ is the Yangian $Y(\mathfrak{g})$.

The differential decomposes as $d_{\bar{B}} = d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}}$: the internal differential of $\mathcal{A}$, the OPE residue extraction along codimension-one boundary strata of $\overline{\mathrm{FM}}_n(X)$, and the de Rham differential on logarithmic forms.

1.3. $d^2 = 0$: Arnold kills the form, Borcherds kills the algebra. Expand $(d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}})^2$. The diagonal terms: $d^2_{\mathrm{int}} = 0$, $d^2_{\mathrm{dR}} = 0$; the cross-terms $[d_{\mathrm{int}}, d_{\mathrm{dR}}] = 0$ by Leibniz. The remaining terms pair into three cancellation mechanisms:

Disjoint supports: when $\{i, j\} \cap \{k, l\} = \emptyset$ the boundary faces are transverse, and the residue operators commute.

Shared index: when $\{i, j\} \cap \{k, l\} = \{j = k\}$ the triple collision $z_i = z_j = z_l$ lies in a codimension-two stratum. The form contribution $\eta_{ij} \wedge \eta_{jl}$ paired with its cyclic partners vanishes by the Arnold relation (0.1); the corresponding algebraic contractions vanish by Borcherds. Arnold kills the form; Borcherds kills the algebra; both are consumed.

Same pair: $d^{(ij)}_{\mathrm{res}}\, d^{(ij)}_{\mathrm{res}} = 0$ by the desuspension sign rule.

Result: $d^2_{\bar{B}} = 0$ iff the n-th products satisfy the Borcherds identity. The ordered bar construction accepts exactly chiral algebras.

All of this is genus 0. At genus $g \geq 1$ the propagator acquires quasi-periodicity from the prime form, and $d^2_{\mathrm{fib}} \neq 0$: the curvature $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g$ is a single scalar times the Hodge class, governing the entire genus expansion.

1.4. The collision residue and the R-matrix. The ordered bar differential extracts OPE data at each collision stratum. At tensor degree 2 the extraction is the *collision residue*: the residue of the OPE integrated against the propagator $\eta_{12} = d\log(z_1 - z_2)$. Since $d\log$ has a simple pole, it absorbs one pole order from the OPE:

$$\text{pole order of } r(z) \;=\; \text{pole order of OPE} - 1.$$

The result is the *classical r-matrix*

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta^{E_1}_{\mathcal{A}}),$$

a matrix-valued rational function encoding the full binary OPE data of the ordered bar. The Yang–Baxter equation $[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0$ is the Arnold relation (0.1) evaluated on the Casimir: three boundary divisors of $\overline{\mathrm{FM}}_3(\mathbb{C})$, three terms.

The r-matrix is an E_1 invariant. The symmetric bar sees only its Σ_2-coinvariant scalar shadow: $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$ for abelian and Virasoro-type families, while non-abelian affine Kac–Moody adds the Sugawara shift $\dim(\mathfrak{g})/2$. This lossy projection is the subject of §1.7.

1.5. Chain-level: the Heisenberg ordered bar. The Heisenberg algebra $\mathcal{H}_k$: one field α of conformal weight 1, $\alpha(z)\alpha(w) \sim k/(z-w)^2$. Only $\alpha_{(1)}\alpha = k \cdot \mathbf{1}$ is nonzero.

Degree 2. The ordered bar at tensor degree 2 is $\bar{B}_2^{\mathrm{ord}} = \mathbb{C} \cdot [s^{-1}\alpha | s^{-1}\alpha] \otimes \Gamma(\overline{\mathrm{FM}}_2, \Omega^1_{\log})$. The residue differential acts by

$$d_{\mathrm{res}}\big(s^{-1}\alpha \otimes s^{-1}\alpha \otimes \eta_{12}\big) \;=\; k \cdot \mathbf{1}.$$

The OPE has pole order 2; the $d\log$ kernel absorbs one order, producing a simple-pole residue. This is the complete degree-2 ordered bar data.

Degree 3. At tensor degree 3, $\bar{B}_3^{\mathrm{ord}} = (s^{-1}\alpha)^{\otimes 3} \otimes \Omega^2(\overline{\mathrm{FM}}_3)$. Every collision stratum extracts $\alpha_{(1)}\alpha = k \cdot \mathbf{1}$, leaving $n-2$ copies of α. Since $\mathbf{1}_{(p)}\alpha = 0$ for all $p \geq 0$, no further contraction occurs: the degree-3 differential vanishes identically. The Arnold relation kills the form-level contribution independently. All higher degrees vanish by the same mechanism.

Bar cohomology. The bar complex is quadratic: all structure is captured at tensor degree 2. Bar cohomology concentrates in degree 1: $H^1(\bar{B}(\mathcal{H}_k)) = \mathbb{C} \cdot \alpha^*$, $H^n = 0$ for $n \geq 2$. The conformal-weight-graded bar Hilbert series is rational. The partition-graded generating function is transcendental: $\prod_{m \geq 1}(1-x^m)^{-1}$, reflecting the Fock-space grading of the single generator. $\mathcal{H}_k$ is chirally Koszul.

R-matrix and averaging. The ordered r-matrix: $r^{\mathcal{H}}(z) = k/z$, a scalar (the OPE double pole $k/(z-w)^2$ drops to a simple pole under $d\log$ absorption). The spectral R-matrix $R(z) = \exp(k/z)$ is already Σ_2-invariant: averaging loses nothing.

$$\mathrm{av}\big(r^{\mathcal{H}}(z)\big) = k = \kappa(\mathcal{H}_k).$$

Koszul dual: $\mathcal{H}_k^! = (\mathrm{Sym}^{\mathrm{ch}}(V^*), m_0 = -k\,\omega)$, a curved symmetric chiral algebra with $m_1^2(a) = [m_0, a]$. Complementarity (free-field branch, sum = 0; contrast Virasoro where sum = 13): $\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = k + (-k) = 0$.

1.6. Chain-level: the Kac–Moody ordered bar. The affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ for simple $\mathfrak{g}$ of rank ℓ and dual Coxeter number $h^\vee$:

$$J^a(z)J^b(w) \sim \frac{f^{ab}{}_c J^c(w)}{z-w} + \frac{k\kappa^{ab}}{(z-w)^2}.$$

Two pole orders. The first carries the Lie bracket $J^a_{(0)}J^b = f^{ab}{}_c J^c$; the second carries the invariant form $J^a_{(1)}J^b = k\kappa^{ab} \cdot \mathbf{1}$.

Degree 2: two components. The ordered bar differential at tensor degree 2 decomposes as

$$d_{\bar{B}}\big([s^{-1}J^a | s^{-1}J^b] \otimes \eta_{12}\big) \;=\; \underbrace{f^{ab}{}_c\,[s^{-1}J^c]}_{\text{Lie bracket (simple pole)}} \;+\; \underbrace{k\kappa^{ab} \cdot \mathbf{1}}_{\text{curvature (double pole)}}.$$

The Lie bracket contributes a Chevalley–Eilenberg-type differential; the inner product contributes the curvature. The coexistence of both poles, absent in the pure double-pole Heisenberg, makes $\widehat{\mathfrak{g}}_k$ the first nontrivial example.

Degree 3: Serre relations. For $\mathfrak{g} = \mathfrak{sl}_3$ with simple roots α_1, α_2 and generators E_1, E_2, the degree-3 ordered bar complex $\bar{B}_3^{\mathrm{ord}} = \mathfrak{g}^{\otimes 3} \otimes \Omega^2(\overline{\mathrm{FM}}_3)$ has dimension $8^3 \times 2 = 1024$. The Serre element

$$S = [E_1|E_1|E_2] - 2[E_1|E_2|E_1] + [E_2|E_1|E_1]$$

is a bar cycle: $d_{\bar{B}}(S \otimes \eta_{12} \wedge \eta_{13}) = 0$. This is the Serre relation $(\mathrm{ad}\,E_1)^2(E_2) = 0$ manifesting as a degree-3 bar cocycle. The two independent Serre relations (for $\{i,j\} = \{1,2\}$) generate $H^3(\bar{B}(\widehat{\mathfrak{sl}}_{3,k}))$ at generic level.

Bar cohomology across Kac–Moody. The bar cohomology of affine Kac–Moody algebras exhibits the full range of algebraic complexity:

Algebra	$\dim H^n(\bar{B})$	Growth	Generating function
$\widehat{\mathfrak{sl}}_2$	$2n+1$	linear	rational: $x(3-x)/(1-x)^2$
$\widehat{\mathfrak{sl}}_3$	$\{8, 36, 204, \ldots\}$	exponential	rational: $4x(2-13x-2x^2)/((1-8x)(1-3x-x^2))$
$\widehat{\mathfrak{g}}_k$ (general)	Garland–Lepowsky	polynomial ($\mathfrak{g}$ simple)	D-finite

For $\widehat{\mathfrak{sl}}_2$, the linear growth $\dim H^n = 2n+1$ is the Garland–Lepowsky concentration: bar cohomology is generated in degree 1 by the 3-dimensional adjoint representation. For $\widehat{\mathfrak{sl}}_3$, the rational generating function with denominator $(1-8x)(1-3x-x^2)$ reflects the Serre relations at degree 3; the PBW spectral sequence collapses at E_2.

R-matrix and averaging. The ordered r-matrix is the level-prefixed Casimir:

$$r^{\mathrm{KM}}(z) \;=\; k\,\Omega/z, \qquad \Omega = \sum_a J^a \otimes J_a$$

(at $k=0$ the r-matrix vanishes; at $k=-h^\vee$ it is critical). Averaging collapses the Casimir tensor to its trace, with a Sugawara shift from simple-pole self-contraction:

$$\mathrm{av}(k\,\Omega/z) \;=\; \frac{k\dim\mathfrak{g}}{2h^\vee} \;=\; \kappa_{\mathrm{dp}}, \qquad \kappa(\widehat{\mathfrak{g}}_k) \;=\; \kappa_{\mathrm{dp}} + \frac{\dim\mathfrak{g}}{2} \;=\; \frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}.$$

The $\dim(\mathfrak{g})/2$ term is the Sugawara shift: the simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$. For $\widehat{\mathfrak{sl}}_2$: $\kappa = 3(k+2)/4$; for $\widehat{\mathfrak{sl}}_3$: $\kappa = 4(k+3)/3$.

The Casimir tensor structure $\Omega = \sum_a J^a \otimes J_a$, invisible to κ, is the data that builds the Yangian $Y(\mathfrak{g})$. Koszul dual: $\widehat{\mathfrak{g}}_k^! = \widehat{\mathfrak{g}}_{-k-2h^\vee}$ (Feigin–Frenkel duality). Complementarity: $\kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}_k^!) = 0$ (KM branch: sum $= 0$; Virasoro branch: sum $= 13$).

1.7. The averaging map: from ordered to symmetric. The symmetric bar $\bar{B}^{\Sigma}(\mathcal{A}) = \mathrm{Sym}^c(s^{-1}\bar{\mathcal{A}})$ is the Σ_n-coinvariant quotient of the ordered bar, with the coshuffle coproduct: 2^n bipartitions of an unordered set rather than $n+1$ consecutive splittings. This is an E_∞-coalgebra, the Beilinson–Drinfeld factorization coalgebra on $\mathrm{Ran}(X)$. Its Koszul dual $\mathcal{A}^!_{\mathrm{ch}}$ is the Francis–Gaitsgory chiral Koszul dual via Verdier duality on $\mathrm{Ran}(X)$.

The averaging map

$$\mathrm{av}\colon \bar{B}^{\mathrm{ord}}(\mathcal{A}) \longrightarrow \bar{B}^{\Sigma}(\mathcal{A})$$

takes Σ_n-coinvariants. For a chiral algebra with OPE poles, this descent uses the spectral R-matrix as twisting datum. Applying av at each degree:

$$\begin{aligned}
&\text{degree } 2: && \mathrm{av}(r(z)) + \kappa_{\mathrm{sp}} = \kappa(\mathcal{A}) && (\text{Theorem D; } \kappa_{\mathrm{sp}} = 0 \text{ for abelian, } \dim\mathfrak{g}/2 \text{ for KM}),\\
&\text{degree } 3: && \mathrm{av}(\text{KZ associator}) = \mathfrak{C} && (\text{cubic shadow}),\\
&\text{degree } 4: && \mathrm{av}(\text{Yangian coproduct}) = \mathfrak{Q} && (\text{quartic resonance class}).
\end{aligned}$$

Each degree-n component of the ordered Maurer–Cartan element $\Theta^{E_1}_{\mathcal{A}}$ projects under av to the degree-n shadow. The full obstruction tower is the coinvariant image of the ordered tower.

What averaging loses. For Heisenberg, $r(z) = k/z$ is already symmetric: averaging loses nothing. For Kac–Moody, $r(z) = k\,\Omega/z$ carries the Casimir tensor $\Omega = \sum_a J^a \otimes J_a$; averaging collapses Ω to $\dim(\mathfrak{g})/(2h^\vee)$ and discards the tensor structure that generates the Yangian $Y(\mathfrak{g})$, the braided monoidal structure on line operators, and the RTT relations. For Virasoro, $r(z) = (c/2)/z^3 + 2T/z$ carries the stress-tensor coupling at degree 2; averaging collapses it to $\kappa = c/2$ and discards the higher-spin data.

In each case: the ordered bar carries the R-matrix, the Yangian lineage, the braiding. The symmetric bar carries κ, the shadow tower, the genus expansion. The first is the full algebra; the second is its scalar fingerprint.

1.8. The five theorems as averaged invariants. The main theorems are the invariants of $\Theta_{\mathcal{A}}^{E_1}$ that survive Σ_n-coinvariant projection.

Theorem A (bar-cobar adjunction and Verdier intertwining). The cobar functor $\Omega(\bar{B}(\mathcal{A}))$ inverts the bar construction on the Koszul locus: $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$. Verdier duality on $\mathrm{Ran}(X)$ sends the factorization coalgebra $\bar{B}_X^{\Sigma}(\mathcal{A})$ to the homotopy Koszul dual $\mathcal{A}^!_\infty$, a factorization algebra whose cohomology recovers $\mathcal{A}^!$. This identification intertwines bar-cobar inversion with Verdier duality.

Theorem B (bar-cobar inversion). On the Koszul locus, the counit $\varepsilon\colon \Omega(\bar{B}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism at every genus. Koszulness is characterised by twelve equivalent conditions (§9).

Theorem C (complementarity). The ordered partition functions $Q_{g,n}^{E_1}(\mathcal{A})$ and $Q_{g,n}^{E_1}(\mathcal{A}^!)$ sum to a topological invariant. The coinvariant projection yields the scalar complementarity $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = \varkappa(\mathcal{A})$, where $\varkappa = 0$ for Kac–Moody and free fields, $\varkappa = 13$ for Virasoro. The scalar $\varkappa$ is distinct from the Trinity ghost-charge conductor $K(\mathcal{A}) := c(\mathcal{A}) + c(\mathcal{A}^!) = -c_{\mathrm{ghost}}(\mathrm{BRST})$, values $K(\mathrm{Vir}) = 26$, $K(\mathcal{W}_3) = 100$, $K(\mathrm{BP}) = 196$, $K(V_k(\mathfrak{g})) = 2\dim\mathfrak{g}$.

Theorem D (modular characteristic). The genus-1 formula $\mathrm{obs}_1(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_1$ is unconditional for every family. On the proved uniform-weight lane, the all-genera extension $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ holds via clutching morphisms on $\partial\overline{\mathcal{M}}_g$ (the clutching-uniqueness step is the principal conditional element; an independent GRR derivation provides an alternative path). Multi-weight $g \geq 2$ requires the cross-channel correction $\delta F_g^{\mathrm{cross}}$.

Theorem H (chiral Hochschild). $\mathrm{ChirHoch}^*(\mathcal{A})$ has polynomial Hilbert series, concentrated in cohomological degrees $\{0, 1, 2\}$.

1.9. Bar chain models on curve geometries. To understand a chiral algebra is to construct its bar complex on every relevant curve geometry. Each geometry reveals a different algebraic structure.

The formal disk D. The local model. The chiral endomorphism operad $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$ is $\mathrm{Aut}(\mathcal{O})$-equivariant, and the bar differential encodes the OPE via mode products $a_{(n)}b$. For Heisenberg: $B_D(\mathcal{H}_k) = T^c(s^{-1}\Bbbk[J])$ with $d[s^{-1}J|s^{-1}J] = k$.

The punctured disk D^*. At homology level, $H_*^{\mathrm{ch}}(D^*, \mathcal{A}) \cong \mathrm{HH}_*(\mathcal{A})$ (state-field correspondence). At *chain level*, $C_*^{\mathrm{ch}}(D^*, \mathcal{A})$ carries the full holomorphic structure (OPE poles, spectral parameters, E_2 data), while $C_*^{\mathrm{top}}(S^1, \mathcal{A})$ carries only topological E_1 data. The chain-level comparison D^* versus S^1 is the $E_2 - E_1$ gap.

The annulus A_{r_1,r_2}. Two boundary circles: bimodule structure. The annular bar $B^{\mathrm{ann}}(\mathcal{A})$ computes sewing and the topological Hochschild bimodule. Pinching the waist gives the pair-of-pants degeneration. For Heisenberg: $B^{\mathrm{ann}}(\mathcal{H}_k)$ gives the partition function $\mathrm{tr}(q^{L_0 - c/24}) = 1/\eta(\tau)$.

Nodal curves. On boundary strata of $\overline{\mathcal{M}}_{g,n}$: each stable graph vertex contributes a component bar complex, each edge contributes the sewing kernel (thin-annulus limit of B^{ann}). Curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ arises from failure to extend across nodes.

The pair of pants. The multiplication cobordism $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its bar chain model encodes fusion: three collision channels, the Arnold relation forcing consistency. For Kac–Moody, the pair-of-pants bar encodes the fusion tensor product via the KZ equation on three-point conformal blocks.

1.10. The E_n-chiral hierarchy and derived centres. E_n-chiral algebras are factorisation algebras and are inherently geometric. E_∞-chiral algebras (all standard vertex algebras) live on $\mathrm{Ran}(X)$; the symmetric bar $\bar{B}^{\Sigma}$ is their bar complex. E_1-chiral algebras live on $\mathrm{Ran}^{\mathrm{ord}}(X)$; the ordered bar $\bar{B}^{\mathrm{ord}}$ carries the R-matrix. The formality bridge (Theorem **??**): for E_∞-input, ordered and symmetric bars are quasi-isomorphic. The formality failure (Theorem **??**): for genuinely E_1-input (Yangians, Etingof–Kazhdan quantum vertex algebras), ordered chiral homology has dimension $n!$ at degree n, while the symmetric quotient collapses to dimension 1.

The chiral quantum group equivalence (Theorem **??**): on the Koszul locus, three structures determine each other–-the spectral R-matrix, the chiral A_∞ structure, and the chiral coproduct. Concrete verifications through $\mathfrak{gl}_N$ for all $N \geq 1$ (Theorem **??**).

The derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ carries E_2 structure from E_1 input (Deligne conjecture) and E_3 from E_2 input (higher Deligne). Theorem H: concentration in degrees $\{0, 1, 2\}$ with polynomial Hilbert series. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure lives on the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\ \mathcal{A})$: bulk acting on boundary. The bar complex $\bar{B}(\mathcal{A})$ is a single-coloured E_1 coalgebra, NOT an $\mathrm{SC}^{\mathrm{ch,top}}$-coalgebra.

The E_3 identification (Theorem **??**): for simple $\mathfrak{g}$, $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})) \cong A^{\lambda}$ as E_3-families over $\lambda \cdot H^3(\mathfrak{g})[[\lambda]]$. The topologisation theorem (Theorem **??**): $\mathrm{SC}^{\mathrm{ch,top}}$ + conformal vector $\simeq E_3^{\mathrm{top}}$.

All of this is algebraic-geometric: configuration spaces, FM compactifications, Mok's logarithmic spaces. Volume II interprets these constructions physically: the derived centre IS the bulk of a 3d gauge theory.

1.11. The E_n operadic circle. The E_n circle $E_3^{\mathrm{top}} \to E_2 \to E_1 \to E_2 \to E_3^{\mathrm{top}}$ is an algebraic-geometric structure: five functorial operations between categories of factorisation algebras on configuration spaces.

Arrow 1 ($E_3 \to E_2$): restriction from the bulk to a codimension-2 defect. Arrow 2 ($E_2 \to E_1$): passage to the ordered bar. Arrow 3 ($E_1 \to E_2$): the Drinfeld centre $\mathcal{Z}\colon E_1\text{-}\mathsf{Cat} \to E_2\text{-}\mathsf{Cat}$, the sole source of non-trivial braiding ($\pi_1(\mathrm{Conf}_2(\mathbb{R}^3)) = 0$ makes direct $E_3 \to E_2$ restriction always symmetric). Arrow 4 ($E_2 \to E_3$): derived centre via higher Deligne. Arrow 5 ($E_3 = E_3$): the closing, conjectural (Conjecture **??**). The circle partly closes: the E_3 identification (Theorem **??**) proves $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})) \cong A^{\lambda}$ for simple $\mathfrak{g}$.

Volume II interprets the circle physically: the derived centre IS the bulk of a 3d gauge theory. Volume III constructs concrete E_1-chiral quantum groups from CY geometry (CoHA, quantum toroidal algebras) that feed the circle at Arrow 2.

Remark 0.1 (The zero-dimensional shadow). When $X = \mathrm{Spec}\,\Bbbk$ is a point, configuration spaces collapse, logarithmic forms vanish, and the chiral bar complex reduces to the classical bar construction of an associative algebra. Classical Koszul duality (Priddy, Beilinson–Ginzburg–Soergel, Loday–Vallette) embeds on $X = \mathbb{A}^1$: the affine line deformation-retracts to a point (but the retraction is additional data, not a canonical identification; on $\mathbb{P}^1$ the global topology has no counterpart over a bare point).

Everything above takes place at genus 0: the propagator is globally defined on $\mathbb{P}^1$, and the Arnold relation holds without correction. On a curve of genus $g \geq 1$, $z_1 - z_2$ has no global meaning; the Arnold relation acquires a defect, the bar differential no longer squares to zero, and the defect is controlled by a single scalar.

2. Curvature and the genus tower

The genus-zero bar complex is exact: $d^2_{\mathrm{fib}} = 0$ by Arnold and Borcherds. At genus $g \geq 1$ the bar differential acquires curvature from the Hodge bundle, and the resulting genus tower carries the scalar modular characteristic $\kappa(\mathcal{A})$, the $\hat{A}$-genus generating function, complementarity, and the Feigin–Frenkel involution. Every formula in this section begins with the Heisenberg computation ($\kappa = k$, the level) and deforms through Kac–Moody ($\kappa = (k + h^\vee)\dim\mathfrak{g}/(2h^\vee)$) to Virasoro ($\kappa = c/2$).

2.1. The genus-g propagator. The genus-0 propagator $\eta_{12} = d\log(z_1 - z_2)$ relied on $z_1 - z_2$ being a globally defined function on $\mathbb{P}^1$. On a compact curve of genus $g \geq 1$, any meromorphic function with a simple zero must also have a pole: $z_1 - z_2$ has no global meaning. The replacement is the *prime form* $E(z_1, z_2)$: a section of $K^{-1/2} \boxtimes K^{-1/2}$ vanishing to first order along the diagonal with no other zeros. The bundle is $K^{-1/2} \boxtimes K^{-1/2}$; the prime form is a $(-\frac{1}{2}, -\frac{1}{2})$-differential. In a local coordinate with diagonal $z_1 = z_2$,

$$E(z_1, z_2) = (z_1 - z_2)\big(1 + O(z_1 - z_2)\big) \cdot (dz_1)^{-1/2}(dz_2)^{-1/2}.$$

Analytic continuation around the B-cycle B_j in z_1 gives

$$E(z_1 + B_j, z_2) = -\exp\big(-\pi i \Omega_{jj} - 2\pi i \textstyle\int_{z_2}^{z_1} \omega_j\big) \cdot E(z_1, z_2),$$

where $\Omega = (\Omega_{jk})$ is the period matrix and $\omega_1, \dots, \omega_g$ the holomorphic differentials normalised by $\oint_{A_j} \omega_k = \delta_{jk}$. The A-cycle monodromy is trivial.

The logarithmic derivative $\omega(z_1, z_2) := d_{z_1} \log E(z_1, z_2)$ is the fundamental meromorphic differential of the second kind. The *genus-g propagator*:

$$\eta_{12}^{(g)} = d\log E(z_1, z_2) + \pi \sum_{j,k=1}^{g} (\operatorname{Im}\Omega)^{-1}_{jk} \operatorname{Im}\Big(\int_{z_2}^{z_1} \omega_j\Big)\omega_k(z_1).$$

The period correction $(\operatorname{Im}\Omega)^{-1}_{jk}$ compensates B-cycle monodromy, rendering $\eta_{12}^{(g)}$ single-valued; this is a full $g \times g$ matrix at genus $g \geq 2$, never collapsed to a scalar. The price: $\eta_{12}^{(g)}$ is no longer holomorphic. The propagator couples to the Hodge bundle $\mathbb{E} \to \overline{\mathcal{M}}_g$ through Ω.

The prime form at genus 1. At genus 1, the curve is the elliptic curve $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ with $\operatorname{Im}\tau > 0$, and the prime form reduces to the Jacobi theta function:

$$E(z_1, z_2) = \frac{\theta_1(z_1 - z_2 \mid \tau)}{\theta_1'(0 \mid \tau)} \cdot (dz_1)^{-1/2}(dz_2)^{-1/2}.$$

The period matrix is the scalar $\Omega = \tau$; the genus-1 propagator

$$\eta_{12}^{(1)} = d_{z_1} \log\theta_1(z_1 - z_2 \mid \tau) + \frac{\pi}{\operatorname{Im}\tau} \operatorname{Im}(z_1 - z_2)\, dz_1$$

has a single simple pole along the diagonal $z_1 = z_2$ (from the zero of θ_1) and is single-valued under both A- and B-cycle translation. The logarithmic derivative $\partial_{z_1} \log\theta_1(z \mid \tau) = \wp_1(z \mid \tau)$ is the Weierstrass zeta variant (the ratio θ_1'/θ_1, periodic under $z \mapsto z + 1$, quasi-periodic under $z \mapsto z + \tau$ with increment $-2\pi i$). The period correction renders the propagator single-valued at the cost of holomorphicity.

The prime form at genus 2. At genus 2, the period matrix $\Omega = \left(\begin{smallmatrix} \tau_1 & \tau_3 \\ \tau_3 & \tau_2 \end{smallmatrix}\right)$ is a 2×2 symmetric matrix in the Siegel upper half-space $\mathfrak{H}_2$. The prime form is constructed from the genus-2 theta function with an odd half-characteristic. The period correction matrix $(\operatorname{Im}\Omega)^{-1}$ is now a genuine 2×2 inverse:

$$(\operatorname{Im}\Omega)^{-1} = \frac{1}{\operatorname{Im}\tau_1 \operatorname{Im}\tau_2 - (\operatorname{Im}\tau_3)^2} \begin{pmatrix} \operatorname{Im}\tau_2 & -\operatorname{Im}\tau_3 \\ -\operatorname{Im}\tau_3 & \operatorname{Im}\tau_1 \end{pmatrix}.$$

This matrix never degenerates to a scalar; formulas written at genus 1 by replacing $1/\operatorname{Im}\tau$ with $(\operatorname{Im}\Omega)^{-1}_{jk}$ must be verified at $g = 2$ in full matrix form.

The propagator has weight 1. The bar propagator $\eta_{12}^{(g)} = d\log E(z_1, z_2) +$ (period correction) is a $(1,0)$-form in z_1 for every genus. It has weight 1, never weight h (the conformal weight of a field). The $d\log$ absorbs one power of the local coordinate from the prime form, producing a meromorphic differential with a simple pole along the diagonal. This weight-1 property is genus-independent: the period correction is smooth and does not change the pole structure. Any formula with a bar propagator of weight $\neq 1$ is wrong.

2.2. Curvature. The Arnold relation at genus g acquires a defect. The period corrections produce a residual $(1,1)$-form, and the fibrewise bar differential satisfies

$$d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g, \tag{0.2}$$

where ω_g is the Arakelov $(1,1)$-form on the fibre of the universal curve $\pi\colon \mathcal{C}_g \to \overline{\mathcal{M}}_g$, and $\kappa(\mathcal{A}) \in \Bbbk$ is the *modular characteristic*, determined entirely by the leading OPE singularity. The class ω_g lives on the fibre Σ_g, not on the moduli space $\overline{\mathcal{M}}_g$; its pushforward $\pi_*\omega_g = c_1(\mathbb{E}) = \lambda_1$ lands in $H^2(\overline{\mathcal{M}}_g)$.

Two conventions coexist. The bar differential $d_{\mathrm{bar}}^2 = 0$ *always*, at every genus: the bar construction is a dg coalgebra by definition. The fiberwise differential $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ is a statement about the fibrewise restriction of the bar complex to a single curve Σ_g: on that fibre, the Arnold relations fail by a $(1,1)$-correction from the period matrix. The total differential $D_{\mathcal{A}}$ incorporates the Gauss–Manin connection and squares to zero globally. This three-level structure (bar = exact; fibrewise = curved; total = exact) controls every formula in the genus tower.

Curvature computation: Heisenberg. The Heisenberg OPE $J(z)J(w) \sim k/(z-w)^2$ has r-matrix $r^{\mathrm{Heis}}(z) = k/z$ (trace-form convention; at $k=0$, $r=0$). The curvature arises from the failure of $\eta^{(g)} \wedge \eta^{(g)}$ to vanish: at genus 0, Arnold gives $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$; at genus $g \geq 1$, the period correction in $\eta^{(g)}$ contributes a $(1,1)$-residual proportional to ω_g.

Every residue extraction from the double-pole OPE gives $\alpha_{(1)}\alpha = k$, with no structure-constant contributions (the algebra is abelian). The curvature is the trace of this operation over the one-dimensional bar cohomology:

$$\kappa(\mathcal{H}_k) = \mathrm{tr}\left(\frac{r^{\mathrm{Heis}}(z)}{z}\bigg|_{z=0}\right) = k.$$

Sanity checks: $k=0$ gives $\kappa=0$ (uncurved, the free boson with no kinetic term). $k=1$ gives $\kappa=1$, matching the central charge $c_{\mathrm{Heis}}(k{=}1) = 1$. The formula $\kappa(\mathcal{H}_k) = k$ is the simplest instance of the modular characteristic: the level *is* the curvature.

Curvature computation: Kac–Moody. For $\widehat{\mathfrak{g}}_k$, the curvature receives the trace over the adjoint representation:

$$\kappa(\widehat{\mathfrak{g}}_k) = \frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}.$$

The derivation proceeds by the averaging map identity. In the trace-form convention, the r-matrix is $r^{\mathrm{KM}}(z) = k\Omega/z$, where $\Omega = \sum_a t^a \otimes t^a$ is the Casimir of the Killing form (at $k=0$, $r=0$; correct). The double-pole channel gives $\mathrm{av}(r(z)) = k\dim\mathfrak{g}/(2h^\vee)$. The simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$ contributes an additional $\dim\mathfrak{g}/2$. Together:

$$\kappa(\widehat{\mathfrak{g}}_k) = \underbrace{\frac{k\dim\mathfrak{g}}{2h^\vee}}_{\text{double-pole}} + \underbrace{\frac{\dim\mathfrak{g}}{2}}_{\text{simple-pole Sugawara shift}} = \frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}.$$

Sanity checks: $k = 0$ gives $\kappa = \dim \mathfrak{g}/2$ (not zero; the Lie bracket persists and the simple-pole channel contributes). $k = -h^\vee$ gives $\kappa = 0$ (critical level: Sugawara undefined, curvature vanishes, bar complex uncurved).

For $\mathfrak{sl}_2$: $\dim \mathfrak{g} = 3$, $h^\vee = 2$, so $\kappa(\widehat{\mathfrak{sl}}_{2,k}) = 3(k+2)/4$. At $k = 1$: $\kappa = 9/4$. At $k = -2$ (critical): $\kappa = 0$.

At the critical level $k = -h^\vee$ the shifted level vanishes and the bar complex is uncurved. The Sugawara construction is undefined (the central charge $c = k \dim \mathfrak{g}/(k + h^\vee)$ has a pole) but the curvature parameter vanishes.

Curvature computation: Virasoro. For Virasoro at central charge c, the OPE is $T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}$. The bar propagator absorbs one pole order (the $d\log$ shifts pole order by one: $\mathrm{pole}_r = \mathrm{pole}_{\mathrm{OPE}} - 1$), so the collision residue is $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ (cubic plus simple, not quartic). The averaging map extracts the scalar from the simple-pole channel and the cubic-pole channel:

$$\kappa(\mathrm{Vir}_c) = \mathrm{av}(r^{\mathrm{Vir}}(z)) = c/2.$$

Sanity checks: $c = 0$ gives $\kappa = 0$ (uncurved). $c = 1$ gives $\kappa = 1/2$. $c = 13$ gives $\kappa = 13/2$ (the self-dual point under Koszul duality).

The Koszul dual is Vir_{26-c}, with $\kappa(\mathrm{Vir}_{26-c}) = (26-c)/2$; the two sum to 13, not zero. At $c = 13$, Virasoro is self-dual under Koszul duality.

Curvature computation: $\mathcal{W}_N$. The principal $\mathcal{W}_N$-algebra has generators at conformal weights $\{2, 3, \ldots, N\}$ (that is, $N-1$ generators; the highest weight is N, not $N+1$). The modular characteristic is

$$\kappa(\mathcal{W}_{N,c}) = c \cdot (H_N - 1), \qquad H_N = \sum_{j=1}^{N} \frac{1}{j},$$

where H_N is the harmonic number (with upper limit N, not $N-1$). Sanity check: at $N = 2$, $H_2 - 1 = 1/2$, so $\kappa(\mathcal{W}_2) = c/2$, matching the Virasoro formula ($\mathcal{W}_2 = \mathrm{Vir}$). At $N = 3$, $H_3 - 1 = 5/6$, so $\kappa(\mathcal{W}_3) = 5c/6$.

2.3. Period integrals restore nilpotence. The curvature (0.2) obstructs $d_{\mathrm{fib}}^2 = 0$ but not the total $D_g^2 = 0$. Integration against A-cycle periods defines a connection on the family of bar complexes parametrised by $\overline{\mathcal{M}}_g$. The corrected total differential

$$D_g = d_{\mathrm{fib}} + \nabla^{\mathrm{GM}}$$

incorporates the Gauss–Manin connection on the Hodge bundle, and $D_g^2 = 0$: the $(1,1)$-curvature of the fibrewise differential is absorbed by the $(0,1)$-part of the base connection. The fibrewise class $\kappa \cdot \omega_g \in H^{1,1}(C_g)$ is identified under the period map $\overline{\mathcal{M}}_g \to \mathcal{A}_g$ with the $(0,1)$-curvature of ∇^{GM} on $R^1\pi_*\mathbb{C}$, and the two curvatures cancel.

The cancellation mechanism. Write the total differential as $D_g = d_{\mathrm{fib}} + \nabla^{\mathrm{GM}}$. Then

$$\begin{aligned} D_g^2 &= d_{\mathrm{fib}}^2 + \{d_{\mathrm{fib}}, \nabla^{\mathrm{GM}}\} + (\nabla^{\mathrm{GM}})^2 \\ &= \kappa \cdot \omega_g + 0 + (-\kappa \cdot \omega_g) = 0. \end{aligned}$$

The cross-term $\{d_{\mathrm{fib}}, \nabla^{\mathrm{GM}}\} = 0$ holds because d_{fib} acts on the fibre and ∇^{GM} acts on the base; the two commute by the product structure of the universal curve (locally). The Gauss–Manin curvature $(\nabla^{\mathrm{GM}})^2 = -\kappa \cdot \omega_g$ follows from the Kodaira–Spencer identification: the obstruction to extending a flat section of $\mathbb{E}$ across moduli is precisely the Hodge class $c_1(\mathbb{E})$, with coefficient κ.

Heisenberg: period integrals as Eisenstein series. For $\mathcal{H}_k$ at genus 1, the propagator is $\eta_{12}^{(1)} = d_{z_1} \log \theta_1(z_1 - z_2|\tau) + (\pi/\operatorname{Im}\tau)\operatorname{Im}(z_1 - z_2) dz_1$. The period integral $\oint_A (\eta^{(1)})^{2n}$ produces the Eisenstein series $G_{2n}(\tau) = (2n-1)!! \cdot E_{2n}(\tau)$ as Taylor coefficients of the Green's function expansion near the diagonal: the bar complex at genus 1 knows the ring of quasi-modular forms $\mathbb{C}[E_2^*, E_4, E_6]$. The quasi-modular $E_2^*(\tau) = E_2(\tau) - 3/(\pi \operatorname{Im}\tau)$ is the genus-1 propagator itself; the almost-holomorphic correction is the period term.

Kac–Moody: the KZB connection. For $\widehat{\mathfrak{g}}_k$, the Gauss–Manin connection on the bar complex over $\overline{\mathcal{M}}_{1,n}$ is the Knizhnik–Zamolodchikov–Bernard (KZB) connection. The heat equation

$$\frac{\partial}{\partial\tau}\Phi = \frac{1}{4\pi i(k+h^\vee)} \sum_a \frac{\partial^2}{\partial z_a^2}\Phi + (\text{interaction terms})$$

expresses the flatness $D_g^2 = 0$ as the joint system of KZ (in z) and heat (in τ) equations. The Sugawara denominator $k + h^\vee$ in the prefactor is the same factor that appears in $\kappa = (k+h^\vee)\dim\mathfrak{g}/(2h^\vee)$: the curvature and the connection are controlled by a single parameter.

2.4. The genus tower (uniform weight). The bar complex traces a family of curved cochain complexes over $\overline{\mathcal{M}}_g$. For *uniform-weight* algebras (single-generator, or multi-generator with all generators of the same conformal weight), the obstruction to flatness is the characteristic class

$$\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g \in H^*(\overline{\mathcal{M}}_g), \qquad \lambda_g = c_g(\mathbb{E}),$$

where $\mathbb{E} = \pi_*\omega_\pi$ is the Hodge bundle: the vector bundle on $\overline{\mathcal{M}}_g$ whose fibre over $[C]$ is the g-dimensional space $H^0(C, \omega_C)$. At genus 1 the factorization holds unconditionally for all families. For *multi-weight* families at genus $g \geq 2$ the scalar formula *fails*: a cross-channel correction from mixed-propagator boundary graphs appears,

$$\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g + \delta F_g^{\mathrm{cross}}, \qquad (\text{all-weight})$$

(multi-weight genus expansion theorem, full monograph). Every appearance of obs_g, F_g, or λ_g in this monograph carries explicit tagging: (UNIFORM-WEIGHT) or (ALL-WEIGHT, WITH CROSS-CHANNEL CORRECTION).

Grothendieck–Riemann–Roch applied to the universal curve $\pi\colon \mathcal{C}_g \to \overline{\mathcal{M}}_g$ computes the total obstruction: the Chern character of $R\pi_*\omega_\pi$ involves the Todd class of the relative tangent, the $\hat{A}$-genus. The generating function in the uniform-weight lane:

$$\sum_{g\geq 1} \mathrm{obs}_g(\mathcal{A}) \cdot x^{2g} = \kappa(\mathcal{A}) \cdot \big(\hat{A}(ix) - 1\big).$$

Since $\hat{A}(ix) = (x/2)/\sin(x/2) = 1 + x^2/24 + 7x^4/5760 + \cdots$, the first Faber–Pandharipande coefficients are

$$F_1 = \tfrac{\kappa}{24}, \qquad F_2 = \tfrac{7\kappa}{5760}, \qquad F_3 = \tfrac{31\kappa}{967680},$$

with the general term $\kappa \cdot \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$. Each of these is the scalar-lane projection; for multi-weight families at $g \geq 2$, add $\delta F_g^{\mathrm{cross}}$. Verify $F_1 = \kappa/24$ as sanity check (Cauchy normalization $1/(2\pi i)$, not $1/(2\pi)$).

The scalar $\kappa(\mathcal{A})$ is only the leading term. The bar complex carries higher-degree data: a universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ whose projection to degree r yields a shadow invariant $S_r(\mathcal{A})$. Write $\alpha := S_3$ for the cubic shadow. On each primary deformation line L (a one-parameter subfamily at fixed spin content), the shadow generating function $H(t) = \sum_{r\geq 2} r S_r t^r$ satisfies the algebraic relation

$$H(t)^2 = t^4\, Q_L(t), \qquad Q_L(t) = 4\kappa^2 + 12\kappa S_3\, t + (9S_3^2 + 16\kappa S_4)\, t^2, \tag{0.3}$$

a quadratic polynomial in three genus-0 invariants (Riccati algebraicity theorem, full monograph). The infinite tower is algebraic of degree 2, governed by the discriminant $\Delta = 8\kappa S_4$.

Heisenberg: the genus-1 partition function. The genus-1 Heisenberg partition function on the elliptic curve $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ is the functional integral over the free boson. The bar complex computes:

$$Z_1(\mathcal{H}_k; \tau) = \frac{1}{\eta(\tau)^k}, \qquad \eta(\tau) = q^{1/24} \prod_{n\geq 1} (1 - q^n), \quad q = e^{2\pi i \tau}.$$

The prefactor $q^{1/24}$ in the Dedekind eta is essential for modular weight 1/2. The free energy is $F_1(\mathcal{H}_k) = -k \log \eta(\tau)$. Expanding:

$$F_1(\mathcal{H}_k) = -k\left(\frac{2\pi i\tau}{24} + \sum_{n\geq 1} \log(1 - q^n)\right) = \frac{\kappa}{24} \cdot 2\pi i\tau + (\text{instanton corrections}).$$

The leading coefficient $\kappa/24 = k/24$ matches the $\hat{A}$-genus prediction $F_1 = \kappa/24$ (with the Cauchy normalization $1/(2\pi i)$, not $1/(2\pi)$). The instanton corrections $\sum_{n\geq 1} \log(1 - q^n)$ are the genus-1 bar complex contributions from states at conformal weight n; these are captured by the full Feynman-transform differential, not just the scalar curvature.

For the rank-d Heisenberg $\mathcal{H}_k^{\oplus d}$: $Z_1 = 1/\eta(\tau)^{dk}$ and $F_1 = dk/24 \cdot 2\pi i\tau + \cdots$, consistent with $\kappa(\mathcal{H}_k^{\oplus d}) = dk$.

Kac–Moody: the genus-1 character. For $\widehat{\mathfrak{g}}_k$ at genus 1, the partition function is the vacuum character $\chi_0(\widehat{\mathfrak{g}}_k; \tau) = \mathrm{tr}_{\mathcal{V}_k(\mathfrak{g})}\, q^{L_0 - c/24}$. The Sugawara central charge is $c = k \dim \mathfrak{g}/(k + h^\vee)$, and the leading asymptotics give $F_1(\widehat{\mathfrak{g}}_k) \sim \kappa(\widehat{\mathfrak{g}}_k)/24 \cdot 2\pi i\tau$ with $\kappa = (k + h^\vee) \dim \mathfrak{g}/(2h^\vee)$, matching the $\hat{A}$-genus prediction (UNIFORM-WEIGHT).

Virasoro: $F_1 = c/48$. For the Virasoro algebra at central charge c, the genus-1 free energy is

$$F_1(\mathrm{Vir}_c) = \frac{\kappa(\mathrm{Vir}_c)}{24} = \frac{c/2}{24} = \frac{c}{48}.$$

The Virasoro vacuum character on the torus is $\chi_0(\mathrm{Vir}_c; \tau) = q^{-c/24}/\prod_{n\geq 2}(1 - q^n)$; the leading term $-c/24$ in the exponent gives $F_1 = c/48$ after normalization. This matches: $\hat{A}(ix) - 1 = x^2/24 + \cdots$, so $F_1 = \kappa/24 = (c/2)/24 = c/48$.

The higher Faber–Pandharipande coefficients. The formula $F_g = \kappa \cdot \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$ (UNIFORM-WEIGHT) involves the Bernoulli numbers B_{2g}. Numerical values:

g	B_{2g}	F_g/κ	Decimal
1	1/6	1/24	0.04167
2	−1/30	7/5760	0.001215
3	1/42	31/967680	0.00003203
4	−1/30	127/154828800	8.202×10^{-7}
5	5/66	511/20437401600	2.501×10^{-8}

The factorial decay $F_g \sim (2g)!^{-1}$ is *not* the full story: the factor $2^{2g-1} - 1$ grows exponentially, and the Bernoulli numbers $|B_{2g}| \sim 2(2g)!/(2\pi)^{2g}$ produce a Gevrey-1 sequence $F_g \sim \kappa \cdot (2\pi)^{-2g}$. The shadow obstruction tower sums these contributions; the Borel summability of the F_g series is controlled by the shadow growth rate $\rho(\mathcal{A})$ (§6.9).

The shadow obstruction tower is not merely formal. Its projections yield: the Pixton ideal of the modular curve, the integrable hierarchies of Gelfand–Dickey type, the Hitchin spectral curve of the shadow connection, the Baxter Q-operator, and the Bekenstein–Hawking entropy formula of three-dimensional gravity (twisted holography chapter, full monograph). These are projections of one object, not separate results.

2.5. Complementarity. The genus tower is not merely a sequence of scalar invariants; it carries a duality structure that pairs each genus-g contribution of $\mathcal{A}$ with the corresponding contribution of the Koszul dual $\mathcal{A}^!$. This duality is the genus-tower incarnation of Verdier duality on the Ran space.

Verdier duality on the Ran space splits the center local system into two halves, one controlled by $\mathcal{A}$ and one by $\mathcal{A}^!$, carrying a shifted-symplectic structure:

$$R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}}) \simeq \mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^!).$$

What $\mathcal{A}$ sees as deformation, $\mathcal{A}^!$ sees as obstruction. The natural pairing

$$\mathbf{Q}_g(\mathcal{A}) \otimes \mathbf{Q}_g(\mathcal{A}^!) \to R\Gamma(\overline{\mathcal{M}}_g, \omega_{\overline{\mathcal{M}}_g}) \simeq \Bbbk[-(3g-3)]$$

is a $(-(3g-3))$-shifted symplectic form, and the two summands are Lagrangian: isotropic of half the total dimension.

Heisenberg: complementarity is antisymmetry. For $\mathcal{H}_k$, the Koszul dual is $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ with $\kappa(\mathcal{H}_k^!) = -k$. The complementarity is $\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = k + (-k) = 0$ (Kac–Moody/free-field type). At genus 1: $\mathbf{Q}_1(\mathcal{H}_k)$ is one-dimensional (the image of λ_1 under the Hodge map), and $\mathbf{Q}_1(\mathcal{H}_k^!)$ is one-dimensional (the complementary image). The shifted symplectic form on $H^*(\overline{\mathcal{M}}_{1,1}) \cong \mathbb{Q}^2$ identifies these as dual lines. The antisymmetry $\kappa' = -\kappa$ means the Koszul dual sees the opposite curvature: the deformations of $\mathcal{H}_k$ are the obstructions of $\mathcal{H}_k^!$ and conversely.

Virasoro: complementarity is asymmetric. For Vir_c, the Koszul dual is Vir_{26-c} with $\kappa(\mathrm{Vir}_{26-c}) = (26-c)/2$. The complementarity conductor is $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = c/2 + (26-c)/2 = 13$, which is nonzero for all c. At the self-dual point $c = 13$: $\kappa = \kappa' = 13/2$, and the Lagrangian decomposition becomes an eigenspace decomposition with equal eigenvalues. At $c = 26$: $\kappa = 13$, $\kappa' = 0$; the dual Vir_0 is uncurved, and all of the deformation data lies on the $\mathcal{A}$-side. The nonzero conductor 13 is the scalar obstruction to recovering the genus-0 Koszul duality pattern at higher genus.

The scalar sum rule is family-dependent (never universal):

$$\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = \begin{cases} 0 & \text{(Kac–Moody, free field, lattice)}, \\ 13 & \text{(Virasoro)}, \\ 250/3 & (\mathcal{W}_3), \\ K_N \cdot (H_N - 1) & (\mathcal{W}_N), \end{cases}$$

where $K_N = c(\mathcal{W}_N) + c(\mathcal{W}_N^!)$ is the central-charge conductor and $H_N = \sum_{j=1}^N 1/j$ the harmonic number. The scalar $\kappa + \kappa'$ is the kappa-level Koszul conductor; it is zero for KM and free fields, nonzero for $\mathcal{W}$-algebras. Self-dual at $c = 13$ for Virasoro, $c = 50$ for $\mathcal{W}_3$; qualify "self-dual at $c = 13$ under Koszul duality," never unqualified.

2.6. Feigin–Frenkel duality. The Sugawara field $T(z) = \frac{1}{2(k+h^\vee)} \sum_a :J^a(z)J^a(z):$ has central charge $c = k \dim \mathfrak{g}/(k + h^\vee)$. At the critical level $k = -h^\vee$ the denominator vanishes: $T(z)$ is undefined, not "c diverges." The chiral algebra acquires the Feigin–Frenkel center $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee})$, isomorphic to functions on the space of ${}^L\mathfrak{g}$-opers on the disc.

The Feigin–Frenkel involution $k \leftrightarrow -k - 2h^\vee$ is Koszul duality on configuration spaces: it interchanges $\mathcal{A}$ and $\mathcal{A}^!$ while preserving the bar-cobar adjunction. At the critical level, the Koszul dual is the Langlands-dual critical-level algebra $\widehat{{}^L\mathfrak{g}}_{-{}^Lh^\vee}$, not a renormalised copy of the original.

The Feigin–Frenkel involution in coordinates. For $\mathfrak{g} = \mathfrak{sl}_2$, $h^\vee = 2$, and the involution is $k \mapsto -k-4$. The Koszul dual of $\widehat{\mathfrak{sl}}_{2,k}$ is $\widehat{\mathfrak{sl}}_{2,-k-4}$. The curvature transforms as

$$\kappa(\widehat{\mathfrak{sl}}_{2,-k-4}) = \frac{3(-k-4+2)}{4} = \frac{-3(k+2)}{4} = -\kappa(\widehat{\mathfrak{sl}}_{2,k}),$$

confirming the antisymmetry $\kappa + \kappa' = 0$ (for affine Kac–Moody). At $k = -2$ (critical): both κ and κ' vanish; the bar complex is uncurved on both sides. At $k = 0$ (abelian limit): $\kappa = 3/2$, $\kappa' = -3/2$.

For $\mathfrak{g} = \mathfrak{sl}_N$, $h^\vee = N$, and the involution is $k \mapsto -k - 2N$. The fixed point $k = -N$ is the self-dual level; the Sugawara central charge at $k = -N$ is $c = (-N) \cdot (N^2 - 1)/(-N + N)$, which is undefined (the fixed point of the involution lies at the critical level for $\mathfrak{sl}_N$). This is not a coincidence: the involution has no fixed point at generic level, and the critical level is precisely where duality collapses.

The critical level jump. At $k = -h^\vee$, several discontinuities occur simultaneously:

(i) $\kappa(\widehat{\mathfrak{g}}_{-h^\vee}) = 0$: the curvature vanishes, the bar complex becomes uncurved, and all monodromy becomes trivial (all KZ exponents become integers).

(ii) The Sugawara field $T(z)$ is undefined: no conformal vector exists. The vertex algebra has no Virasoro subalgebra.

(iii) The center jumps from finite-dimensional to $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee}) \cong \mathrm{Fun}(\mathrm{Op}_{^L\mathfrak{g}}(D))$: the space of ${}^L\mathfrak{g}$-opers on the formal disc.

(iv) The degree-2 ordered chiral homology $H^1_{\mathrm{ord},2}(\widehat{\mathfrak{g}}_{-h^\vee})$ on $V \otimes V$ doubles in dimension: for $\mathfrak{sl}_2$, from 4 (at generic k) to 8 (at $k = -2$). This is a finite computation on $V \otimes V$ conserving $\chi_{\mathrm{ord},2} = -4$. The FULL symmetric bar cohomology at critical level becomes $\Omega^*(\mathrm{Op}_{^L\mathfrak{g}}(D))$, which is infinite-dimensional in every degree (Frenkel–Teleman 2006).

(v) Koszulness fails: the PBW spectral sequence has nontrivial differentials.

For $\mathfrak{sl}_2$ at $k = -2$: the KZ connection ∇_{KZ} has regular singularities with integer exponents, so all solutions are single-valued on $\mathrm{Conf}_n(\mathbb{P}^1)$. The genus spectral sequence has $d_1 = \lambda \cdot [\delta]$ (where $\lambda = (k + h^\vee)$ is the shifted level); at $\lambda = 0$ this differential vanishes, and the E_1 page does not degenerate.

2.7. The total space. The universal configuration fibration $\pi_n \colon \overline{C}_n(C_g) \to \overline{\mathcal{M}}_g$ packages the bar complex into a single global object. Its boundary divisors carry two distinct geometric roles. *Collision divisors* (vertical): the fibrewise boundary of $\overline{\mathrm{FM}}_n(X_s)$, producing the bar differential d_{res} (OPE residues). *Degeneration divisors* (horizontal): the boundary of $\overline{\mathcal{M}}_g$ itself (nodal curves), producing the period corrections and Gauss–Manin terms.

Algebraic boundary (Borcherds) meets geometric boundary (period map and clutching). Their interaction is the curvature κ: the fibrewise bar differential fails to be nilpotent precisely because the vertical and horizontal boundaries do not commute.

2.8. The E_n hierarchy and the genus tower. The curvature $d^2_{\mathrm{fib}} = \kappa \cdot \omega_g$ is a statement about the E_∞-chiral bar complex. At the E_1 (ordered) level, the same curvature appears but the additional structure of the R-matrix produces genus-dependent corrections invisible to the symmetric bar.

E_∞-chiral (standard VAs): the curvature is scalar. For any E_∞-chiral algebra (all standard VAs: Heisenberg, Kac–Moody, Virasoro, $\mathcal{W}_N$, $\beta\gamma$, lattice), the symmetric bar complex $\bar{B}^\Sigma(\mathcal{A})$ is a coalgebra on the Ran space $\mathrm{Ran}(X)$ (unordered configurations). The curvature $d^2_{\mathrm{fib}} = \kappa \cdot \omega_g$ is a scalar multiple of the Hodge class: the Σ_n-averaging has projected out all non-scalar data. The genus tower is controlled by κ alone (UNIFORM-WEIGHT).

E_1-chiral (Yangians): the curvature carries R-matrix corrections. For a genuinely E_1-chiral algebra (e.g., a chiral Yangian $Y(\mathfrak{g})^{\mathrm{ch}}$), the ordered bar $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ is a coalgebra on ordered configurations $\mathrm{Conf}_n^{\mathrm{ord}}(X)$. The curvature d^2_{fib} at genus g is *not* scalar: it carries R-matrix corrections from the ordering. The genus-1 curvature for $Y(\mathfrak{g})^{\mathrm{ch}}$ involves the elliptic R-matrix (the KZB R-matrix on the torus), which is a function of the spectral parameter z and the modular parameter τ. The genus tower for E_1-chiral algebras carries strictly more data than the symmetric bar: the KZB connection, the elliptic monodromy, and the Drinfeld–Kohno relations.

The derived chiral centre and the E_n output. The bar complex $\bar{B}(\mathcal{A})$ is an E_1-chiral coassociative coalgebra (differential plus deconcatenation coproduct). It does *not* carry $\mathrm{SC}^{\mathrm{ch,top}}$ structure. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure (Swiss-cheese: two colours, bulk and boundary) emerges on the *derived chiral centre* $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{ChirHoch}^*(\mathcal{A}, \mathcal{A})$, computed using the bar complex as a resolution:

$$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = R\mathrm{Hom}_{\mathcal{A}\text{-bimod}}(\mathcal{A}, \mathcal{A}) \simeq \mathrm{Hom}(\bar{B}(\mathcal{A}), \mathcal{A}).$$

The pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ is the $\mathrm{SC}^{\mathrm{ch,top}}$-datum: $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ is the bulk (closed colour), $\mathcal{A}$ is the boundary (open colour), and the action of the bulk on the boundary is by chiral Hochschild evaluation.

For E_∞-chiral input: Theorem H gives $\mathrm{ChirHoch}^*(\mathcal{A}) \in \{0, 1, 2\}$ (concentrated, polynomial Hilbert series). With conformal vector at non-critical level (Sugawara): $\mathrm{SC}^{\mathrm{ch,top}}$ + conformal vector = E_3-topological; the conformal vector kills the chiral direction. Proved for affine KM at non-critical level; conjectural for general chiral algebras with conformal vector.

For E_1-chiral input (Yangians): the symmetric bar $\bar{B}^{\Sigma}$ does not exist (the D-module does not descend to symmetric configurations). The derived centre gives E_2 only (via classical Deligne, not Higher Deligne), and the passage to E_3 requires the Drinfeld centre construction.

2.9. Bar chain models on curve geometries. The bar complex changes character as the underlying geometry changes. Three geometric inputs produce three different bar models, each carrying progressively richer structure:

Over the formal disc $D = \mathrm{Spec}\, \mathbb{C}[[z]]$: The bar complex $\bar{B}(\mathcal{A}|_D) = (T^c(s^{-1}\bar{\mathcal{A}}), d_B)$ is the classical bar construction with deconcatenation coproduct. The propagator is $d\log z$, the Arnold relation is trivial (one variable), and $d_B^2 = 0$ unconditionally. The bar cohomology $H^*(\bar{B}(\mathcal{A}|_D))$ is the Koszul dual in the classical sense. Restricting to D recovers the formal-disk Koszul duality of Beilinson–Ginzburg–Soergel; the passage $D \to X$ adds configuration-space geometry.

Over the affine line $\mathbb{A}^1$: The bar complex $\bar{B}(\mathcal{A}|_{\mathbb{A}^1})$ uses the propagator $d\log(z_1 - z_2)$ on ordered configurations $\mathrm{Conf}_n(\mathbb{A}^1)$. The Arnold relation $\eta_{12} \wedge \eta_{23} + \mathrm{cyc} = 0$ is nontrivial and provides the codimension-two cancellation $d_B^2 = 0$. The FM compactification $\overline{\overline{C}}_n(\mathbb{A}^1)$ is the real Stasheff associahedron K_n; its chain complex $C_*(K_n)$ carries the A_∞-operad. The ordered bar on $\mathbb{A}^1$ produces the A_∞-structure on $H^*(\bar{B}(\mathcal{A}))$: the transferred operations m_n are the Stasheff products, computed by integrating η_{ij} over the faces of K_n.

Over a compact curve X of genus g: The bar complex $\bar{B}^{(g)}(\mathcal{A})$ uses the genus-g propagator $\eta^{(g)}_{12} = d\log E(z_1, z_2)$ + (period correction). The Arnold relation acquires a $(1,1)$-defect proportional to ω_g, producing the curvature $d^2_{\mathrm{fib}} = \kappa \cdot \omega_g$. The configuration space $\mathrm{Conf}_n(X)$ compactifies to $\overline{\overline{C}}_n(X)$ (smooth, with normal-crossings boundary), and the chain complex $C_*(\overline{\overline{C}}_n(X))$ carries the operadic structure of $C_*(\overline{\mathcal{M}}_{0,n+1})$ plus higher-genus corrections from the period matrix. The bar complex over X sees the Hodge bundle $\mathbb{E}$, the clutching morphisms, and the planted-forest strata of Mok's compactification.

The chain of refinements. The passage from D to $\mathbb{A}^1$ to X is a chain of geometric refinements, each introducing new structure:

Space	**Propagator**	**New feature**	**Bar model**
Formal disc D	$d\log z$	none	classical bar
Affine line $\mathbb{A}^1$	$d\log(z_1 - z_2)$	Arnold relation	A_∞-bar
Smooth curve X_g	$d\log E$ + period	curvature $\kappa \cdot \omega_g$	modular bar
Nodal curve X_0	degenerate limit	clutching morphisms	Feynman transform
Universal family C_g	fibred propagator	Gauss–Manin	total bar $D_{\mathcal{A}}$

No step in this chain is trivial. The passage $D \to \mathbb{A}^1$ already introduces configuration-space topology, FM compactifications, and the E_1/E_∞ distinction. The passage $\mathbb{A}^1 \to X$ adds curvature from the Hodge bundle. The passage to nodal curves adds clutching from boundary strata. The passage to the universal family assembles everything into the total differential $D_{\mathcal{A}}$ with $D_{\mathcal{A}}^2 = 0$.

3. Modular homotopy theory

The genus tower demands an organising framework. The fibrewise differential, the Gauss–Manin correction, and the clutching morphisms at boundary strata satisfy compatibility relations dictated by the stratification of $\overline{\mathcal{M}}_{g,n}$. The governing algebraic structure is the *modular operad*: an operad whose operations are indexed by connected stable graphs of arbitrary genus.

The *modular homotopy type* of $\mathcal{A}$ is the formal moduli problem classifying the inverse system of bar complexes,

$$\mathcal{M}_{\mathcal{A}}^{\mathrm{mod}}(R) := \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} \widehat{\otimes}_{\mathfrak{m}} R)/\sim$$

for local Artin rings R. Here $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is the *modular convolution algebra*: a complete filtered cyclic L_∞-algebra. Three inputs enter: a homotopy chiral algebra on the curve, a cooperadic chain complex on logarithmic configuration spaces, and a convolution L_∞-structure mediating between them. The bar complex, the Koszul dual, the genus tower, the complementarity splitting, and the shadow obstruction tower are all representations of $\mathcal{M}_{\mathcal{A}}^{\mathrm{mod}}$.

The primitive story lives on ordered configurations. The E_1 convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},E_1}$, built on ribbon graphs from $F\mathrm{Ass}$, is the primary object; the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is its image under the av map from ordered to symmetric. Every construction in this section begins with the E_1 structure and descends.

3.1. The modular operad. At genus 0, the combinatorics is governed by trees; at higher genus, by stable graphs with genus labels at their vertices. A *modular operad* (Getzler–Kapranov [8]) has operations indexed by connected stable graphs. A *connected stable graph* Γ has a label $g(v) \in \mathbb{Z}_{\geq 0}$ at each vertex satisfying $2g(v) - 2 + \mathrm{val}(v) > 0$ and total genus $g(\Gamma) = b_1(\Gamma) + \sum_v g(v)$; trees are the genus-0 case. The structure maps are *separating gluing* $\xi_{\mathrm{sep}} : \mathcal{O}(g_1, n_1{+}1) \otimes \mathcal{O}(g_2, n_2{+}1) \to \mathcal{O}(g_1{+}g_2, n_1{+}n_2)$ and *non-separating gluing* $\xi_{\mathrm{ns}} : \mathcal{O}(g, n{+}2) \to \mathcal{O}(g{+}1, n)$.

A stable graph records a degeneration of a smooth curve into a nodal curve: vertices are irreducible components, edges are nodes, genus labels are component genera, half-edges are marked points. The set $\mathsf{Gr}_{g,n}^{\mathrm{st}}$ parametrises the boundary stratification of $\overline{\mathcal{M}}_{g,n}$. Every codimension-one boundary divisor is a one-edge degeneration; every codimension-two stratum is the intersection of exactly two codimension-one divisors, appearing with opposite orientations.

Stable graph counts. The number of stable graph types grows with genus and marked points. At genus 1: $\overline{\mathcal{M}}_{1,1}$ has one boundary divisor δ_{irr} (the single self-loop graph with one genus-0 vertex and one non-separating edge). $\overline{\mathcal{M}}_{1,2}$ has three boundary divisors: one non-separating (δ_{irr}, the self-loop with two external legs at the same vertex) and two separating (δ_0, splitting into a genus-1 vertex with one leg and a genus-0 vertex with two legs; counted once for each labeling). At genus 2: $\overline{\mathcal{M}}_{2,0}$ has 3 boundary divisors (the figure-eight self-loop, the separating barbell, and the sunset/banana graph); $\overline{\mathcal{M}}_{2,1}$ has 7 boundary strata including codimension-two intersections. The count of stable graph types at genus g with n marked points is the number of cells in the orbicell decomposition of $\overline{\mathcal{M}}_{g,n}$, which grows polynomially in n at fixed g and exponentially in g at fixed n.

The E_1 modular operad: ribbon graphs. The E_1-primitive version uses the Feynman transform of Ass: the operations are indexed by *ribbon graphs* (fatgraphs), connected graphs with a cyclic ordering of the half-edges at each vertex. A ribbon graph Γ determines a compact surface with boundary by thickening each vertex to a polygon and each edge to a ribbon. The genus is $g(\Gamma) = (2 - |V| + |E| - |F|)/2$ where F counts the boundary components. Ribbon graphs are the combinatorial model of $\overline{\mathcal{M}}_{g,n}$ via the Harer–Mumford–Penner cell decomposition: each top cell corresponds to a trivalent ribbon graph, and the cell complex computes $H_*(\mathcal{M}_{g,n})$.

The coinvariant map av: $F\mathrm{Ass} \to F\mathrm{Com}$ forgets the cyclic ordering. At each vertex, the $|\mathrm{val}(v)|!$ orderings of incoming half-edges are identified; the resulting Euler characteristic is the count of unordered stable graphs, weighted by $1/|\mathrm{Aut}(\Gamma)|$. This projection is lossy: the R-matrix data at each vertex, which requires an ordering of inputs, is destroyed.

The Feynman transform replaces trees by stable graphs,

$$FO(g,n) = \bigoplus_{\Gamma \in \mathsf{Gr}^{\mathrm{st}}_{g,n}} \frac{g(\Gamma)}{|\operatorname{Aut}(\Gamma)|} \bigotimes_{v \in V(\Gamma)} O\big(g(v), \mathrm{In}(v)\big). \tag{0.4}$$

3.2. The Feynman transform differential. The differential contracts one edge at a time,

$$d_{FO}(\gamma_\Gamma) = \sum_{e \in E(\Gamma)} (-1)^{\epsilon(e)} \xi_e(\gamma_{\Gamma/e}),$$

with the sign from the position of e in the ordering on $E(\Gamma)$ recorded by $\det(E(\Gamma))$. The identity $d^2 = 0$ is the codimension-2 cancellation: each pair of edges contracts in either order, and the two terms cancel by associativity of O. At genus 0, FO reduces to the operadic bar construction.

When $O = \mathrm{Com}$, each vertex factor in (0.4) is one-dimensional, and the Feynman transform becomes the stable graph complex

$$F\mathrm{Com}(g,n) = \bigoplus_{\Gamma \in \mathsf{Gr}^{\mathrm{st}}_{g,n}} \frac{g(\Gamma)}{|\operatorname{Aut}(\Gamma)|} \det(E(\Gamma)).$$

When $O = \mathrm{Ass}$, the Feynman transform $F\mathrm{Ass}$ sums over ribbon graphs (fatgraphs); this is the E_1-primitive input for the ordered convolution algebra.

Explicit graph computation: genus 1, $n = 1$. At $(g,n) = (1,1)$, there is a single stable graph type: the graph Γ_{irr} with one vertex of genus 0, one external half-edge (the marked point), and one internal edge forming a self-loop. The genus formula: $g(\Gamma) = b_1(\Gamma) + g(v) = 1 + 0 = 1$. The stability condition: $2g(v) - 2 + \mathrm{val}(v) = 0 - 2 + 3 = 1 > 0$ (the vertex has valence 3: one external half-edge plus two half-edges of the self-loop). The automorphism group has order 2 (swapping the two half-edges of the self-loop). The Feynman transform gives:

$$F\mathrm{Com}(1,1) = \frac{}{2} \cdot \mathrm{Com}(0,3) \otimes \det(\Bbbk \cdot e) = \frac{}{2} \cdot \Bbbk \otimes \Bbbk.$$

The differential from $(1,1)$: contracting the self-loop sends Γ_{irr} to the corolla $(1,1)$ (a single genus-1 vertex with one external leg). The differential computes: $d_{F\mathrm{Com}}(\gamma_{\Gamma_{\mathrm{irr}}}) = \pm\xi_{\mathrm{ns}}(\gamma)$.

Explicit graph computation: genus 2, $n = 0$. At $(g, n) = (2, 0)$, there are 3 stable graph types:

(a) The *figure-eight* Γ_8: one genus-0 vertex with two self-loops. $b_1 = 2$, $g(v) = 0$, $\mathrm{val}(v) = 4$, stability $0 - 2 + 4 = 2 > 0$. $|\mathrm{Aut}| = 8$ (swap each loop $\times$ swap the two loops: $2 \times 2 \times 2$).

(b) The *sunset* (banana) Γ_{sun}: two genus-0 vertices connected by three edges. $b_1 = 2$, $g(v_1) = g(v_2) = 0$, val $= 3$ at each vertex, stability $0 - 2 + 3 = 1 > 0$ at each. $|\mathrm{Aut}| = 12$ (S_3 permuting three edges $\times$ $\mathbb{Z}/2$ swapping the two vertices).

(c) The *separating barbell* Γ_{bar}: two genus-1 vertices connected by one edge. $b_1 = 0$, $g(v_1) = g(v_2) = 1$, total $g = 0 + 1 + 1 = 2$. val $= 1$ at each, stability $2 - 2 + 1 = 1 > 0$. $|\mathrm{Aut}| = 2$ (swapping the two vertices).

These three types exhaust $\mathsf{Gr}^{\mathrm{st}}_{2,0}$; no other connected stable graph of genus 2 with 0 marked points exists. The three types correspond to the three boundary divisors of $\overline{\mathcal{M}}_2$: δ_0 (irreducible, the figure-eight and sunset), δ_1 (the separating barbell).

3.3. The convolution ladder. At each level of generality, the bar-type construction is a convolution dg Lie algebra whose Maurer–Cartan elements classify algebraic structures:

$$\begin{array}{rll} \text{algebra}: & \mathrm{Hom}(B(\mathrm{Ass}), \mathrm{End}_A) & A_\infty\text{-structures on } A \\ \text{operad}: & \mathrm{Hom}_\Sigma(B(\mathcal{P}), \mathrm{End}_V) & \mathcal{P}_\infty\text{-structures on } V \\ \text{modular}: & \mathrm{Hom}_\Sigma(F\mathcal{O}, \mathrm{End}_{\bar{B}(\mathcal{A})}) & \text{consistent genus towers} \end{array}$$

The first line is topological Hochschild; the second is the Ginzburg–Kapranov deformation complex; the third governs this book. The Lie bracket is the convolution bracket

$$[f, g] = \mu \circ (f \otimes g) \circ \Delta - (-1)^{|f||g|} \mu \circ (g \otimes f) \circ \Delta,$$

with Δ the cooperadic decomposition and μ the operadic composition.

The *primitive E_1 convolution*

$$\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},E_1} := \mathrm{Hom}_\Sigma\big(F\mathrm{Ass}, \mathrm{End}_{\bar{B}^{E_1}(\mathcal{A})}\big)$$

is the ordered, ribbon-graph input: the algebraic form of the 't Hooft $1/N$ expansion. Setting $\;= 1/N^2$, the MC element $\Theta_{\mathcal{A}}^{E_1} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},E_1})$ is the genus expansion $\sum_g N^{2-2g} F_g(\lambda)$ summed over ribbon graphs weighted by the genus of their embedding surface. Planar ($g = 0$) diagrams dominate at large N; the shadow obstruction tower $\Theta^{E_1,\le r}$ is the truncation to r-point 't Hooft interactions.

The *modular convolution algebra* is the av-image,

$$\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} := \mathrm{Hom}_\Sigma\big(F\mathrm{Com}, \mathrm{End}_{\bar{B}(\mathcal{A})}\big).$$

The coinvariant map $F\mathrm{Ass} \to F\mathrm{Com}$ projects $\Theta^{E_1} \mapsto \Theta$: colour-ordered amplitudes to full amplitudes. On each primary line, the shadow tower is algebraic of degree 2: the entire infinite sequence is the Taylor expansion of $\sqrt{Q_L}$ for a single quadratic Q_L, the shadow metric (§6 below).

Heisenberg: the convolution algebra is abelian. For $\mathcal{H}_k$, the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}[\mathcal{H}_k]$ is abelian: $[f, g] = 0$ for all f, g. The bracket vanishes because the only nonzero amplitude is $\Phi_{\Gamma_2} = k$ (single-edge graph), and $\Delta(\Gamma_2)$ has no non-trivial splitting. Hence $\Theta_{\mathcal{H}_k} = k \cdot \eta \otimes \Lambda$ is the unique MC element (no gauge freedom, no moduli). The genus tower is determined entirely by the graph enumeration of $F\mathrm{Com}$: for the Heisenberg, the bar construction reduces to the combinatorics of stable graphs weighted by a single scalar.

Kac–Moody: the bracket is the Lie algebra. For $\widehat{\mathfrak{g}}_k$, the degree-3 convolution bracket $\ell_2(\Theta, \Theta)$ encodes the Lie bracket $[\cdot, \cdot]$ of $\mathfrak{g}$ through the cubic shadow $S_3 = 2h^\vee/(3\kappa)$. The Jacobi identity

$[[\cdot,\cdot],\cdot] + \text{cyc} = 0$ kills the degree-4 shadow: $S_4 = 0$, $\Delta = 0$, and the tower terminates. The convolution algebra of $\widehat{\mathfrak{g}}_k$ is a non-abelian Lie algebra at degree 3 that becomes abelian at degree 4 by Jacobi. Class **L**: Lie structure controls and then terminates the tower.

Remark (Homotopy types in three stages). The three levels of the convolution ladder mirror three levels of homotopy theory. *Over a field* (the bar complex of a graded algebra): $B(A) = (T^c(s^{-1}\bar{A}), d_B)$ determines the homotopy type of an associative algebra; formality corresponds to intrinsic formality; Massey products measure the obstruction. *Over a manifold*: the Sullivan minimal model $(\Lambda V, d)$ determines the rational homotopy type; Kähler manifolds are formal. The Heisenberg algebra is the vertex-algebraic analogue: its shadow obstruction tower terminates at degree 2, the Gaussian archetype. *Over a curve at all genera*: the A_∞-products m_k of the first stage become the modular L_∞-brackets $\ell_k^{(g)}$; formality becomes the Gaussian archetype; the shadow obstruction tower is the modular Massey product tower. The shadow depth classification (§6) makes this analogy precise: the four classes G, L, C, M correspond to four levels of modular formality.

Virasoro: the convolution algebra is infinite-dimensional. For Vir_c, the convolution bracket $\ell_2(\Theta,\Theta)$ produces a nonzero cubic shadow from the simple-pole OPE $T(z)T(w) \sim 2T(w)/(z-w)^2 + \cdots$. But the bar cohomology $H^*(\bar{B}(\mathrm{Vir}_c))$ is one-dimensional in each degree: spanned by the class x^r (the r-th symmetric power of the dual to T). The convolution bracket $[\cdot,\cdot]$ on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[\mathrm{Vir}_c]$ is *not* a Lie algebra (it violates Jacobi at degree 4, where the quartic shadow $S_4 \neq 0$). Instead, the convolution algebra carries the full L_∞ structure with all higher brackets ℓ_k nonvanishing; the MC equation $D\Theta + \frac{1}{2}[\Theta,\Theta] = 0$ holds in the L_∞ *sense*, not the strict Lie sense. The shadow obstruction tower for Virasoro is governed by this L_∞ structure: the depth-∞ property is a direct reflection of the infinite sequence of nontrivial L_∞ brackets.

Koszulness as central question. Within this formality classification, the Koszulness property is the central organizing question of the programme. A chiral algebra is *chirally Koszul* iff its bar cohomology concentrates in diagonal degrees and the cobar-bar counit $\Omega(\bar{B}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism. Twelve equivalent conditions (eight genuinely independent bidirectional equivalences, one listed Barr–Beck–Lurie consequence, one one-way chiral Hochschild consequence on the Koszul locus, one conditional on perfectness, one one-directional) characterise this locus. The Koszulness programme (§9) proves the core equivalences from PBW concentration, bar-cobar inversion, chiral Hochschild polynomial growth, and Koszul-complex acyclicity. All standard families are Koszul; shadow depth classifies the complexity of Koszul algebras, not Koszulness itself.

The twelve Koszul characterisations (overview). The characterisation of chirally Koszul algebras involves twelve numbered conditions, organised as $8+1+1+1+1$: eight genuinely independent bidirectional equivalences, one Barr–Beck–Lurie monadicity listed as a consequence of the bar-cobar counit, one one-way chiral Hochschild consequence on the Koszul locus, one conditional Lagrangian criterion, and one one-directional $\mathcal{D}$-module purity:

(i) Bar cohomology concentrates on the diagonal (PBW).
(ii) The cobar-bar counit is a quasi-isomorphism (inversion).
(iii) The Koszul complex is acyclic.
(iv) The chiral Hochschild cohomology $\mathrm{ChirHoch}^*(\mathcal{A})$ has polynomial Hilbert series, concentrated in degrees $\{0,1,2\}$.
(v) The bar Hilbert series $h_{\mathcal{A}^!}(q)$ is rational (holonomic).
(vi) The spectral discriminant $\Delta_{\mathcal{A}}(t)$ is a polynomial.

(vii) The genus expansion $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ holds at all genera (UNIFORM-WEIGHT); for multi-weight: $F_g = \kappa \cdot \lambda_g + \delta F_g^{\mathrm{cross}}$ at genus 0 only.

(viii) $\mathrm{ChirHoch}^*(\mathcal{A})$ is a free polynomial algebra (conditional on Massey vanishing from FM formality).

(ix) The vertex algebra admits an E_∞-chiral structure compatible with the bar filtration.

(x) The Koszul dual $\mathcal{A}^!$ is well-defined as a strict chiral algebra.

Equivalences (i)⇔(ii)⇔(iii) are classical (Priddy, Beilinson–Ginzburg–Soergel); their chiral extensions are Theorem B. Equivalences (iv)⇔(v)⇔(vi) are Theorem H. Equivalence (vii) is Theorem D (genus 1 unconditional for all families; uniform-weight at all genera via clutching/GRR; multi-weight $g \geq 2$ with explicit cross-channel correction).

3.4. Homotopy chiral algebras. The convolution algebra requires a cochain-level algebraic input, not merely the cohomology-level chiral bracket. The Čech totalisation provides this: it replaces the strict chiral algebra with a homotopy-coherent model carrying all derived OPE data.

Fix an affine cover $\mathfrak{U} = \{U_\alpha\}$ of X. The Čech totalisation

$$A_\infty^{\mathrm{ch}} := \check{C}^\bullet(\mathfrak{U}; \mathcal{A}) = \bigoplus_{p \geq 0} \prod_{\alpha_0 < \cdots < \alpha_p} \mathcal{A}(U_{\alpha_0} \cap \cdots \cap U_{\alpha_p})$$

carries a Ch_∞-algebra structure (Malikov–Schechtman): the chiral bracket restricted to overlaps satisfies Jacobi up to a coherent system of higher homotopies, the *secondary Borcherds operations*

$$F_n\colon\ \mathrm{Lie}(n) \otimes \mathcal{Y}(n) \to P_n(\{A_\infty^{\mathrm{ch}}\}, A_\infty^{\mathrm{ch}}), \qquad n \geq 2,$$

encoding derived OPE data at all pole orders. F_2 is the chiral bracket; F_3 is the Jacobiator homotopy arising from $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, whose boundary has three points corresponding to OPE compositions $(a_{(m)}b)_{(n)}c$ and cyclic permutations. Higher operations F_n are determined inductively by $C_*(\overline{\mathcal{M}}_{0,n+1})$; the transferred operations are the associahedral A_∞-corrections.

Different affine covers yield quasi-isomorphic Ch_∞-algebras. The strict chiral algebra $\mathcal{A}$ is the H^0 of this structure: the primitive local object is the homotopy chiral algebra, and the strict algebra appears after rectification. Vallette rectification: the homotopy category of Ch_∞-algebras is equivalent to the homotopy category of strict chiral algebras, via the bar-cobar adjunction.

Heisenberg: the A_∞ model is strict. For $\mathcal{H}_k$, the OPE is purely quadratic ($J(z)J(w) \sim k/(z-w)^2$), so all transferred operations F_n with $n \geq 3$ vanish: the Jacobiator homotopy $F_3 = 0$ (there is no Jacobi identity to correct for a single generator with no simple pole), and all higher homotopies vanish. The Ch_∞-model is *strict*: $A_\infty^{\mathrm{ch}}(\mathcal{H}_k) = \mathcal{H}_k$ with $m_2 = \mu$ (the chiral product) and $m_n = 0$ for $n \geq 3$. This strictness reflects the Gaussian archetype: the bar complex $\bar{B}(\mathcal{H}_k) = (T^c(s^{-1}\bar{\mathcal{H}}_k), d_B)$ has d_B determined entirely by m_2, with no higher corrections.

Kac–Moody: a single nontrivial homotopy. For $\widehat{\mathfrak{g}}_k$, the OPE has both a double pole ($J^a(z)J^b(w) \sim k(a,b)/(z-w)^2$) and a simple pole ($J^a(z)J^b(w) \sim [a,b](w)/(z-w)$). The Jacobiator homotopy F_3 is nontrivial: it corrects the failure of strict Jacobi in the Čech model. However, $F_4 = 0$ (and all higher $F_n = 0$) because the Jacobi identity for $\mathfrak{g}$ forces the higher obstruction classes to be exact. The A_∞-model has m_2 (the chiral bracket), m_3 (the Jacobiator homotopy), and $m_n = 0$ for $n \geq 4$. This reflects the class-L structure: the shadow obstruction tower terminates at degree 3.

Virasoro: all homotopies nonvanishing. For Vir_c, the Jacobiator homotopy $F_3 \neq 0$ (from the quartic OPE pole, after $d\log$ absorption), the quaternary homotopy $F_4 \neq 0$ (from the failure of the Jacobi identity at the quartic level), and all higher $F_n \neq 0$ by induction (the quintic forcing of §6.9 generates nonvanishing homotopies at every order). The A_∞-model has all $m_n \neq 0$: this is the class-M structure, the infinite A_∞-depth.

3.5. Logarithmic Fulton–MacPherson spaces. The classical FM compactification records collisions on a smooth variety. Its logarithmic extension (Mok) records collisions along a divisor and provides the normal-crossings boundary structure on which $D^2 = 0$ depends.

For a simple-normal-crossing pair (X, D), Mok constructs $\mathrm{FM}_n(X|D)$: a smooth variety with corners compactifying $\mathrm{Conf}_n(X \setminus D)$, with boundary stratification indexed by *rigid planted-forest types*. A planted-forest type ρ consists of a partition of $\{1, \ldots, n\}$ into blocks (the roots), a rooted tree per block recording the hierarchy of collisions, and a grid-depth assignment $w_v \in \mathbb{Z}_{\geq 0}$ at each vertex recording the order of tangency with D. The ordinary FM is the case $D = \emptyset$.

The *degeneration formula*: for a semistable $W \to B$ with singular fibre $W_0 = \bigcup Y_v$ and a rigid type ρ,

$$\mathrm{FM}_n(W/B)(\rho) \longrightarrow \prod_{v\in V(S_\rho)} \mathrm{FM}_{I_v}(Y_v|D_v)$$

is a proper birational correspondence factoring the global FM geometry into a fibre product of local pieces. Tropicalisation: $\mathrm{Trop}(\mathrm{FM}_n(\mathbb{C}|D)) = G_{\mathrm{pf}}$, the planted-forest poset with grafting as partial order. The planted forest poset is the combinatorial skeleton of the geometric input, just as the stable graph complex is the skeleton of the Feynman transform.

The *Mok codimension formula*: for a type with grid depths $(w_1, \ldots, w_r)$ and tree vertex counts $(|V(T_j)|)_{j=1,\ldots,s}$,

$$\mathrm{codim}(W_\rho) = \sum_{i=1}^{r} w_i + \sum_{j=1}^{s}(|V(T_j)| - 1).$$

For FM_3: four codimension-one strata ($\{12\}$, $\{23\}$, $\{13\}$, $\{123\}$), three codimension-two strata ($\{12<123\}$, $\{23<123\}$, $\{13<123\}$).

The chain complexes $C^{\log \mathrm{FM}}_{X|D}(n) := C_\bullet(\mathrm{FM}_n(X|D))$ carry cooperadic structure: the inclusion of boundary strata defines cocomposition maps. For a rigid stable graph Γ,

$$\Delta^{\log}_\Gamma := (\nu_\Gamma)_* \circ \mathrm{Res}_{D^{\log}_\Gamma} : C^{\log \mathrm{FM}}_{\mathrm{mod}}(g, n) \to \bigotimes_{v\in V(\Gamma)} C^{\log \mathrm{FM}}_{\mathrm{mod}}\big(g(v), I_v \sqcup H(v)\big)[-|E_{\mathrm{int}}(\Gamma)|],$$

where ν_Γ is the Mok birational correspondence and $\mathrm{Res}_{D^{\log}_\Gamma}$ is the residue along the logarithmic boundary divisor.

3.6. The convolution L_∞-algebra. The cooperadic and operadic inputs of §§3.4–3.5 must be coupled. The coupling datum is a twisting morphism: a linear map satisfying a Maurer–Cartan equation in the convolution complex.

A twisting morphism $\alpha\colon C \to \mathcal{P}$ (satisfying $\partial\alpha + \alpha \star \alpha = 0$) determines a convolution sL_∞-algebra $\hom_\alpha(C, \mathcal{P})$ (Robert-Nicoud–Wierstra, after Vallette) with underlying space $\prod_{n\geq 1} \mathrm{Hom}_{\Sigma_n}(C(n), \mathcal{P}(n))$. The brackets

$$\ell_k(f_1, \ldots, f_k)(x) = \sum \gamma_{\mathcal{P}}\big((\alpha \circ 1)(f_1 \otimes \cdots \otimes f_k)\Delta^{(k)}_C(x)\big)$$

come from iterated cooperadic decomposition. ℓ_1 is the convolution differential; ℓ_2 is the Lie bracket of coderivations; higher ℓ_k, $k \geq 3$, arise whenever C is not concentrated in a single degree.

The sL_∞-algebra extends to ∞-morphisms in either slot separately but not both simultaneously: the convolution is functorial in each slot, but does not assemble into a bifunctor. The MC3 categorical lift proceeds one slot at a time. The bar-cobar adjunction is a Quillen equivalence (Vallette); different Ch_∞ presentations yield quasi-isomorphic deformation theories.

The three pillar constraints. Three structural limitations govern the convolution construction:

(i) The convolution sL_∞-algebra $\hom_\alpha(C, \mathcal{A})$ is *not* strict Lie (it carries higher brackets ℓ_k for $k \geq 3$). The strict chart $\mathrm{Conv}_{\mathrm{str}}$ is a strict Lie algebra with the same MC moduli, but it is a chart, not the full structure: transfer and formality require the L_∞-data.

(ii) The convolution $\hom_\alpha$ is functorial in each slot by ∞-morphisms, but fails as a *bifunctor*: one cannot simultaneously vary both the cooperadic and operadic inputs. The MC3 categorical lift (thick generation of the homotopy category) proceeds one slot at a time.

(iii) The log-FM cooperad $C^{\log \overline{C}}$ requires the snc pair (X, D): classical FM spaces $\overline{\overline{C}}_n(X)$ do not have the log structure needed for the planted-forest boundary strata. Without the logarithmic extension, the codimension-two cancellation mechanism (Mechanism 3 of §3.10) would fail.

These constraints are not deficiencies but structural features: they encode the geometry of the moduli space in the algebraic formalism.

Heisenberg: the strict chart suffices. For $\mathcal{H}_k$, the convolution sL_∞-algebra has $\ell_k = 0$ for $k \geq 3$ (the homotopy chiral algebra is strict, so all higher cooperadic decompositions produce zero). The strict chart equals the full L_∞ structure: $\mathrm{Conv}_{\mathrm{str}}(C, \mathrm{End}(\mathcal{H}_k)) = \mathrm{Conv}_\infty(C, \mathrm{End}(\mathcal{H}_k))$. Consequently, gauge equivalence in the strict chart is gauge equivalence in the full L_∞ structure, and the MC moduli is computed by the strict dg Lie algebra alone. This is the Gaussian simplification: the deformation theory of $\mathcal{H}_k$ is controlled by a classical (non-homotopical) Lie algebra.

Virasoro: the full L_∞ is essential. For Vir_c, the higher brackets ℓ_k for $k \geq 3$ are nonzero (reflecting the infinite A_∞-depth). The strict chart and the L_∞ structure have the same MC moduli (MC elements are the same), but different notions of gauge equivalence: two MC elements that are L_∞-gauge-equivalent may not be strictly gauge-equivalent. The full L_∞ structure is needed for the formality theorem (the shadow obstruction tower is a formality obstruction tower in the L_∞ sense) and for the transfer theorem (transferring the MC element to cohomology requires the ∞-morphism data).

3.7. Assembly. The modular convolution L_∞-algebra:

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} := \mathrm{Conv}_\infty\Big(C^{\log \mathrm{FM}}_{\mathrm{mod}}, \mathrm{End}_{\mathrm{Ch}_\infty}(A^{\mathrm{ch}}_\infty)\Big).$$

The cooperadic input is the log-FM chain cooperad $C^{\log \mathrm{FM}}_{\mathrm{mod}}$; the operadic input is the Ch_∞-endomorphism operad of A^{ch}_∞. The E_1 cousin $\mathfrak{g}^{\mathrm{mod},E_1}_{\mathcal{A}}$ uses the ribbon-graph-indexed cooperad and the ordered bar endomorphism operad.

One-slot convention: fix the log-FM (resp. ribbon) cooperad as the "geometric" slot; vary the chiral algebra input as the "algebraic" slot by ∞-morphisms.

The *strict chart*: the coderivation dg Lie algebra

$$\mathfrak{g}^{\mathrm{mod,log}}_{\mathcal{A}}(\mathfrak{U}) := \mathrm{Conv}_{\mathrm{str}}(C^{\log \mathrm{FM}}_{\mathrm{mod}}, \mathrm{End}_{\mathrm{Ch}_\infty}(A^{\mathrm{ch}}_\infty))$$

has classical commutator bracket, no higher L_∞ brackets, but the same Maurer–Cartan moduli. The strict chart suffices for constructing $\Theta_{\mathcal{A}}$ and computing shadows; the full L_∞ structure is needed for transfer, formality, and gauge equivalence.

3.8. The modular bar functor. In log-FM coordinates,

$$\mathrm{B}^{\log}_{\mathrm{mod}}(\mathcal{A})(g,n) = \bigoplus_{\Gamma \in \mathsf{Gr}^{\mathrm{st}}_{g,n}} \left(\bigotimes_{v \in V(\Gamma)} \bar{B}^{\mathrm{ch}}(A^{\mathrm{ch}}_\infty; \mathrm{In}(v)) \otimes C_\bullet(\mathrm{FM}^{\log}_\Gamma) \otimes \mathfrak{o}_{E_{\mathrm{int}}(\Gamma)} \right)_{\mathrm{Aut}(\Gamma)}.$$

The ingredients: at each vertex the transferred Ch_∞-bar contribution on its incoming half-edges; at each internal edge the propagator $P_{\mathcal{A}} = H_{\mathcal{A}}^{-1}$; the geometric factor $C_\bullet(\mathrm{FM}_\Gamma^{\log})$; the orientation line $\mathfrak{o}_{E_{\mathrm{int}}(\Gamma)} = \det(\Bbbk E(\Gamma)) \otimes \det(H_1(\Gamma;\Bbbk))^{-1}$; the automorphism quotient.

3.9. The six-component differential. The total differential has six components, whose mutual cancellations constitute $D^2 = 0$:

$$D^{\log}_{\mathrm{mod}} = \underbrace{d_{\check{C}}}_{\text{Čech}} + \underbrace{d_{\mathrm{Ch}_\infty}}_{\text{transferred chiral}} + \underbrace{d_{\mathrm{coll}}}_{\text{collision}} + \underbrace{d_{\mathrm{sew}}}_{\text{separating}} + \underbrace{d_{\mathrm{pf}}}_{\text{planted-forest}} + \underbrace{\Delta}_{\text{genus-raising}} .$$

$d_{\check{C}}$ is the Čech differential of the cover, acting on Čech indices. d_{Ch_∞} is the transferred chiral differential on each vertex, from secondary Borcherds operations. d_{coll} contracts an internal edge by extracting the OPE residue at the corresponding codimension-one collision stratum of $\overline{\mathrm{FM}}_n(X)$. The three positive-genus boundary operators act on a pure tensor $a \otimes \eta$ indexed by Γ:

$$\begin{aligned} d_{\mathrm{sew}}(a \otimes \eta) &= \sum_{e \in E_{\mathrm{sep}}(\Gamma)} \pm \mu_e(a) \otimes (\xi_e)_* \operatorname{Res}_{D_e^{\log}}(\eta), \\ d_{\mathrm{pf}}(a \otimes \eta) &= \sum_{F \in \mathsf{PF}^{\mathrm{rig}}(\Gamma)} \pm \mu_F(a) \otimes (\xi_F)_* \operatorname{Res}_{D_F^{\log}}(\eta), \\ \Delta(a \otimes \eta) &= \sum_{e \in E_{\mathrm{ns}}(\Gamma)} \pm \operatorname{Tr}_e(a) \otimes (\xi_e)_* \operatorname{Res}_{D_e^{\log}}(\eta). \end{aligned}$$

Separating edges give the clutching morphism (gluing two lower-genus curves); non-separating edges give the genus-raising self-gluing (BV operator); planted-forest strata give the genuinely logarithmic corrections from Mok. Explicitly,

$$d_{\mathrm{pf}} = \sum_{\rho \in \mathsf{PF}^{\mathrm{rig}}} \epsilon_\rho (\kappa_\rho)_* \mathrm{pr}_\rho^* \otimes \mu_\rho :$$

Mok's degeneration formula internalised as a summand of the total differential.

3.10. $D^2 = 0$. The nilpotence $(D^{\log}_{\mathrm{mod}})^2 = 0$ holds by three cancellation mechanisms. *Vertex labels*: $d_{\check{C}}^2 = 0$, $d_{\mathrm{Ch}_\infty}^2 = 0$, and $\{d_{\check{C}}, d_{\mathrm{Ch}_\infty}\} = 0$ by the Ch_∞-algebra axioms on A_∞^{ch}. *One-edge expansions*: anticommutators of a vertex differential with an edge-contraction differential vanish by the Leibniz rule for coderivations. *Codimension-two cancellation*: the sum of all products of two distinct edge-contraction operators vanishes, because the codimension-two boundary of Mok's log-FM compactification is a normal-crossings divisor (Mok, Theorem 3.3.1) and each codimension-two face appears as the intersection of exactly two codimension-one faces with opposite orientations.

The third mechanism is the nontrivial one. The normal-crossings property is essential: without it, codimension-two faces could have excess intersection, and the cancellation would fail.

The three mechanisms in detail. Expand D^2 into a 6×6 matrix of anticommutators:

$$D^2 = \sum_{1 \le i \le j \le 6} \{d_i, d_j\},$$

where $d_1, \ldots, d_6$ are the six components of §3.9.

Mechanism 1. **Vertex labels** ($i, j \in \{1, 2\}$): six terms. $d_{\check{C}}^2 = 0$ is the Čech identity. $d_{\mathrm{Ch}_\infty}^2 = 0$ is the A_∞-axiom. $\{d_{\check{C}}, d_{\mathrm{Ch}_\infty}\} = 0$ is the compatibility of the Čech differential with the transferred chiral structure. These three vanish independently of the genus data.

Mechanism 2. **One-edge expansions** ($i \in \{1,2\}$, $j \in \{3,4,5,6\}$): eight terms. Each computes $\{d_{\mathrm{vertex}}, d_{\mathrm{edge}}\}$. Because the edge-contraction operators act as coderivations on the coalgebra underlying the bar complex, the Leibniz rule gives $\{d_{\mathrm{vertex}}, d_{\mathrm{edge}}\}(\gamma) = d_{\mathrm{edge}}(d_{\mathrm{vertex}}\gamma) + (-1)^{|d_{\mathrm{vertex}}|} d_{\mathrm{vertex}}(d_{\mathrm{edge}}\gamma)$, and the two terms cancel by the vertex axioms applied to each half of the coderivation split.

Mechanism 3. **Codimension-two cancellation** ($i, j \in \{3,4,5,6\}$): ten terms. Each computes $\{d_{\mathrm{edge}_i}, d_{\mathrm{edge}_j}\}$ where the two edge operators correspond to two distinct codimension-one boundary strata. Their product is supported on the codimension-two intersection. The normal-crossings property of $\overline{\overline{C}}{}_n^{\log}(X|D)$ ensures that this intersection is transversal, and each codimension-two face is reached from exactly two codimension-one faces with opposite orientation signs. The cancellation in pairs is the higher-genus analogue of the genus-0 identity $\partial^2 = 0$ on the associahedron.

$D^2 = 0$ **at genus 1: explicit check.** At $(g,n) = (1,1)$, the total differential has three terms: d_{Ch_∞} (from the vertex), d_{coll} (from collision, a tree-level edge contraction within the self-loop), and $\ \Delta$ (the genus-raising self-gluing). The identity $D^2 = 0$ at $(1,1)$ reduces to

$$d^2_{\mathrm{Ch}_\infty} + \{d_{\mathrm{Ch}_\infty}, \ \Delta\} + (\ \Delta)^2 = 0.$$

The first term vanishes by A_∞. The cross-term $\{d_{\mathrm{Ch}_\infty}, \ \Delta\} = 0$ by the one-edge expansion mechanism. The last term $(\ \Delta)^2$ computes: the BV operator applied twice contracts two pairs of legs, but at $(1,1)$ there is only one self-loop, so $\Delta^2|_{(1,1)} = 0$ by degree count (no codimension-two stratum). Each mechanism kills its terms independently; $D^2 = 0$ holds.

$D^2 = 0$ **at genus 2: the nontrivial cancellation.** At $(g,n) = (2,0)$, the codimension-two cancellation mechanism is first exercised nontrivially. The three boundary divisors of $\overline{\mathcal{M}}_2$ (figure-eight, sunset, separating barbell from §3.2) meet pairwise in codimension-two strata. The figure-eight and sunset share a codimension-two stratum (contracting the figure-eight to a point that lies on one of the three sunset edges). The two edge-contraction operators produce opposite signs at this intersection because the normal-crossings structure on $\overline{\mathcal{M}}_2$ gives opposite orientations to the two orderings of the contracted edges. This is the first instance where the third cancellation mechanism is genuinely needed, and where the normal-crossings property of the Mok compactification (not just the Deligne–Mumford compactification) plays a role.

3.11. The E_n hierarchy in modular homotopy theory. The modular homotopy theory of §§3.1–3.10 has been presented in the E_∞ (symmetric) setting. The E_1 (ordered) setting is more fundamental: every construction above has a primitive E_1 version from which the E_∞ version is derived by averaging.

E_1**-modular operads: ribbon graphs.** The E_1-modular operad uses the Feynman transform of Ass (instead of Com): operations indexed by ribbon graphs (fatgraphs) with cyclic ordering at each vertex. The E_1 convolution algebra

$$\mathfrak{g}^{\mathrm{mod},E_1}_{\mathcal{A}} := \mathrm{Conv}_\infty\big(F\mathrm{Ass}, \mathrm{End}_{\bar{B}^{E_1}(\mathcal{A})}\big)$$

carries the R-matrix data: the genus-0 binary shadow $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta^{E_1}_{\mathcal{A}})$ is a matrix-valued function of the spectral parameter z, encoding the ordering of OPE insertions. The coinvariant map $\mathrm{av}\colon F\mathrm{Ass} \longrightarrow F\mathrm{Com}$ projects $\Theta^{E_1}_{\mathcal{A}} \mapsto \Theta_{\mathcal{A}}$: colour-ordered amplitudes to full amplitudes, destroying the R-matrix.

E_2**-chiral: the intermediate level.** The chiral Hochschild complex $\mathrm{ChirHoch}^*(\mathcal{A},\mathcal{A})$ of an E_∞-chiral algebra $\mathcal{A}$ carries an E_2-algebra structure (chiral Gerstenhaber bracket from OPE residues on $\overline{\overline{C}}_k(\mathbb{C})$).

This E_2 is the boundary algebra of the Swiss-cheese structure; the pair $(\mathrm{ChirHoch}^*(\mathcal{A},\mathcal{A}),\mathcal{A})$ is the $\mathrm{SC}^{\mathrm{ch,top}}$-datum. The E_2-chiral structure on Hochschild is the intermediate level between E_1 (ordered bar) and E_3 (topologized derived centre).

E_3-topological: the topologized output. When $\mathcal{A}$ has a conformal vector at non-critical level (Sugawara $T(z)=\{Q,G(z)\}$), the chiral direction becomes Q-exact in cohomology: the complex structure on $\mathbb{C}$ becomes irrelevant, and $\mathrm{SC}^{\mathrm{ch,top}}$ collapses to E_3-topological. This is proved for affine Kac–Moody $\widehat{\mathfrak{g}}_k$ at $k\neq -h^\vee$ (where Sugawara is explicit); conjectural for general chiral algebras with conformal vector. The proof is cohomological: it works on Q-cohomology, not on cochains. For class-M algebras where chain-level data is essential, the E_3-topological structure may exist only on cohomology.

The E_n hierarchy in modular homotopy theory is thus: E_1 (ordered bar, R-matrix, Yangian) $\to E_2$ (chiral Hochschild, Gerstenhaber) $\to E_3$ (topologized derived centre, requires conformal vector). Each step is a genuine loss of structure (the averaging map $E_1\to E_\infty$ loses the R-matrix; the topologization $\mathrm{SC}^{\mathrm{ch,top}}\to E_3$ loses the chiral direction). The ordered bar $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ sits at the E_1 level and is the primary object of the theory.

4. The universal Maurer–Cartan element

The modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is a complete filtered L_∞-algebra (§3.7). Its total differential $D_{\mathcal{A}}$ already squares to zero. Decomposing $D_{\mathcal{A}}=d_0+\Theta_{\mathcal{A}}$ into a tree-level local part and its complement produces a universal element $\Theta_{\mathcal{A}}\in\mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ satisfying the Maurer–Cartan equation as a formal consequence of $D^2_{\mathcal{A}}=0$. This element requires no equation to solve and no convergence hypothesis. Every invariant in this survey is a projection of $\Theta_{\mathcal{A}}$: the five theorems A–D and H are its five Σ_n-coinvariant shadows.

4.1. Γ-amplitudes and Taylor coefficients. Each stable graph produces a multilinear operation on the convolution algebra, built from chiral operations at vertices, propagators along edges, and geometric factors from log-FM strata. For $\Gamma\in\mathsf{Gr}^{\mathrm{st}}$ with $|V(\Gamma)|=k$ vertices and elements $f_1,\dots,f_k\in\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$, the Γ-*amplitude* assembles

$$\ell_\Gamma(f_1,\dots,f_k)=\mu_\Gamma\circ(\alpha^{\mathrm{mod}}_\Gamma\otimes f_1\otimes\cdots\otimes f_k)\circ\Delta^{\log}_\Gamma,$$

a k-linear map of cohomological degree $2-2g(\Gamma)-k$. Here μ_Γ is the graph composition (transferred chiral operations at vertices, propagators along edges), $\alpha^{\mathrm{mod}}_\Gamma$ is the modular twisting morphism component, and $\Delta^{\log}_\Gamma$ is the cooperadic cocomposition.

The Taylor coefficients of the modular L_∞ structure sum over stable graphs of fixed genus and vertex count:

$$\ell^{(g)}_k=\sum_{\substack{\Gamma\in\mathsf{Gr}^{\mathrm{st}},\ |V(\Gamma)|=k\\ b_1(\Gamma)+\sum_v g(v)=g}}\frac{1}{|\mathrm{Aut}(\Gamma)|}\ell_\Gamma.$$

The homotopy Jacobi identity is encoded by $D^2=0$ on the total convolution algebra.

Low-order Taylor coefficients. At tree level, the binary bracket is the transferred Lie bracket: $\ell^{(0)}_2(\alpha,\beta)=\pi[m_2(\iota(\alpha),\iota(\beta))]$. The ternary bracket sums over the three binary trees in Trees_3 (channels s, t, u), each inserting a homotopy h at the internal edge. The quaternary bracket sums over the fifteen labeled trees in Trees_4 (two unlabeled topologies: three balanced $((ab)(cd))$ and twelve caterpillar $(((ab)c)d)$), each with two propagator insertions; this is the classical homological perturbation lemma in its L_∞ incarnation.

The first genus-raised bracket is unary, the BV operator on the single graph with one vertex and one self-loop:

$$\ell_1^{(1)}(\alpha) = \mathrm{tr}(P_{\mathcal{A}} \circ \alpha \circ \partial^{\mathrm{ns}}) = \sum_i \langle e^i, \alpha(e_i)\rangle_{\mathrm{Sh}},$$

trace over the propagator contracting two half-edges at the non-separating node of $\overline{\mathcal{M}}_{1,1}$. The genus-one binary bracket

$$\ell_2^{(1)}(\alpha,\beta) = \mathrm{tr}\big(P_{\mathcal{A}} \circ m_2(\iota\alpha, -) \circ \partial^{\mathrm{ns}} \circ \iota\beta\big) + (\alpha \leftrightarrow \beta)$$

accounts for the separating node of $\overline{\mathcal{M}}_{1,2}$. These low-order brackets are the only data needed for shadow computations through degree 4.

4.2. The bar-intrinsic construction. $\Theta_{\mathcal{A}}$ requires no equation to solve: it is extracted from a differential that already squares to zero.

Decompose the total bar differential $D_{\mathcal{A}} : \mathrm{B}^{\mathrm{mod}}(\mathcal{A}) \to \mathrm{B}^{\mathrm{mod}}(\mathcal{A})$ as $D_{\mathcal{A}} = d_0 + \Theta_{\mathcal{A}}$, where $d_0 = d_{\mathrm{int}} + [\tau, -]$ is the tree-level local part. The *universal Maurer–Cartan element* is the remainder,

$$\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 = \sum_{\Gamma \in \mathsf{Gr}^{\mathrm{st}}} \frac{W_\Gamma^{\log \mathrm{FM}}}{|\mathrm{Aut}(\Gamma)|} \otimes \Phi_\Gamma^{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}),$$

with $W_\Gamma^{\log \mathrm{FM}} \in H^*(\overline{\mathcal{M}}^{\log}_{g(\Gamma),n(\Gamma)})$ the log-FM fundamental class of the stratum and $\Phi_\Gamma^{\mathcal{A}} \in \mathrm{Hom}(\bar{\mathcal{A}}^{\otimes n(\Gamma)}, \Bbbk)$ the chiral amplitude (propagators on internal edges, operations at vertices, cyclic trace at the output).

The MC property is immediate. $D^2_{\mathcal{A}} = 0$ by the normal-crossings property of $\overline{\mathcal{M}}^{\log}_{g,n}$; $d_0^2 = 0$ by construction; expanding $(d_0 + \Theta_{\mathcal{A}})^2 = 0$ gives

$$[d_0, \Theta_{\mathcal{A}}] + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

The MC property is a formal consequence of $\partial^2 = 0$ on the moduli space of curves, requiring no simple-Lie restriction, no convergence hypothesis, and no choice of basis. This is MC2 (conjecture-turned-theorem) in its cleanest form.

Proof sketch. Write $D_{\mathcal{A}} = d_0 + \Theta_{\mathcal{A}}$. Then $0 = D^2_{\mathcal{A}} = d_0^2 + d_0\Theta_{\mathcal{A}} + \Theta_{\mathcal{A}} d_0 + \Theta^2_{\mathcal{A}} = d_0^2 + [d_0, \Theta_{\mathcal{A}}] + \Theta^2_{\mathcal{A}}$. Now $d_0^2 = 0$ and $\Theta^2_{\mathcal{A}} = \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}]$ (since $\Theta_{\mathcal{A}}$ has odd cohomological degree); so $[d_0, \Theta_{\mathcal{A}}] + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$. This is the MC equation. The proof requires: (a) $D^2_{\mathcal{A}} = 0$, which is §3.10; (b) $d_0^2 = 0$, which is the genus-0 bar differential; (c) the decomposition $D_{\mathcal{A}} = d_0 + \Theta_{\mathcal{A}}$, which is the definition of $\Theta_{\mathcal{A}}$ as the non-tree-level part. No further input is needed.

The E_1 analogue $\Theta^{E_1}_{\mathcal{A}}$ is constructed identically on the ordered, ribbon-graph-indexed convolution algebra, and $\Theta_{\mathcal{A}} = \mathrm{av}(\Theta^{E_1}_{\mathcal{A}})$ under the coinvariant map $F\mathrm{Ass} \to F\mathrm{Com}$.

Heisenberg: $\Theta_{\mathcal{H}_k}$ collapses to a scalar. For the Heisenberg algebra $\mathcal{H}_k$, the only nonvanishing graph amplitude is on the single-edge graph Γ_2 (two vertices, one edge): $\Phi^{\mathcal{H}_k}_{\Gamma_2} = k$ (the level). Every graph with $|E| \geq 2$ vanishes because the only OPE mode is $\alpha_{(1)}\alpha = k$, and two propagator insertions on the same vertex yield zero by degree. Hence $\Theta_{\mathcal{H}_k} = k \cdot \eta_{\mathrm{univ}} \otimes \Lambda_{\mathrm{univ}}$: the MC element is a scalar multiple of the universal class. The genus-g projection is $F_g(\mathcal{H}_k) = k \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT). The modular characteristic $\kappa(\mathcal{H}_k) = k$ is the sole surviving datum.

Kac–Moody: $\Theta_{\widehat{\mathfrak{g}}_k}$ terminates at degree 3. For the affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$, the graph amplitudes at degree $r \leq 3$ are:

- Degree 2: the single-edge graph gives $\Phi_{\Gamma_2} = k\,\Omega = k \sum_a t^a \otimes t^a$ (the Casimir, with level prefix; at $k = 0$, $\Phi = 0$). Averaging: $\mathrm{av}(\Phi_{\Gamma_2}) = \kappa = (k + h^\vee)\dim\mathfrak{g}/(2h^\vee)$.

- Degree 3: the trivalent tree gives $\Phi_{\Gamma_3}(a,b,c) = \kappa([a,b],c)$, where $[\cdot,\cdot]$ is the Lie bracket and $(\cdot,\cdot)$ the Killing form. This is the cubic shadow $S_3 = \mathfrak{C}$.
- Degree 4: the quartic amplitude vanishes on bar cohomology by the Jacobi identity. The transferred quartic bracket $\ell_4^{(0)}$ on $H^*(\bar{B}(\widehat{\mathfrak{g}}_k))$ is zero because Jacobi forces the quartic obstruction class to be exact.

$\Theta_{\widehat{\mathfrak{g}}_k}$ is determined by $(\kappa, \mathfrak{C})$ with $S_r = 0$ for $r \geq 4$: class L, $r_{\max} = 3$.

Virasoro: Θ_{Vir_c} is infinite. For Virasoro at central charge c:

- Degree 2: $\kappa = c/2$.
- Degree 3: the cubic shadow is nonzero; S_3 arises from the simple-pole OPE $T(z)T(w) \sim 2T(w)/(z-w)^2 + \cdots$, which produces a Lie-bracket-like structure on the one-dimensional bar cohomology spanned by the class dual to T.
- Degree 4: $S_4 = 10/[c(5c+22)]$ (the quartic contact invariant). Nonzero for all $c \neq 0, -22/5$. No Jacobi identity kills it.
- Degree $r \geq 5$: the Poisson bracket $\{S_3, S_4\}_H$ generates S_5, and induction produces nonvanishing S_r at all orders.

Θ_{Vir_c} is an infinite MC element: class M, $r_{\max} = \infty$. The entire infinite tower is algebraic of degree 2, governed by (0.3).

4.3. $\Theta_{\mathcal{A}}$ as universal modular twisting morphism. The Maurer–Cartan space of the modular convolution L_∞ algebra is identified with the space of modular twisting morphisms,

$$\mathrm{MC}_{\bullet}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}) \simeq \mathrm{Tw}_{\alpha^{\mathrm{mod}}}\Big(C_{\mathrm{mod}}^{\log \mathrm{FM}}, \mathrm{End}_{\mathrm{Ch}_\infty}(A_\infty^{\mathrm{ch}})\Big).$$

Under this identification $\Theta_{\mathcal{A}}$ is the *universal modular twisting morphism*, the operadic analogue of the canonical flat connection on a principal bundle. A twisting morphism $\alpha \colon \mathcal{C} \to \mathcal{P}$ determines a twisted composition product $\mathcal{C} \circ_\alpha \mathcal{P}$ whose differential squares to zero iff α satisfies the MC equation. $\Theta_{\mathcal{A}}$ is the unique twisting morphism that simultaneously twists the modular bar complex and intertwines with the log-FM boundary structure.

The principal-bundle analogy is exact. The modular bar bundle $\mathrm{B}_{\mathrm{mod}}^{\log}(\mathcal{A}) \to \overline{\mathcal{M}}_{g,n}$ is the principal bundle; $\Theta_{\mathcal{A}}$ is the connection; the MC equation is flatness; the shadow tower (§6) is the Chern–Weil theory. The structure group is the pro-unipotent MC gauge group $\exp(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},0})$.

The twisting morphism in low degrees. At genus 0, degree 2: the twisting morphism component $\alpha_{0,2}^{\mathrm{mod}} \colon \mathcal{C}^{\log \overline{C}}(0,2) \to \mathrm{End}(A)$ is the chiral product $\mu_{\mathcal{A}}$. The twisted composition gives the bar differential $d_B = \sum_k m_k$ (sum of all A_∞-operations). At genus 0, degree 3: the component $\alpha_{0,3}^{\mathrm{mod}} \colon \mathcal{C}^{\log \overline{C}}(0,3) \to \mathrm{End}(A)$ uses the chain map $C_*(\overline{\mathcal{M}}_{0,4}) \to \mathrm{End}(A)$ from the three boundary points of $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ to the three OPE channels. At genus 1, degree 1: the component $\alpha_{1,1}^{\mathrm{mod}} \colon \mathcal{C}^{\log \overline{C}}(1,1) \to \mathrm{End}(A)$ is the BV operator Δ, using the chain of the boundary divisor $\delta_{\mathrm{irr}} \in \overline{\mathcal{M}}_{1,1}$.

Uniqueness of the twisting morphism. On the Koszul locus, $\Theta_{\mathcal{A}}$ is unique up to MC gauge equivalence. The tree-level complex carries a contracting homotopy (from PBW degeneration), so the obstruction to existence at each genus is exact, and the solution is determined up to gauge by induction from genus 0. Different choices of contracting homotopy yield gauge-equivalent MC elements; no free parameters appear at any genus. Off the Koszul locus: the twisting morphism still exists (bar-intrinsic), but uniqueness up to gauge may fail (the contracting homotopy may not exist, and the obstruction classes may have nontrivial cohomology).

4.4. Tridegree, depth, and convergence. Each coefficient of $\Theta_{\mathcal{A}}$ carries a natural tridegree

$$(g, n, d) = (\text{loop genus, external degree, log boundary depth}),$$

with $d(\rho) = \#E_{\mathrm{sep}}(\Gamma_\rho) + \#E_{\mathrm{ns}}(\Gamma_\rho) + \mathrm{depth}_{\mathrm{pf}}(\Gamma_\rho)$ the codimension of the log-FM stratum. The depth filtration $F^d\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \bigoplus_{d'\geq d}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})_{*,*,d'}$ makes the convolution algebra complete:

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \varprojlim_{d} \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}/F^d\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}.$$

The inverse limit $\Theta_{\mathcal{A}} = \varprojlim_r \Theta^{\leq r}_{\mathcal{A}}$ converges unconditionally: no analytic input, no growth estimate, no restriction on the algebra. Convergence is purely algebraic, guaranteed by completeness of the depth filtration. This is MC4 (completion tower): proved. Generating depth (the degree at which higher operations are determined by lower ones) is distinct from algebraic depth (whether the tower terminates).

At fixed (g, n), only finitely many stable graph types contribute. The depth filtration refines this further: at depth d, the stratum has codimension d in $\overline{\mathcal{M}}^{\log}_{g,n}$, and the amplitude has internal-edge count bounded by d.

Generating depth versus algebraic depth. Two distinct notions of depth govern $\Theta_{\mathcal{A}}$:

- *Generating depth* $d_{\mathrm{gen}}(\mathcal{A})$: the smallest r such that all m_s for $s > r$ in the minimal A_∞-model are determined (as algebraic consequences) by $m_2, \ldots, m_r$. For Virasoro: $d_{\mathrm{gen}} = 3$ (the ternary operation m_3 generates all higher m_s recursively).
- *Algebraic depth* $d_{\mathrm{alg}}(\mathcal{A})$: the smallest r such that $m_s = 0$ for all $s > r$ in the minimal model, or ∞ if no such r exists. For Virasoro: $d_{\mathrm{alg}} = \infty$ (all m_s are nonzero, class M).

The two are independent: $d_{\mathrm{gen}} = 3$ and $d_{\mathrm{alg}} = \infty$ coexist for Virasoro. The shadow depth $r_{\max}$ equals d_{alg}: it records whether the tower terminates, not whether the tower is finitely generated.

Convergence is unconditional. The inverse limit $\Theta_{\mathcal{A}} = \varprojlim_r \Theta^{\leq r}_{\mathcal{A}}$ exists for *every* chiral algebra $\mathcal{A}$, without any analytic hypothesis. The argument:

(i) The convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is complete with respect to the depth filtration F^d: $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \varprojlim_d \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}/F^d$.

(ii) At each depth d, the truncation $\Theta^{\leq d}_{\mathcal{A}} \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}/F^{d+1}$ is a finite sum over stable graphs with at most d internal edges.

(iii) The transition maps $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}/F^{d+1} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}/F^d$ are surjective (by the bar-intrinsic construction, which solves all obstruction classes simultaneously).

(iv) The Mittag-Leffler condition holds: the inverse system has surjective transitions, so $\varprojlim^1 = 0$, and the inverse limit is exact.

No growth estimate, no radius of convergence, no restriction on κ or the depth class is needed. The convergence is algebraic, not analytic: it is a consequence of the completeness of a pro-nilpotent filtration on a graded vector space, with no metric structure required.

4.5. The weight filtration. Decompose the MC equation by internal-edge count,

$$\Theta_{\mathcal{A}} = \sum_{r\geq 1} \Theta^{[r]}_{\mathcal{A}}, \qquad \Theta^{[r]}_{\mathcal{A}} = \sum_{\substack{\Gamma\in\mathsf{Gr}^{\mathrm{st}} \\ |E_{\mathrm{int}}(\Gamma)|=r}} \frac{W^{\log\mathrm{FM}}_{\Gamma}}{|\mathrm{Aut}(\Gamma)|} \otimes \Phi^{\mathcal{A}}_{\Gamma}.$$

Weight 1: $[D_{\mathrm{loc}}, \Theta^{[1]}_{\mathcal{A}}] = 0$. Weight N: $[D_{\mathrm{loc}}, \Theta^{[N]}_{\mathcal{A}}] + \frac{1}{2}\sum_{i+j=N}[\Theta^{[i]}_{\mathcal{A}}, \Theta^{[j]}_{\mathcal{A}}] = 0$. The obstruction to extending from weight $N-1$ to weight N lies in $H^2(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}/F^{N+1}, D_{\mathrm{loc}})$. The bar-intrinsic construction solves all stages simultaneously.

Weight 1: boundary divisors. In degree $(0,4)$: the three boundary points of $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ correspond to the three channels $(12|34)$, $(13|24)$, $(14|23)$,

$$\Theta^{[1]}_{\mathcal{A}}|_{(0,4)} = [\delta_s] \otimes \Phi_s + [\delta_t] \otimes \Phi_t + [\delta_u] \otimes \Phi_u.$$

The weight-one equation $\Phi_s + \Phi_t + \Phi_u = F_3$ identifies the Jacobiator homotopy with a boundary in the bar complex: the first Borcherds identity. In degree $(1,1)$: the single boundary divisor δ_{irr} gives $\Theta^{[1]}_{\mathcal{A}}|_{(1,1)} = [\delta_{\mathrm{irr}}] \otimes \Delta$ with $\Delta = \sum_i e_i \otimes e^i$ the Casimir.

In degree $(0,5)$: the ten boundary divisors of $\overline{\mathcal{M}}_{0,5}$ form the Petersen graph (each divisor is a product of two $\overline{\mathcal{M}}_{0,4}$'s meeting at a node). The weight-one equation at $(0,5)$ produces the *pentagon identity*: the five ways to parenthesise $(((ab)c)d)e, \dots, a(b(c(de)))$ are related by a cycle of four associator homotopies, closing by the fifth. In the A_∞ framework, this is the m_4-equation:

$$\partial m_4 + m_2(m_3 \otimes \mathrm{id} + \mathrm{id} \otimes m_3) + m_3(m_2 \otimes \mathrm{id} \otimes \mathrm{id} + \cdots) = 0.$$

The pentagon identity is the degree-$(0,5)$ component of the MC equation; the entire A_∞ hierarchy is the tree-level projection of a single MC element.

Heisenberg: all weights collapse. For $\mathcal{H}_k$, the weight-one component $\Theta^{[1]}_{\mathcal{H}_k}$ at $(0,4)$ has $\Phi_s = \Phi_t = \Phi_u = 0$ (no cubic OPE pole, so no Jacobiator), and $\Theta^{[1]}|_{(1,1)} = k \cdot [\delta_{\mathrm{irr}}]$ (the curvature $\kappa = k$). All higher weight components $\Theta^{[r]}_{\mathcal{H}_k} = 0$ for $r \geq 2$. The MC element is entirely determined by the weight-one data, which is a single scalar.

Kac–Moody: weight 1 carries the Lie bracket. For $\widehat{\mathfrak{g}}_k$, the weight-one component at $(0,4)$ has $\Phi_s(a,b,c,d) = \kappa([a,b],[c,d])$ (the structure constants of $\mathfrak{g}$ contracted through the Killing form). The weight-one equation at $(0,4)$ is the Jacobi identity. At $(1,1)$: $\Theta^{[1]}|_{(1,1)} = \kappa \cdot [\delta_{\mathrm{irr}}]$. Weight 2: the planted-forest correction enters through the log-FM degeneration, but it is killed by the Jacobi identity at degree 4. For $\widehat{\mathfrak{g}}_k$, weights ≥ 2 contribute only through the genus-raising BV operator; no planted-forest or degree-raising corrections survive.

Weight 2: the first planted-forest layer. At weight 2, codimension-two strata enter: classical products of two boundary divisors plus genuinely logarithmic planted-forest strata with no DM counterpart. The weight-two equation is $[D_{\mathrm{loc}}, \Theta^{[2]}_{\mathcal{A}}] = -\frac{1}{2}[\Theta^{[1]}, \Theta^{[1]}]$, and the planted-forest correction $\Theta^{[2]}_{\mathrm{pf}}$ records how collision limits in $\mathrm{FM}_n(X|D)$ correct the naive square. For the $\mathcal{W}_3$ central-parameter line, cubic gauge triviality kills the weight-2 obstruction; the first genuine invariant appears at weight 3.

4.6. The all-genus recursion. Projecting the MC equation to fixed (g,n),

$$d\Theta^{(g,n)}_{\mathcal{A}} + \frac{1}{2}\sum_{\substack{g_1+g_2=g\\ n_1+n_2=n+2}} \ell_2^{(0)}(\Theta^{(g_1,n_1)}_{\mathcal{A}}, \Theta^{(g_2,n_2)}_{\mathcal{A}}) + \Delta_{\mathrm{ns}}\Theta^{(g-1,n+2)}_{\mathcal{A}} + \sum_{k\geq 3}\frac{1}{k!}\ell_k^{(\mathrm{tree})}(\Theta_{\mathcal{A}}, \dots, \Theta_{\mathcal{A}})^{(g,n)} = 0.$$

Four terms: the internal differential, separating clutching, non-separating BV, and planted-forest corrections. This recursion determines $\Theta^{(g,n)}_{\mathcal{A}}$ inductively from lower (g',n') in lexicographic order. The base cases are $\Theta^{(0,3)}_{\mathcal{A}}$ (the chiral bracket) and $\Theta^{(0,4)}_{\mathcal{A}}$ (the Jacobiator data). The recursion terminates at each (g,n).

The clutching identity. On each rigid boundary stratum ρ,

$$\Theta_{\mathcal{A}}\Big|_{\rho} = (\kappa_\rho)_*\left(\bigotimes_{v\in V(S_\rho)} \Theta_{\mathcal{A},v}\right).$$

The amplitude on the degenerate curve equals the product of amplitudes on the irreducible components, sewn by propagators along the nodes. This is the MC-level sewing axiom, and the starting point for the clutching law of the quartic shadow (§6).

Recursion at genus 1. At $(g, n) = (1, 0)$ (genus 1, no marked points), the recursion simplifies to

$$d\Theta^{(1,0)}_{\mathcal{A}} + \Delta_{\mathrm{ns}}\Theta^{(0,2)}_{\mathcal{A}} = 0.$$

The separating clutching term vanishes (no way to split $(g_1, n_1) \sqcup (g_2, n_2)$ with $g_1 + g_2 = 1$ and $n_1 + n_2 = 2$ using only tree-level data; the base case $(0, 2)$ contributes through the BV operator). The non-separating term computes $\Delta_{\mathrm{ns}}\Theta^{(0,2)}_{\mathcal{A}} = \sum_i \langle e^i, \Theta^{(0,2)}_{\mathcal{A}}(e_i)\rangle$: the contraction of the genus-0 binary amplitude (the chiral product) by the propagator. The result is $\Theta^{(1,0)} = \kappa \cdot \lambda_1$ (UNIFORM-WEIGHT), the genus-1 amplitude. The computation reduces to computing $\mathrm{tr}(P_{\mathcal{A}} \circ m_2) = \kappa$, which is the definition of the modular characteristic.

For Heisenberg: $\Theta^{(0,2)}_{\mathcal{H}_k} = k$ (the level), $P_{\mathcal{H}_k} = 1/k$, so $\Delta_{\mathrm{ns}}\Theta^{(0,2)} = k \cdot (1/k) = 1$. Hence $\Theta^{(1,0)}_{\mathcal{H}_k} = \lambda_1$, and $F_1(\mathcal{H}_k) = \kappa/24 = k/24$. The single stable graph at $(1, 0)$ (the self-loop graph Γ_{irr}) contributes $(\ /2) \cdot \mathrm{tr}(P \circ m_2) = \kappa \ \ /2$.

Recursion at genus 2. At $(g, n) = (2, 0)$, the recursion has three terms:

$$\begin{aligned} d\Theta^{(2,0)}_{\mathcal{A}} &+ \tfrac{1}{2}\ell_2^{(0)}(\Theta^{(1,0)}_{\mathcal{A}}, \Theta^{(1,0)}_{\mathcal{A}}) \\ &+ \Delta_{\mathrm{ns}}\Theta^{(1,2)}_{\mathcal{A}} + (\text{planted-forest}) = 0. \end{aligned}$$

The separating clutching $\frac{1}{2}\ell_2^{(0)}(\Theta^{(1,0)}, \Theta^{(1,0)})$ produces the separating barbell graph contribution: two genus-1 vertices connected by one edge. The non-separating BV $\Delta_{\mathrm{ns}}\Theta^{(1,2)}$ produces the iterated BV and the clutching-times-BV terms. These correspond to the three stable graph types of $\overline{\mathcal{M}}_2$ enumerated in §3.2. The planted-forest correction is genuinely logarithmic.

For Heisenberg at genus 2: only the iterated BV contributes (the separating term involves $\ell_2^{(0)}(\Theta^{(1,0)}, \Theta^{(1,0)})$, which vanishes because the Heisenberg convolution bracket is abelian). The result is $F_2(\mathcal{H}_k) = 7k/5760$.

For affine KM at genus 2: the separating term is nonzero (the Lie bracket gives a nontrivial $\ell_2^{(0)}$), and the planted-forest correction enters through the log-FM degeneration formula. The Jacobi identity ensures that all terms combine to give $F_2(\widehat{\mathfrak{g}}_k) = 7\kappa(\widehat{\mathfrak{g}}_k)/5760$ (UNIFORM-WEIGHT).

Recursion at genus 3: five graph types. At $(g, n) = (3, 0)$, the recursion involves five stable graph types: (a) the iterated figure-eight with three self-loops; (b) the barbell with one genus-1 and one genus-2 vertex; (c) the theta graph (three parallel edges between two genus-0 vertices, with one self-loop at one vertex); (d) the triple-banana (three edges, each with one self-loop); (e) the complete graph K_4 on four genus-0 vertices. For Heisenberg, only type (a) contributes (all other types involve Lie brackets or cubic shadows that vanish). The result: $F_3(\mathcal{H}_k) = 31k/967680$ from the $\hat{A}$-genus.

For Virasoro at genus 3: all five types contribute, with the planted-forest corrections carrying the quartic and quintic shadow data. The genus-3 free energy $F_3(\mathrm{Vir}_c) = 31c/(2 \cdot 967680) = 31c/1935360$ (UNIFORM-WEIGHT).

The recursion as a Feynman diagram expansion. The recursion at (g, n) is a sum over Feynman diagrams with g loops and n external legs, weighted by the chiral amplitudes at each vertex and the propagator $P_{\mathcal{A}} = H_{\mathcal{A}}^{-1}$ at each internal edge. The structure is identical to the perturbative expansion of a quantum field theory on the moduli space $\overline{\mathcal{M}}_{g,n}$, with the bar complex as the kinetic term and the shadow tower as the interaction vertices. The 't Hooft large-N expansion corresponds to the genus filtration: the colour-ordered amplitudes at genus g are weighted by $\ ^{g} = N^{-2g}$, and the planar $(g = 0)$ diagrams dominate.

The recursion terminates at each (g, n) because the set of stable graph types $\mathsf{Gr}_{g,n}^{\mathrm{st}}$ is finite. No convergence hypothesis is needed: the recursion is finite at each step, and the inverse limit over all (g, n) converges by completeness of the depth filtration (MC4).

Every invariant in this monograph is a projection of $\Theta_{\mathcal{A}}$: the curvature κ, the spectral discriminant $\Delta_{\mathcal{A}}$, the shadow metric Q_L, the sewing amplitudes, the $\hat{A}$-genus, the Fredholm determinant, and the quartic contact invariant Q^{contact}. The five main theorems are five projections.

Projection of $\Theta_{\mathcal{A}}$	**Invariant**	**Section**
Scalar ($r = 2$, genus tower)	$\kappa(\mathcal{A})$, F_g, $\hat{A}$-genus	§2
Binary genus-0 residue	classical r-matrix, Yangian seed	§7.7
Binary Verdier dual	complementarity $Q_g \oplus Q_g^!$	§2
Bar-cobar counit	inversion $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$	§1
Chiral Hochschild	$\mathrm{ChirHoch}^*(\mathcal{A})$, deformation ring	§9
Spectral discriminant	$\Delta_{\mathcal{A}}(x)$, genus-1 Hessian	§5
Cubic shadow ($r = 3$)	gauge-trivial obstruction	§6
Quartic resonance ($r = 4$)	$\mathfrak{Q}^{\mathrm{contact}}$, clutching law	§6
MC tautological descent	shadow CohFT, $R^*(\overline{\mathcal{M}}_{g,n})$	§5
Primitive kernel	shell equations, branch BV	§5
Dirichlet–sewing lift	$S_{\mathcal{A}}(u)$, Euler–Koszul class	§8
Swiss-cheese colour decomposition	α_T, six projections (Vol. II)	§10

These five theorems are five projections of $\Theta_{\mathcal{A}}$. Theorem A constructs the arena in which $\Theta_{\mathcal{A}}$ lives. Theorem B recovers $\mathcal{A}$ from $\Theta_{\mathcal{A}}$ on the Koszul locus. Theorem C splits the genus tower into Lagrangian summands (C1 unconditional, C2 (UNIFORM-WEIGHT)). Theorem D extracts the scalar κ with $\hat{A}$-genus generating function (uniform-weight; multi-weight adds $\delta F_g^{\mathrm{cross}}$). Theorem H identifies the deformation ring via $\mathrm{ChirHoch}^*$ concentration.

The finite-order projections form an obstruction tower $\Theta_{\mathcal{A}}^{\leq 2} \to \Theta_{\mathcal{A}}^{\leq 3} \to \Theta_{\mathcal{A}}^{\leq 4} \to \cdots$ whose generating function is algebraic of degree 2. On each primary line, $H(t) = t^2\sqrt{Q_L(t)}$ is determined by three seed invariants (κ, α, S_4). The inverse limit exists and is automatically Maurer–Cartan because $D_{\mathcal{A}}^2 = 0$ (MC4).

In three dimensions, $\Theta_{\mathcal{A}}$ extends to an open/closed Maurer–Cartan element $\Theta_{\mathcal{A}}^{\mathrm{oc}} = \Theta_{\mathcal{A}} + \sum_j \mu^{M_j}$ packaging the holomorphic-topological QFT on $\mathbb{C}_z \times \mathbb{R}_t$: the bar complex $B(\mathcal{A})$, coassociative over $(\mathrm{ChirAss})^!$, supplies the holomorphic factorization data; the derived center pair $(\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}), \mathcal{A})$ carries the $\mathrm{SC}^{\mathrm{ch,top}}$-algebra structure encoding both closed and open colours; and $\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$ is the universal bulk (Volume II).

The bar complex thus organises five mathematical structures as projections of one object: the genus expansion $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT) is a GUE matrix model at $N^2 = \kappa$; the ordered bar

$\bar{B}^{\mathrm{ord}}(\mathcal{A})$ produces $U_q(\mathfrak{g})$ via FRT reconstruction; the quantum-group R-matrix yields knot invariants; the shadow metric Q_L defines a Bridgeland stability condition with universal wall phase $\pi/4$; and the Stokes constants $S_n/S_1 = (-1)^{n+1}n$ encode the resurgent structure.

Corollary (MC replaces the sum over topologies). On the Koszul locus, the tree-level complex carries a contracting homotopy (from the PBW degeneration), so $\Theta^{(g)}$ is determined up to gauge-equivalent MC elements by induction from $\Theta^{(0)}$. No free parameters appear at any genus; distinct choices of contracting homotopy yield gauge-equivalent MC elements; no choice of bulk topologies is required. The genus expansion is computed algebraically from the genus-0 chiral bracket by the clutching recursion of §4.6, not approximated perturbatively. The Maloney–Witten problem (negative Fourier coefficients in the Poincaré series) does not arise: the MC equation determines each F_g uniquely by algebraic induction, without summing over bulk topologies at fixed genus. The non-perturbative sum over distinct three-manifold topologies lies outside the MC framework.

5. The modular tangent complex and Chern–Weil theory

The MC element $\Theta_{\mathcal{A}}$ exists unconditionally and satisfies the MC equation. Its deformation theory is controlled by the modular tangent complex, obtained by twisting the convolution algebra by $\Theta_{\mathcal{A}}$ itself.

5.1. The L_∞-twisted differential. The MC element twists the convolution differential. For $x \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$,

$$d_{\Theta_{\mathcal{A}}}(x) := \sum_{n\geq 0}\sum_{g\geq 0} \frac{\hbar^g}{n!}\ell^{(g)}_{n+1}(\Theta_{\mathcal{A}}^{\otimes n}, x) = d(x) + [\Theta_{\mathcal{A}}, x] + \tfrac{1}{2!}\ell^{(0)}_3(\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}, x) + \hbar\,\ell^{(1)}_1(x) + \cdots$$

In the strict chart this collapses to $d_{\Theta_{\mathcal{A}}}(x) = d(x) + [\Theta_{\mathcal{A}}, x]$. The identity $d^2_{\Theta_{\mathcal{A}}} = 0$ is the MC equation for $\Theta_{\mathcal{A}}$.

5.2. The modular tangent complex. The *modular tangent complex at* $\Theta_{\mathcal{A}}$:

$$T^{\mathrm{mod}}_{\Theta_{\mathcal{A}}} := (\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, d_{\Theta_{\mathcal{A}}}).$$

Its cohomology governs deformations: H^0 infinitesimal automorphisms (stabiliser), H^1 first-order deformations (tangent space), H^2 obstructions. $H^2 = 0$ makes the MC moduli smooth at $\Theta_{\mathcal{A}}$; $H^1 = 0$ makes $\Theta_{\mathcal{A}}$ rigid.

For a one-parameter family $\mathcal{A}_t$ varying in $k(t)$ or $c(t)$, the tangent complex forms a local system over the parameter space and the obstruction class traces a path in H^2. The vanishing locus is the moduli of liftable deformations.

Tangent cohomology: Heisenberg. For $\mathcal{H}_k$, the MC element $\Theta_{\mathcal{H}_k} = k \cdot \eta \otimes \Lambda$ is a scalar, so the twisted differential $d_\Theta(x) = d(x) + [k\eta\Lambda, x]$ acts on the (abelian) convolution algebra with vanishing bracket. The tangent complex reduces to the untwisted convolution complex: $H^0(T^{\mathrm{mod}}_{\Theta_{\mathcal{H}_k}}) = \Bbbk$ (the identity automorphism), $H^1 = \Bbbk$ (the tangent direction $\partial/\partial k$), $H^2 = 0$ (no obstructions). The MC moduli is smooth and one-dimensional: the level k is a free parameter with no obstruction to deformation.

Tangent cohomology: Virasoro. For Vir_c, the tangent complex is nontrivial. The twisted differential d_Θ acts on the one-dimensional bar cohomology spanned by the class x dual to T. The tangent cohomology $H^1(T^{\mathrm{mod}}_{\Theta_{\mathrm{Vir}_c}})$ is one-dimensional (the direction $\partial/\partial c$). At $c = 0$ and $c = -22/5$ the quartic contact invariant $Q^{\mathrm{contact}} = 10/[c(5c+22)]$ has poles; these are *not* singularities of the tangent complex itself (which is well-defined for all c) but singularities of the shadow projection. The tangent complex detects: the central charge c is a free deformation parameter; the shadow invariants S_r are *not* independent deformation parameters but are algebraically determined functions of c.

Tangent cohomology: affine KM at critical level. At the critical level $k = -h^\vee$, the tangent complex undergoes a jump discontinuity. The curvature $\kappa = 0$ means the MC element is uncurved; the tangent cohomology H^1 jumps in dimension from 1 at generic k to $1 + \dim \mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee})$ at critical level. Since $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee}) \cong \mathrm{Fun}(\mathrm{Op}_{^L\mathfrak{g}}(D))$ is infinite-dimensional, this is not a "doubling" but a jump from finite to infinite dimension; the additional tangent directions correspond to the Feigin–Frenkel center, so the deformation theory at $k = -h^\vee$ sees the full space of opers.

5.3. Three characteristic projections. Three successive projections of $\Theta_{\mathcal{A}}$ capture progressively finer structure: $\kappa(\mathcal{A})$ (leading scalar curvature), $\Delta_{\mathcal{A}}(x)$ (characteristic polynomial of a finite-rank operator), $\mathfrak{R}_4^{\mathrm{mod}}(\mathcal{A})$ (first nonlinear characteristic class).

The scalar curvature.

$$\kappa(\mathcal{A}) = \mathrm{tr}(\Theta_{\mathcal{A}})_{2,0} = \mathrm{tr}(\pi \circ m_2 \circ \iota^{\otimes 2} \circ \Delta_{\mathrm{sep}}) \in \Bbbk.$$

This is the first Chern class of the modular bar bundle: $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ (UNIFORM-WEIGHT); multi-weight at $g \geq 2$ adds $\delta F_g^{\mathrm{cross}}$ (ALL-WEIGHT). The Mumford relation $\lambda_g^{g+1} = 0$ confines genus-g scalar obstructions to dimension at most g.

Explicit values across the standard landscape:

Algebra $\mathcal{A}$	$\kappa(\mathcal{A})$	$\kappa(\mathcal{A}^!)$	**Sum rule**
Heisenberg $\mathcal{H}_k$	k	$-k$	0
Affine $\widehat{\mathfrak{g}}_k$	$\frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}$	$-\frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}$	0
Virasoro Vir_c	$c/2$	$(26-c)/2$	13
$\beta\gamma$ system (λ=1)	1	-1	0
Lattice V_Λ	$\mathrm{rank}(\Lambda)$	$-\mathrm{rank}(\Lambda)$	0
$\mathcal{W}_N(\widehat{\mathfrak{sl}}_N)$	$c \cdot (H_N - 1)$	$c' \cdot (H_N - 1)$	$K_N \cdot (H_N - 1)$
Free fermion ψ	$1/4$	$-1/4$	0

$H_N = \sum_{j=1}^N 1/j$ is the harmonic number ($H_N - 1 \neq H_{N-1}$: at $N = 2$, $H_2 - 1 = 1/2$ but $H_1 = 1$). The sum rule $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = 0$ holds for Kac–Moody, free fields, lattice algebras. For $\mathcal{W}$-algebras the sum is a nonzero constant: $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$ and $\kappa(\mathcal{W}_{N,c}) + \kappa(\mathcal{W}_{N,c'}) = K_N(H_N - 1)$ with $K_N = c + c'$. Self-duality occurs at $c = 13$ for Virasoro and $c = 50$ for $\mathcal{W}_3$.

The spectral discriminant. The Koszul dual Hilbert series

$$h_{\mathcal{A}^!}(q) = \sum_{n \geq 0} \dim H^n(\bar{B}(\mathcal{A}))\, q^n$$

is rational for every standard family: it satisfies a holonomic recurrence of finite order d, and the transfer matrix T_{rec} is a $d \times d$ matrix. Define the *spectral branch object* as a finite-rank perfect complex $V_{\mathcal{A}}^{\mathrm{br}}$ with *branch transport operator* $T_{\mathrm{br},\mathcal{A}} : V_{\mathcal{A}}^{\mathrm{br}} \to V_{\mathcal{A}}^{\mathrm{br}}$ and

$$\Delta_{\mathcal{A}}(x) := \mathrm{sdet}(1 - xT_{\mathrm{br},\mathcal{A}}).$$

The rank of $V_{\mathcal{A}}^{\mathrm{br}}$ equals the degree of algebraicity of $h_{\mathcal{A}^!}$. Three properties: $\Delta_{\mathcal{A}}$ is self-dual ($\Delta_{\mathcal{A}^!} = \Delta_{\mathcal{A}}$), unlike κ which is anti-symmetric for KM/free; exact factorization functors preserving quadratic OPE data preserve T_{br}; the spectral radius equals $\dim \mathfrak{g}$ for affine KM at generic level.

Spectral discriminant: family-by-family. The spectral discriminant is computed explicitly for each standard family:

Heisenberg $\mathcal{H}_k$: the bar cohomology is $H^*(\bar{B}(\mathcal{H}_k)) = \Bbbk \oplus \Bbbk \cdot x$ (two-dimensional, concentrated in degrees 0 and 1). The Hilbert series is $h(q) = 1 + q$, which is polynomial (degree of algebraicity 0). The transfer matrix is trivial: $T_{\mathrm{br}} = 0$, and $\Delta_{\mathcal{H}_k}(x) = 1$. The spectral discriminant carries no data beyond κ. This is the Gaussian archetype: the second characteristic projection is trivial.

$\beta\gamma$ system: the bar cohomology has Hilbert series $h(q) = \sqrt{(1+q)/(1-3q)}$, algebraic of degree 2. The branch transport operator is a 2×2 matrix with eigenvalues 3 and -1, giving $\Delta_{\beta\gamma}(x) = (1-3x)(1+x)$. The two roots $x = 1/3$ and $x = -1$ are the branch points of the algebraic function $h(q)$. The spectral radius is 3 (the larger eigenvalue); the spectral gap is $3/1 = 3$.

Affine $\mathfrak{sl}_2$: the bar cohomology dimensions grow as $\dim H^n(\bar{B}(\widehat{\mathfrak{sl}}_{2,k})) = \binom{n+2}{2}$ at generic level, and $h(q) = 1/(1-q)^3$ (rational, degree of algebraicity 3). The transfer matrix eigenvalues are 1, 1, 1 (all equal), and $\Delta_{\widehat{\mathfrak{sl}}_2}(x) = (1-x)^3$. The spectral radius is 1; the degenerate eigenvalue structure reflects the $\mathfrak{sl}_2$ Weyl group symmetry.

For $\mathfrak{sl}_3$: the degree-3 recurrence has transfer matrix eigenvalues 8, $(3 \pm \sqrt{13})/2$, and $\Delta_{\widehat{\mathfrak{sl}}_3}(x) = (1 - 8x)(1 - 3x - x^2)$. For Heisenberg: $h_{\mathcal{H}_k^!}(q) = 1 + q$ and $\Delta_{\mathcal{H}_k}(x) = 1$. For $\beta\gamma$: $h = \sqrt{(1+q)/(1-3q)}$ and $\Delta_{\beta\gamma}(x) = (1-3x)(1+x)$.

In the Chern–Weil dictionary, $\Delta_{\mathcal{A}}$ plays the role of the Chern character: $\mathrm{ch}_r(\mathcal{A}) = \frac{1}{r!}\frac{d^r}{dt^r}|_{t=0}[-\log \Delta_{\mathcal{A}}(t)] = \frac{1}{r}\,\mathrm{tr}(T_{\mathrm{br}}^r)$.

The quartic shadow.

$$\mathfrak{R}^{\mathrm{mod}}_{4,g,n}(\mathcal{A}) = \mathrm{pr}_{4,g,n}(\Theta_{\mathcal{A}}) \in H^*(\overline{\mathcal{M}}_{g,n}) \otimes (\bar{\mathcal{A}}^\vee)^{\otimes n}_{\mathrm{cyc}}.$$

This is the first *nonlinear* characteristic class: the first invariant not recoverable from κ and $\Delta_{\mathcal{A}}$. Its genus-zero, four-point specialisation is the quartic contact invariant

$$Q^{\mathrm{contact}}_{\mathrm{Vir}} = \frac{10}{c(5c+22)}, \qquad Q^{\mathrm{contact}}_{\mathcal{H}} = 0, \qquad Q^{\mathrm{contact}}_{\beta\gamma} \neq 0.$$

(Verify: at $c = 1$, $Q_{\mathrm{Vir}} = 10/27$.) For Heisenberg the quartic vanishes (no cubic OPE pole). For $\beta\gamma$ the quartic is the first gauge-nontrivial shadow (the cubic is gauge-removable). For Virasoro, both cubic and quartic are nonvanishing.

5.4. The Chern–Weil dictionary. The analogy between $\Theta_{\mathcal{A}}$ and a connection 1-form is exact. The modular bar bundle $\mathrm{B}^{\log}_{\mathrm{mod}}(\mathcal{A}) \to \overline{\mathcal{M}}_{g,n}$ is a principal bundle for the gauge group of the convolution algebra; $\Theta_{\mathcal{A}}$ is its flat connection (flatness automatic); the shadow tower is its Chern–Weil theory. Holonomy becomes factorisation homology, gauge equivalence becomes MC gauge, and the Weil homomorphism becomes the shadow algebra $\mathcal{A}^{\mathrm{sh}} = H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}})$.

Classical Chern–Weil	**Modular factorization**
Principal G-bundle $P \to M$	Modular bar bundle $\mathrm{B}^{\log}_{\mathrm{mod}}(\mathcal{A}) \to \overline{\mathcal{M}}_{g,n}$
Connection $\omega \in \Omega^1(P, \mathfrak{g})$	MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$
Curvature $\Omega = d\omega + \frac{1}{2}[\omega, \omega]$	$D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$
Flatness $\Omega = 0$	Automatic (bar-intrinsic)
First Chern c_1	$\kappa(\mathcal{A}) = \mathrm{tr}(\Theta_{\mathcal{A}})_{2,0}$
Chern character	Spectral discriminant $\Delta_{\mathcal{A}}(t)$
Higher Chern classes c_r	Shadows $\mathfrak{R}^{\mathrm{mod}}_{4,g,n}$, Sh_r $(r \geq 5)$
Weil homomorphism	Shadow algebra $\mathcal{A}^{\mathrm{sh}} = H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}})$
ad-invariant polynomial	Cyclic cocycle in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$
Cyclic pairing $\mathrm{tr}(XY)$	Verdier duality on $\mathrm{Ran}(X)$
Group multiplication	Operadic composition in $\mathcal{P}^{\mathrm{ch}}$
Coproduct on $\mathcal{O}(G)$	Operadic cocomposition / clutching
Lagrangian splitting $T^*G = \mathfrak{g} \oplus \mathfrak{g}^*$	Complementarity $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!)$
Holonomy $\mathrm{Hol}(\gamma)$	Factorisation homology $\int_\Sigma \mathcal{A}$
Gauge equivalence	MC gauge in $\mathrm{MC}_\bullet(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$
Transgression $\mathrm{CS}(\omega)$	Shadow obstruction tower (§6)
Flat sections $\nabla s = 0$	Twisted cochains $d_{\Theta_{\mathcal{A}}}(x) = 0$

The Chern–Weil dictionary in practice. The table above is not merely an analogy: each entry is a precise mathematical identification. Three examples:

Scalar curvature = first Chern class: the classical first Chern class $c_1(P) = \frac{1}{2\pi i}\mathrm{tr}(\Omega)$ computes the curvature of a line bundle. The modular analogue: $\kappa(\mathcal{A})$ is the scalar curvature of the modular bar bundle, computed as $\mathrm{tr}(\Theta_{\mathcal{A}})_{2,0}$. The Hodge class $\lambda_g = c_g(\mathbb{E})$ plays the role of the Chern class polynomial.

Flatness = MC equation: a connection is flat iff $\Omega = d\omega + \frac{1}{2}[\omega, \omega] = 0$. The MC element $\Theta_{\mathcal{A}}$ satisfies $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ automatically (bar-intrinsic). In the Chern–Weil setting, flat connections are rare; in the modular setting, flatness is guaranteed. The modular bar bundle is always flat; the shadow tower records the characteristic classes of this flat connection, which are secondary (Chern–Simons-type), not primary.

Holonomy = factorisation homology: the holonomy of a flat connection around a loop γ is a group element $\mathrm{Hol}(\gamma) \in G$. The modular analogue: the factorisation homology $\int_\Sigma \mathcal{A}$ of the chiral algebra over a surface Σ is the "holonomy" of the modular bar bundle around the surface. For a closed surface of genus g: the factorisation homology computes the genus-g conformal blocks; the MC projection to genus g is the shadow of this computation.

5.5. The genus spectral sequence. The genus filtration $F^g L_{\mathrm{mod}} := \bigoplus_{g' \geq g}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})_{g',*,*}$ yields a convergent spectral sequence. Write $D_0 = D_{\mathrm{int}} + D_{\mathrm{sep}}$ for the genus-preserving part and $D_1 = D_{\mathrm{nsep}}$ for the genus-raising part; these satisfy $D_0^2 = 0$, $\{D_0, D_1\} = 0$, $D_1^2 = 0$ (a bicomplex).

The spectral sequence: $E_0^{p,q} = \mathrm{gr}^p_F L^{p+q}_{\mathrm{mod}}$ with $d_0 = [D_0, -]$; $E_1^{p,*}$ is the associated graded cohomology, genus by genus; the differential d_r at page r encodes the obstruction Ob_r; $E_\infty^{p,q} \cong \mathrm{gr}^p_F H^{p+q}(L_{\mathrm{mod}})$. Convergence follows from completeness of the depth filtration and pronilpotence of the weight filtration; the $\varprojlim^1$ term vanishes by Mittag-Leffler.

This spectral sequence is distinct from the PBW spectral sequence (which decomposes by conformal weight within fixed genus). PBW computes *what* bar cohomology is; the genus spectral sequence determines *whether* the MC element extends across genera. For Heisenberg it degenerates at E_1. For Virasoro, $d_1 \neq 0$ and nontrivial differentials persist at every page.

Heisenberg: E_1-degeneration. For $\mathcal{H}_k$, the genus-preserving differential D_0 acts on each genus-g stratum independently. The genus-raising differential D_1 is the BV operator Δ. Because the convolution bracket of $\mathcal{H}_k$ is abelian, D_1 acts by scalar multiplication: $\Delta(\Theta^{(g)}) = \kappa \cdot \Theta^{(g+1)}$ (the genus loop operator contracts two legs of a genus-g amplitude and produces a genus-$(g+1)$ amplitude). The E_1 page is one-dimensional at each genus, generated by $\kappa^g \cdot [\lambda_g]$ (UNIFORM-WEIGHT). The differential d_1 on the E_1 page vanishes because there is no room: the target $E_1^{g+1,*}$ is generated by $\kappa^{g+1} \cdot [\lambda_{g+1}]$, and the map $d_1 \colon \kappa^g[\lambda_g] \mapsto 0$ by degree counting (the BV operator lowers degree by 2, and $\dim \overline{\mathcal{M}}_g = 3g - 3$, so the degree mismatch forces $d_1 = 0$). The spectral sequence degenerates at E_1: the full cohomology is the direct sum of the genus-by-genus contributions.

Affine KM: E_2-degeneration. For $\widehat{\mathfrak{g}}_k$, the E_1 page is nontrivial at each genus (the convolution bracket is nonabelian), and the differential $d_1 \neq 0$. However, the Jacobi identity on $\mathfrak{g}$ implies that $d_2 = 0$: the quartic shadow vanishes ($S_4 = 0$), and the obstruction to extending the MC element from the E_1 page to E_2 is trivial. The spectral sequence degenerates at E_2. For $\mathfrak{sl}_2$ at level $k = 1$: the $E_1^{1,*}$ page has dimension 4; the differential d_1 reduces this to dimension 2 at $E_2^{1,*}$; then $d_r = 0$ for all $r \geq 2$.

Virasoro: non-degeneration. For Vir_c, the spectral sequence does not degenerate at any finite page. The differentials d_r are nonzero for infinitely many r, reflecting the infinite shadow depth. At each page, the quartic contact invariant and its descendants produce nontrivial differentials. The E_∞ page is computed by the full infinite tower; no finite truncation suffices.

5.6. Homotopy invariance and functoriality. The shadow algebra $\mathcal{A}^{\mathrm{sh}} = H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$ is a homotopy invariant. For any ∞_α-quasi-isomorphism $f \colon \mathcal{A} \xrightarrow{\sim} \mathcal{A}'$,

$$f_* \colon \mathrm{MC}_\bullet(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}) \xrightarrow{\sim} \mathrm{MC}_\bullet(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}'})$$

is a weak homotopy equivalence. The scalar κ, the spectral determinant $\Delta_{\mathcal{A}}$, and the full shadow obstruction tower $(\kappa, \Delta, \mathfrak{C}, \mathfrak{Q}, \mathrm{Sh}_5, \ldots)$ consist of quasi-isomorphism invariants; $\Theta_{\mathcal{A}}$ itself is a quasi-isomorphism invariant up to gauge. The assignment $\mathcal{A} \mapsto (\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, \Theta_{\mathcal{A}})$ is a functor from chiral algebras with ∞_α-quasi-isos inverted to MC moduli problems. The convolution L_∞-algebra is functorial in each slot separately but not as a bifunctor; functoriality of $\mathcal{A} \mapsto \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ proceeds through the source slot.

5.7. The shadow CohFT and topological recursion. Projected to $\overline{\mathcal{M}}_{g,n}$, the MC equation produces tautological classes and tautological relations. The MC element determines shadow tautological classes $\tau_{g,n}(\mathcal{A}) \in R^*(\overline{\mathcal{M}}_{g,n+1})$ assembling into a cohomological field theory (Theorem **??**): the factorisation axiom holds by the clutching law, the Σ_n-equivariance by the symmetry of stable-graph summation, the flat identity conditionally on the vacuum lying in V.

Projecting the MC equation to the (g,n)-component yields a tautological relation in $R^*(\overline{\mathcal{M}}_{g,n})$ at each (g,n):

$$\sum_{\mathrm{sep/nsep}} \xi_{\mathrm{sep},*}\big(\tau_{g_1,|S|}(\mathcal{A}) \cdot \tau_{g_2,|S^c|}(\mathcal{A})\big) + \delta^{(g,n)}_{\mathrm{pf}} = 0.$$

WDVV and Mumford from MC. At genus 0, degree 3: the MC tautological relation on $\overline{\mathcal{M}}_{0,4}$ reduces to WDVV. At degree 2, genus $g \geq 2$: substituting $\tau_{g,2}(\mathcal{A}) = \kappa \cdot \pi^*\lambda_g$ (UNIFORM-WEIGHT) recovers the Mumford λ-class clutching relation on $\overline{\mathcal{M}}_g$.

The complementarity propagator $P_{\mathcal{A}}\colon H^*(\mathcal{A}) \otimes H^*(\mathcal{A}^!) \to \mathbb{C}$ is the Givental R-matrix of the shadow CohFT; when it carries a flat unit (all standard families with vacuum in V), Teleman reconstruction recovers $\{\tau_{g,n}\}$ at all (g,n) from genus-0 data and $P_{\mathcal{A}}$. Eynard–Orantin topological recursion is the MC equation applied to the shadow obstruction tower.

MC datum	**Tautological output**
$\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$	shadow CohFT $\Omega^{\mathcal{A}}_{g,n}$
$D^2 = 0$ on $\mathrm{FM}^{\log}$	CohFT axioms
$D\Theta + \frac{1}{2}[\Theta,\Theta] = 0$ at (g,n)	tautological relation
$[\Theta,\Theta]\vert_{\mathrm{sep}}$	WDVV / associativity
$\Delta(\Theta)\vert_{\mathrm{nsep}}$	Mumford λ-class relations
$P_{\mathcal{A}}$	Givental R-matrix
MC recursion	EO topological recursion

The Pixton shadow bridge. For class-M algebras, $\Theta_{\mathcal{A}}$ generates tautological classes at every genus. At genus 2–3, the relations $\tau_{g,n}(\mathrm{Vir}_c)$ for generic c produce elements of the Pixton ideal in $R^*(\overline{\mathcal{M}}_{g,n})$: Pixton's tautological ring relations arise as shadows of the Virasoro MC equation. On the semisimple locus, MC-descended relations together with Mumford relations generate the Pixton ideal (Theorem **??**); the non-semisimple frontier remains open.

Pixton relations from MC: the mechanism. The MC equation at (g,n) produces a tautological relation in $R^*(\overline{\mathcal{M}}_{g,n})$ because each term in the MC equation is a pushforward of a boundary class. At genus 2, $n=0$: the MC relation involves three terms (iterated BV, clutching-times-BV, planted-forest) as in the genus-2 shell table of §6.7. For Virasoro at generic c, these three terms are nonzero and produce a nontrivial relation in $R^2(\overline{\mathcal{M}}_2)$: the Pixton relation $\kappa_1^2 - 12\lambda_1\kappa_1 + 48\lambda_1^2 = 0$ (in the notation of the tautological ring). The class-M property guarantees that the relation is nontrivial: if the tower terminated, the relation would be vacuous. The Pixton relations are shadows of the infinite tower.

At genus 3: the MC equation at $(3,0)$ involves five stable graph types and produces relations in $R^3(\overline{\mathcal{M}}_3)$. These reproduce the Faber–Zagier relations, which are known to generate the Pixton ideal at $g=3$. At higher genus, the MC-descended relations become progressively more complex but always land in the Pixton ideal by the semisimplicity theorem.

The Eynard–Orantin connection. Eynard–Orantin topological recursion computes the correlators $\omega_{g,n}(z_1,\dots,z_n)$ of a spectral curve from the genus-0 two-point function $\omega_{0,2}$ and the Bergman kernel. In the MC framework, the spectral curve is the shadow metric $Q_L(t)$ of (0.3), the Bergman kernel is the complementarity propagator $P_{\mathcal{A}}$, and the recursion kernel is the MC clutching law. The correspondence is:

Eynard–Orantin	**MC framework**
Spectral curve (C,x,y)	Shadow metric (L,Q_L,H)
Bergman kernel $B(z_1,z_2)$	Complementarity propagator $P_{\mathcal{A}}$
Recursion kernel $K(z_0,z)$	MC clutching law at (g,n)
Branch points	Zeros of Q_L: $\Delta = 0$ locus
Residue at branch point	Shadow obstruction class o_{r+1}
Free energy F_g	Shadow amplitude $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (uniform-weight)

The recursion is the MC equation projected to fixed (g,n) and restricted to the primary line L. For class-G algebras, the spectral curve is rational and the recursion terminates. For class-M algebras, the spectral curve is hyperelliptic (the algebraic curve $y^2 = Q_L(t)$ of degree 2) and the recursion produces an infinite family of correlators. This identification is exact on the semisimple locus (where $\Delta \neq 0$); the non-semisimple locus $\Delta = 0$ requires a separate analysis.

The Böcherer bridge. At genus 2, the shadow tautological class $\tau_{2,0}(\mathrm{Vir}_c)$ for lattice vertex algebras computes a central L-value via the Böcherer conjecture (Furusawa–Morimoto). For the Leech lattice, $\chi_{12} = \mathrm{SK}(f_{22})$, and the MC-derived second Fourier coefficient $c_2 \approx -1.918 \times 10^{-6} \neq 0$ is the first nonzero central L-value produced by the shadow obstruction tower.

5.8. The primitive kernel and cofree reduction. $\Theta_{\mathcal{A}}$ lives in the cofree conilpotent coalgebra underlying the modular bar construction. By Milnor–Moore for modular operads, $\Theta_{\mathcal{A}}$ is determined by its restriction to primitive corollas and rigid cutting strata, the *primitive logarithmic modular kernel*

$$\mathfrak{K}_{\mathcal{A}} = \sum_{g \geq 0, n \geq 2} K^{\mathcal{A}}_{g,n} + \sum_{\rho \in \mathrm{Rig}} K^{\mathcal{A}}_{\rho},$$

and the equivalence (Theorem ??)

$$(Q^{\mathrm{mod}}_{\mathcal{A}})^2 = 0 \iff d\mathfrak{K}_{\mathcal{A}} + \mathfrak{K}_{\mathcal{A}} \star \mathfrak{K}_{\mathcal{A}} + \quad \Delta_{\mathrm{ns}}(\mathfrak{K}_{\mathcal{A}}) = 0,$$

with $\star$ the primitive pre-Lie product from log-FM residues, replaces the genus-by-genus obstruction programme by a single equation on primitive data.

Shell equations. Projecting to genus 2 and 3:

$$R_2(\mathfrak{K}_{\mathcal{A}}) = \Delta_{\mathrm{ns}}(K_1) + \tfrac{1}{2}[K_1, K_1] + \sum_{\rho \in \mathrm{Rig}_2} R^{\mathcal{A}}_{\rho}(K^{\mathcal{A}}_{0,2}),$$
$$R_3(\mathfrak{K}_{\mathcal{A}}) = \Delta_{\mathrm{ns}}(K_2) + [K_1, K_2] + \tfrac{1}{3!}\ell_3(K_1, K_1, K_1) + \sum_{\rho} R^{\mathcal{A}}_{\rho}(\ldots).$$

For Heisenberg, only Δ_{ns} survives. For affine Kac–Moody, $[K_1, K_1]$ contributes the cubic shadow. For $\beta\gamma$, cubic gauge triviality kills the bracket and the rigid stratum contributes the quartic contact invariant. For Virasoro and $\mathcal{W}_N$, all three terms are active at every genus.

The branch BV action. The finite-rank branch-active quotient $V^{\mathrm{br}}_{\mathcal{A}} = V^{+}_{\mathcal{A}} \oplus V^{-}_{\mathcal{A}}$ carries a BV master action

$$S^{\mathrm{br}}_{\mathcal{A}}(x, \xi) = \tfrac{1}{2}\Omega_{\mathcal{A}}(K^{\mathrm{br}}_{\mathcal{A}}x, x) + \sum_{m+n \geq 3} \frac{1}{m!n!}\mu^{\mathcal{A}}_{m,n}(x^{\otimes m}; \xi^{\otimes n}),$$

satisfying $\Delta^{\mathrm{br}}_{\mathcal{A}} S^{\mathrm{br}}_{\mathcal{A}} + \frac{1}{2}\{S^{\mathrm{br}}_{\mathcal{A}}, S^{\mathrm{br}}_{\mathcal{A}}\}_{\mathrm{BV}} = 0$. The metaplectic half-density $\eth_{\mathcal{A}} \in 1 + x\Bbbk[[x]]$ satisfies $\eth^2_{\mathcal{A}} = \Delta_{\mathcal{A}} = \det(1 - xT^{\mathrm{br,red}}_{\mathcal{A}})$: the spectral discriminant is the square of a canonical half-density.

Shell equations: Heisenberg worked example. For $\mathcal{H}_k$, the primitive kernel reduces to $\mathfrak{K}_{\mathcal{H}_k} = K_1$ (genus 1, the BV operator applied once). The genus-2 shell: $R_2 = \Delta_{\mathrm{ns}}(K_1) = \Lambda_P(K_1)$, the genus loop operator applied to the genus-1 amplitude. The bracket $[K_1, K_1] = 0$ (abelian) and the rigid stratum contribution is zero (no higher shadows). So $R_2 = \Lambda_P(\kappa \cdot \lambda_1)$. The loop operator contracts two legs with the propagator $P = 1/k$, giving $R_2 = k \cdot (1/k) \cdot \lambda_2 = \lambda_2$ (a cohomology class on $\overline{\mathcal{M}}_2$). The genus-3 shell: $R_3 = \Delta_{\mathrm{ns}}(K_2) = \Lambda_P(R_2) = \lambda_3$. By induction: $R_g = \lambda_g$ for all g, and $F_g(\mathcal{H}_k) = \kappa \cdot \lambda^{\mathrm{FP}}_g$ (UNIFORM-WEIGHT): the $\hat{A}$-genus is reproduced by iterated BV alone.

Shell equations: affine KM worked example. For $\widehat{\mathfrak{sl}}_{2,k}$, the genus-2 shell has two nonzero terms: $R_2 = \Delta_{\mathrm{ns}}(K_1) + \frac{1}{2}[K_1, K_1]$. The bracket $[K_1, K_1]$ is nonzero (the Lie bracket of $\mathfrak{sl}_2$ via the cubic shadow S_3), but the Jacobi identity forces $\frac{1}{3!}\ell_3(K_1, K_1, K_1) = 0$ at genus 3: the Lie bracket contribution terminates. The genus-2 shell simplifies to $R_2 = \Lambda_P(\kappa\lambda_1) + \frac{1}{2}\kappa S_3 \cdot \lambda_2$, and the result is $F_2 = 7\kappa/5760$ (UNIFORM-WEIGHT): the scalar formula holds because $\mathfrak{sl}_2$ is uniform-weight (single generator at conformal weight 1). The cubic shadow S_3 enters the shell equation but cancels in the uniform-weight projection.

5.9. Entanglement from the modular characteristic. The bar coalgebra carries a natural bipartite structure: its coproduct splits a configuration of points into two halves, one on each side of a spatial cut. For a spatial bipartition of an interval of length L with UV cutoff ε, the scalar projection of $\Theta_{\mathcal{A}}$ restricted to the bipartite cut yields

$$S_{\mathrm{EE}}(\mathcal{A}) = \frac{c(\mathcal{A})}{3} \log \frac{L}{\varepsilon}.$$

For Virasoro $c = 2\kappa$; the relationship between c and κ is family-dependent. Complementarity gives the entanglement sum rule $S_{\mathrm{EE}}(\mathcal{A}) + S_{\mathrm{EE}}(\mathcal{A}^!) = \frac{c(\mathcal{A})+c(\mathcal{A}^!)}{3} \log(L/\varepsilon)$; for the Virasoro family $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$ this gives $c + c' = 26$ and the universal sum $26/3 \cdot \log(L/\varepsilon)$.

The four shadow-depth classes (G/L/C/M) classify entanglement complexity: class G (Gaussian, $r_{\max} = 2$) exhibits area-law entanglement only; class L ($r_{\max} = 3$) adds subleading logarithmic corrections from the cubic shadow; class C ($r_{\max} = 4$) adds a contact correction from the quartic invariant; class M ($r_{\max} = \infty$) carries an infinite tower of corrections.

The entanglement programme (G11–G16). Six theorem targets extend the scalar result. G11: Ryu–Takayanagi from the scalar projection of $\Theta_{\mathcal{A}}$. G12: Koszulness as exact quantum error correction; the Knill–Laflamme condition is the Lagrangian isotropy condition from the complementarity pairing; shadow depth equals the number of redundancy channels; the twelve Koszulness characterisations translate into a twelve-fold code dictionary; failure of Koszulness is failure of the code. G13: replica trick from the MC equation. G14: modular flow from the tangent complex. G15: QES as the shadow connection Ward identity. G16: Page curve at $c = 13$ from the shadow connection at the self-dual point; the scrambling time is controlled by $\Delta_{\mathcal{A}}$.

6. The shadow obstruction tower

6.1. The extension tower. Can $\Theta_{\mathcal{A}}$ be approximated by finitely many degrees, and what controls the passage from one finite approximation to the next? The total weight filtration answers both.

Define the *total weight* $w(g, r, d) := 2g - 2 + r + d$. The filtration $F^N \mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}} := \bigoplus_{w \geq N} (\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}})_{g,r,d}$ is exhaustive, separated, complete. $\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}}$ is the *ambient* (planted-forest-extended) modular convolution algebra. The *modular extension tower*:

$$\mathcal{E}_{\mathcal{A}}(N) := \mathrm{MC}(\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}}/F^{N+1}), \qquad N \geq 1,$$

with restriction maps $\mathcal{E}_{\mathcal{A}}(N+1) \to \mathcal{E}_{\mathcal{A}}(N)$. A full lift is a compatible point of the inverse system; the bar-intrinsic construction provides one unconditionally. The shadow obstruction tower studies $\mathcal{E}_{\mathcal{A}}(1) \leftarrow \mathcal{E}_{\mathcal{A}}(2) \leftarrow \mathcal{E}_{\mathcal{A}}(3) \leftarrow \cdots$.

6.2. Obstruction classes and the all-degree master equation. Define $\Theta^{\leq r}_{\mathcal{A}}$ by projecting to total weight $\leq r$. The *shadow obstruction tower* is the sequence $\Theta^{\leq 2}_{\mathcal{A}}, \Theta^{\leq 3}_{\mathcal{A}}, \Theta^{\leq 4}_{\mathcal{A}}, \ldots$ At each truncation the MC equation fails by a remainder,

$$D\Theta^{\leq r}_{\mathcal{A}} + \tfrac{1}{2}[\Theta^{\leq r}_{\mathcal{A}}, \Theta^{\leq r}_{\mathcal{A}}] = -o_{r+1}(\mathcal{A}),$$

with obstruction class $o_{r+1}(\mathcal{A}) \in Z^2(F^{r+1}/F^{r+2}, d_{\leq r})$. Extension is possible iff $[o_{r+1}] = 0$ in $H^2(F^{r+1}/F^{r+2}, d_{\leq r})$. The bar-intrinsic construction guarantees extension, so the obstruction class is always exact; the shadow obstruction tower records the gauge choices at each stage.

The all-degree master equation at degree r:

$$\nabla_H(\mathrm{Sh}_r(\mathcal{A})) + o^{(r)}(\mathcal{A}) = 0, \qquad r \geq 3,$$

with ∇_H the Hodge covariant derivative. This equation determines Sh_r uniquely modulo gauge if $H^1(F^r/F^{r+1}, d_{\leq r-1}) = 0$.

Obstruction classes: explicit computation. The obstruction class o_{r+1} at degree $r+1$ has a concrete graph formula:

$$o_{r+1}(\mathcal{A}) = \sum_{\substack{\Gamma \in \mathsf{Gr}^{\mathrm{st}} \\ w(\Gamma)=r+1}} \frac{1}{|\mathrm{Aut}(\Gamma)|} \cdot \partial_\Gamma(\Phi_{\mathcal{A}}^{\leq r}),$$

where ∂_Γ is the boundary operator that contracts one edge of Γ and evaluates the amplitude on the contraction, and $\Phi_{\mathcal{A}}^{\leq r}$ is the graph amplitude truncated to weight $\leq r$. The sum runs over all stable graphs of total weight $r+1$; at each graph, the obstruction is the failure of the truncated amplitude to satisfy the MC equation on that graph.

At degree 3: the obstruction $o_3(\mathcal{A})$ is the Jacobiator. It lies in $H^2(\mathcal{A}^{\mathrm{sh}}_{3,0})$, which for a Lie algebra input is $H^2(\mathfrak{g}, \mathfrak{g})$ (the Lie algebra cohomology governing deformations). For $\widehat{\mathfrak{g}}_k$: $o_3 = 0$ because the Lie bracket satisfies Jacobi. For Vir_c: $o_3 \neq 0$ but is gauge-trivial (cubic gauge triviality, §6.3 below).

At degree 4: the obstruction $o_4(\mathcal{A})$ is the quartic contact invariant (after the cubic gauge is fixed). For $\widehat{\mathfrak{g}}_k$: $o_4 = 0$ by Jacobi. For Vir_c: $o_4 = 10/[c(5c+22)] \neq 0$. For $\beta\gamma$: $o_4 \neq 0$ (the first nonvanishing obstruction, since $o_3 = 0$ for $\beta\gamma$).

At degree 5: the obstruction $o_5(\mathcal{A}) = \{\mathfrak{C}, \mathfrak{Q}\}_H$ is the Poisson bracket of the cubic and quartic shadows. For Virasoro: $o_5 = 480/[c^2(5c+22)] \cdot x^5 \neq 0$ (§6.9).

6.3. Cubic gauge triviality and the canonical quartic.

> **Cubic gauge triviality.** When $H^1(F^3/F^4, d_2) = 0$, the cubic MC term $\Theta_{\mathcal{A}}^{[3]}$ is gauge-trivial: there exists $\exp(\xi_3)$ with $\xi_3 \in F^3$ such that $\exp(\xi_3) \cdot \Theta_{\mathcal{A}}^{\leq 3}$ has vanishing weight-3 component.

After this gauge transformation, the quartic obstruction class $[\mathfrak{Q}(\mathcal{A})] \in H^2(F^4/F^5, d_2)$ is canonical: independent of homotopy retraction data, cubic gauge, and MC gauge orbit representative. This is the first characteristic class not polynomial in κ. The mechanism is standard Postnikov: trivial H^1 at level n makes the H^2 class at level $n+1$ canonical.

Cubic gauge triviality: proof. The obstruction to extending $\Theta_{\mathcal{A}}^{\leq 2}$ to degree 3 lies in $H^2(F^3/F^4, d_2)$. The bar-intrinsic construction provides a specific extension (so the obstruction class is exact), but the extension is not unique: two extensions differ by an element $\xi_3 \in \ker(d_2) \cap F^3/F^4 = H^1(F^3/F^4, d_2)$. When $H^1(F^3/F^4, d_2) = 0$, the cubic extension is unique (up to d_2-exact terms), and the exponential gauge action $\exp(\xi_3) \cdot \Theta^{\leq 3}$ has no gauge freedom at weight 3. After fixing this gauge, the quartic class $[\mathfrak{Q}] = [\Theta^{[4]} + \frac{1}{2}\{\Theta^{[3]}, \Theta^{[3]}\}]$ is canonically determined.

The condition $H^1(F^3/F^4, d_2) = 0$ holds for all standard families on the Koszul locus. Off the Koszul locus (e.g., at the critical level $k = -h^\vee$ for affine KM), H^1 may be nonzero and the quartic class acquires a choice.

Family-by-family quartic invariant.

Algebra	Q^{contact}	**Comment**
$\mathcal{H}_k$	0	no cubic OPE pole
$\widehat{\mathfrak{g}}_k$	0	Jacobi kills quartic
$\beta\gamma$	$\neq 0$	first nontrivial shadow
Vir_c	$10/[c(5c+22)]$	both cubic and quartic nonzero
$\mathcal{W}_3$	$\neq 0$	DS-escalated from $\mathfrak{sl}_3$

The clutching law for the quartic follows from Mok's log FM degeneration formula. For $C \rightsquigarrow C_0 = C_1 \cup_p C_2$,

$$\mathfrak{Q}(\mathcal{A})|_{C_0} = \sum_{a+b=4} \mathfrak{Q}^{(a)}(\mathcal{A})|_{C_1} \cdot \mathfrak{Q}^{(b)}(\mathcal{A})|_{C_2} + \text{planted-forest correction}.$$

The correction is genuinely logarithmic, with no DM counterpart.

The quartic contact invariant $Q^{\text{contact}}_{\text{Vir}} = 10/[c(5c+22)] \in H^2(F^4/F^5, d_2)$. The denominator has physical meaning: $c = 0$ is the logarithmic threshold; $c = -22/5$ is the minimal model accumulation point $\mathcal{M}(2,5)$. The numerator $10 = \binom{5}{2}$ counts degree-four Catalan channels.

Numerical values of the Virasoro quartic.

c	$Q^{\text{contact}}_{\text{Vir}}$	**Significance**
1	$10/27 \approx 0.370$	free boson Virasoro
$1/2$	$10/(1/2 \cdot 49/10) = 200/49 \approx 4.08$	Ising model
13	$10/(13 \cdot 87) = 10/1131 \approx 0.00884$	self-dual
25	$10/(25 \cdot 147) = 10/3675 \approx 0.00272$	bosonic string
26	$10/(26 \cdot 152) = 10/3952 \approx 0.00253$	critical string

The quartic decreases monotonically for $c > 0$ and diverges as $c \to 0^+$ (logarithmic threshold) and $c \to (-22/5)^+$ (minimal model accumulation).

6.4. The Hamilton–Jacobi generating function. For a multi-generator algebra ($\mathcal{W}_3$ with generators T (weight 2) and W (weight 3)), the deformation space is two-dimensional. The Riccati ODE lifts to a Hamilton–Jacobi PDE. Write $U = \sum_{s \geq 3} \mathrm{Sh}_s(x_T, x_W)$; the master equation becomes

$$2E(U) + \tfrac{1}{2}\|\nabla U\|_H^2 = R(x_T, x_W),$$

with Euler operator $E = x_T \partial_T + x_W \partial_W$, kinetic energy $\|\nabla U\|_H^2 = (2/c)(\partial_T U)^2 + (3/c)(\partial_W U)^2$, and R a polynomial of degree ≤ 4 in $(\kappa, \mathfrak{C}, \mathfrak{Q})$. The gradient $dU : X \to T^*X$ embeds the deformation space as a Lagrangian submanifold of T^*X. Whether the genus expansion admits a quantisation with U as leading semiclassical term is open: it would require a quantum Hamiltonian whose WKB expansion has leading term U and whose quantum corrections reproduce the genus-g shadow amplitudes.

On the T-line the conformal weight forces the transverse momentum to vanish and the PDE collapses to the Riccati ODE. On the W-line the cubic shadow $\mathrm{Sh}_3 = 2x_T^3 + 3x_T x_W^2$ has nonzero normal derivative $(\partial_T \mathrm{Sh}_3)|_{x_T=0} = 3x_W^2$, and the first inter-channel coupling correction enters at degree 6:

$$\delta_6^{W\text{-line}} = \frac{576(5c+38)}{c^3(5c+22)^3}.$$

The T-line is autonomous; the W-line is not. The asymmetry is intrinsic to the weight difference between T (weight 2) and W (weight 3).

6.5. The four depth classes. The *shadow depth* $r_{\max}(\mathcal{A})$ is the smallest r such that $o_{r+1}(\mathcal{A}) = 0$ for all subsequent levels, or ∞ if the tower never terminates. The standard landscape partitions into four depth classes; no other values occur on any one-dimensional primary slice. The single-line dichotomy: the shadow metric $Q_L(t)$ is quadratic in (κ, α, S_4), and its factorisation type (perfect square or irreducible) determines whether the tower terminates.

Class	**Symbol**	$r_{\max}$	$\ell_3^{(0)}, \ell_4^{(0)}$	**Potential**	**Examples**
Gaussian	G	2	$0, 0$	quadratic	Heisenberg $\mathcal{H}_k$, free fermion ψ
Lie/tree	L	3	$\neq 0, 0$	cubic	Affine $\widehat{\mathfrak{sl}}_N$, currents
Contact	C	4	$0, \neq 0$	quartic	$\beta\gamma$, bc ghosts
Mixed	M	∞	$\neq 0, \neq 0$	non-polynomial	Virasoro, $\mathcal{W}_N$, $\mathcal{W}_\infty$

Gaussian ($r_{\max} = 2$): transferred brackets $\ell_k^{(0)}$ vanish for $k \geq 3$ on bar cohomology, so the tower terminates at κ. Two mechanisms: Heisenberg has no simple pole; the free fermion has a simple pole but fermionic antisymmetry forces $S_3 = 0$. Sewing is by Fredholm determinant (Heisenberg) or Pfaffian (fermion). **Lie/tree** ($r_{\max} = 3$): the OPE has a simple pole generating a Lie bracket, but $\ell_4^{(0)}$ vanishes on bar cohomology. **Contact** ($r_{\max} = 4$): the simple-pole bracket vanishes on bar cohomology, but the quaternary bracket survives; the first nontrivial shadow is $\mathfrak{Q}(\mathcal{A})$. **Mixed** ($r_{\max} = \infty$): both cubic and quartic shadows are nontrivial, and their Poisson bracket $\{\mathfrak{C}, \mathfrak{Q}\}_H$ generates a nonvanishing quintic, then sextic, and so on.

Shadow depth is not Koszulness. All four depth classes contain chirally Koszul algebras. Heisenberg (depth 2), affine at generic level (depth 3), $\beta\gamma$ (depth 4), and Virasoro (depth ∞) are all chirally Koszul. Shadow depth classifies the complexity of Koszul algebras, not Koszulness itself.

The *operadic complexity theorem* (Theorem **??**):

$$r_{\max}(\mathcal{A}) = A_\infty\text{-depth}(\mathcal{A}) = L_\infty\text{-formality level}(\mathcal{A}).$$

A_∞-depth is the largest r with $m_r \neq 0$ in the minimal A_∞ model; L_∞-formality level is the largest r with a nontrivial r-ary bracket on bar cohomology not removable by L_∞-isomorphism.

The depth gap theorem. The algebraic depth $d_{\mathrm{alg}}(\mathcal{A})$ takes values in $\{0, 1, 2, \infty\}$; the value 3 is impossible. This is a structural constraint, not an empirical observation. The proof:

At depth 3 in the shadow obstruction tower, the MC equation at degree 4 reads $d(S_4) + \frac{1}{2}\{S_3, S_3\}_H = 0$. If $d_{\mathrm{alg}} = 3$, then $S_r = 0$ for $r \geq 4$; in particular $S_4 = 0$. But the Poisson bracket $\{S_3, S_3\}_H$ of the cubic shadow with itself is computed by the shadow Lie algebra Jacobi identity: $\{S_3, S_3\}_H = 0$ iff the graded Jacobi identity holds on the cubic shadow viewed as a Lie bracket on bar cohomology. For a genuine Lie bracket (class L), Jacobi holds, so $S_4 = 0$ and the tower terminates at $r_{\max} = 3$. For a bracket that is *not* Lie (violates Jacobi), $\{S_3, S_3\}_H \neq 0$ and $S_4 \neq 0$, forcing $d_{\mathrm{alg}} \geq 4$. The MC relation at degree 5 then forces $\{S_3, S_4\}_H \neq 0$ and the induction continues: $d_{\mathrm{alg}} = \infty$.

The transition $3 \to 4$ is impossible: if $S_3 \neq 0$, then either Jacobi kills S_4 (and the tower stops at 3) or Jacobi fails and the tower runs to ∞. The value $d_{\mathrm{alg}} = 4$ requires $S_3 = 0$ and $S_4 \neq 0$, which is the contact class C ($\beta\gamma$). The four values $\{0, 1, 2, \infty\}$ map to: 0=trivial, 1=free field (linear differential), 2=Gaussian (class G), ∞=mixed (class M). The value 3 would require a non-Lie cubic shadow that magically satisfies Jacobi, which is a contradiction.

Depth classification: extended table.

Algebra	**Class**	$r_{\max}$	κ	S_3	S_4	Δ
$\mathcal{H}_k$	G	2	k	0	0	0
Free fermion ψ	G	2	$1/4$	0	0	0
$\widehat{\mathfrak{sl}}_{2,k}$	L	3	$3(k+2)/4$	$\neq 0$	0	0
$\widehat{\mathfrak{sl}}_{3,k}$	L	3	$4(k+3)/3$	$\neq 0$	0	0
$\beta\gamma$ (λ=1)	C	4	1	0	$\neq 0$	$\neq 0$
bc (λ=2)	C	4	$-26/2$	0	$\neq 0$	$\neq 0$
Vir_c	M	∞	$c/2$	$\neq 0$	$10/[c(5c+22)]$	$\neq 0$
$\mathcal{W}_3$	M	∞	$5c/6$	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_\infty$	M	∞	$c/2$	$\neq 0$	$\neq 0$	$\neq 0$

The discriminant $\Delta = 8\kappa S_4$ controls the single-line dichotomy: $\Delta = 0$ forces finite tower termination; $\Delta \neq 0$ forces an infinite tower. The discriminant is *linear* in κ (not quadratic); for $\kappa \neq 0$, $\Delta = 0$ iff $S_4 = 0$.

6.6. The genus–degree bigrading. The MC equation decomposes along two orthogonal axes. The *genus axis* extends $\Theta_{\mathcal{A}}$ from genus 0 to 1 to 2, controlled by the genus spectral sequence. The *degree axis* extends $\Theta^{\leq 2}_{\mathcal{A}} \to \Theta^{\leq 3}_{\mathcal{A}} \to \Theta^{\leq 4}_{\mathcal{A}}$, controlled by cyclic obstruction classes. At fixed degree 2 the genus tower is governed by κ alone (UNIFORM-WEIGHT); at fixed genus 0 each new shadow coefficient introduces independent data. The axes couple through the visibility formula $g_{\min}(S_r) = \lfloor r/2 \rfloor + 1$: S_r first contributes at genus $\lfloor r/2 \rfloor + 1$ and two new shadow coefficients enter at each genus.

6.7. Genus-two shells. Genus 2 is the first level where all three boundary strata interact:

$$\Theta^{(2)}_{\mathcal{A}} = \underbrace{\Theta^{(2)}_{\mathrm{loop}\circ\mathrm{loop}}}_{\text{iterated BV}} + \underbrace{\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}}}_{\text{clutching} \times \text{BV}} + \underbrace{\Theta^{(2)}_{\mathrm{pf}}}_{\text{planted-forest}} .$$

The iterated-BV term involves the figure-eight graph:

$$\Theta^{(2)}_{\mathrm{loop}^2} = \tfrac{1}{2}\,\mathrm{tr}_1\,\mathrm{tr}_2\big(P^{\otimes 2}_{\mathcal{A}} \circ m^{(h)}_{0,4} \circ \Delta^{\otimes 2}\big).$$

The separating-times-loop term:

$$\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}} = \mathrm{tr}_{\mathrm{ns}}\big(P_{\mathcal{A}} \circ \ell^{(0)}_2(\Theta^{(1,\bullet)}, \Theta^{(1,\bullet)})\big).$$

The planted-forest term is the genuinely logarithmic contribution, present only when cubic or quartic shadows are nontrivial.

Family-by-family structure:

	$\Theta^{(2)}_{\mathrm{loop}^2}$	$\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}}$	$\Theta^{(2)}_{\mathrm{pf}}$	**Depth**
Heisenberg $\mathcal{H}_k$	✓	--	--	G
Affine $\widehat{\mathfrak{g}}_k$	✓	✓	✓	L
$\beta\gamma$ system	✓	--	--	C
Virasoro Vir_c	✓	✓	✓	M
$\mathcal{W}_N$	✓	✓	✓	M

Heisenberg: only iterated BV. Affine: adds separating-times-loop. Virasoro and $\mathcal{W}_N$: all three active, with planted-forest carrying the quartic resonance. $\beta\gamma$: only iterated BV survives; the separating term vanishes because the Lie bracket is trivial on bar cohomology, and the planted-forest vanishes because the planted-forest coupling $\mu^{\mathrm{pf}}_{\beta\gamma} = 0$ (the planted-forest stratum requires coupling between cubic and quartic shadows, and the cubic $\mathfrak{C}_{\beta\gamma} = 0$). Contact termination via the iterated BV shell, not Gaussian collapse.

Heisenberg genus-2 computation. For $\mathcal{H}_k$, the only active term is the iterated BV $\Theta^{(2)}_{\mathrm{loop}^2}$, which contracts two self-loops from the figure-eight graph Γ_8. The bar cohomology is one-dimensional (spanned by the class x dual to the current J), and the propagator is $P_{\mathcal{H}_k} = 1/k$. The computation:

$$\Theta^{(2)}_{\mathcal{H}_k} = \frac{1}{2} \cdot \frac{1}{k^2} \cdot k^2 \cdot \lambda_2 = \frac{1}{2}\lambda_2.$$

Wait: the factor k^2 from two propagator insertions cancels against the $1/k^2$ from two applications of the inverse Killing form. The numerical coefficient $1/2$ is the $|\mathrm{Aut}(\Gamma_8)|^{-1} = 1/8$ times the combinatorial factor $\binom{4}{2} = 6$ (choosing which two of the four half-edges form the first loop) times a sign correction. Sanity check: $F_2(\mathcal{H}_k) = 7\kappa/5760 = 7k/5760$ from the $\hat{A}$-genus; the genus-2 amplitude should be proportional to $k \cdot \lambda_2$ (UNIFORM-WEIGHT), consistent.

Affine KM genus-2: the Jacobi mechanism. For $\widehat{\mathfrak{g}}_k$, the separating-times-loop term $\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}}$ involves a genus-1 vertex (contributing $\Theta^{(1)}$) connected by a separating edge to a genus-1 vertex (also contributing $\Theta^{(1)}$). The sewing propagator contracts one pair of legs. The Lie bracket enters through the cubic shadow S_3, which multiplies the genus-1 amplitude $\Theta^{(1)}$ at each vertex. The Jacobi identity constrains the result: at degree 4, the quartic shadow $S_4 = 0$ forces the planted-forest term to vanish independently (not by Jacobi, but by the vanishing of its coefficient). The genus-2 amplitude for $\widehat{\mathfrak{g}}_k$ is thus

$$\Theta^{(2)}_{\widehat{\mathfrak{g}}_k} = \Theta^{(2)}_{\mathrm{loop}^2} + \Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}},$$

with no planted-forest correction; the tower terminates cleanly.

Virasoro genus-2: all three shells active. For Vir_c at genus 2, all three terms contribute. The planted-forest correction carries the quartic contact invariant $Q^{\mathrm{contact}}_{\mathrm{Vir}} = 10/[c(5c + 22)]$ (the first nonlinear characteristic class, §5.3). The genus-2 amplitude is:

$$\Theta^{(2)}_{\mathrm{Vir}_c} = \left(\frac{c}{2}\right)^2 \cdot a_{\mathrm{loop}^2} \cdot \lambda_2 + \frac{c}{2} \cdot S_3 \cdot a_{\mathrm{sep}\circ\mathrm{loop}} \cdot \lambda_2 + \frac{10}{c(5c+22)} \cdot a_{\mathrm{pf}} \cdot \lambda_2,$$

where a_{loop^2}, $a_{\mathrm{sep}\circ\mathrm{loop}}$, a_{pf} are numerical combinatorial factors from the graph enumeration. The presence of all three terms is the signature of class M: the infinite tower contributes already at genus 2. At $c = 0$: $\kappa = 0$ and the first two terms vanish; the planted-forest term has a pole at $c = 0$ (the logarithmic threshold). At $c = -22/5$: the third term has a pole (the minimal model accumulation point $\mathcal{M}(2,5)$). At $c = 13$ (the self-dual point): all three terms are finite and nonzero.

6.8. The genus loop operator. The *genus loop operator* $\Lambda_P \colon \mathrm{Sym}^r(\bar{\mathcal{A}}^\vee)_{\mathrm{cyc}} \to \mathrm{Sym}^{r-2}(\bar{\mathcal{A}}^\vee)_{\mathrm{cyc}}$ contracts two legs with the propagator:

$$\Lambda_P(\phi)(a_1, \ldots, a_{r-2}) = \sum_i \phi(a_1, \ldots, a_{r-2}, e_i, e^i).$$

This raises genus by 1 and lowers degree by 2: the shadow-level incarnation of Δ_{ns}.

Explicit Virasoro computation. On the Virasoro primary line, the conformal-weight-2 bar cohomology is one-dimensional, spanned by the class x dual to T. The Killing form is $H = (c/2)x^2$, the propagator $P = 2/c$, and the quartic contact invariant is $Q^{\mathrm{contact}}_{\mathrm{Vir}} \cdot x^4$ with $Q^{\mathrm{contact}}_{\mathrm{Vir}} = 10/[c(5c + 22)]$. Applying Λ_P:

$$\Lambda_P\!\left(\frac{10}{c(5c+22)} \cdot x^4\right) = \binom{4}{2} \cdot \frac{2}{c} \cdot \frac{10}{c(5c+22)} \cdot x^2 = \frac{120}{c^2(5c+22)}x^2,$$

the genus-one Hessian correction. The loop ratio

$$\rho^{(1)}_{\mathrm{Vir}} = \frac{\Lambda_P(Q \cdot x^4)}{\kappa \cdot x^2} = \frac{120/[c^2(5c+22)]}{c/2} = \frac{240}{c^3(5c+22)}$$

is not determined by $\kappa = c/2$: it detects the quartic structure through its genus-one projection. Asymptotically $\rho^{(1)} \sim 48/c^4$ for $c \to \infty$.

6.9. Quintic forcing and infinite towers. The quintic obstruction determines whether the tower terminates at degree 4.

Virasoro. Both $\mathfrak{C}_{\mathrm{Vir}}$ and $Q^{\mathrm{contact}}_{\mathrm{Vir}}$ are nonzero. Their Poisson bracket

$$o^{(5)}_{\mathrm{Vir}} = \{\mathfrak{C}_{\mathrm{Vir}}, Q^{\mathrm{contact}}_{\mathrm{Vir}}\}_H = \frac{480}{c^2(5c+22)} x^5 \neq 0.$$

For all $c \neq 0, -22/5$, the Virasoro shadow tower does not terminate; $r_{\max}(\mathrm{Vir}_c) = \infty$. Same for every $\mathcal{W}_N$.

Heisenberg. $\mathfrak{C}_{\mathcal{H}} = 0$, $Q^{\mathrm{contact}}_{\mathcal{H}} = 0$, hence $o^{(5)} = 0$ and all higher obstructions vanish. $r_{\max} = 2$.

Affine. $\mathfrak{C}_{\widehat{\mathfrak{g}}} \neq 0$ but $Q^{\mathrm{contact}}_{\widehat{\mathfrak{g}}} = 0$ (quaternary bracket vanishes on bar cohomology by the PBW theorem). $o^{(5)} = 0$; $r_{\max} = 3$.

***βγ*.** $\mathfrak{C}_{\beta\gamma} = 0$ (Lie bracket trivial on bar cohomology) but $Q^{\mathrm{contact}}_{\beta\gamma} \neq 0$. $o^{(5)} = 0$; $r_{\max} = 4$.

	$\mathfrak{C}$	$\mathfrak{Q}$	$o^{(5)} = \{\mathfrak{C}, \mathfrak{Q}\}_H$	$r_{\max}$
Heisenberg	0	0	0	2
Affine	$\neq 0$	0	0	3
$\beta\gamma$	0	$\neq 0$	0	4
Virasoro	$\neq 0$	$\neq 0$	$\neq 0$	∞

Finite termination requires $\mathfrak{C} = 0$ or $\mathfrak{Q} = 0$; when both are nonzero, their bracket generates the quintic, and the induction continues without bound.

The shadow growth rate. For class M algebras, the Taylor coefficients S_r of $\sqrt{Q_L}$ grow as $S_r \sim A \cdot \rho(\mathcal{A})^r \cdot r^{-5/2} \cos(r\theta + \varphi)$ with shadow growth rate $\rho(\mathcal{A}) = \sqrt{9\alpha^2 + 2\Delta}/(2|\kappa|)$, a continuous invariant. Self-dual Virasoro ($c = 13$) has $\rho \approx 0.467$. The critical cubic $5c^3 + 22c^2 - 180c - 872 = 0$ has root $c^* \approx 6.125$: the tower converges for $c > c^*$ and diverges for $c < c^*$.

The shadow partition function. The double sum $Z^{\mathrm{sh}}_{\mathcal{A}} = \sum_{g,r} \ {}^{2g-2} Z^{(r)}_g$ converges absolutely. Genus decay at rate $1/(2\pi)^{2g}$ (Faber–Pandharipande); degree decay at rate $\rho^r r^{-5/2}$ (Darboux analysis). The string partition function diverges as $(2g-2)!$. For Virasoro at $c = 26$, $\rho \approx 0.234$ and Z^{sh} has positive radius of convergence in both and degree.

6.10. Independent sum factorisation. For a direct sum $L = L_1 \oplus L_2$ with vanishing mixed OPE, all shadow invariants separate:

$$\begin{aligned}
\kappa(L_1 \oplus L_2) &= \kappa(L_1) + \kappa(L_2), \\
T^{\mathrm{br}}(L_1 \oplus L_2) &= T^{\mathrm{br}}(L_1) \oplus T^{\mathrm{br}}(L_2), \\
\Delta_{L_1 \oplus L_2}(t) &= \Delta_{L_1}(t) \cdot \Delta_{L_2}(t), \\
R^{\mathrm{mod}}_4(L_1 \oplus L_2) &= R^{\mathrm{mod}}_4(L_1) + R^{\mathrm{mod}}_4(L_2).
\end{aligned}$$

The full MC element factorises: $\Theta_{L_1 \oplus L_2} = \Theta_{L_1} \hat{\otimes} 1 + 1 \hat{\otimes} \Theta_{L_2}$, and $Z_{L_1 \oplus L_2} = Z_{L_1} \cdot Z_{L_2}$.

Every algebra in the standard landscape is either primitive (Heisenberg, Virasoro, $\beta\gamma$) or built from primitive pieces (affine = current algebra over simple Lie, lattice = exponential extension of Heisenberg, $\mathcal{W}_N$ = quantum Drinfeld–Sokolov reduction of affine). Drinfeld–Sokolov reduction is the *shadow-depth escalator* (Principle ??): it sends current algebras (class L, $r_{\max} = 3$, finite towers) to $\mathcal{W}$-algebras (class M, $r_{\max} = \infty$, infinite towers). The nonlinearity of the Dirac bracket on the Slodowy slice creates the quartic and all higher shadows.

Factorisation proof. The factorisation $\Theta_{L_1 \oplus L_2} = \Theta_{L_1} \hat{\otimes} 1 + 1 \hat{\otimes} \Theta_{L_2}$ follows from the tensor structure of the bar complex: $\bar{B}(L_1 \oplus L_2) \cong \bar{B}(L_1) \hat{\otimes} \bar{B}(L_2)$ as cofree coalgebras, where $\hat{\otimes}$ is the completed tensor product respecting the depth filtration. The modular convolution algebra decomposes as a direct sum $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[L_1 \oplus L_2] \cong \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[L_1] \oplus \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[L_2]$ when the mixed OPE vanishes, and the MC equation decouples. The kappa additivity $\kappa(L_1 \oplus L_2) = \kappa(L_1) + \kappa(L_2)$ is the trace of the decomposed curvature; the spectral discriminant product $\Delta_{L_1 \oplus L_2} = \Delta_{L_1} \cdot \Delta_{L_2}$ follows from the block-diagonal structure of the branch transport operator.

Heisenberg example. The rank-d Heisenberg at level k decomposes as $\mathcal{H}_k^{\oplus d} = \mathcal{H}_k \oplus \cdots \oplus \mathcal{H}_k$ (d copies). By independent sum factorisation: $\kappa(\mathcal{H}_k^{\oplus d}) = d \cdot k$, $\Delta_{\mathcal{H}_k^{\oplus d}}(t) = \Delta_{\mathcal{H}_k}(t)^d = 1$ (since $\Delta_{\mathcal{H}_k} = 1$ for each copy), $Z(\mathcal{H}_k^{\oplus d}) = Z(\mathcal{H}_k)^d = \eta(\tau)^{-dk}$. The shadow depth remains class G for all d: the direct sum of Gaussian algebras is Gaussian.

The shadow-depth escalator: Drinfeld–Sokolov reduction. Why does DS reduction escalate shadow depth from 3 (class L) to ∞ (class M)? The mechanism: DS reduction of $\widehat{\mathfrak{g}}_k$ by a nilpotent element f produces the $\mathcal{W}$-algebra $\mathcal{W}(\mathfrak{g}, f, k)$ via the BRST complex

$$\mathcal{W}(\mathfrak{g}, f, k) = H^0_{\mathrm{BRST}}(\widehat{\mathfrak{g}}_k \otimes \mathcal{F}^{\mathrm{gh}}),$$

where $\mathcal{F}^{\mathrm{gh}}$ is the ghost system of central charge $c_{\mathrm{gh}} = c(\widehat{\mathfrak{g}}_k) - c(\mathcal{W}(\mathfrak{g}, f, k))$. The ghost system is of bc-type with conformal weight λ determined by the $\mathfrak{sl}_2$-grading on $\mathfrak{g}$ induced by f.

The quartic shadow of the $\mathcal{W}$-algebra receives contributions from the nonlinear terms in the DS reduction: the Dirac bracket on the Slodowy slice $\mathcal{S}_f \subset \mathfrak{g}^*$ is nonlinear (unless f is principal and $\mathfrak{g} = \mathfrak{sl}_2$, i.e., $\mathcal{W}_2 = \mathrm{Vir}$). The cubic terms in the Dirac bracket produce a nonzero S_3; the quartic terms produce a nonzero S_4; the depth gap theorem (§6.5) then forces $r_{\max} = \infty$. The "escalation" is the passage from the linear Lie bracket of $\mathfrak{g}$ (which satisfies Jacobi and kills S_4) to the nonlinear Dirac bracket of $\mathcal{S}_f$ (which does not satisfy Jacobi in degree 4 and generates an infinite tower).

For $\mathfrak{sl}_2 \to \mathrm{Vir}$: the DS reduction sends $\kappa(\widehat{\mathfrak{sl}}_{2,k}) = 3(k+2)/4$ to $\kappa(\mathrm{Vir}_c) = c/2$ with $c = 1 - 6(k+1)^2/(k+2)$. The depth escalates from $r_{\max} = 3$ (affine $\mathfrak{sl}_2$) to $r_{\max} = \infty$ (Virasoro) because the Slodowy slice for $\mathfrak{sl}_2$ with principal nilpotent is zero-dimensional (no Dirac bracket), but the BRST complex introduces the bc-ghost whose quartic OPE gives $S_4 \neq 0$.

For $\mathfrak{sl}_3 \to \mathcal{W}_3$: the Slodowy slice is one-dimensional, the Dirac bracket is nonlinear, and the quartic shadow is $S_4(\mathcal{W}_3) \neq 0$. The depth escalates from 3 to ∞.

6.11. The representation-theoretic perspective. The categorical logarithm $\eta_{ij} = d\log(z_i - z_j)$ of §1 is the integral kernel of a tautology: $D^2_{\mathcal{A}} = 0$ on the modular bar complex. Three structure theorems emerge. *Algebraicity*: on each primary slice L, $H(t)^2 = t^4 Q_L(t)$ with Q_L quadratic in κ, α, S_4. *Formality*: the genus-0 shadow obstruction tower is the L_∞ formality obstruction tower; the four depth classes G/L/C/M are the corresponding formality types, while positive-genus corrections form a separate quantum layer. *Complementarity*: for a Koszul pair $(\mathcal{A}, \mathcal{A}^!)$, the summands $\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^!)$ are Lagrangian (Theorem C).

The discriminant $\Delta = 8\kappa S_4$ governs the single-line dichotomy: $\Delta = 0$ forces tower termination (L_∞-formality); $\Delta \neq 0$ forces an infinite tower (intrinsic non-formality). The contact class C ($\beta\gamma$) escapes via stratum separation. No invariant of the classical bar complex of a graded algebra sees Δ: it is visible only on curves of positive genus, where the Arnold relation acquires a defect.

Classical Koszul duality (Priddy, Beilinson–Ginzburg–Soergel) embeds into the genus 0, degree 2, $\Delta = 0$ stratum via formal-disk restriction (the formal, class-G case); the E_1/E_∞ distinction and Fulton–MacPherson geometry are already absent over a bare point. The R-matrix of quantum groups arises as the genus-0 shadow $\mathrm{Sh}_{0,2}(\Theta_{\widehat{\mathfrak{g}}_k})$ (§7.7). At the critical level $k = -h^\vee$, the scalar class vanishes ($\kappa = 0$) and bar cohomology recovers the Feigin–Frenkel center $\mathrm{Fun}(\mathrm{Op}_{{}^L\mathfrak{g}})$; the Langlands programme lives on this orthogonal axis, not in the MC element but in the cohomology that emerges on the uncurved scalar locus.

Summary: the shadow obstruction tower in one table. The entire content of this section is captured by the following table, which records the shadow data for each standard family:

Family	**Class**	$r_{\max}$	κ	S_3	S_4	Δ	**Mechanism**
$\mathcal{H}_k$	G	2	k	0	0	0	no simple pole
Free fermion	G	2	$1/4$	0	0	0	antisymmetry
$\widehat{\mathfrak{sl}}_{2,k}$	L	3	$3(k+2)/4$	$\neq 0$	0	0	Jacobi
$\widehat{\mathfrak{g}}_k$ (any)	L	3	$\frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}$	$\neq 0$	0	0	Jacobi
$\beta\gamma$	C	4	1	0	$\neq 0$	$\neq 0$	no Lie bracket
bc (λ=2)	C	4	-13	0	$\neq 0$	$\neq 0$	no Lie bracket
Vir_c	M	∞	$c/2$	$\neq 0$	$\frac{10}{c(5c+22)}$	$\neq 0$	Jacobi fails
$\mathcal{W}_3$	M	∞	$5c/6$	$\neq 0$	$\neq 0$	$\neq 0$	DS escalation
$\mathcal{W}_N$ ($N \geq 3$)	M	∞	$c(H_N-1)$	$\neq 0$	$\neq 0$	$\neq 0$	DS escalation
Lattice V_Λ	G	2	$\mathrm{rk}(\Lambda)$	0	0	0	no simple pole

Every row is computable from the OPE data of the corresponding algebra via the bar machine of §§1–5. The depth class is a theorem about each family (proved by explicit computation of the cubic and quartic shadows), not a classification assumption. The depth gap theorem (§6.5) constrains the values of $r_{\max}$ to $\{2, 3, 4, \infty\}$; no other values can occur.

The four axes of the programme. The shadow obstruction tower organises the programme along four orthogonal axes:

(i) *Genus axis*: extends the MC element from $g = 0$ to $g = 1$ to $g = 2$, controlled by the genus spectral sequence.
(ii) *Degree axis*: extends the truncation $\Theta^{\leq 2} \to \Theta^{\leq 3} \to \Theta^{\leq 4} \to \cdots$, controlled by cyclic obstruction classes.
(iii) E_n *axis*: lifts from E_∞ (symmetric bar) to E_1 (ordered bar), adding the R-matrix and quantum group data. The E_1 structure is primitive; the symmetric bar is the coinvariant shadow.
(iv) *Family axis*: parametrises the standard landscape by the OPE data (central charge, level, conformal weights). The shadow tower is a universal machine; the family axis selects the input.

Sections 1–6 develop axes (i)–(ii). Section 7 runs the family axis. The E_n axis runs throughout, with the E_1 ordered bar as the primary construction at every stage.

7. The standard landscape: E_1 chain-level census

The bar machine of Sections 1–6 accepts any chiral algebra as input. For each standard family, we construct the ordered bar $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ at the chain level, compute the differential explicitly at low tensor degrees, extract the r-matrix and bar cohomology, and then average to recover κ and the symmetric shadow data. The E_1 ordered construction is primary; every symmetric result below is derived from it by Σ_n-coinvariant projection.

7.0. Three-tier R-matrix classification. The genus-0 binary shadow $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ sorts chiral algebras into three tiers:

- *Tier (a): scalar R-matrix.* The Heisenberg algebra $\mathcal{H}_k$ and lattice extensions. The spectral R-matrix is scalar, $R(z) = \exp(k/z)$, and the Yang–Baxter equation is trivial. Shadow class **G**.
- *Tier (b): matrix R-matrix from OPE poles.* Affine Kac–Moody $\widehat{\mathfrak{g}}_k$, Virasoro, principal $\mathcal{W}_N$, $\beta\gamma$, non-principal $\mathcal{W}$-algebras. The classical r-matrix is $k\Omega/z$ (affine) or $(c/2)/z^3 + 2T/z$ (Virasoro) or the DS-reduced analogue. The level prefix is load-bearing: at $k = 0$, the affine r-matrix vanishes. Shadow class **L**, **C**, or **M** depending on the depth of the OPE.
- *Tier (c): genuinely E_1.* The dg-shifted Yangian $Y^{\mathrm{dg}}(\mathfrak{g})$, Etingof–Kazhdan quantum vertex algebras, line-operator atoms of 3d HT theories. The R-matrix carries genuine topological direction data: nonsymmetric braiding, meromorphic poles of higher order, RTT relations that do not factor through a commutative vertex algebra. Every standard VOA is E_∞ (local); E_1 primacy requires passing to the tier (c) atoms.

Tier (a) is the free-field archetype, tier (b) supplies the standard landscape, tier (c) is the genuine E_1 frontier occupied in Volumes II and III.

The seven families below exhaust the tier (a)/(b) landscape. Each is presented in the E_1 order: ordered bar differential, bar cohomology, R-matrix, then averaged invariants (κ, complementarity, shadow class). No two families share all invariants; taken together they realise every depth class **G**/**L**/**C**/**M** and every genus-2 shell profile.

Master table of E_1 bar cohomology.

Family	$r(z)$	$\dim H^n(\bar{B})$	Growth	GF type	κ	Class
$\mathcal{H}_k$	k/z	$\delta_{n,1}$	0	rational	k	**G**
$\widehat{\mathfrak{sl}}_2$	$k\Omega/z$	$2n+1$	linear	rational	$\frac{3(k+2)}{4}$	**L**
$\widehat{\mathfrak{sl}}_3$	$k\Omega/z$	$\{8, 36, 204, \ldots\}$	exponential	rational	$\frac{4(k+3)}{3}$	**L**
Vir_c	$\frac{c/2}{z^3}+\frac{2T}{z}$	$\{1, 2, 5, \ldots\}$	3^n	algebraic	$c/2$	**M**
$\mathcal{W}_{3,c}$	$r_c^{W_3}(z)$	$\{2, 5, 16, \ldots\}$	3^n	algebraic	$5c/6$	**M**
$\beta\gamma$	$r_\lambda(z)$	$\{2, 4, 10, \ldots\}$	3^n	algebraic	$6\lambda^2-6\lambda+1$	**C**
V_Λ	N/z	$\{N, \ldots\}$	sub-exp	transcendental	N	**G**

7.1. Heisenberg. The Gaussian archetype: shadow depth 2, tower terminates at degree 2, every higher-genus invariant is determined by κ alone. One bosonic field α of conformal weight 1, $\alpha(z)\alpha(w) \sim k/(z-w)^2$, $c = 1$.

Ordered bar differential. The ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{H}_k) = T^c(s^{-1}\overline{\mathcal{H}_k})$ has a single generator $s^{-1}\alpha$ in the augmentation ideal.

Degree 2: the residue differential extracts the unique OPE coefficient:

$$d_{\mathrm{res}}\big([s^{-1}\alpha|s^{-1}\alpha] \otimes \eta_{12}\big) \;=\; k \cdot \mathbf{1}.$$

The double-pole OPE $k/(z-w)^2$ drops by one order under $d\log$ absorption, producing a simple-pole residue.

Degree 3: the collision at any boundary stratum extracts $\alpha_{(1)}\alpha = k \cdot \mathbf{1}$, leaving $n-2$ copies of α; since $\mathbf{1}_{(p)}\alpha = 0$ for $p \geq 0$, no further contraction occurs. The Arnold relation kills the form contribution independently. The degree-3 differential vanishes identically. All higher degrees vanish by the same mechanism.

Bar cohomology. The complex is quadratic: all structure lives at tensor degree 2.

$$H^n(\bar{B}(\mathcal{H}_k)) \;=\; \begin{cases} \mathbb{C}\cdot\alpha^* & n=1,\\ 0 & n\geq 2.\end{cases}$$

Conformal-weight-graded bar Hilbert series: rational. Partition-graded generating function: transcendental, $\prod_{m\geq 1}(1-x^m)^{-1}$ (Fock-space grading). The PBW spectral sequence collapses at E_1. $\mathcal{H}_k$ is chirally Koszul.

R**-matrix.** $r^{\mathcal{H}}(z)=k/z$, scalar (Tier (a)). Spectral R-matrix $R(z)=\exp(k/z)$; Yang–Baxter trivial.

Averaging. $r(z)=k/z$ is already Σ_2-invariant: $\mathrm{av}(r(z))=k=\kappa(\mathcal{H}_k)$. Averaging loses nothing. Koszul dual: $\mathcal{H}_k^!=(\mathrm{Sym}^{\mathrm{ch}}(V^*), m_0=-k\omega)$ (curved). Complementarity (free-field branch, sum = 0; Virasoro branch gives 13): $\kappa(\mathcal{H}_k)+\kappa(\mathcal{H}_k^!)=k+(-k)=0$. Shadow class **G**; genus-2 shell: loop∘loop only.

Higher-genus. The $\hat{A}$-genus controls every F_g:

$$\textstyle\sum_{g\geq 1} F_g\,x^{2g} \;=\; k\big(\hat{A}(ix)-1\big) \;=\; k\Big(\frac{x/2}{\sin(x/2)}-1\Big),$$

so $F_1=k/24$, $F_2=7k/5760$, $F_3=31k/967680$. The series converges exponentially, $|F_g|\sim 2|k|/(2\pi)^{2g}$.

7.2. Affine Kac–Moody. The Lie/tree archetype: shadow depth 3, cubic shadow is the Lie bracket, Jacobi kills everything higher. For simple $\mathfrak{g}$ of rank ℓ and dual Coxeter $h^\vee$, currents J^a of weight 1 with $J^a(z)J^b(w)\sim f^{ab}{}_cJ^c(w)/(z-w)+k\kappa^{ab}/(z-w)^2$. Two pole orders: simple pole carries the bracket, double pole carries the level.

Ordered bar differential.

Degree 2: the differential decomposes into two components, one per pole order:

$$d_{\bar{B}}\big([s^{-1}J^a|s^{-1}J^b]\otimes\eta_{12}\big) \;=\; \underbrace{f^{ab}{}_c\,[s^{-1}J^c]}_{\text{CE (simple pole)}} \;+\; \underbrace{k\kappa^{ab}\cdot\mathbf{1}}_{\text{curvature (double pole)}}\;.$$

The Chevalley–Eilenberg component comes from the Lie bracket $J^a_{(0)}J^b=f^{ab}{}_cJ^c$; the curvature component from the invariant form $J^a_{(1)}J^b=k\kappa^{ab}\cdot\mathbf{1}$.

Degree 3 *($\mathfrak{sl}_3$)*: the bar complex $\bar{B}^{\mathrm{ord}}_3=\mathfrak{g}^{\otimes 3}\otimes\Omega^2(\overline{\mathrm{FM}}_3)$ has dimension $8^3\times 2=1024$. The Serre element

$$S \;=\; [E_1|E_1|E_2]-2[E_1|E_2|E_1]+[E_2|E_1|E_1]$$

is a bar 3-cocycle: $d_{\bar{B}}(S\otimes\eta_{12}\wedge\eta_{13})=0$. This is the Serre relation $(\mathrm{ad}\,E_1)^2(E_2)=0$ manifesting at degree 3. The two independent Serre relations (for $\{i,j\}=\{1,2\}$) generate $H^3(\bar{B}(\widehat{\mathfrak{sl}}_{3,k}))$ at generic level.

Degree 4: for $\mathfrak{sl}_3$, $\dim\bar{B}^{\mathrm{ord}}_4=4096\times 6=24576$. The bracket differential $d\colon\bar{B}_4\to\bar{B}_3$ is surjective (rank 1024); degree-4 cohomology is conjectured to be 1352-dimensional. Full Borcherds (not Jacobi alone) is required for $d^2_{\bar{B}}=0$.

Bar cohomology.

$\mathfrak{g}$	$\dim H^n$	Growth	GF	Source
$\mathfrak{sl}_2$	$2n+1$	linear	$x(3-x)/(1-x)^2$	Garland–Lepowsky
$\mathfrak{sl}_3$	$\{8,36,204,\ldots\}$	exponential	$4x(2-13x-2x^2)/((1-8x)(1-3x-x^2))$	Serre at deg 3
$\mathfrak{g}$ (general)	polynomial	$\leq$ exponential	D-finite	PBW collapse at E_2

For $\widehat{\mathfrak{sl}}_2$, the linear growth $\dim H^n = 2n + 1$ is the Garland–Lepowsky concentration: bar cohomology is generated in degree 1 by the 3-dimensional adjoint. For $\widehat{\mathfrak{sl}}_3$, the rational generating function with denominator $(1 - 8x)(1 - 3x - x^2)$ reflects the Serre relations; PBW concentration at all genera is proved.

R-matrix. $r^{\mathrm{KM}}(z) = k\,\Omega/z$ with $\Omega = \sum_a J^a \otimes J_a$ (Tier (b)); at $k = 0$ the r-matrix vanishes, at $k = -h^\vee$ it is critical. The CYBE is the Arnold relation on $\overline{\mathrm{FM}}_3(\mathbb{C})$ evaluated on the Casimir.

Averaging. Averaging collapses the Casimir tensor to its trace, with Sugawara shift:

$$\mathrm{av}(k\,\Omega/z) = \frac{k \dim \mathfrak{g}}{2h^\vee} = \kappa_{\mathrm{dp}}, \qquad \kappa(\widehat{\mathfrak{g}}_k) = \kappa_{\mathrm{dp}} + \frac{\dim \mathfrak{g}}{2} = \frac{(k + h^\vee)\dim \mathfrak{g}}{2h^\vee}.$$

The $\dim(\mathfrak{g})/2$ is the simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$. The Casimir tensor structure, invisible to κ, is the data that builds the Yangian $Y(\mathfrak{g})$.

Koszul dual: $\widehat{\mathfrak{g}}_k^! = \widehat{\mathfrak{g}}_{-k-2h^\vee}$ (Feigin–Frenkel duality, $k + h^\vee \mapsto -(k + h^\vee)$). Complementarity (KM branch, sum = 0; Virasoro branch gives 13): $\kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}_k^!) = 0$. Shadow class $\mathbf{L}$; genus-2 shells: loop∘loop and sep∘loop active, planted-forest absent.

Critical level. At $k = -h^\vee$: $\kappa = 0$, the Sugawara construction is undefined, and the Feigin–Frenkel centre $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee}) \cong \mathrm{Fun}(\mathrm{Op}_{^L\mathfrak{g}}(D))$ is a commutative VA of $^L\mathfrak{g}$-opers on the formal disk. Critical-level chiral Koszul duality is bar-level geometric Langlands.

7.3. Virasoro.

The mixed archetype: shadow depth infinity, tower never terminates. One generator T of weight 2, $T(z)T(w) \sim (c/2)/(z-w)^4 + 2T/(z-w)^2 + \partial T/(z-w)$. Three pole orders. The self-coupling $T_{(1)}T = 2T$ drives all nonlinear complexity.

Ordered bar differential.

Degree 2: the bar differential mixes both pole channels:

$$d_{\bar{B}}\big([s^{-1}T|s^{-1}T] \otimes \eta_{12}\big) = \underbrace{\tfrac{c}{2}\cdot \mathbf{1}}_{\substack{\text{curvature}\\ \text{(order-4 pole)}}} + \underbrace{2\,[s^{-1}T]}_{\substack{\text{self-coupling}\\ \text{(order-2 pole)}}} + \underbrace{[\partial T]}_{\substack{\text{translation}\\ \text{(order-1 pole)}}}.$$

Three contributions from three OPE pole orders, each reduced by one under $d\log$ absorption: order-$4 \to 3$, $2 \to 1$, $1 \to 0$. The curvature element $m_0 = c/2 \in \bar{B}^0 = \mathbb{C}$ is the bar-dual of the quartic OPE pole.

Degree 3: the ordered bar $\bar{B}_3 = (s^{-1}T)^{\otimes 3} \otimes \Omega^2(\overline{\mathrm{FM}}_3)$ has differential

$$\begin{aligned} &d_{\bar{B}}\big([s^{-1}T|s^{-1}T|s^{-1}T] \otimes \eta_{12} \wedge \eta_{13}\big) \\ &\qquad = \tfrac{c}{2}\big([s^{-1}T] \otimes (\eta_{12} + \eta_{13} + \eta_{23})\big) + \text{self-coupling terms.} \end{aligned}$$

The vacuum term $\frac{c}{2}[s^{-1}T] \otimes (\eta_{12} + \eta_{13} + \eta_{23})$ vanishes by the Arnold relation: three boundary divisors, three forms, zero sum. The surviving self-coupling terms $[\partial T|T] \otimes \eta_{13} + [T|\partial T] \otimes (\eta_{12} + \eta_{23})$ are nontrivial: this is where infinite depth begins.

Bar cohomology. The bar cohomology of Vir_c exhibits exponential growth 3^n, in contrast to the linear growth of Kac–Moody. The *Motzkin path model*: $\dim H^n(\bar{B}(\mathrm{Vir}_c)) = M(n{+}1) - M(n)$, where $M(n)$ is the n-th Motzkin number (paths in $\mathbb{Z}_{\geq 0}$ of length n with steps +1, 0, −1). The generating function is algebraic:

$$\sum_{n\geq 1} \dim H^n \cdot x^n = x + 2x^2 + 5x^3 + 12x^4 + 30x^5 + \cdots.$$

The conformal-weight-graded spectral sequence collapses at E_2; PBW concentration holds.

R-matrix. $r_c(z) = (c/2)/z^3 + 2T/z$: the quartic and quadratic OPE poles drop by one order each under d log absorption. Tier (b), class **M**.

Averaging. $\mathrm{av}(r_c(z)) = c/2 = \kappa(\mathrm{Vir}_c)$. The stress-tensor coupling $2T/z$, invisible to κ, drives the infinite shadow tower. Koszul dual: $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$ (self-dual at $c = 13$). Complementarity (Virasoro branch, sum = 13; KM branch gives 0):

$$\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_c^!) \;=\; \tfrac{c}{2} + \tfrac{26-c}{2} \;=\; 13.$$

Shadow class **M**; genus-2: all three shells active including planted-forest (first family to see log-FM boundary strata of $\overline{\mathcal{M}}_{2,n}$).

Quartic shadow and infinite tower. The quartic contact shadow

$$Q^{\mathrm{contact}}_{\mathrm{Vir}} \;=\; \frac{10}{c(5c+22)},$$

has poles at $c = 0$ and $c = -22/5$ (the $(2,5)$ minimal model). The quintic obstruction $o^{(5)} = \{C_{\mathrm{Vir}}, Q^{\mathrm{contact}}\}_H = 480/[c^2(5c+22)]\,x^5$ is nonzero: no finite truncation captures Virasoro.

Critical dimension $c = 26$. The shadow connection $\nabla^{\mathrm{hol}}_c = d - (c/2)\omega_g - [10/(c(5c+22))]\delta_4 - \cdots$ is flat for every c by the MC equation. The critical dimension arises from the *dual* curvature vanishing: $\kappa(\mathrm{Vir}_{26-c}) = (26-c)/2 = 0$ forces $c = 26$. At $c = 26$ the Koszul dual Vir_0 is uncurved; the algebra itself retains $\kappa = 13$.

Polyakov identification. The Polyakov formula $\log(\det'\Delta_{g_1}/\det'\Delta_{g_0}) = -(1/6\pi)\int(|\nabla\sigma|^2 + R\sigma)$ is the degree-2 shadow evaluation at $\kappa = 1$; the higher-degree corrections δ_4, δ_H are inaccessible to the free-field path integral.

7.4. $\mathcal{W}_3$. First multi-generator family: two generators T (weight 2) and W (weight 3). The multi-weight structure produces cross-channel mixing that the Virasoro single-generator story cannot see.

Ordered bar differential. The TT OPE contributes the same three-pole structure as Virasoro. The WW OPE introduces a sixth-order pole:

$$\begin{aligned} W(z)W(w) \sim{}& \frac{c/3}{(z-w)^6} + \frac{2T}{(z-w)^4} + \frac{\partial T}{(z-w)^3} \\ &+ \frac{1}{(z-w)^2}\Big[\tfrac{16}{22+5c}\Lambda + \tfrac{3}{10}\partial^2 T\Big] + \cdots, \end{aligned}$$

where $\Lambda =\!: TT : -(3/10)\partial^2 T$ is the weight-4 quasi-primary composite field. At degree 2 the ordered bar differential on $[s^{-1}W|s^{-1}W]$ produces five terms from five pole orders (order $6 \to 5$, $4 \to 3$, $3 \to 2$, $2 \to 1$, $1 \to 0$ under d log absorption), including the composite field Λ at the second-order pole. The coefficient $\alpha(c) = 16/(22+5c)$ is fixed uniquely by quasi-primarity ($L_1\Lambda = 0$) and the Jacobi identity. At $c = -22/5$, Λ becomes null and $\mathcal{W}_3$ is not finitely generated by T, W alone.

Cross-channel at degree 3: the degree-3 bar differential mixes TT, TW, and WW channels. The TW cross-channel produces terms involving Λ that have no analogue in the single-generator Virasoro; these drive the multi-weight genus-2 correction.

Bar cohomology. Bar cohomology growth rate 3^n (same as Virasoro: the self-coupling $T_{(1)}T = 2T$ generates the same exponential mechanism). The generating function is algebraic. The two-generator PBW spectral sequence collapses at E_2 with a 2×2 spectral discriminant: $\Delta_{\mathcal{W}_3}(x) = (1 - \frac{c}{2}x)(1 - \frac{c}{3}x)$, branch points $2/c$ and $3/c$.

R-matrix. The classical r-matrix is the Casimir piece plus T- and W-mediated corrections; the per-channel eigenvalues are $\kappa_T = c/2$ (from T) and $\kappa_W = c/3$ (from W). Tier (b), class **M**.

Averaging. $\mathrm{av}(r_c^{W_3}(z)) = c/2 + c/3 = 5c/6 = \kappa(\mathcal{W}_3)$: the weight-2 generator contributes $c/2$, the weight-3 generator contributes $c/3$, additively. Koszul dual: $\mathcal{W}_3^! = \mathcal{W}_{3,100-c}$ (self-dual at $c = 50$).

$$\kappa(\mathcal{W}_{3,c}) + \kappa(\mathcal{W}_{3,100-c}) \;=\; \tfrac{250}{3}.$$

Shadow class **M**; genus-2: all three shells active, planted-forest sensitive to $WW\Lambda$ channel.

Genus-2 cross-channel. (all-weight, with cross-channel correction.)

$$F_2(\mathcal{W}_3) \;=\; \underbrace{\tfrac{7c}{6912}}_{\kappa\cdot\lambda_2^{\mathrm{FP}}} \;+\; \underbrace{\tfrac{c+204}{16c}}_{\delta F_2^{\mathrm{cross}}},$$

with $\delta F_2^{\mathrm{cross}}$ computed graph-by-graph over the seven genus-2 stable graphs. The scalar uniform-weight formula fails at $g \geq 2$ for multi-weight algebras. A universal closed form for principal $\mathcal{W}_N$:

$$\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) \;=\; \frac{(N-2)(N+3)}{96} + \frac{(N-2)(3N^3+14N^2+22N+33)}{24c}.$$

Holographic datum. $\mathcal{H}(\mathcal{W}_{3,c}) = (\mathcal{W}_{3,c}, \mathcal{W}_{3,100-c}, C_c^{W_3}, r_c^{W_3}(z), \Theta_c, \nabla_c^{\mathrm{hol}})$.

7.5. $\beta\gamma$ system. The contact archetype: shadow depth 4, first nonlinearity at degree 4 (not 3). Fields β of weight λ, γ of weight $1-\lambda$, with $\beta(z)\gamma(w) \sim 1/(z-w)$ and both self-OPEs vanishing. Simple pole only.

Ordered bar differential. The ordered bar has two generators $s^{-1}\beta, s^{-1}\gamma$ in the augmentation ideal. At degree 2:

$$d_{\bar B}\big([s^{-1}\beta|s^{-1}\gamma]\otimes\eta_{12}\big) \;=\; \mathbf{1}, \qquad d_{\bar B}\big([s^{-1}\beta|s^{-1}\beta]\otimes\eta_{12}\big) \;=\; 0,$$

since $\beta_{(0)}\gamma = \mathbf{1}$ (simple pole) and $\beta_{(0)}\beta = 0$ (self-OPE vanishes). The simple-pole-only structure means $d\log$ absorption produces a degree-0 residue: the OPE pole is order 1, absorbed to order 0.

At degree 3: the cubic shadow $\ell_3^{(0)}$ vanishes because β and γ carry different conformal weights; the U(1) ghost-number grading forces all odd-degree ordered shadows to zero. At degree 4: the quartic $\ell_4^{(0)} \neq 0$ from $\langle\beta\gamma\beta\gamma\rangle$ cross-contractions, but the quartic contact invariant $\mu_{\beta\gamma} = \langle\eta, m_3(\eta,\eta,\eta)\rangle = 0$ by ghost-number rigidity. The tower terminates: depth 4, not by Lie cancellation (Jacobi) but by an abelian grading mechanism.

Bar cohomology. Growth rate 3^n (same exponential class as Virasoro). Generating function: algebraic. The two-generator PBW spectral sequence collapses at E_2. Central charge: $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$; at $\lambda = 1$, $c = 2$. Partner: the fermionic bc system has $c_{bc}(\lambda) = 1 - 3(2\lambda-1)^2$, and $c_{\beta\gamma} + c_{bc} = 0$ (verify: $\lambda = 1$ gives $2 + (-2) = 0$).

R-matrix. Proportional to the identity Casimir on the $\beta\gamma$-state space; the spectral R-matrix factors through the ghost-number grading. Tier (b), class **C**.

Averaging. $\kappa(\beta\gamma) = c_{\beta\gamma}/2 = 6\lambda^2 - 6\lambda + 1$. Koszul dual: $(\beta\gamma)^! = bc$ (bosonic/fermionic exchange). Complementarity (free-field branch, sum = 0; Virasoro branch gives 13): $\kappa(\beta\gamma) + \kappa(bc) = 0$. Shadow class **C**; genus-2: loop∘loop only (contact termination, distinct from Gaussian collapse).

7.5b. Fermionic bc system. The fermionic partner of $\beta\gamma$: fields b of weight λ, c of weight $1-\lambda$, both fermionic, $b(z)c(w) \sim 1/(z-w)$ with anticommuting statistics. Central charge $c_{bc}(\lambda) = 1 - 3(2\lambda-1)^2$; verify $c_{bc} + c_{\beta\gamma} = 0$ at $\lambda = 1$: $-2 + 2 = 0$; at $\lambda = 2$: $-26 + 26 = 0$ (string ghost cancellation). At $\lambda = 1/2$: $c_{bc} = 1$ (single Dirac fermion); at $\lambda = 2$: $c_{bc} = -26$ (reparametrisation ghost).

Ordered bar differential. The ordered bar has two generators $s^{-1}b$, $s^{-1}c$ in the augmentation ideal. Fermionic statistics change the sign structure: $[s^{-1}b|s^{-1}c] = -[s^{-1}c|s^{-1}b]$ at the bar level because both generators are odd after desuspension ($|s^{-1}b| = |b| - 1 = \lambda - 1$). At degree 2:

$$d_{\bar{B}}\big([s^{-1}b|s^{-1}c] \otimes \eta_{12}\big) \;=\; \mathbf{1},$$

identical to the $\beta\gamma$ formula (simple-pole OPE). Higher-degree differentials vanish by the same ghost-number grading mechanism as $\beta\gamma$.

Bar cohomology and R-matrix. Growth rate 3^n, generating function algebraic (same exponential class). $r^{bc}(z) = r^{bc}_{\lambda}(z)$; Tier (b), class **C**.

Averaging. $\kappa(bc_\lambda) = c_{bc}(\lambda)/2 = (1 - 3(2\lambda - 1)^2)/2 = -(6\lambda^2 - 6\lambda + 1) = -\kappa(\beta\gamma_\lambda)$. Koszul dual: $(bc)^! = \beta\gamma$ (fermionic/bosonic exchange, inverse of the $\beta\gamma$ duality). Complementarity (free-field branch): $\kappa(bc) + \kappa(\beta\gamma) = 0$. The $bc/\beta\gamma$ duality is the simplest instance of bosonic/fermionic Koszul complementarity: it interchanges commuting and anticommuting statistics while preserving the OPE structure.

7.6. Lattice vertex algebras. Arithmetic laboratory: the shadow tower inherits the Gaussian archetype from the Heisenberg factor; the braiding carries the full arithmetic of the lattice. Curvature sees only rank; braiding sees the quadratic form.

Ordered bar differential. For an even lattice Λ of rank N, quadratic form q, 2-cocycle ε: $V_\Lambda^{N,q} = \mathcal{H}_1^{\otimes N} \otimes_{\mathbb{C}} \mathbb{C}[\Lambda]_\varepsilon$ (level $k = 1$ from the lattice bilinear form). The ordered bar decomposes as a tensor product of N copies of the Heisenberg ordered bar, tensored with the group algebra of the lattice. At degree 2, the differential acts on each Heisenberg factor independently:

$$d_{\bar{B}}\big([s^{-1}\alpha^i|s^{-1}\alpha^j] \otimes \eta_{12}\big) \;=\; \delta^{ij} \cdot \mathbf{1},$$

producing the identity Casimir in the rank-N current space. The vertex operators $e^{\lambda\cdot\phi}$ contribute only through the lattice cocycle $\varepsilon(\lambda, \mu)$, which enters the ordering of tensor factors but not the collision residue. All structure is captured at tensor degree 2: the bar complex is quadratic.

Bar cohomology. $H^1(\bar{B}(V_\Lambda))$ has dimension N (the N current generators α^{i*}); higher cohomology inherits the Heisenberg vanishing $H^n = 0$ for $n \geq 2$ in each tensor factor, but the lattice group algebra contributes additional classes through the vertex operator sector. The partition-graded generating function is transcendental: $\prod_{m\geq 1}(1 - x^m)^{-N}$, the N-fold Fock-space grading.

R-matrix. $r^\Lambda(z) = N/z$ (scalar, rank copies of Heisenberg r-matrix at level $k = 1$). Tier (a); spectral R-matrix $R(z) = \exp(N/z)$; Yang–Baxter trivial. The braiding $\pi_1(\mathrm{Conf}_n(X)) \to \mathrm{Aut}(V_\Lambda^{\otimes n})$ from the lattice 2-cocycle is invisible to the r-matrix.

Averaging. $\kappa(V_\Lambda^{N,q}) = \mathrm{rank}(\Lambda) = N$, independent of ε and q. Curvature-braiding orthogonality: the curvature class $[\kappa \cdot \omega_g] \in H^{1,1}(\overline{\mathcal{M}}_g)$ and the braiding representation act on complementary factors of the state space. Shadow class **G**; depth 2; genus-2: loop∘loop only. Self-dual lattices (E_8: $N = 8$; Leech: $N = 24$) give holomorphic VAs with trivial modular functor and theta-function partition Θ_Λ/η^N.

7.7. Yangians and quantum groups. The line-operator algebra. Yangian $Y(\mathfrak{g})$ governs the topological direction of a 3d HT theory; its R-matrix is the genus-0 binary shadow of Θ. Yang–Baxter is the degree-3 MC equation; RTT encodes all degrees.

R-matrix as genus-0 binary shadow.

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}), \qquad R(z) \;=\; 1 + r/z + r^{(2)}/z^2 + \cdots .$$

The classical Yang–Baxter equation $[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0$ is the Arnold relation on $\overline{C}_3(\mathbb{C})$ evaluated on the Casimir, and the three terms correspond to the three boundary divisors of $\overline{C}_3(\mathbb{C})$.

MC3 and all-types CG. On the type-A Yangian surface the post-CG problem reduces to a single compact/completed extension packet by (a) chromatic filtration of $K_0(\mathrm{Rep})$ by q-character support, (b) prefundamental Clebsch–Gordan closure $V_n \otimes L^- = \bigoplus_{j=0}^{n} L^-(n-2j)$ at the character level, and (c) Francis–Gaitsgory pro-nilpotent completion. The CG decomposition is extended to all simple types via the Chari–Moura multiplicity-free ℓ-weight property (Theorem **??**, Corollary **??**); the remaining post-CG completion packets are open beyond type A.

dg-shifted Yangian. The modular dg-shifted Yangian $Y_T^{\mathrm{mod}} = \varprojlim_r Y_T^{\mathrm{mod},\le r}$ is the pro-nilpotent completion of the convolution dg Lie algebra of the 3d HT chiral algebra for T, and the modular R-matrix $R_T^{\mathrm{mod}}(z;\ \ ) \in \mathrm{MC}(Y_T^{\mathrm{mod}})$ is an MC element. The Yang–Baxter equation is the degree-3 projection; the higher RTT relations encode all degrees; the inverse limit converges by Mittag-Leffler stabilisation.

Drinfeld–Kohno. The KZ connection $\nabla_{\mathrm{KZ}} = d - \sum_{i<j} \Omega_{ij}\, dz_{ij}/z_{ij}$ on $\mathrm{Conf}_n(\mathbb{C})$ has monodromy that agrees with the quantum-group representation category by the affine Drinfeld–Kohno theorem; shadow-side this is the statement $\nabla_{\mathrm{KZ}} = \mathrm{Sh}_{0,n}(\Theta_{\widehat{\mathfrak{g}}_k})$ on the affine comparison surface.

7.8. Drinfeld–Sokolov reduction as package morphism. $\mathrm{DS}\colon \widehat{\mathfrak{sl}}_{3k} \to \mathcal{W}_{3,c(k)}$ with $c(k) = 2 - 24(k+2)^2/(k+3)$ extends to a morphism of holographic data:

	$\mathcal{H}(\widehat{\mathfrak{sl}}_{3k})$	$\mathrm{DS}(\mathcal{H})$
$\mathcal{A}$	$\widehat{\mathfrak{sl}}_{3k}$	$\mathcal{W}_{3,c(k)}$
$\mathcal{A}^!$	$\widehat{\mathfrak{sl}}_{3-k-6}$	$\mathcal{W}_{3,100-c(k)}$
$r(z)$	$k\,\Omega_{\mathfrak{sl}_3}/z$	$r_c^{W_3}(z)$
κ	$4(k+3)/3$	$5c(k)/6$
depth	3	∞
$\Theta_{\mathrm{pf}}^{(2)}$	0	$\neq 0$

The Casimir r-matrix $k\Omega/z$ specialises to $r_c^{W_3}(z)$; the curvature shifts; the shadow depth jumps $3 \to \infty$; the planted-forest shell activates. At the critical level $k = -3$: $\kappa \to 0$ on the affine side, $c \to \infty$ on the $\mathcal{W}_3$ side (Sugawara pole).

The DS-HPL transfer theorem (Volume II) makes the chain-level mechanism precise. Homological perturbation through the BRST SDR transfers the affine dg-shifted Yangian triple (m, r, Δ) to the $\mathcal{W}$-algebra side: the A_∞ products transfer with nontrivial quartic pole (infinite degree); the r-matrix transfers with the expected pole-order shift; the coproduct is annihilated by a ghost-number obstruction, so $\Delta_z^{\mathcal{W}} = 0$ at all degrees. Gravity does not produce particles, only braids them. Central-charge additivity: the central charge difference $c(\widehat{\mathfrak{sl}}_{3k}) - c(\mathcal{W}_3, k) = 6(4k+5)$ is k-dependent (it includes both the free ghost bc central charge and the BRST stress-tensor improvement).

7.9. Geometric localization principle. Drinfeld–Sokolov reduction is Hamiltonian reduction from $J_\infty(\mathfrak{g}^*)$ to $J_\infty(S_f)$, where $S_f = f + \mathfrak{g}^e$ is the Slodowy transverse slice to the nilpotent orbit of f. The Slodowy slice meets every orbit transversally, carries a Kirillov–Kostant Poisson structure, and its quantisation is the finite W-algebra (Gan–Ginzburg, Premet). The chiral upgrade replaces S_f by its arc space, and $\mathrm{gr}_F\, \mathcal{W}^k(\mathfrak{g}, f) \cong \mathbb{C}[J_\infty(S_f)]$.

In this reading: the PBW–Slodowy collapse is smoothness of the Slodowy slice; the shadow depth jump $3 \to \infty$ under DS records the nonlinearity of the Dirac bracket on $J_\infty(S_f)$; Barbasch–Vogan orbit duality $f \leftrightarrow f^D$ is Springer duality of Slodowy slices, and curvature complementarity is an algebraic identity under the Feigin–Frenkel involution; the master commutative square (Theorem **??**) organises the whole picture, with bar complexes on the upper row and Hamiltonian reduction on arc spaces on the lower row.

7.10. The $\mathfrak{gl}_N$ chiral quantum group tower. The chiral quantum group programme covers the full $\mathfrak{gl}_N$ tower. For every $N \geq 1$, the principal $\mathcal{W}$-algebra $\mathcal{W}_N$ carries a chiral quantum group datum: the Yang R-matrix $R(u) = u\,I + \Psi\,P$ (where Ψ is the deformation parameter and P is the permutation matrix), the Drinfeld coproduct $\Delta_z(T(u)) = T(u) \cdot T(u-z)$ as $N \times N$ matrix multiplication, and nontrivial RTT relations for $N \geq 2$. The universal Miura coefficient $(\Psi-1)/\Psi$ on all cross-terms $J \otimes W_{s-1}$ persists at every spin $s \geq 2$: the coproduct corrections are determined by a single structural coefficient from the Prochazka–Rapcak Miura factorization. Concrete verifications: $N = 2$ (explicit 4×4 R-matrix), $N = 3$ (9×9, all 81 component RTT relations, quantum determinant centrality with decreasing column ordering).

At integer level k, ordered chiral homology recovers the Verlinde formula: $Z_g = \sum_j S_{0j}^{2-2g}$ as the dimension of ordered chiral homology at level k with $n = k + 2$. Closed-form polynomials $P_g(n) = n^{g-1}(n^2-1) \cdot R_{g-2}(n^2)$ are proved through genus 6; the leading coefficient $[n^{3g-3}]P_g = \zeta(2g-2)/(2^{g-2}\pi^{2g-2})$ connects the Verlinde tower to values of the Riemann zeta function at even positive integers.

The $\mathfrak{gl}_N$ tower in detail. The tower is controlled by the Miura factorisation $T(u) = \prod_{i=1}^{N}(u + \Lambda_i)$, where Λ_i are the free-boson momenta in the Miura representation. The elementary symmetric polynomials $\psi_s = e_s(\Lambda_1, \ldots, \Lambda_N)$ generate $\mathcal{W}_N$; the coproduct is multiplicative in $T(u)$:

$$\Delta_z(T(u)) \;=\; T(u) \cdot T(u-z),$$

where the product is $N \times N$ matrix multiplication in the generating-matrix formalism of RTT. The coefficient of $:J \cdot W_{s-1}:$ in the coproduct at spin s is $(\Psi - 1)/\Psi$ at all $s \geq 2$: this universality is proved by a three-step argument from the Miura decomposition: (1) the single-J sector produces $1/\Psi$ from the elementary symmetric expansion; (2) the Drinfeld shift $u \mapsto u - z$ adds $+1$; (3) lower sectors with ≥ 2 factors of J do not contribute at linear order. The coefficient $(\Psi - 1)/\Psi$ agrees with $1/\Psi$ only at $\Psi = 2$; three-point verification at $\Psi = 1, 2, 3$ eliminates the coincidence.

The RTT relations encode commutation of T-matrix entries: $R(u_1 - u_2)T_1(u_1)T_2(u_2) = T_2(u_2)T_1(u_1)R(u_1 - u_2)$, with $R(u) = u\,I + \Psi\,P$ (Yang's rational solution). At $N = 1$: $T(u) = u + J$, the RTT relation is the Heisenberg commutation $[J, J] = k$, and the R-matrix is scalar. At $N = 2$: the 4×4 R-matrix produces four independent commutation relations among $\{J, T\}$; the quantum determinant $\operatorname{qdet} T(u) = T_{11}(u)T_{22}(u-\Psi) - T_{12}(u)T_{21}(u-\Psi)$ is central. At $N = 3$: the quantum determinant with *decreasing* column ordering

$$\operatorname{qdet} T(u) \;=\; \sum_{\sigma \in S_3} (-1)^{|\sigma|} T_{1,\sigma(1)}(u) T_{2,\sigma(2)}(u-\Psi) T_{3,\sigma(3)}(u-2\Psi)$$

is central; increasing column ordering fails centrality at $N \geq 3$ (coincidental agreement at $N = 2$ masks the distinction). All 81 component RTT relations are verified for $\mathfrak{gl}_3$.

The Drinfeld coproduct Δ_z is *spectral coassociative*:

$$(\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_1+z_2} \;=\; (\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1},$$

with *shifted* parameters (not $\Delta_z \circ \Delta_z$). The Yang–Baxter equation for $R(u)$ is the degree-3 shadow of coassociativity; the full RTT tower encodes all degrees.

Stokes geometry. The KZ connection for $\mathcal{W}_N$ has irregular singularity at $z = 0$ of Poincaré rank $2N - 2$, producing $4N - 4$ Stokes rays (linear in N): the W–W channel (pole order $2N$) dominates the Stokes structure. At $N = 2$: 4 Stokes rays (Virasoro). At $N = 3$: 8 rays. At $N = 4$: 12 rays. The Stokes data encode the connection matrices between asymptotic solutions of the differential equation $\nabla_{\mathrm{KZ}}\Phi = 0$ and constitute the nonperturbative completion of the perturbative shadow series.

Antipode obstruction. The formal antipode $S(T(u)) = T(u)^{-1}$ (matrix inverse as formal power series in u^{-1}) does NOT lift to a vertex-algebraic antipode. Two independent obstructions: (i) the OPE obstruction: $S(T)_{(3)}S(T) \neq c/2$ at generic Ψ, violating the Virasoro Ward identity for the stress tensor; (ii) the Hopf axiom obstruction: a residual $z \cdot J$ term prevents $m \circ (S \otimes \mathrm{id}) \circ \Delta_z = \eta \circ \epsilon$ from holding at the vertex-algebraic level. Source: the Miura factorisation is *nonlinear* (the product $T(u) \cdot T(u - z)$ mixes spins), and the antipode would require inverting this nonlinearity, which the vertex algebra does not support. This is a genuine structural obstruction, not a computational difficulty.

7.11. Verlinde recovery from ordered chiral homology. The Verlinde formula, the enumeration of conformal blocks at integer level, is recovered from the ordered bar complex. For $\mathcal{A} = V_k(\mathfrak{sl}_2)$ at positive integer level k, set $n = k + 2$. The ordered chiral homology on a genus-g surface Σ_g has dimension

$$Z_g = \dim \int_{\Sigma_g}^{\mathrm{ord}} V_k(\mathfrak{sl}_2) = \sum_{j=0}^{k} S_{0j}^{2-2g},$$

where $S_{0j} = \sqrt{2/n}\,\sin(\pi(j + 1)/n)$ is the modular S-matrix entry. The three sewing mechanisms (handle attachment, separating factorisation, nonseparating factorisation) are verified independently:

- *Handle attachment*: gluing a handle to Σ_{g-1} multiplies the trace by $\sum_j S_{0j}^{-2}$, producing the correct genus recursion $Z_g = (\sum_j S_{0j}^{-2}) \cdot Z_{g-1}$.
- *Separating factorisation*: cutting Σ_g along a separating circle recovers $Z_g = \sum_j Z_{g_1}^{(j)} Z_{g_2}^{(j)}$ with $g_1 + g_2 = g$.
- *Nonseparating factorisation*: cutting along a nonseparating circle recovers the trace with insertion of the Casimir.

The closed-form polynomials:

$$\begin{aligned}
Z_2 &= n(n^2 - 1)/6,\\
Z_3 &= n^2(n^2 - 1)(n^2 + 11)/180,\\
Z_4 &= n^3(n^2 - 1)(2n^4 + 23n^2 + 191)/7560,\\
Z_5 &= n^4(n^2 - 1)(n^2 + 11)(3n^4 + 10n^2 + 227)/226800,\\
Z_6 &= n^5(n^2 - 1)(2n^8 + 35n^6 + 321n^4 + 2125n^2 + 14797)/2993760.
\end{aligned}$$

General form: $P_g(n) = n^{g-1}(n^2 - 1) \cdot R_{g-2}(n^2)$ with R_{g-2} a polynomial of degree $g - 2$ in n^2. At $k = 1$ $(n = 3)$: $Z_2 = 4$, $Z_3 = 40$, $Z_4 = 560$.

The rational generating function $G_n(x) = \sum_{j=1}^{n-1} 1/(1 - a_j x)$ with eigenvalues $a_j = n/(2\sin^2(\pi j/n))$ encodes all Z_g simultaneously: $Z_g = [x^{g-1}]\, G_n(x)$ for $g \geq 2$. The eigenvalues a_j are the inverse squares of the modular S-matrix entries (up to normalisation), providing a spectral decomposition of the Verlinde numbers.

The leading coefficient: $[n^{3g-3}]P_g = \zeta(2g - 2)/(2^{g-2}\pi^{2g-2})$. At $g = 2$: $\zeta(2)/(2^0\pi^2) = (\pi^2/6)/\pi^2 = 1/6$, matching the leading coefficient of $Z_2 = n^3/6 - n/6$. The Verlinde tower is the integer shadow of the shadow obstruction tower: Z_g is a polynomial in the level (an integer parameter); F_g is a rational function of the central charge (a continuous parameter). The two towers share the leading asymptotic but differ at subleading order.

7.12. Depth escalation: from class L to class M. Drinfeld–Sokolov reduction is the depth escalation mechanism. For $\mathfrak{g} = \mathfrak{sl}_N$ with principal nilpotent $f = f_{\mathrm{prin}}$, the BRST reduction $\mathcal{W}_{N,c(k)} = H^0_{Q_{\mathrm{DS}}}(V_k(\mathfrak{sl}_N) \otimes F_{\mathrm{gh}})$ takes an affine Kac–Moody algebra (class **L**, shadow depth 3) to a $\mathcal{W}$-algebra (class **M**, shadow depth ∞). The depth jumps because the Dirac bracket on the Slodowy slice $S_f = f + \mathfrak{g}^e$ is nonlinear: the Poisson structure on $J_\infty(S_f)$ has polynomial brackets of unbounded degree, generating the infinite shadow tower.

The escalation is controlled by the homological perturbation lemma (HPL). The DS strong deformation retract

$$(V_k(\mathfrak{g}) \otimes F_{\mathrm{gh}}, Q_{\mathrm{DS}}) \underset{\iota}{\overset{\pi}{\rightleftarrows}} (\mathcal{W}_{N,c(k)}, 0)$$

with homotopy h transfers the bar-complex data through the BRST complex. The transferred structures:

(i) The A_∞ products transfer with pole-order shift: $m_k^{\mathcal{W}}$ acquires contributions from k-fold nested BRST contractions, generating the infinite tower. The quartic pole from $:TT:$ (absent on the affine side) appears as $m_4^{\mathcal{W}} = \pi \circ m_2^{\mathfrak{g}} \circ (h \circ m_2^{\mathfrak{g}})^2 \circ \iota^{\otimes 4}$.

(ii) The r-matrix transfers with expected pole-order shift: the Casimir r-matrix $k\,\Omega/z$ specialises to the $\mathcal{W}_N$ r-matrix, gaining higher-order poles from the nonlinear field redefinition.

(iii) The coproduct is annihilated: a ghost-number obstruction kills every HPL tree correction to Δ_z, so the transferred coproduct $\Delta_z^{\mathcal{W}} = 0$ at all degrees. Gravity does not produce particles, only braids them.

Central charge shift. The central charge transforms as $c(\mathcal{W}_{N,k}) = (N-1)(1 - N(N+1)(k+N)^{-2})$ (Fateev–Lukyanov). The ghost central charge $c_{\mathrm{gh}}(N,k) = c(\widehat{\mathfrak{sl}}_{N\,k}) - c(\mathcal{W}_{N,k})$ includes both the free ghost bc contribution and the BRST stress-tensor improvement; it is k-dependent (not k-independent as one might expect from the free ghost alone).

The modular characteristic shifts accordingly: $\kappa(\mathcal{W}_{N,c}) = c \cdot (H_N - 1)$, where $H_N = \sum_{j=1}^{N} 1/j$ is the N-th harmonic number. At $N=2$: $H_2 - 1 = 1/2$, so $\kappa(\mathcal{W}_2) = c/2 = \kappa(\mathrm{Vir})$. At $N=3$: $H_3 - 1 = 5/6$, so $\kappa(\mathcal{W}_3) = 5c/6$.

Complementarity under DS. The Koszul conductor transforms: $K(\mathcal{W}_{N,c}) = \kappa(\mathcal{W}_{N,c}) + \kappa(\mathcal{W}_{N,c^!})$ where $c^! = c(k^!)$ with $k^! = -k - 2h^\vee$. For principal $\mathcal{W}_N$:

N	$K(\mathcal{W}_N)$	Self-dual c	Shadow class
2	13	$c = 13$	**M**
3	250/3	$c = 50$	**M**
4	?	?	**M**
BP	196	$k = -3$	**M**

The Koszul conductor is a global duality invariant, not a local ghost constant: $K_{\mathrm{BP}} = 196$ records the full $c(k) + c(-k-6)$ sum for the Bershadsky–Polyakov algebra, not any single ghost number or grading shift.

7.13. Master invariant table. The complete family-by-family census of E_1 bar invariants:

Family	κ	$\varkappa := \kappa + \kappa^!$	$r(z)$	Class	d_{alg}	d_{gen}
$\mathcal{H}_k$	k	0	k/z	**G**	2	2
$V_k(\mathfrak{g})$	$\frac{\dim\mathfrak{g}(k+h^\vee)}{2h^\vee}$	0	$k\,\Omega/z$	**L**	3	3
Vir_c	$c/2$	13	$\frac{c/2}{z^3}+\frac{2T}{z}$	**M**	∞	3
$\mathcal{W}_{3,c}$	$5c/6$	$250/3$	$r_c^{W_3}(z)$	**M**	∞	3
$\mathcal{W}_{N,c}$	$c(H_N{-}1)$	family-dep	$r_c^{W_N}(z)$	**M**	∞	3
$\beta\gamma_\lambda$	$6\lambda^2{-}6\lambda{+}1$	0	$r_\lambda(z)$	**C**	4	4
bc_λ	$-(6\lambda^2{-}6\lambda{+}1)$	0	$r_\lambda^{bc}(z)$	**C**	4	4
V_Λ	$\mathrm{rk}(\Lambda)$	0	N/z	**G**	2	2
BP_k	$\mathrm{BP}(k)$	$98/3$	$r_k^{\mathrm{BP}}(z)$	**M**	∞	?
SVir_c	$(3c-2)/4$	?	$r_c^{\mathrm{SVir}}(z)$	**M**	∞	?

Two depths are recorded: the algebraic depth d_{alg} (whether the shadow tower terminates) and the generating depth d_{gen} (the degree at which all higher operations are determined by lower ones). For Virasoro: $d_{\mathrm{gen}} = 3$ (the operation m_3 generates all m_k recursively), but $d_{\mathrm{alg}} = \infty$ (all m_k are nonzero, so the tower never terminates). The depth gap theorem: $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ with gap at 3 (no standard family has $d_{\mathrm{alg}} = 3$ because the Jacobi identity on the shadow Lie algebra forces either termination at degree 4 or propagation to all degrees).

The Bershadsky–Polyakov algebra $\mathrm{BP}_k = \mathcal{W}^k(\mathfrak{sl}_3, f_{\mathrm{min}})$ is the simplest non-principal $\mathcal{W}$-algebra: the level involution $k \mapsto -k-6$ has fixed point $k = -3$, which coincides with the critical level $-h^\vee(\mathfrak{sl}_3) = -3$ where $c(\mathrm{BP}_k)$ has a simple pole; the value $\kappa(-3) = 49/3$ is the *principal-value symmetric limit* across the pole, not a regular function value. The central-charge conductor is a polynomial identity $c(k) + c(-k-6) \equiv 196$ in $\mathbb{Q}(k)$, and the induced κ-conductor is $K = \varrho \cdot 196 = 98/3$ with $\varrho = 1/6$. The conformal weight range is $\{1, 3/2, 2\}$ (a weight-3/2 field from the minimal nilpotent orbit, absent in principal reductions). The super Virasoro algebra SVir_c has $\kappa = (3c-2)/4$ and an infinite shadow tower driven by the superconformal self-coupling.

8. Arithmetic of the shadow obstruction tower

The shadow obstruction coefficients for the standard families are rational functions of the central charge. Their denominators, zeros, and growth rates carry number-theoretic information connecting the shadow calculus to L-functions. What arithmetic does the MC equation force?

The Dirichlet–sewing lift. The genus-1 connected free energy $F_1^{\mathrm{conn}}(\mathcal{A}; q) = -\log\det(1 - K_q)$ encodes connected sewing amplitudes $a_{\mathcal{A}}(N)$ as Fourier coefficients. The Dirichlet–sewing lift

$$S_{\mathcal{A}}(u) = \sum_{N\ge 1} a_{\mathcal{A}}(N) N^{-u} = \frac{(2\pi)^u}{\Gamma(u)} \int_0^\infty F_{\mathcal{A}}^{\mathrm{conn}}(e^{-2\pi y})\, y^{u-1}\, dy$$

is a Dirichlet series whose analytic structure is controlled by $\Theta_{\mathcal{A}}$. The sewing amplitudes define an inner product $\langle f, g\rangle_{\mathcal{A}} = \sum_N a_{\mathcal{A}}(N) f(N) g(N)$ with reproducing kernel $K_{\mathcal{A}}(s,t) = S_{\mathcal{A}}(s+t)$; the surface moment matrix $\mathscr{M}_{ij} = K_{\mathcal{A}}(\alpha/2 + i, \alpha/2 + j)$ is positive definite for all $\alpha \ge 2$ (Theorem **??**), a Gram matrix $V^T D V$ with Vandermonde V.

Explicit values:

$$\begin{aligned} S_{\mathcal{H}}(u) &= \zeta(u)\zeta(u+1), \\ S_{V_k(\mathfrak{g})}(u) &= \dim(\mathfrak{g}) \cdot \zeta(u)\zeta(u+1), \\ S_{\mathrm{Vir}}(u) &= \zeta(u+1)(\zeta(u)-1), \\ S_{\mathcal{W}_N}(u) &= \zeta(u+1)\Big((N-1)\zeta(u) - \textstyle\sum_{j=1}^{N-1} H_j(u)\Big), \end{aligned}$$

where $H_j(u) = \sum_{n=1}^{j} n^{-u}$ are partial harmonic sums.

The three-tier Euler–Koszul classification. The *Euler defect* $D_{\mathcal{A}}(u) := S_{\mathcal{A}}(u)/(|W(\mathcal{A})|\,\zeta(u)\,\zeta(u+1))$, where $|W(\mathcal{A})|$ is the number of strong generators ($|W(\mathcal{H})| = 1$, $|W(\widehat{\mathfrak{g}}_k)| = \dim\mathfrak{g}$, $|W(\beta\gamma)| = 2$), sorts chiral algebras:

Tier	Condition	Families
Exact Euler–Koszul	$D_{\mathcal{A}}(u) \equiv 1$	$\mathcal{H}, V_k(\mathfrak{g}), \beta\gamma$
Finitely defective	$1 - D_{\mathcal{A}}(u)$ finite harmonic ratio	Virasoro, $\mathcal{W}_N$
Non-Euler–Koszul	$S_{\mathcal{A}}$ decomposes into Hecke L-functions	lattice VOAs with $h(D) \geq 2$

The Euler–Koszul class $\mathrm{ek}(\mathcal{A}) = \max_i(w_i - 1)$ and the shadow depth $r_{\max}(\mathcal{A})$ are independent invariants (Theorem ??): character arithmetic and OPE interaction are logically independent axes.

The two-variable L-object. Rankin–Selberg unfolding against $E^*(\tau, s)$ gives

$$L_{\mathcal{A}}(s,u) \;=\; \int_{\mathcal{F}}^{\mathrm{reg}} F_{\mathcal{A}}^{\mathrm{conn}}(\tau) E^*(\tau,s)\, y^{u-2}\, dx\, dy.$$

The *sewing–Hecke reciprocity theorem* collapses both variables to one: $L_{\mathcal{A}}(s,u) = \Gamma(v)/(2\pi)^v\, S_{\mathcal{A}}(v)$ with $v = s+u-1$. The scattering poles of E^* at $s = \rho/2$ (nontrivial zeros ρ of ζ) are *not* poles of $L_{\mathcal{A}}$; the nontrivial zeros appear as structural obstructions to the Euler product, not as poles of the L-object.

Shadow-moduli resolution. The shadow-moduli map $t\colon \mathfrak{H} \to \mathbb{C}$ solves $\prod_j (1 - \lambda_j\, t(q))^{c_j} = \det(1 - K_q^{\mathrm{conn}})$ with spectral atoms λ_j and multiplicities c_j. The connected free energy factorises as $F_1^{\mathrm{conn}}(q) = G_{\mathcal{A}}(t(q))$ with

$$G_{\mathcal{A}}(t) \;=\; \int \log(1-\lambda t)\, d\rho_{\mathcal{A}}(\lambda),$$

determined by the spectral measure alone (Theorem ??). The moments $\mu_r = \int \lambda^r\, d\rho_{\mathcal{A}}$ satisfy the MC recursion

$$\mu_{r+1} \;=\; (r+1)\,\mathrm{Br}_{r+1}(\mu_2, \ldots, \mu_r; \mathcal{A})$$

(the bar-recursion polynomial determined by the MC structure) degree by degree. For two Satake atoms α, β, the Newton relations give the power sums $p_r = e_1 p_{r-1} - e_2 p_{r-2}$, encoding the MC recursion in spectral coordinates.

Interacting Gram matrix and resonance. The shadow obstruction tower provides corrections to Gram positivity. The interacting weight $w_{\mathcal{A}}(N;\alpha) = a_{\mathcal{A}}(N) \cdot \prod_j (1 - \lambda_j N^{-\alpha/2})^{c_j}$ is unconditionally positive when all atoms λ_j are negative. For Virasoro with $c > 0$: $\lambda_{\mathrm{eff}} = -6/c < 0$, so positivity holds unconditionally for $\alpha \geq 2$. The resonance locus $\mathcal{R}$ is empty for $c > 0$ and lies in $c < 0$, with the Lee–Yang point $c = -22/5$ as the first entry.

The Weil analogy.

Weil proof for $C/\mathbb{F}_q$	MC equation for $\mathcal{A}$
Curve $C/\mathbb{F}_q$	chirally Koszul $\mathcal{A}$
Frobenius Frob_q	MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$
$\mathrm{Frob}^2 = q$	$D_{\mathcal{A}}^2 = 0$
Étale cohomology $H^i_{\text{ét}}$	cyclic deformation $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$
Weight filtration	degree filtration $\Theta_{\mathcal{A}}^{\leq r}$
Eigenvalues α_i on H^1	spectral atoms λ_j
$\|\alpha_i\| = q^{1/2}$	$\|a_{f_j}(p)\| \leq 2p^{(k-1)/2}$
Lefschetz L	bracket with $\kappa(\mathcal{A})$

For even unimodular lattice VOAs V_Λ with $\Theta_\Lambda = c_E E_{r/2} + \sum_j c_j f_j$, the Hecke–Newton closure gives: MC equation ⇒ prime-local CG closure ⇒ CPS automorphy ⇒ $\mathrm{Sym}^{r-1}(f)$ ⇒ Ramanujan bound. Deligne's theorem supplies the bound itself; the MC framework supplies the systematic Hecke decomposition.

The key disanalogy. The Weil analogy is structural, not literal. The principal difference: Weil's proof uses the Riemann hypothesis for curves ($|\alpha_i| = q^{1/2}$ as *equalities*); the MC framework uses Ramanujan-type bounds ($|a_f(p)| \leq 2p^{(k-1)/2}$ as *inequalities*). The passage from bound to equality would require a deeper structure: a Langlands-type functoriality that promotes the MC framework from an analogy to an equivalence. The shadow obstruction tower provides the systematic input; the promotion is the open problem.

A second disanalogy: the weight filtration in the Weil setting is a canonical, geometric invariant of the motive. The degree filtration $\Theta_{\mathcal{A}}^{\leq r}$ is a choice of truncation, not a canonical structure on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$. Whether $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ carries a natural weight filtration (in the motivic sense) is a deep question connecting the programme to mixed Hodge theory on moduli spaces of stable curves.

A third disanalogy: in the Weil setting, the zeta function is a single rational function. In the MC setting, the two-variable L-object $L_{\mathcal{A}}(s,u)$ is not manifestly rational in either variable. Rationality in u at fixed s (or vice versa) would be a structural theorem about the MC equation.

Polyakov–bar-cobar dictionary.

Polyakov	Bar-cobar
Conformal anomaly $(c/12)R$	modular characteristic κ
$\det'_\zeta \Delta$	Fredholm $\det(1-K)$
Critical dimension $c = 26$	dual curvature $\kappa(\mathrm{Vir}_{26-c}) = 0$
Ghost system bc	Koszul dual $\mathcal{A}^!$ (BRST pairing)
Genus expansion $Z = \sum g_s^{2g-2} Z_g$	MC5 sewing, $F_g = \kappa\lambda_g^{\mathrm{FP}}$ (uniform)
BPZ equations	shadow connection on comparison surface
Instanton sectors	Novikov-completed bar $B^\Lambda(\mathcal{A})$
Wilson line $\langle W(C)\rangle$	Swiss-cheese boundary module

Anomaly, genus expansion, duality, critical dimension, and non-perturbative completions all descend from $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$. The four depth classes **G**/**L**/**C**/**M** correspond to increasing departures from the Gaussian path integral.

Arithmetic packet connection. Hecke operators T_p act on the space $M_{\mathcal{A}}$ of activated graph amplitudes $\ell_\Gamma^{(g)}$. Decompose $M_{\mathcal{A}} = \bigoplus_\chi M_\chi$ into Hecke generalised eigenspaces with eigenvalue $a_\chi(p)$ and nilpotent part N_χ; attach to each eigenspace its packet function $\Lambda_\chi(s)$ (completed L-function of the eigenform). The *arithmetic packet connection*

$$\nabla^{\mathrm{arith}}_{\mathcal{A}} = d - \bigoplus_{\chi} \frac{\Lambda'_\chi(s)}{\Lambda_\chi(s)}(\mathrm{id}_{M_\chi} + N_\chi)\, ds$$

is flat (each block is scalar form times constant endomorphism). The singular divisor $D_{\mathcal{A}}$ is the zero-pole set of the packet functions, depending only on the arithmetic skeleton $M^{\mathrm{ss}}_{\mathcal{A}}$.

The Eisenstein scattering function on $\mathcal{M}_{1,1}$ is $\varphi(s) = \Lambda^*(1-s)/\Lambda^*(s)$ with $\Lambda^*(s) = \pi^{-s}\Gamma(s)\zeta(2s)$. The frontier defect form $\Omega_{\mathcal{A}} = d\log\Lambda_{\mathrm{Eis}}(s) - d\log\varphi(s)$ vanishes iff the Eisenstein block is gauge-equivalent to the scattering connection. For lattice VOAs at rank r, $\Omega_{V_\Lambda} \neq 0$ in general because $\zeta(s)$ and $\zeta(2s)$ have different zero sets.

Finite Miura defect. The principal $\mathcal{W}_N$ character is $\chi_{\mathcal{W}_N}(q) = \prod_{n\geq 2}(1-q^n)^{-\min(n-1,N-1)}$; the Heisenberg character is $\chi_{\mathcal{H}}(q) = \prod_{n\geq 1}(1-q^n)^{-1}$. The ratio is finite:

$$\chi_{\mathcal{W}_N}(q) = \chi_{\mathcal{H}}(q)^{N-1} \cdot \prod_{m=1}^{N-1}(1-q^m)^{N-m}.$$

The finite product is the Miura defect; all genus-1 arithmetic content in $\mathcal{W}_N$ beyond Heisenberg is carried by it. The prime-side Dirichlet analogue is $D^{\mathrm{prime}}_{\mathcal{W}_N}(u) = -\zeta(u+1)\sum_{m=1}^{N-1}(N-m)m^{-u}$; the packet connection splits into $(N-1)$ Heisenberg copies plus a defect governing modes $m = 1, \ldots, N-1$.

Quartic closure programme. The quartic resonance class $\mathfrak{R}^{\mathrm{mod}}_{4,g,n}(\mathcal{A})$ satisfies a clutching law along every separating stratum

$$\xi^*\mathfrak{R}^{\mathrm{mod}}_{4,g,n} = p_1^*\mathfrak{R}^{\mathrm{mod}}_{4,g_1,n_1+1} + p_2^*\mathfrak{R}^{\mathrm{mod}}_{4,g_2,n_2+1} + \mathfrak{T}_{3,\xi}$$

(Theorem ??). The 3×3 Schur complement of the mixed (s,u)-transform extracts the quartic contact potential $S_4(\mathcal{A})$ on the potential side and its reciprocal on the Gram side. For Virasoro: $S_4(\mathrm{Vir}_c) = 10/[c(5c+22)]$.

Three closure conjectures form a hierarchy (each implies its predecessors): *quartic closure* (intersection of Gram loci equals $\{\Re\rho = 1/2\}$, Conjecture ??), *Beilinson closure* (global discrepancy $\mathcal{B}^{\mathrm{glob}}(\rho)$ vanishes iff $\Re\rho = 1/2$, Conjecture ??), and *clutching closure* (residue clutching defect $K_{\mathcal{A},\rho,\xi}$ vanishes for all families iff $\Re\rho = 1/2$, Conjecture ??). The chain of constructions $\Theta_{\mathcal{A}} \to w_{c,\rho,u_0} \to$ Gram matrix $\to$ Schur complement $\to$ quartic resonance divisor $\to$ residue clutching defect $\to$ closure conjecture is proved up to the final link, which is conjectural.

Constrained Epstein zeta. For a lattice VOA V_Λ of rank r, the constrained Epstein zeta $\varepsilon^c_s(V_\Lambda) = \sum'_{\ell\in\Lambda^{\mathrm{bar}}} \|\ell\|^{-2s}$ restricts the Epstein series to the sublattice $\Lambda^{\mathrm{bar}} \subset \Lambda$ determined by the bar complex. For the rank-1 lattice at self-dual radius $R = 1$, every nonzero integer contributes: $\varepsilon^1_s(R{=}1) = 4\zeta(2s)$. The simplest lattice VOA's constrained Epstein zeta is the Riemann zeta.

The Hecke decomposition: $\varepsilon^r_s(V_\Lambda) = \sum_j c_j L(s,\chi_j)$. Shadow depth d contributes $d-1$ critical lines to the meromorphic continuation, one for each degree stratum beyond the Gaussian base. The nontrivial zeros of the component L-functions are the scattering resonances of the hyperbolic Laplacian on $\mathcal{M}_{1,1}$: frequencies at which the genus-1 sewing amplitude fails to decay.

Genus-2 Böcherer bridge. For the Leech lattice VOA, the Siegel modular form $\chi_{12} = \mathrm{SK}(f_{22})$ is the Saito–Kurokawa lift of the unique weight-22 cuspidal eigenform, and the MC-derived second Fourier coefficient $c_2 \approx -1.918\times 10^{-6} \neq 0$ is the first nonzero central L-value produced by the shadow obstruction tower, connecting $\Theta_{\mathcal{A}}$ to genus-2 Siegel modular forms via the Furusawa–Morimoto theorem.

8.8. Verlinde polynomials from ordered chiral homology. At integer level k, ordered chiral homology recovers the Verlinde formula $Z_g = \sum_j S_{0j}^{2-2g}$ as the dimension of ordered chiral homology of $V_k(\mathfrak{sl}_2)$ with $n = k+2$. The Verlinde numbers are polynomials in n:

$$\begin{aligned} Z_2 &= n(n^2-1)/6, \\ Z_3 &= n^2(n^2-1)(n^2+11)/180, \\ Z_4 &= n^3(n^2-1)(2n^4+23n^2+191)/7560, \end{aligned}$$

with general form $P_g(n) = n^{g-1}(n^2-1)\cdot R_{g-2}(n^2)$ proved through genus 6. The leading coefficient $[n^{3g-3}]P_g = \zeta(2g-2)/(2^{g-2}\pi^{2g-2})$ connects the Verlinde tower to values of the Riemann zeta function at even positive integers: the arithmetic of level-k conformal blocks encodes zeta values. The rational generating function $G_n(x) = \sum_j 1/(1-a_j x)$ with $a_j = n/(2\sin^2(\pi j/n))$ gives a closed form for all Z_g simultaneously.

The Verlinde numbers Z_g (integer, bundle ranks) are distinct from the shadow free energies F_g (rational, shadow obstruction coefficients). At $k=1$, $g=2$: $Z_2 = 4$ versus $F_2 = 7/2560$. The two towers have different analytic behaviour: Z_g is polynomial in n (the level); F_g is rational in the central charge c (a continuous parameter).

8.9. The depth decomposition of the Dirichlet series. The Dirichlet–sewing lift $S_{\mathcal{A}}(u)$ admits a decomposition indexed by the shadow depth classes:

(i) *Gaussian stratum* ($r=2$): $S_{\mathcal{A}}^{(2)}(u) = \kappa(\mathcal{A})\cdot\zeta(u)\zeta(u+1)$. This stratum is universal: it depends only on the modular characteristic κ and reproduces the Heisenberg answer for all families. For $\mathcal{H}_k$: $S_{\mathcal{H}} = S_{\mathcal{H}}^{(2)}$, the full series is exhausted by the Gaussian stratum.
(ii) *Lie stratum* ($r=3$): $S_{\mathcal{A}}^{(3)}(u)$ carries the Chevalley–Eilenberg contribution from the cubic shadow $\mathfrak{C}(\mathcal{A})$. For affine Kac–Moody: $S_{\widehat{\mathfrak{g}}_k}^{(3)}(u)$ encodes the Serre-relation corrections to the Gaussian base.
(iii) *Contact stratum* ($r=4$): $S_{\mathcal{A}}^{(4)}(u)$ records the quartic contact shadow $\mathfrak{Q}(\mathcal{A})$. For $\beta\gamma$: the series terminates here.
(iv) *Mixed strata* ($r\geq 5$): for Virasoro and $\mathcal{W}_N$, infinitely many strata contribute. Each adds $d-1$ critical lines to the meromorphic continuation.

The decomposition $S_{\mathcal{A}}(u) = \sum_{r\geq 2} S_{\mathcal{A}}^{(r)}(u)$ converges absolutely for $\Re(u) > 2$; the individual strata have distinct analytic continuations, and their interaction is governed by the MC equation: the compatibility $[D_0, D_1] = 0$ between the genus-0 and genus-1 differentials forces cancellations among strata that would individually produce spurious poles.

The Euler defect inherits this stratification: $D_{\mathcal{A}}(u) = 1 + \sum_{r\geq 3} D_{\mathcal{A}}^{(r)}(u)$, with $D_{\mathcal{A}}^{(r)}$ the depth-r correction. For Virasoro: $D_{\mathrm{Vir}}(u) = 1 - 1/\zeta(u)$, the correction is the reciprocal of the Riemann zeta function. For $\mathcal{W}_3$: $D_{\mathcal{W}_3}(u) = 1-(1+2^{1-u})/\zeta(u)$, with the extra 2^{1-u} from the weight-3 generator.

8.10. Verlinde–shadow bridge. The Verlinde polynomials $P_g(n)$ and the shadow free energies $F_g(c)$ are distinct objects with a precise bridge. At level k with $n = k+2$ and $c(k) = 3k/(k+2)$ for $\mathfrak{sl}_2$:

Verlinde	**Shadow**	**Bridge**
$Z_g = P_g(n)$	$F_g = F_g(c)$	$Z_g = Z_g^{\mathrm{pert}} + Z_g^{\mathrm{nonpert}}$
polynomial in n	rational in c	perturbative + instanton
integer (rank of bundle)	rational (graph sum)	Verlinde $\supset$ shadow
all levels	continuous c	evaluates at discrete k

The perturbative part Z_g^{pert} is the shadow evaluation at $\kappa = 3(k+2)/4$; the nonperturbative part Z_g^{nonpert} comes from the finite sum over integrable representations (a sum over $n-1$ terms, not a continuous integral). The shadow free energy $F_g(c)$ is the perturbative shadow obstruction coefficient and agrees with Z_g^{pert} when c is specialised to $c(k)$. The nonperturbative contribution is invisible to the shadow tower: it requires the full ordered chiral homology, not just its Taylor-coefficient extraction.

The leading asymptotics agree: $P_g(n) \sim n^{3g-3} \cdot \zeta(2g-2)/(2^{g-2}\pi^{2g-2})$ for large n matches $F_g(c) \sim c^{g-1} \cdot \lambda_g^{\mathrm{FP}}$ for large c (where $c \sim 3n/2$ for $\mathfrak{sl}_2$), because the perturbative expansion dominates at large level.

8.11. Constrained Epstein zeta: depth decomposition. The constrained Epstein zeta for rank-r lattice VOAs admits a parallel depth decomposition. For an even unimodular lattice Λ of rank r, the constrained series

$$\varepsilon_s^r(V_\Lambda) = \sum_{\ell \in \Lambda^{\mathrm{bar}}}{}' \|\ell\|^{-2s}$$

splits by shell: $\varepsilon_s^r = \varepsilon_{s,\mathrm{Gauss}}^r + \varepsilon_{s,\mathrm{arith}}^r$. The Gaussian part is $r \cdot \zeta(2s)$ (rank copies of the simplest constrained zeta). The arithmetic part $\varepsilon_{s,\mathrm{arith}}^r$ is the Hecke-eigenvalue decomposition $\sum_j c_j L(s, \chi_j)$, with χ_j running over Hecke characters attached to the lattice.

For the E_8 lattice ($r = 8$): the constrained Epstein zeta is $\varepsilon_s^8(E_8) = 8\zeta(2s) + 480L(s, \Delta_{16})$, where Δ_{16} is the unique normalised cuspidal eigenform of weight 16 for $\mathrm{SL}_2(\mathbb{Z})$. The coefficient 480 is the number of roots of E_8; the cuspidal component carries the arithmetic beyond the Gaussian base.

For the Leech lattice ($r = 24$): the decomposition $\varepsilon_s^{24}(\Lambda_{24}) = 24\zeta(2s) + c_{22}L(s, \Delta_{22}) + \cdots$ involves the weight-22 cusp form Δ_{22} and finitely many additional Hecke eigenspaces. The Böcherer bridge connects the genus-2 coefficient to the central L-value $L(1/2, \Delta_{22} \times \chi_D)$.

The functional equation $\varepsilon_s^r(V_\Lambda) = \varepsilon_{r/2-s}^r(V_\Lambda)$ (from the Poisson summation formula on the lattice) is the Epstein analogue of the MC complementarity $\kappa + \kappa^! = \varrho_{\mathcal{A}} \cdot K$ (with $\varrho_{\mathcal{A}}$ family-specific: $\varrho_{\mathrm{KM}} = \varrho_{\mathrm{free}} = 0$, $\varrho_{\mathrm{Vir}} = 1/2$, $\varrho_{\mathcal{W}_N} = H_N - 1$, $\varrho_{\mathrm{BP}} = 1/6$). At the self-dual point $s = r/4$: the constrained zeta evaluates to $\varepsilon_{r/4}^r = r \cdot \zeta(r/2) +$ cuspidal contribution, connecting the complementarity constant $K = 0$ (for lattice VOAs in the free-field branch) to the trivial functional equation of $\zeta(r/2)$.

9. Scalar saturation and the Koszulness programme

The bar construction works for all chiral algebras. It works best, the cobar inverts the bar, the spectral sequence collapses, the resolution is linear, precisely on the Koszul locus. Characterizing this locus is the central foundational question introduced in §3.3 and anticipated throughout sections 4 and 5. Two complementary results: scalar saturation (the MC element is determined by a single scalar for all standard families) and the Koszulness meta-theorem (fourteen characterisations: among the core (i)–(x), eight are genuinely independent bidirectional equivalences with (vi) a proved consequence of (v) and (viii) a one-way chiral Hochschild consequence on the Koszul locus; one further characterisation is conditional on perfectness; one is one-directional; plus the bifunctor decomposition and the Sklyanin Poisson $H^2 = 0$ criterion).

9.1. Scalar saturation. For a one-parameter algebraic family $\mathcal{A}_k$ with rational OPE coefficients (structure constants rational functions of k), the universal MC element satisfies

$$\Theta_{\mathcal{A}_k} = \kappa(\mathcal{A}_k) \cdot \eta_{\mathrm{univ}} \otimes \Lambda_{\mathrm{univ}} + (\text{terms in } F^3\mathfrak{g}_{\mathcal{A}_k}^{\mathrm{mod}}).$$

Scalar saturation universality: higher-order terms vanish. For every algebraic family with rational OPE coefficients, $\Theta_{\mathcal{A}_k}$ is determined by $\kappa(\mathcal{A}_k)$ modulo terms in the degree-≥ 3 filtration.

Step 1: Whitehead reduction. For a vertex algebra with reductive symmetry at generic level, the primitive cyclic cohomology $H^2_{\mathrm{cyc,prim}}(\mathfrak{g}^{\mathrm{mod}}/F^3, d_2)$ is computed by a Whitehead-type spectral sequence with $E_2^{p,q} = H^p(\mathfrak{g}; \mathrm{Sym}^q V)$. For reductive $\mathfrak{g}$ and generic level, $H^2(\mathfrak{g}; \mathrm{Sym}^q V) = 0$ by the Whitehead lemma, and the degree-2 MC class lifts uniquely.

Step 2: Algebraic semicontinuity. The vanishing locus

$$E = \{k \in \mathbb{C} : H^2_{\mathrm{cyc,prim}}(\mathcal{A}_k) \neq 0\}$$

is Zariski-closed by upper semicontinuity. At $k = \infty$ the algebra is Heisenberg and $H^2_{\mathrm{cyc,prim}} = 0$, so E is a proper Zariski-closed subset. Explicit computation verifies $E = \varnothing$ for $\mathfrak{sl}_N$ ($N = 2, \ldots, 10$): no exceptional level. The result covers the entire standard Lie-theoretic landscape at all non-critical levels, including admissible.

Scope. Scalar saturation applies to every family in the standard landscape: affine Kac–Moody (all types, all non-critical levels), Virasoro (all c), $\mathcal{W}_N$ (all N, all non-degenerate levels), $\beta\gamma$ and lattice algebras (all parameters). The residual conjecture, that $E = \varnothing$ for every positive-energy chiral algebra with rational OPE coefficients, is open only for non-algebraic-family constructions.

9.2. The Koszulness meta-theorem. The Koszul condition admits twelve formulations, spanning homological algebra (PBW collapse, Ext vanishing), homotopy theory (A_∞ formality, E_2 formality), operadic algebra (Barr–Beck–Lurie, FM acyclicity), modular geometry (FH concentration, tropical Koszulness), and symplectic topology (Lagrangian criterion). Ten are unconditionally equivalent; one (the Lagrangian criterion) is conditional on perfectness/nondegeneracy; one ($\mathcal{D}$-module purity) is one-directional (forward direction proved, converse open).

For $\mathcal{A}$ positive-energy and finitely strongly generated, the following are equivalent:

(i) $\mathcal{A}$ is chirally Koszul.
(ii) The PBW spectral sequence on $\bar{B}^{(g)}(\mathcal{A})$ collapses at E_2 for every g.
(iii) The minimal A_∞-model of $\bar{B}(\mathcal{A})$ (the transferred structure on $H^\bullet$) is formal: $m_n = 0$ for $n \geq 3$.
(iv) $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X) = 0$ for $p \neq q$.
(v) The bar-cobar counit $\Omega(\bar{B}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism.
(vi) The Barr–Beck–Lurie comparison for $\bar{B} \dashv \Omega$ is an equivalence.
(vii) $\int_X \mathcal{A}$ is concentrated in degree 0 for every smooth proper X.
(viii) $\mathrm{ChirHoch}^*(\mathcal{A})$ vanishes outside degrees $\{0, 1, 2\}$.
(ix) The Kac–Shapovalov determinant $\det G_h \neq 0$ in the bar-relevant range.
(x) The restriction $i_S^! \bar{B}_n(\mathcal{A})$ to every FM boundary stratum S is acyclic outside degree 0.

Under the additional hypothesis that the ambient tangent complex is perfect and nondegenerate:

(xi) $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians in the (-1)-shifted symplectic deformation space $\mathcal{M}_{\mathrm{comp}}$.

Finally (xii)$\Rightarrow$(x), where:

(xii) Each $\bar{B}_n(\mathcal{A})$ is pure of weight n as a mixed Hodge module, with characteristic variety aligned to FM boundary strata ($\mathcal{D}$-module purity).

The converse of (xii) is open.

The proof circuit. The core equivalence chain is (i)$\Leftrightarrow$(ii)$\Leftrightarrow$(iii)$\Leftrightarrow$(v)$\Leftrightarrow$(viii), proved by PBW concentration $\to$ bar-cobar inversion $\to$ chiral Hochschild polynomial growth $\to$ Koszul-complex acyclicity, with the A_∞ formality branch via HPL transfer and Keller classicality. From this core: (iv) by Ext spectral sequence, (vi) by conservativity with log-FM totalization, (vii) by $\bar{B} = \int_X$, (ix) by non-degeneracy forcing PBW injectivity, (x) by stratum-by-stratum PBW concentration. The Lagrangian criterion (xi) uses PTVV derived intersection theory on the perfectness hypothesis. The $\mathcal{D}$-module purity implication (xii)$\Rightarrow$(x) uses weight filtration on the bar complex; the converse would require concentration to force purity. The cycle closes via (viii)$\Rightarrow$(i): Koszul-complex acyclicity implies the defining characterisation.

Beyond the meta-theorem. Six additional characterisations are proved in the text:

(a) *Shadow-formality at low degree.* The shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$ equals the L_∞-formality obstruction tower of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ at degrees $r = 2, 3, 4$ (Proposition **??**). Full equivalence to (iii) requires all degrees.

(b) *E_2-formality of chiral Hochschild cohomology.* Koszulness implies $\mathrm{ChirHoch}^*(\mathcal{A})$ is formal as an E_2-algebra (Deligne conjecture applied to the bar-cobar resolution).

(c) *Genus-0 curve independence.* Koszulness on $\mathbb{P}^1$ implies Koszulness on every smooth proper curve (Proposition **??**). Proved independently by factorization descent.

(d) *PBW universality* (sufficient). Every freely strongly generated vertex algebra is chirally Koszul (Proposition **??**). Applies to universal algebras: $V_k(\mathfrak{g})$ is Koszul at every level including critical and admissible. For simple quotients $L_k(\mathfrak{g})$ at admissible levels, status is rank-dependent: $L_k(\mathfrak{sl}_2)$ is Koszul at all admissible levels, but at $\mathrm{rk}(\mathfrak{g}) \geq 2$ with denominator $q \geq 3$, Koszulness fails via a $\mathrm{rk}(\mathfrak{g})$-dimensional bar obstruction from the abelian Cartan (Theorem **??** at $\mathfrak{g} = \mathfrak{sl}_3$; Conjecture **??** at $\mathrm{rk}(\mathfrak{g}) \geq 3$). PBW universality is one-directional: not every Koszul algebra is freely strongly generated.

(e) *Tropical Koszulness.* $\mathcal{A}$ is chirally Koszul iff $\bar{B}^{\mathrm{trop}}(\mathcal{A})$ is acyclic. Proved by the weight spectral sequence on the real blowup.

(f) *Bifunctor obstruction decomposition* (one-directional). Koszulness implies the RNW one-slot obstruction decomposes into independent bar-side and cobar-side components (Theorem **??**). The converse is open.

Status summary.

*Meta-theorem (Theorem **??**):*		
(i)–(v), (vii)$\vert_{g=0}$, (ix), (x)	**8 genuinely independent bidirectional**	proved
(vi) Barr–Beck–Lurie	consequence of (v)	proved
(viii) chiral Hochschild $\{0, 1, 2\}$	one-way consequence on Koszul locus	proved
(xi) Lagrangian	conditional on perfectness	proved (unconditional for standard landscap
(xii) $\mathcal{D}$-module purity	(xii)$\Rightarrow$(x) only	converse open
Additional characterisations:		
(a) Shadow-formality	equivalence at degrees 2, 3, 4	proved
(b) E_2-formality	consequence of Koszulness	proved
(c) Curve independence	independent theorem	proved
(d) PBW universality	sufficient condition	proved
(e) Tropical Koszulness	equivalence	proved
(f) Bifunctor decomposition	one-directional	converse open

9.3. Shadow depth and operadic complexity. The four depth classes G/L/C/M admit refinement: in general $r_{\max} \in \{2, 3, 4, 5, \ldots\} \cup \{\infty\}$, with class $\mathbf{F}_r$ (tower terminates at degree r) structurally permitted for all $r \geq 2$; $\mathbf{G} = \mathbf{F}_2$, $\mathbf{L} = \mathbf{F}_3$, $\mathbf{C} = \mathbf{F}_4$ name the standard-landscape representatives.

The operadic complexity theorem (Theorem **??**):

$$r_{\max}(\mathcal{A}) = A_\infty\text{-depth}(\mathcal{A}) = L_\infty\text{-formality level}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}).$$

A_∞-depth is the minimal r with $m_k = 0$ for $k > r$ on $H^\bullet(\bar{B}(\mathcal{A}))$; L_∞-formality level is the minimal r such that the L_∞-structure on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is formal through degree r. Three consequences: the shadow obstruction tower *is* the L_∞-formality obstruction tower; the depth classification becomes a structural invariant of the operadic type of $\mathcal{A}$; the tower of a Gaussian algebra is a quadratic form, of a Lie algebra a Chevalley–Eilenberg complex, of a contact algebra a cyclic A_∞-structure, of a mixed algebra an infinite L_∞-tower with no polynomial truncation.

The programme extends to non-principal $\mathcal{W}$-algebras: the Bershadsky–Polyakov algebra BP_k is chirally Koszul at generic level ($k \neq -3$) with dual BP_{-k-6} and Koszul conductor $K_{\mathrm{BP}} = 196$ (a global duality invariant, not a local ghost constant). Nilpotent transport from hook-type seeds covers all orbits in $\mathfrak{sl}_2$ through $\mathfrak{sl}_8$.

9.4. The PBW spectral sequence. The computational engine for bar cohomology within fixed genus is the PBW (Poincaré–Birkhoff–Witt) spectral sequence. Filter $\bar{B}^{(g)}(\mathcal{A})$ by *conformal weight*: assign each normally ordered monomial its total weight (sum of constituent field weights). The associated graded is the bar complex of a free-field algebra, reducing to symmetric-function combinatorics.

The E_1 page at genus g:

$$E_1^{p,q}(g) = E_1^{p,q}(g{=}0) \oplus \mathcal{E}_g^{p,q},$$

with the first summand the genus-0 bar cohomology determined by Koszulness of the associated graded and $\mathcal{E}_g^{p,q}$ the genus-g enrichment from the Hodge bundle, arising from the regular Eisenstein series coupled to the curvature. *Concentration*: $E_\infty^{p,q}(g) = 0$ for $q \neq 0$ at all $g \geq 1$, for all standard families. PBW degeneration: the spectral sequence collapses at E_2 and bar cohomology concentrates on the diagonal.

The PBW spectral sequence is distinct from the genus spectral sequence of §5.5. PBW uses conformal weight within fixed genus to compute bar cohomology groups; the genus spectral sequence uses the genus filtration on the deformation algebra to control the MC lift across genera. One computes *what*; the other determines *whether*.

9.5. Chiral Hochschild cohomology (Theorem H). The chiral Hochschild complex $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ is the derived endomorphism algebra of the identity functor on $\mathcal{A}$-modules, computed by the bar construction with bimodule coefficients. At genus 0,

$$\mathrm{ChirHoch}^\bullet(\mathcal{A}) \simeq R\mathrm{Hom}_{\mathcal{A}\text{-bimod}}(\mathcal{A},\mathcal{A}) \simeq \mathrm{Ext}^\bullet_{\mathcal{A}^e}(\mathcal{A},\mathcal{A}),$$

where $\mathcal{A}^e = \mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}$ is the enveloping chiral algebra.

For $\mathcal{A} = \mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ a principal $\mathcal{W}$-algebra at generic level, the chiral Hochschild cohomology is concentrated in degrees $\{0, 1, 2\}$:

$$\mathrm{ChirHoch}^n(\mathcal{W}^k(\mathfrak{g})) = 0 \text{ for } n > 2, \qquad P(t) = 1 + t^2.$$

The deformation class in $\mathrm{ChirHoch}^2 = \mathbb{C}$ is the level deformation $k \mapsto k + \epsilon$. The concentration bound is the de Rham amplitude on a curve ($\dim X = 1$); the continuous Lie algebra cohomology $H^*_{\mathrm{cont}}(\mathrm{Lie}(\mathcal{W}), \mathrm{Lie}(\mathcal{W})_0; \mathbb{C}) = \mathbb{C}[\Theta_1, \ldots, \Theta_r]$ (Gelfand–Fuchs, $\deg \Theta_i = m_i + 1$) is a different, unbounded invariant. For $\mathfrak{g} = \mathfrak{sl}_2$ (Virasoro): $P(t) = 1 + t^2$. For $\mathfrak{g} = \mathfrak{sl}_3$ ($\mathcal{W}_3$): $P(t) = 1 + t^2$. Total dimension 2, never $\mathbb{C}[\Theta]$.

On the Koszul locus, $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ carries an E_2-algebra structure that is formal. The Connes periodicity operator $S\colon \mathrm{ChirHoch}^n \to \mathrm{ChirHoch}^{n-2}$ intertwines Koszul duality: $S \circ \mathrm{KD} = \mathrm{KD} \circ S$. At genus 0, the chiral Hochschild complex identifies with the bulk chiral homology complex of Volume II.

The bulk-boundary-line triangle is the genus-0 shadow of this identification: $\mathcal{A}_{\mathrm{bulk}} \simeq Z_{\mathrm{der}}(\mathcal{B}_\partial) \simeq \mathrm{ChirHoch}^\bullet(\mathcal{A}^!)$.

Per-family computation.

Family	ChirHoch^0	ChirHoch^1	ChirHoch^2
$\mathcal{H}_k$	$\mathbb{C}$	0	$\mathbb{C}$
$V_k(\mathfrak{g})$ ($k \neq -h^\vee$)	$\mathbb{C}$	$\mathfrak{g}$	$\mathbb{C}$
Vir_c	$\mathbb{C}$	0	$\mathbb{C}$
$\mathcal{W}_{3,c}$	$\mathbb{C}$	0	$\mathbb{C}$
$\beta\gamma_\lambda$	$\mathbb{C}$	0	$\mathbb{C}$
V_Λ	$\mathbb{C}$	0	$\mathbb{C}$

The Hilbert polynomial is $P(t) = 1 + t^2$ for every family except affine Kac–Moody, where $P(t) = 1 + \dim(\mathfrak{g})\, t + t^2$ (cohomological amplitude $[0, 2]$ in both cases). The degree-1 class for $V_k(\mathfrak{g})$ is the Lie algebra $\mathfrak{g}$ acting by infinitesimal automorphisms of the affine algebra; it has total dimension $\dim(\mathfrak{g}) + 2$. Three invariants that are NOT chiral Hochschild: (i) the Gelfand–Fuchs cohomology $H^*_{\mathrm{cont}}(\mathrm{Lie}(\mathcal{A}))$ is infinite-dimensional for Virasoro and $\mathcal{W}_N$ (polynomial algebra $\mathbb{C}[\Theta_1, \ldots, \Theta_r]$ with $\deg \Theta_i = m_i + 1$); (ii) the naive centre $Z(\mathcal{A}) = \{a \in \mathcal{A} : [a, b] = 0\ \forall b\}$ has dimension 1 for $V_k(\mathfrak{g})$ (the vacuum) but $\mathrm{ChirHoch}^0 = \mathbb{C}$ matches; (iii) the categorical Hochschild of the module category $\mathrm{HH}(\mathcal{A}\text{-mod})$ carries a Calabi–Yau structure that differs from the chiral Hochschild by the CY shift.

9.6. The shadow algebra. The *shadow algebra* $\mathcal{A}^{\mathrm{sh}}$ is the cohomology of the modular cyclic deformation complex as a graded-commutative ring,

$$\mathcal{A}^{\mathrm{sh}} := H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})) = \bigoplus_{r \geq 2, g \geq 0} \mathcal{A}^{\mathrm{sh}}_{r,g}.$$

The single MC element $\Theta_{\mathcal{A}}$ decomposes by degree and genus. Named shadows are projections:

$$\kappa(\mathcal{A}) = \mathcal{A}^{\mathrm{sh}}_{2,0}, \quad \mathfrak{C}(\mathcal{A}) = \mathcal{A}^{\mathrm{sh}}_{3,0}, \quad \mathfrak{Q}(\mathcal{A}) = \mathcal{A}^{\mathrm{sh}}_{4,0}, \quad \mathrm{Sh}_r(\mathcal{A}) = \mathcal{A}^{\mathrm{sh}}_{r,0} \text{ for } r \geq 5.$$

The shadow algebra organises the entire hierarchy; it is a homotopy invariant, computed by the Weil homomorphism of the Chern–Weil dictionary (§5.4).

The all-degree master equation in the shadow algebra: $\nabla_H(\mathrm{Sh}_r) + o^{(r)} = 0$ for all $r \geq 3$. The obstruction class $o^{(r)}$ lies in $\mathcal{A}^{\mathrm{sh}}_{r,0}$ and measures the failure of $\Theta^{\leq r-1}_{\mathcal{A}}$ to extend to degree r.

9.7. BV/BRST identification at genus 0. At genus 0, the bar differential of a chiral algebra on $\mathbb{C}$ is the BRST differential of the associated holomorphic twist, $(d_{\bar{B}})_{g=0} = Q_{\mathrm{BRST}}$. The bar complex $\bar{B}^{(0)}_{\mathbb{C}}(\mathcal{A})$ is the BRST complex of the 2d chiral algebra, and bar-cobar inversion $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ (Theorem B) says the BRST cohomology processed through the cobar functor reconstructs the original algebra.

At genus 1, the identification acquires curvature: $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_1$ is the one-loop anomaly, and the curved bar complex is the curved BRST complex of the theory on an elliptic curve. The complementarity $Q_1(\mathcal{A}) \oplus Q_1(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_{1,1}, \mathcal{Z}_{\mathcal{A}})$ is the genus-1 anomaly cancellation condition.

At genus $g \geq 2$, the identification with Costello–Gwilliam requires renormalisation: higher-loop integrals over $\overline{\mathcal{M}}_{g,n}$ need regularisation beyond the algebraic bar-cobar framework, and the correct receptacle is the coderived category.

For the Heisenberg algebra, the identification is proved at all genera by four independent paths: zeta-regularisation, Selberg zeta function, Grothendieck–Riemann–Roch, direct bar computation. The shadow obstruction tower equals the L_∞-formality obstruction tower at all degrees.

Genus-two shell activation as depth diagnostic. The genus-two shell profile independently confirms the depth classification:

	$\Theta^{(2)}_{\mathrm{loop}^2}$	$\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}}$	$\Theta^{(2)}_{\mathrm{pf}}$
G (Heisenberg, lattice)	✓	--	--
L (affine $\widehat{\mathfrak{g}}_k$)	✓	✓	✓
C ($\beta\gamma$)	✓	--	--
M (Virasoro, $\mathcal{W}_N$)	✓	✓	✓

G and C activate only the loop∘loop shell; L and M activate all three including planted-forest. The diagnostic distinguishing L from M is the critical discriminant $\Delta = 8\kappa S_4$: zero for L ($S_4 = 0$), nonzero for M.

Structural refinements. The three bar complexes on a chiral algebra (ordered tensor T^c, symmetric Sym^c, Chevalley–Eilenberg Lie^c) sit in a chain $\mathrm{Lie}^c \hookrightarrow \mathrm{Sym}^c \hookrightarrow T^c$ with distinct coalgebra structures at each level (Theorem **??**). The averaging map $\mathrm{av}\colon T^c \to \mathrm{Sym}^c$ from ordered to symmetric is surjective but not split: the kernel at degree 3 contains the Drinfeld associator $\Phi_{\mathrm{KZ}}(\mathcal{A})$, and the fibre of splittings is a torsor for GRT_1. The E_1 ordered bar complex is therefore the natural primitive object; the modular/symmetric story is its Σ_n-coinvariant image (Theorem **??**). The genus-3 cross-channel correction for $\mathcal{W}_3$ has been computed by summing over all 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$ (Proposition **??**): the cross-channel tower accelerates with genus and dominates the scalar part $\kappa \cdot \lambda_g^{\mathrm{FP}}$ by genus 4 (ALL-WEIGHT, WITH CROSS-CHANNEL CORRECTION).

On the open/closed side, the Koszul dual cooperad $\mathsf{SC}^{\mathrm{ch,top,!}}$ decomposes into three sectors: closed ($\dim = (n-1)!$, Lie cooperad), open ($\dim = m!$, Ass cooperad), and mixed ($\dim = (k-1)!\binom{k+m}{m}$). The mixed sector encodes the bulk-to-boundary module structure by which the chiral derived centre acts on the boundary algebra.

The BV/BRST = bar identification at higher genus (Conjecture **??**): the chain-level obstruction (harmonic propagator correction) is resolved for classes G, L, and C, for G and L by the Heisenberg sewing theorem and its extension to affine algebras, for C by contact decoupling. For class M (Virasoro, $\mathcal{W}_N$) the identification holds on bar cohomology but fails at the chain level: the harmonic propagator produces a nonvanishing quartic obstruction, so a coderived reformulation is needed.

9.8. The 14-characterisation circuit. The eight genuinely independent bidirectional equivalences among (i)–(x) -- namely (i)–(v), the genus-0 clause of (vii), (ix), and (x) -- together with (vi) (consequence of (v)) and (viii) (one-way chiral Hochschild consequence on the Koszul locus), the Lagrangian criterion (xi), the $\mathcal{D}$-module purity implication (xii), and the six additional characterisations (a)–(f) organise into a circuit with four hubs:

Hub	**Characterisations**
Homological	(i) Koszulness, (ii) PBW collapse, (iv) Ext diagonal, (ix) Kac–Shapovalov
Homotopical	(iii) A_∞ formality, (v) bar-cobar qi, (vi) Barr–Beck–Lurie, (b) E_2-formality
Geometric	(vii) FH concentration, (x) FM acyclicity, (xi) Lagrangian, (xii) $\mathcal{D}$-module purity
Tropical	(e) tropical Koszulness, (c) curve independence, (d) PBW universality

The core chain (i)⇔(ii)⇔(iii)⇔(v)⇔(viii) forms a pentagon connecting all four hubs. Each link is an independent theorem:

- (i)⇔(ii): PBW concentration controls bar resolution by conformal-weight filtration.
- (ii)⇔(iii): HPL transfer converts spectral collapse to A_∞ formality (Keller classicality theorem).
- (iii)⇔(v): A_∞ formality on $H^*(\bar{B})$ is equivalent to the counit $\Omega(\bar{B}(\mathcal{A})) \to \mathcal{A}$ being a qi (a chain-level inversion, not merely a cohomological isomorphism).

- (v)$\Leftrightarrow$(viii): bar-cobar inversion gives $\mathrm{ChirHoch}^*(\mathcal{A}) \simeq R\mathrm{Hom}_{\mathcal{A}^e}(\mathcal{A}, \mathcal{A}) \simeq H^*(B(\mathcal{A}))$; the computation in the bar model yields concentration in degrees $\{0, 1, 2\}$.

From this core: (iv) by the Ext spectral sequence from (ii); (vi) by conservativity of the $\bar{B} \dashv \Omega$ adjunction with log-FM totalisation; (vii) by the identification $\bar{B} = \int_X$ (the bar complex computes factorisation homology on X); (ix) by non-degeneracy of the Kac–Shapovalov form forcing PBW injectivity at the relevant conformal weights; (x) by the FM stratum restriction carrying PBW concentration stratum-by-stratum.

The Lagrangian criterion (xi). Under perfectness and nondegeneracy of the ambient tangent complex, $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are Lagrangians in a (-1)-shifted symplectic stack $\mathcal{M}_{\mathrm{comp}}$. Their transversality is the geometric incarnation of bar-cobar inversion: the Lagrangian intersection $\mathcal{M}_{\mathcal{A}} \cap \mathcal{M}_{\mathcal{A}^!}$ is clean iff $\mathcal{A}$ is Koszul. The proof uses the PTVV derived intersection theory, specifically the (-1)-shifted symplectic structure on the mapping stack $\mathrm{Map}(X, \mathcal{M}_{\mathrm{comp}})$ for smooth proper X. Perfectness ensures the cotangent complex is perfect (bounded coherent), which is automatic for all standard landscape families but not for arbitrary chiral algebras.

The $\mathcal{D}$-module purity direction (xii)$\Rightarrow$(x). Each $\bar{B}_n(\mathcal{A})$ on $\overline{C}_n(X)$ is a $\mathcal{D}$-module. If it is pure of weight n as a mixed Hodge module (meaning the weight filtration has a single nonzero graded piece at weight n), then its restriction to any FM boundary stratum is acyclic outside degree 0, which is characterisation (x). The forward direction uses the weight filtration on nearby cycles; the converse (concentration $\Rightarrow$ purity) would require concentration to *force* purity, which is not known in general.

9.9. Scalar saturation: the proof. The scalar saturation theorem asserts that for a one-parameter algebraic family $\mathcal{A}_k$, the universal MC element $\Theta_{\mathcal{A}_k}$ is determined by the single scalar $\kappa(\mathcal{A}_k)$. The proof has three steps:

Step 1 (Whitehead reduction at generic level). The cyclic deformation complex $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A}_k)$ at generic k has a primitive filtration with associated graded controlled by Lie algebra cohomology. The relevant group is $H^2(\mathfrak{g}; \mathrm{Sym}^q V)$ for the Lie algebra $\mathfrak{g}$ underlying the chiral algebra at the generic point. For reductive $\mathfrak{g}$ and generic k, the Whitehead lemma gives $H^2(\mathfrak{g}; M) = 0$ for any finite-dimensional $\mathfrak{g}$-module M; in particular $H^2(\mathfrak{g}; \mathrm{Sym}^q V) = 0$ for all q. The vanishing at $q = 0$ kills the primitive degree-2 obstruction; the vanishing at all q kills all higher corrections to the degree-2 MC class. The unique lift is $\kappa(\mathcal{A}_k) \cdot \eta_{\mathrm{univ}} \otimes \Lambda_{\mathrm{univ}}$.

Step 2 (semicontinuity). The non-vanishing locus $E = \{k \in \mathbb{C} : H^2_{\mathrm{cyc,prim}}(\mathcal{A}_k) \neq 0\}$ is Zariski-closed by upper semicontinuity of cohomology dimensions in algebraic families. Since the generic point (which is the Heisenberg limit $k \to \infty$) has $H^2_{\mathrm{cyc,prim}} = 0$, the set E is a proper Zariski-closed subset of $\mathbb{C}$, hence finite. If E is empty, scalar saturation holds at all levels; if E is nonempty, it consists of finitely many exceptional levels.

Step 3 (explicit verification). For $\mathfrak{g} = \mathfrak{sl}_N$ with $N = 2, \ldots, 10$, the computation verifies $E = \varnothing$: no exceptional level exists. The computation uses the PBW spectral sequence at each candidate level (roots of the Kac–Shapovalov determinant, admissible levels, critical level) and confirms that $H^2_{\mathrm{cyc,prim}} = 0$ everywhere. The critical level $k = -h^\vee$ is included: $\kappa(-h^\vee) = 0$, so scalar saturation gives $\Theta = 0$, consistent with the Feigin–Frenkel centre (the MC element vanishes when the curvature vanishes).

9.10. The admissible-level question. At admissible levels $k = -h^\vee + p/q$ ($p \geq h^\vee$, $q \geq 1$, $\gcd(p, q) = 1$), the universal vertex algebra $V_k(\mathfrak{g})$ is *not* simple: the maximal ideal J_k produces the simple quotient $L_k(\mathfrak{g}) = V_k(\mathfrak{g})/J_k$. The bar-Koszul theory bifurcates:

(i) $V_k(\mathfrak{g})$ (the universal algebra) is chirally Koszul at every level, including admissible and critical, by PBW universality: it is freely strongly generated by the currents J^a, and PBW universality applies.

(ii) $L_k(\mathfrak{g})$ (the simple quotient) at admissible levels is a different question. The quotient map $V_k(\mathfrak{g}) \twoheadrightarrow L_k(\mathfrak{g})$ kills singular vectors, potentially destroying the PBW property.

The rank-1 case is completely settled: $L_k(\mathfrak{sl}_2)$ is chirally Koszul at all admissible levels $k = -2 + p/q$, because the bar complex of the simple quotient inherits concentration from the universal algebra via the surjection $\bar{B}(V_k) \twoheadrightarrow \bar{B}(L_k)$ (the bar functor preserves surjections).

At rank ≥ 2 with denominator $q \geq 3$, Koszulness of $L_k(\mathfrak{g})$ is expected to fail: the abelian Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ produces a $\mathrm{rk}(\mathfrak{g})$-dimensional bar obstruction in $H^2(\bar{B}(L_k))$ that does not arise for V_k. The obstruction comes from the interaction between singular vectors and the Cartan direction, which is invisible at rank 1. This constitutes a rank-dependent Koszulness failure that distinguishes universal from simple vertex algebras at admissible levels. The precise rank threshold and the structure of the obstruction complex are the subject of an ongoing programme.

The practical consequence: all formulas in this survey apply to *universal* vertex algebras $V_k(\mathfrak{g})$ at all levels without restriction. Specialisation to simple quotients $L_k(\mathfrak{g})$ requires additional analysis that depends on the level denominator and the rank of $\mathfrak{g}$.

10. The open/closed world (Volume II)

Sections 1–9 construct E_n-chiral algebras as algebraic-geometric objects on curves: bar complexes on every curve geometry, R-matrices, Yangian structures, derived chiral centres carrying E_2/E_3 structure, and the $\mathrm{SC}^{\mathrm{ch,top}}$ datum on the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$. Volume II discovers that these constructions are physics: the derived chiral centre IS the bulk algebra of a real 3d holomorphic-topological gauge theory. The passage from algebraic geometry to physics is through the derived centre.

On the product $\mathbb{C}_z \times \mathbb{R}_t$, the holomorphic direction produces the bar differential of Sections 1–9, extracting OPE residues from collisions in the holomorphic plane. The topological direction produces the deconcatenation coproduct, splitting an ordered tensor sequence at a cut point of the real line. The two are unified by the *holomorphic-topological Swiss-cheese operad* $\mathrm{SC}^{\mathrm{ch,top}}$: closed colour ch $= \overline{C}_k(\mathbb{C})$ governs holomorphic factorization, open colour top $= E_1(m)$ governs topological factorization, and the open/closed mixed sector encodes the action of bulk on boundary.

The primitive object of 3d HT quantum field theory on $\mathbb{C}_z \times \mathbb{R}_t$ is not the bar complex but the open/closed factorization dg-category C on the bordified curve $\widetilde{X}_D$. Objects are boundary conditions, morphisms are open-string states, and a vacuum b produces a boundary algebra $A_b = \mathrm{End}_C(b)$ only up to Morita equivalence. The governing operad is $\mathrm{SC}^{\mathrm{ch,top}}$; it is homotopy-Koszul (proved via Kontsevich formality), so the bar-cobar adjunction on $\mathrm{SC}^{\mathrm{ch,top}}$-algebras is a Quillen equivalence.

The bar complex of Volume I is the ordered E_1 coalgebraic engine for C: its differential records holomorphic collision data, and its coproduct records ordered topological splitting. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure itself lives on the derived-center pair $(C^\bullet_{\mathrm{ch}}(A_b, A_b),\ A_b)$. Three objects must never be conflated:

(i) the *bar complex* $\bar{B}(A_b)$ classifies twisting morphisms, universal couplings between A_b and $A_b^!$;
(ii) the *chiral derived centre* $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(C) = R\mathrm{Hom}_{\mathrm{Fun}(C,C)}(\mathrm{Id}, \mathrm{Id})$ is the universal bulk, computed by chiral Hochschild cochains in any chart;
(iii) the Koszul dual $A_b^!$ governs the line-operator category $A_b^!$-mod.

The primitive hierarchy

$$C_{\mathrm{op}} \xrightarrow{\mathrm{End}(b)} A_b \xrightarrow{\bar{B}} \bar{B}(A_b) \xrightarrow{\Theta} \Theta_{\mathcal{A}} \xrightarrow{+\mu^{M_j}} \Theta^{\mathrm{oc}}_{\mathcal{A}}$$

records successive lossy functors: C_{op} is Morita invariant; A_b depends on b; $\bar{B}$ forgets the bulk; $\Theta_{\mathcal{A}}$ is the closed-sector MC element; $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ restores open/closed data. Modularity is determined by the open sector:

a cyclic trace on $\mathrm{End}(b)$ seeds the Calabi–Yau structure, the annulus identification $\int_{S^1} C \simeq \mathrm{HH}_\bullet(C)$ is the first modular shadow, and clutching along boundary circles raises genus.

Volume II proceeds in four stages. Stage 1 (local one-colour theorem: Swiss-cheese universality of the derived centre), Stage 2 (open primitive: C with Morita invariance), and Stage 3 (globalisation on tangential log curves (X, D, τ)) are proved (Stage 3 locally; global extension modest). In Stage 4 (modularisation), the open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ is constructed, the modular MC equation holds at all genera, and the projection principle recovers both the closed-sector genus tower and the open-sector module structures. The annulus trace is proved; the independent higher-genus derivation and chain-level cooperad compatibilities remain open.

10.1. The Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$**.** Two colours $\{\mathrm{ch}, \mathrm{top}\}$, with operation spaces

$$\begin{aligned} \mathrm{SC}(\mathrm{ch}^k; \mathrm{ch}) &= \overline{C}_k(\mathbb{C}), \\ \mathrm{SC}(\mathrm{ch}^k, \mathrm{top}^m; \mathrm{top}) &= \overline{C}_k(\mathbb{C}) \times E_1(m), \\ \mathrm{SC}(\dots, \mathrm{top}, \dots; \mathrm{ch}) &= \varnothing. \end{aligned}$$

Closed-to-closed is the Fulton–MacPherson compactification in $\mathbb{C}$; the mixed direction combines holomorphic configurations acting on the fibre above a topological interval; open-to-closed is forbidden. Operadic composition is the boundary stratification of FM spaces in the holomorphic direction and nested interval inclusion in the topological direction.

A bar element of tensor degree k lives on the product $\overline{C}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$. The bar differential extracts OPE residues along collision divisors of the first factor: when points z_i, z_j collide, the residue of $\eta_{ij} = d\log(z_i - z_j)$ yields the mode $a_{(n)}b$. The coproduct splits the ordered sequence at a cut point of the second factor: for $t_1 < \cdots < t_k$ with $t_p < c < t_{p+1}$, $(a_1 \otimes \cdots \otimes a_k)$ decomposes as $(a_1 \otimes \cdots \otimes a_p) \otimes (a_{p+1} \otimes \cdots \otimes a_k)$. The bar complex with both structures is a $\mathrm{SC}^{\mathrm{ch,top}}$-algebra.

10.2. Homotopy-Koszulity. The classical Swiss-cheese operad SC is Koszul (Livernet–Voronov, Ginzburg–Kapranov), with quadratic presentation and distributive-lattice criterion on the poset of nested disks. Kontsevich formality supplies a quasi-iso $\Phi\colon \mathrm{SC}^{\mathrm{ch,top}} \xrightarrow{\sim} \mathrm{SC}$ from configuration-space integrals on FM spaces with propagators, extended by including $\mathbb{H}$ and $\mathbb{R}$. The bar-cobar adjunction preserves quasi-isos, and two-out-of-three in the operadic model category gives $\Omega\mathbf{B}(\mathrm{SC}^{\mathrm{ch,top}}) \xrightarrow{\sim} \mathrm{SC}^{\mathrm{ch,top}}$. The bar-cobar adjunction on $\mathrm{SC}^{\mathrm{ch,top}}$-algebras is therefore a Quillen equivalence, with three consequences:

(i) every $\mathrm{SC}^{\mathrm{ch,top}}$-algebra admits a cofibrant bar-cobar resolution;
(ii) the homotopy category of $\mathrm{SC}^{\mathrm{ch,top}}$-algebras is equivalent to that of coalgebras over the Koszul dual cooperad;
(iii) the line-operator category $A^!_b$-mod is unconditionally defined without finiteness hypotheses on $\mathcal{A}$.

10.3. The A_∞ chiral algebra structure. The full Swiss-cheese MC element $\alpha_T \in \mathrm{MC}(\mathfrak{g}^{\mathrm{SC}}_T)$ lives in the convolution algebra $\mathfrak{g}^{\mathrm{SC}}_T = \mathrm{Hom}_\Sigma(\mathbf{B}(\mathrm{SC}^{\mathrm{ch,top}}), \mathrm{End}_{(B_\partial, C_{\mathrm{line}})})$. Its closed-colour projection $(m_k)_{k\geq 1}$ encodes the A_∞ chiral algebra on $\mathcal{A}$. Each $m_k\colon \mathcal{A}^{\otimes k} \to \mathcal{A}((\lambda_1)) \cdots ((\lambda_{k-1}))$ is a configuration-space integral on $\overline{C}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$, with the holomorphic factor producing OPE singularities and the topological factor producing the ordering. Summing over trees T with k leaves, $m_T(a_1, \dots, a_k) = \int_{\overline{C}_k(\mathbb{C})} \omega_T \prod_v a_{(n_v)} b_v$. The Maurer–Cartan equation is the spectral Stasheff identity: Stokes' theorem on boundary strata of $\overline{C}_k(\mathbb{C})$ produces the A_∞ relations.

10.4. PVA descent (Theorem G). On cohomology, the operations simplify. The regular part of m_2 is a commutative associative product $\mu(a,b) = \mathrm{Reg}_{z_1 \to z_2} m_2(a(z_1), b(z_2))$; commutativity follows from the $\mathbb{Z}/2$ involution of $\overline{C}_2(\mathbb{C})$, associativity from the three FM_3 boundary strata. The singular part defines the lambda-bracket $\{a_\lambda b\} = \mathrm{Res}_{z_1 \to z_2} e^{\lambda(z_1 - z_2)} m_2(a(z_1), b(z_2))$; the exponential extracts all pole orders. All $m_{k\geq 3} = 0$ on cohomology because $\mathrm{Conf}_k(\mathbb{R})$ is contractible for $k \geq 3$.

The Arnold relation $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$ on $\overline{C}_3(\mathbb{C})$ is equivalent to the three codim-1 boundary divisors of $\overline{M}_{0,4}$ summing to zero in homology; its residue against $a \otimes b \otimes c$ produces the Jacobi identity for $\{-_\lambda-\}$. The three-face Stokes identity applied to $\omega = \eta_{12} \wedge \eta_{23} \cdot a(z_1)b(z_2)c(z_3)$ produces the Leibniz rule $\{a_\lambda (b \cdot c)\} = \{a_\lambda b\}c + b\{a_\lambda c\}$. The exchange cylinder on $\overline{C}_2(\mathbb{C}) \times E_1(2)$ gives skew-symmetry $\{a_\lambda b\} = -\{b_{-\lambda-\partial} a\}$. Thus $(\mu, \{-_\lambda-\})$ is a (-1)-shifted Poisson vertex algebra, each axiom in one line from the geometry of FM.

(Vol I convention: we state the PVA in OPE modes; Vol II uses lambda-brackets throughout, and the two conventions are translated by the standard $\{a_\lambda b\} = \sum_{n\geq 0} \lambda^n a_{(n)} b/n!$.)

10.5. The Heisenberg algebra in Swiss-cheese coordinates. For $\mathcal{H}_k$: $m_2(J,J) = k \cdot \mathbf{1}$, $m_{k\geq 3} = 0$, $\{J_\lambda J\} = k\lambda$. All higher operations vanish because the OPE has no simple pole. The spectral R-matrix is the scalar $R(z) = \exp(k/z)$; YBE holds trivially. The PVA is the free bosonic PVA, the universal enveloping PVA of the rank-one abelian Lie conformal algebra.

10.6. The Koszul triangle. From C on $\widetilde{X}_D$ three independent objects arise:

(i) Boundary $A_b = \mathrm{End}_C(b)$: a chart.

(ii) Bulk $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(C) = R\mathrm{Hom}_{\mathrm{Fun}(C,C)}(\mathrm{Id}, \mathrm{Id})$: the universal bulk, Morita invariant, computed by chiral Hochschild cochains.

(iii) Lines $A^!_{b\infty} = D_{\mathrm{Ran}}(\bar{B}(A_b))$: the homotopy Koszul dual, whose cohomology on the Koszul locus recovers the strict dual $A^!_b = (H^*(\bar{B}(A_b)))^\vee$. The module category $A^!_b$-mod with spectral braiding $R(z)$ is the braided tensor category of line operators.

The triangle of identifications

$$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(C) \simeq \mathrm{HH}^\bullet(A_b) \simeq \mathrm{ChirHoch}^\bullet(A^!_b), \qquad A^!_b\text{-mod} = \{\text{line operators}\}$$

relates three distinct functors: endomorphisms (bulk), Ran-space Verdier duality (lines), and identity-bimodule endomorphisms (centre). The bar complex classifies twisting morphisms, not the bulk.

Spectral braiding. Stokes' theorem on $\overline{C}_3(\mathbb{C})$ applied to $\alpha \star \alpha \star \alpha$ decomposes the boundary into three pairwise collision divisors, one for each side of the Yang–Baxter equation $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$.

Bulk = derived centre. Let $\mathcal{B}_\partial$ be the boundary factorization algebra on $\mathbb{R}$. Its derived centre $Z_{\mathrm{der}}(\mathcal{B}_\partial) = R\mathrm{Hom}_{\mathcal{B}_\partial\text{-bimod}}(\mathcal{B}_\partial, \mathcal{B}_\partial)$ computes endomorphisms of the identity bimodule. The E_1-structure gives $Z_{\mathrm{der}}(\mathcal{B}_\partial) \simeq \mathrm{HH}^\bullet(\mathcal{B}_\partial)$ as E_2-algebra. The Swiss-cheese structure identifies the holomorphic bulk with this E_2-algebra.

From E_2 to E_3: the topologisation mechanism. The E_2-algebra structure on the derived centre is the minimal output. When $\mathcal{A}$ carries a conformal vector $T(z)$ at non-critical level, the Sugawara construction $T(z) = \{Q, G(z)\}$ makes $\mathbb{C}$-translations Q-exact: $L_{-1} = [Q, G_0]$ on the BRST complex. In cohomology, the complex structure on $\mathbb{C}$ becomes irrelevant: the E_2 holomorphic structure degenerates to an E_2 topological structure, and Dunn additivity $E_1^{\mathrm{top}} \otimes E_2^{\mathrm{top}} \simeq E_3^{\mathrm{top}}$ promotes the combined structure to E_3-topological; the conformal vector kills the chiral direction. For affine Kac–Moody $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$: proved cohomologically and chain-level on a qi-equivalent model. For general chiral algebras with conformal vector: conjectural (the 3d HT BRST complex is conditional).

For algebras without conformal vector (Heisenberg at critical level, lattice VOAs without Sugawara): stuck at $\mathrm{SC}^{\mathrm{ch,top}}$.

The E_3 identification theorem: for simple $\mathfrak{g}$ at non-critical level, the derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ is identified as an E_3-topological algebra that forms a family over $\quad \cdot H^3(\mathfrak{g})[[\quad]]$. The identification uses E_3 formality (Kontsevich) and the one-dimensionality of $H^3(\mathfrak{g})$ for simple $\mathfrak{g}$ to force order-by-order uniqueness. For non-simple $\mathfrak{g}$: $H^3(\mathfrak{g})$ may be higher-dimensional, and the identification is open.

The Koszul dual cooperad. The homotopy-Koszul dual cooperad $\mathsf{SC}^{\mathrm{ch,top,!}} = (\mathrm{Lie}^c, \mathrm{Ass}^c, \text{shuffle-mixed})$ has:

- Closed sector: $\dim \mathsf{SC}^{\mathrm{ch,top,!}}(\mathrm{cl}^n; \mathrm{cl}) = (n-1)!$ (Lie cooperad). The factorial growth distinguishes it from $\mathrm{SC}^{\mathrm{ch,top}}$ itself, which has $\dim \mathrm{SC}^{\mathrm{ch,top}}(\mathrm{cl}^n; \mathrm{cl}) = 1$ (commutative).
- Open sector: $\dim \mathsf{SC}^{\mathrm{ch,top,!}}(\mathrm{op}^m; \mathrm{op}) = m!$ (associative cooperad, self-dual).
- Mixed sector: $\dim \mathsf{SC}^{\mathrm{ch,top,!}}(\mathrm{cl}^k, \mathrm{op}^m; \mathrm{op}) = (k-1)!\binom{k+m}{m}$, encoding shuffle-mixed compositions.

$\mathrm{SC}^{\mathrm{ch,top}}$ is NOT Koszul self-dual: the closed dimensions are $(n-1)! \neq 1$. What IS true: the bar-cobar duality functor is an involution ($(P^!)^! \simeq P$ for Koszul operads), but the operad $P = \mathrm{SC}^{\mathrm{ch,top}}$ is not isomorphic to $P^!$. The Koszul duality exchanges Com (closed) with Lie (closed); Ass (open) is self-dual.

10.7. Curved Swiss-cheese at $g \geq 1$. At genus $g \geq 1$, the bar complex carries curvature:

$$d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g,$$

with ω_g the Arakelov $(1,1)$-form representing the first Chern class of the relative dualising sheaf. The coproduct remains strictly coassociative: the interval $\mathbb{R}$ has no topology, so the curvature lives entirely in the holomorphic direction. The total corrected differential $D_g = d_{\mathrm{fib}} + \nabla^{\mathrm{GM}}_g$ restores flatness by coupling the ordering to the Hodge bundle: $d^2_{\mathrm{fib}} + [d_{\mathrm{fib}}, \nabla^{\mathrm{GM}}] + (\nabla^{\mathrm{GM}})^2 = \kappa\omega_g - \kappa\omega_g = 0$.

The derived category must be replaced by the coderived category $\mathsf{D}^{\mathrm{co}}(\mathcal{A}^{(g)})$: ordinary $d^2 = 0$ fails at $g \geq 1$, every object is acyclic in the derived sense, and the correct homotopy theory comes from totalising the curvature bicomplex. Passage from derived to coderived is the categorified form of passage from $d^2 = 0$ to curved MC $d\alpha + \alpha^2 + \kappa\omega = 0$.

10.8. Lagrangian self-intersection. Volume II identifies the geometric substrate of the bar construction: the bar complex is the structure sheaf of the Lagrangian self-intersection $\mathfrak{S} = \mathcal{L} \times_{\mathcal{M}} \mathcal{L}$ in a (-2)-shifted symplectic stack. Every theorem is a face of that geometry: Theorem A is the groupoid comodule-module adjunction for $\mathfrak{S}$; Theorem B (bar-cobar inversion) reconstructs the Lagrangian from its clean self-intersection; Theorem C (complementarity) is two Lagrangians whose intersection carries a (-1)-shifted symplectic structure; Theorem D is the Kodaira–Spencer class; Theorem H (chiral Hochschild polynomial growth) is the HKR theorem for the Lagrangian embedding.

10.9. Factorization-primary hierarchy. $\mathrm{SC}^{\mathrm{ch,top}}$ is a *local model*: the formal completion of a factorisation structure at collision points. The global truth is a factorisable $\mathcal{D}$-module on $\mathrm{Ran}(\Sigma_g)$ over $\overline{\mathcal{M}}_g$. Six layers:

(i) Geometric substrate: smooth and stable nodal curves, Ran space, $\overline{\mathcal{M}}_{g,n}$.
(ii) Factorizable $\mathcal{D}$-modules (Beilinson–Drinfeld).
(iii) HT product geometry $\Sigma_g \times \mathbb{R}$.
(iv) Bar as factorization coalgebra on $\mathrm{Ran}(X)$.
(v) Koszul duality as factorization duality; Feynman involutivity $\mathrm{FT}^2 \simeq \mathrm{id}$.
(vi) Three models as three gauges: flat associated graded ($D_0^2 = 0$, operadic/local), corrected holomorphic ($D^{(g)2} = 0$, Fay trisecant, still local), curved geometric ($d^2_{\mathrm{fib}} = \kappa \cdot \omega_g$, Arakelov, global).

The curvature measures what the local extraction loses: it is the monodromy of the $\mathcal{D}$-module connection around B-cycles of Σ_g. The modular bar complex $\tilde{B}_{\mathrm{mod}}(\mathcal{A})$ is flat because it couples all genera simultaneously via D_1; the genus-fixed bar complex is curved because restriction decouples this connection.

The relative Feynman transform $\mathrm{FT}_{\mathsf{Com}_{\mathrm{mod}}/\mathrm{SC}^{\mathrm{ch,top}}}$ mediates between factorisation (global) and operadic (local) viewpoints. The modular bar coalgebra is an algebra over this transform; homotopy-involutivity plus factorisation Koszul duality gives the flat-derived / curved-coderived comparison at all genera.

Three models compared. The three models of the bar complex correspond to three levels of geometric approximation:

Model	Differential	Domain	Status
Flat (operadic)	$D_0^2 = 0$	formal disk	proved, all families
Corrected (holomorphic)	$D^{(g)2} = 0$	smooth curve	proved, uses Fay trisecant
Curved (geometric)	$d_{\mathrm{fib}}^2 = \kappa\omega_g$	Arakelov	proved, coderived category

The flat model is the operadic bar complex on a formal disk; it sees only the OPE data and produces $D_0^2 = 0$ by Arnold plus Borcherds. The corrected model on a smooth curve restores the A-cycle periods by the Fay trisecant identity; the B-cycle periods produce quasi-periodicity factors that the corrected model absorbs by the KZB connection. The curved model on a stable curve over $\overline{\mathcal{M}}_g$ has $d_{\mathrm{fib}}^2 = \kappa\omega_g$; the curvature is the Chern class of the Hodge bundle, and flatness is restored by the Gauss–Manin connection across moduli. The three models are related by restriction: curved $\longrightarrow$ corrected (restricting to a smooth fiber) $\longrightarrow$ flat (restricting to a formal neighborhood of a point).

10.10. Four-stage architecture. *Stage 1.* $\mathrm{SC}^{\mathrm{ch,top}}$ homotopy-Koszul; bar-cobar is Quillen-equivalent; chiral derived centre $\simeq C^\bullet_{\mathrm{ch}}(A_b, A_b)$ is the universal bulk. PROVED.

Stage 2. Primitive object is the open factorisation dg-category $\mathcal{C}$; A_b is a chart; bulk is Morita invariant. PROVED.

Stage 3. Tangential log curves (X, D, τ) provide bordered FM compactification; local-global bridge lifts operadic structure to factorisable $\mathcal{D}$-module on $\mathrm{Ran}(X)$. PROVED LOCALLY; MODEST GLOBALLY.

Stage 4. Trace and clutching on the open sector produce the full open/closed MC element Θ^{oc}. Annulus trace is proved; full chain-level clutching is programme.

Five negative principles: (N1) bar $\neq$ bulk; (N2) boundary algebra is a chart; (N3) tangential log geometry required; (N4) modularity is trace plus clutching on the open sector; (N5) dg-shifted Yangian is proved for affine lineage only, extension to non-standard families is programme.

Stage 1 in detail: the local one-colour theorem. The chiral derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ is computed by chiral Hochschild cochains: the cochain complex $\prod_{k\geq 0} \mathrm{Hom}(\mathcal{A}^{\otimes k}, \mathcal{A})$ with differential from the bar resolution. On the Koszul locus, the complex concentrates in cohomological degrees $\{0, 1, 2\}$ by Theorem H; the chiral Gerstenhaber bracket $[-,-]_G\colon C^p \otimes C^q \longrightarrow C^{p+q-1}$ makes $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ into an E_2-algebra. The $\mathrm{SC}^{\mathrm{ch,top}}$ structure emerges: the pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}), \mathcal{A})$ carries an $\mathrm{SC}^{\mathrm{ch,top}}$-algebra structure where the closed colour acts on $\mathcal{A}$ (bulk acts on boundary) and the open-to-closed map is zero (boundary does not produce bulk).

The passage from E_2 to E_3 requires a conformal vector: the Sugawara construction $T(z) = \{Q, G(z)\}$ makes $\mathbb{C}$-translations Q-exact, killing the holomorphic direction in cohomology. For affine Kac–Moody $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$: the Sugawara T is explicit, the topologisation is proved (cohomologically, and chain-level on a qi-equivalent model), and the

E_3-topological structure on $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ is identified as a family over $\cdot H^3(\mathfrak{g})[[\ \ ]]$. For Virasoro and $\mathcal{W}$-algebras with conformal vector: conjectural (the 3d HT BRST complex is conditional). Without conformal vector (Heisenberg at $k = 0$, critical level, lattice VOAs without Sugawara): stuck at $\mathrm{SC}^{\mathrm{ch,top}}$. The Swiss-cheese structure is the generic case; E_3-topological is special.

Stage 2 in detail: Morita invariance. The primitive object $\mathcal{C}$ is a dg-category on the bordified curve $\widetilde{X}_D$. Morita equivalence is the statement that $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C}) \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C}')$ whenever $\mathcal{C}$ and $\mathcal{C}'$ are derived Morita equivalent. The boundary algebra $A_b = \mathrm{End}_{\mathcal{C}}(b)$ depends on the choice of boundary condition b; the bulk does not. Two boundary conditions b, b' produce Morita equivalent boundary algebras iff the bimodule ${}_b\mathcal{C}_{b'} = \mathrm{Hom}_{\mathcal{C}}(b, b')$ is an invertible bimodule. The holographic datum $\mathcal{H}(T)$ is therefore boundary-dependent: the R-matrix, the Koszul dual, and the line-operator category all depend on the choice of b.

Stage 3 in detail: tangential log globalisation. A tangential log curve (X, D, τ) consists of a smooth curve X, a reduced effective divisor $D \subset X$ (the marked points), and a tangential base point $\tau \in T_D X/\mathbb{C}^*$ at each component of D (a ray in the tangent space, specifying the ordering direction). The log-FM compactification $\overline{C}^{\log}_n(X, D)$ extends $\overline{C}_n(X)$ by allowing points to collide with boundary divisors; the log-differential forms $\Omega^*_{\log}(\overline{C}^{\log}_n)$ carry both holomorphic and tangential data.

The local-global bridge: operadic $\mathrm{SC}^{\mathrm{ch,top}}$ data at each point of D assemble into a factorisable $\mathcal{D}$-module on $\mathrm{Ran}(X \setminus D)$ via the Beilinson–Drinfeld factorisation construction. The modular extension couples the factorisable data to $\overline{\mathcal{M}}_{g,n}$ via the universal curve.

10.11. The 3d MC colour decomposition. The 3d MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ decomposes into three colour sectors:

$$\Theta^{\mathrm{oc}}_{\mathcal{A}} = \Theta^{\mathrm{cl}}_{\mathcal{A}} + \Theta^{\mathrm{op}}_{\mathcal{A}} + \Theta^{\mathrm{mix}}_{\mathcal{A}},$$

with the closed sector Θ^{cl} encoding bulk data, the open sector Θ^{op} encoding boundary data, and the mixed sector Θ^{mix} encoding the bulk-to-boundary action. The MC equation $D\Theta^{\mathrm{oc}} + (1/2)[\Theta^{\mathrm{oc}}, \Theta^{\mathrm{oc}}] = 0$ decomposes by colour:

$$\begin{aligned}
&\text{closed:} \quad D_{\mathrm{cl}}\Theta^{\mathrm{cl}} + \tfrac{1}{2}[\Theta^{\mathrm{cl}}, \Theta^{\mathrm{cl}}]_{\mathrm{cl}} = 0,\\
&\text{open:} \quad D_{\mathrm{op}}\Theta^{\mathrm{op}} + \tfrac{1}{2}[\Theta^{\mathrm{op}}, \Theta^{\mathrm{op}}]_{\mathrm{op}} + [\Theta^{\mathrm{mix}}, \Theta^{\mathrm{mix}}]_{\mathrm{op}} = 0,\\
&\text{mixed:} \quad D_{\mathrm{mix}}\Theta^{\mathrm{mix}} + [\Theta^{\mathrm{cl}}, \Theta^{\mathrm{mix}}]_{\mathrm{mix}} + [\Theta^{\mathrm{op}}, \Theta^{\mathrm{mix}}]_{\mathrm{mix}} = 0.
\end{aligned}$$

The closed equation is the genus tower of Sections 1–9; the open equation determines the boundary A_∞ structure with corrections from mixed-sector interactions; the mixed equation is the compatibility between bulk and boundary.

Example 1: Abelian CS ($U(1)$, level k). $\Theta^{\mathrm{cl}} = k \cdot \eta \otimes \Lambda$ (scalar, the Heisenberg MC element); $\Theta^{\mathrm{op}} = 0$ (the boundary algebra $\mathcal{H}_k$ is strict, no A_∞ corrections); $\Theta^{\mathrm{mix}} = 0$ (no bulk-to-boundary action beyond the identity). Class **G**: the full 3d MC element collapses to its closed sector.

Example 2: Non-abelian CS (SL_2, level k). Θ^{cl} contains the Casimir r-matrix $k\,\Omega/z$ at degree 2 and the Lie bracket at degree 3; Θ^{op} encodes the A_∞ structure on $Y^{\mathrm{dg}}(\mathfrak{sl}_2)$; Θ^{mix} encodes the evaluation map $\mathrm{ev}\colon Y(\mathfrak{sl}_2) \otimes V \to V$ by which the Yangian acts on evaluation modules. Class **L**: the shadow tower terminates at degree 3 in the closed sector, but the open sector carries the full Yangian A_∞ structure to all degrees. The Yangian side is infinite even when the shadow tower is finite.

Example 3: 3d gravity (Vir_c). Θ^{cl} is the infinite shadow tower: $\kappa = c/2$ at degree 2, the Virasoro bracket at degree 3, $10/[c(5c + 22)]$ at degree 4, and infinitely many higher corrections. Θ^{op} encodes the gravitational A_∞ structure, which is infinite (class **M**); Θ^{mix} encodes the action of the gravitational bulk on the Virasoro boundary. The coproduct $\Delta_z^{\mathrm{Vir}} = 0$ at all degrees: the ghost-number obstruction from DS reduction kills every HPL tree correction. Class **M**: all three sectors are infinite.

10.12. Three holographic datums. *Abelian CS* ($U(1)$, level k): boundary $\mathcal{H}_k$, bulk = $\mathcal{H}_k$, $R(z) = e^{k\ /z}$ (scalar, YBE trivial), $\kappa = k$. Class **G**: the entire holographic datum collapses to the single scalar k.

Non-abelian CS (SL_2, level k): boundary $\widehat{\mathfrak{sl}}_{2k}$, line dual $\mathcal{Y}^{\mathrm{dg}}(\mathfrak{sl}_2)$, $r(z) = k\,\Omega/z$ (vanishing at $k = 0$), $\kappa = 3(k{+}2)/4$. Evaluation-module spectral braiding = $U_q(\mathfrak{sl}_2)$ at $q = e^{i\pi/(k+2)}$. Class **L**: the Jacobi identity kills the shadow tower at degree 3.

3d gravity ($SL(2,\mathbb{R}) \times SL(2,\mathbb{R})$ CS, $c = 3\ell/(2G)$): boundary Vir_c, Koszul dual Vir_{26-c} (self-dual at $c = 13$), $r(z) = (c/2)/z^3 + 2T/z$, $\kappa = c/2$. Quartic contact $10/[c(5c{+}22)]$; infinite tower (class **M**); the shadow reads as the gravitational perturbation series. Page time $t_P = 3S_{\mathrm{BH}}/13$ is c-independent because $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$. The climax of Volume II.

11. Modular PVA quantization

The open/closed world supplies a genus-zero datum: the Swiss-cheese MC element α_T, the PVA $(\mu, \{-_\lambda-\})$ on cohomology, the KZ/BPZ shadow connections at degree n. All are tree-level. How does this structure extend across the genus filtration? The answer is a genus-by-genus obstruction theory whose input is the genus-0 PVA and whose output is the full modular MC element. The nonseparating degeneration operator D_1 is the primitive genus-raising differential; it squares to zero, anticommutes with the tree-level differential D_0, and defines a spectral sequence whose obstructions are the quantisation anomalies.

11.1. One-edge principle. Every nontrivial modular obstruction theory must contain a primitive genus-raising operator; otherwise any genus-0 MC element lifts tautologically to all genera. The one-edge expansion

$$D_{\exp} \;=\; D_{\mathrm{sep}} + D_{\mathrm{nsep}}, \qquad D_0 \;=\; D_{\mathrm{int}} + D_{\mathrm{sep}}, \qquad D_1 \;=\; D_{\mathrm{nsep}}$$

separates genus-preserving from genus-raising degenerations. $(D_0 + D_1)^2 = 0$ stratifies by genus shift: $D_0^2 = 0$, $D_0D_1 + D_1D_0 = 0$, $D_1^2 = 0$. Trees govern chirality; loops govern modularity. The three identities are codim-2 cancellation stratified by genus.

11.2. Modular bar datum. A modular bar datum on a reduced complete graded collection $F \mapsto C(F)$ consists of (i) an internal differential $d\colon C(F) \to C(F)$ with $d^2 = 0$; (ii) for every one-edge expansion $\epsilon \in \mathrm{OE}(F)$ (separating or nonseparating), a degree-+1 map

$$\Delta_\epsilon\colon sC(F) \;\longrightarrow\; \mathfrak{o}_(\epsilon) \otimes \bigotimes_{u \in V(\epsilon)} sC(\mathrm{Fl}_\epsilon(u));$$

(iii) the codim-2 cancellation $\sum_{\epsilon_1,\epsilon_2} \operatorname{sgn} \Delta_{\epsilon_2} \circ \Delta_{\epsilon_1} = 0$ for every two-edge graph; and (iv) $d \circ \Delta_\epsilon + \Delta_\epsilon \circ d = 0$. The total $D = d + \sum_\epsilon \Delta_\epsilon$ satisfies $D^2 = 0$ by three mechanisms: $d^2 = 0$, anticommutation, and the codim-2 axiom. This is the algebraic incarnation of the cell decomposition of $\overline{\mathcal{M}}_{g,n}$ into stable-graph strata.

11.3. Complete modular bar coalgebra. For a connected stable graph Γ of genus g, set

$$C[\Gamma] \;=\; \mathfrak{o}_(\Gamma) \otimes \bigotimes_{v \in V(\Gamma)} sC(\mathrm{Fl}(v)), \qquad \mathfrak{o}_(\Gamma) = \bigotimes_e \mathfrak{o}_(e).$$

The complete modular bar coalgebra is

$$\mathbf{B}_{\mathrm{mod}}(C) \;=\; \prod_{[\Gamma] \in \mathsf{StGraph}^{\mathrm{conn}}} (C[\Gamma])_{\mathrm{Aut}(\Gamma)},$$

completed over total genus $g(\Gamma) = b_1 + \sum_v g(v)$. Edge contraction is the reduced coproduct; the coalgebra is conilpotent.

11.4. Strict modular deformation algebra. $L_{\mathrm{mod}}(C) := \mathrm{Coder}(\mathbf{B}_{\mathrm{mod}}(C))[-1]$ is complete filtered by genus, with convolution $L_{\mathrm{mod}}(C) \cong \mathrm{Hom}(\mathbf{B}_{\mathrm{mod}}(C), sC)[-1]$ under commutator bracket. The algebra is pro-nilpotent in F^1, and the associated graded inherits a dg Lie structure at each genus.

11.5. Genus-1 obstruction theorem. For a genus-0 MC element $\Theta_0 \in \mathrm{MC}(\mathrm{gr}^0_F L)$, the genus-1 obstruction is

$$\mathrm{Ob}_1(\Theta_0) = [D_1\Theta_0] \in H^2((\mathrm{gr}^1_F L)^{\Theta_0}, d_0^{\Theta_0}).$$

$D_1\Theta_0$ is a $d_0^{\Theta_0}$-cocycle by $D_0D_1 + D_1D_0 = 0$ and the MC equation for Θ_0. The class vanishes iff Θ_0 lifts to $\Theta_0 + \quad \Theta_1$ satisfying the modular MC equation modulo $\quad ^2$.

In PVA coordinates, the nonseparating one-edge operator acts as the reduced odd Laplacian: for cyclic $(A, m_k, \langle -, - \rangle)$ with basis $\{e^i\}$ and dual basis $\{e_i\}$,

$$\ell_1^{(1)}(a) = \Delta_{\mathrm{cyc}}(a) = \sum_i m_2(e^i, m_2(a, e_i)).$$

The full Ob_1 is the cohomology class of $\Delta_{\mathrm{cyc}}(\Theta_0)$.

Heisenberg: $\mathrm{Ob}_1 = 0$ **trivially.** For $\mathcal{H}_k$: $m_k = 0$ for $k \geq 3$, so $\Delta_{\mathrm{cyc}}(\Theta_{\mathcal{H}_k}) = \quad \sum_i m_2(e^i, m_2(\Theta_0, e_i))$. But $\Theta_0 = k \cdot \eta_{\mathrm{univ}} \otimes \Lambda_{\mathrm{univ}}$ is a scalar, so $m_2(\Theta_0, e_i) = k \cdot e_i$ and $m_2(e^i, k \cdot e_i) = k^2 \cdot \delta^i_i$. The trace gives $\Delta_{\mathrm{cyc}}(\Theta_0) = \quad k^2 \cdot \dim(V)$: a scalar, hence d_0-closed and d_0-exact (it is the curvature $\kappa \cdot \omega_1$ in the bar complex, which is absorbed by the Gauss–Manin connection). Obstruction zero; the Heisenberg lifts to all genera with no quantum correction. Class **G**: the simplest possible modular behaviour.

Kac–Moody: $\mathrm{Ob}_1 = 0$ **by Whitehead.** For $\widehat{\mathfrak{g}}_k$: $\Delta_{\mathrm{cyc}}(\Theta_0)$ involves m_3 and the Lie bracket $f^{ab}{}_c$. The obstruction class lives in $H^2(\mathfrak{g}; \mathrm{Sym}^q V)$ for reductive $\mathfrak{g}$; Whitehead's lemma kills H^2 for semisimple $\mathfrak{g}$. The genus-1 lift exists unconditionally for all simple types. Class **L**: Whitehead is the mechanism.

Virasoro: $\mathrm{Ob}_1 = 0$ **by central direction.** For Vir_c: $\Delta_{\mathrm{cyc}}(\Theta_0)$ involves the self-coupling $T_{(1)}T = 2T$ and the quartic pole $T_{(3)}T = c/2$. The contraction $\sum_i m_2(T^i, m_2(\Theta_0, T_i))$ produces a linear combination of $T\partial T$, $\partial^3 T$, and $c \cdot \mathbf{1}$. The quasi-primary projection kills the total-derivative terms ($\partial^3 T$ and $T\partial T$ project to Λ and a quasi-primary correction); what survives is a scalar proportional to c times the central direction ∂_c. Since $H^1((\mathrm{gr}^1 L)^{\Theta_0}, d_0^{\Theta_0}) \cong \mathbb{C} \cdot \partial_c$, the class is d_0-exact: $\mathrm{Ob}_1(\Theta_0) = 0$. The genus-1 lift exists as a renormalisation of the central charge: $c \mapsto c + \quad c_1$ with c_1 determined by the one-loop computation. Class **M**: the central direction absorbs the obstruction at every genus individually, but the full tower is infinite.

$\beta\gamma$**:** $\mathrm{Ob}_1 = 0$ **by ghost-number grading.** For $\beta\gamma_\lambda$: the fields β and γ carry opposite ghost numbers (+1 and −1). The cyclic Laplacian $\Delta_{\mathrm{cyc}}(\Theta_0)$ involves contracting β with γ through the cyclic pairing; the ghost-number selection rule forces the contraction to land in ghost number 0, which is the scalar sector. The obstruction is a scalar multiple of $\mathbf{1}$, hence d_0-exact. Class **C**: the ghost-number grading is the mechanism, distinct from both the scalar absorption (class **G**) and the Whitehead vanishing (class **L**). The $\beta\gamma$ system provides the third independent vanishing mechanism for the genus-1 obstruction.

11.6. The quantisation theorem.

Theorem 0.2 (Modular PVA quantisation)**.** *For a chirally Koszul, positive-energy, finitely strongly generated chiral algebra $\mathcal{A}$ with $\kappa(\mathcal{A}) \neq 0$, the genus-0 MC element $\Theta_0 \in \mathrm{MC}(\mathrm{gr}^0_F L_{\mathrm{mod}})$ lifts to a full modular MC element $\Theta = \sum_{g \geq 0} \quad ^g \Theta_g \in \mathrm{MC}(L_{\mathrm{mod}})$ satisfying the quantum master equation at all genera. The lift is unique up to gauge equivalence in* $\exp(F^1 L_{\mathrm{mod}})$.

The proof proceeds genus-by-genus. At each genus g, the obstruction to extending $\Theta^{<g}$ to genus g is a class $[\mathrm{Ob}_g] \in H^2(\mathrm{gr}^g_F L^{\Theta^{<g}}_{\mathrm{mod}}, d_0^{\Theta^{<g}})$. The class is computed by the formula

$$\mathrm{Ob}_g(\Theta^{<g}) \;=\; \sum_{\substack{\Gamma\in\mathsf{StGraph}_{g,0}\\ b_1(\Gamma)=g}} \frac{1}{|\mathrm{Aut}(\Gamma)|} W_\Gamma(\Theta^{<g}),$$

where the sum runs over connected stable graphs of genus g with no legs and first Betti number g. The vanishing of $[\mathrm{Ob}_g]$ is proved by three independent mechanisms depending on the depth class:

- Class **G** (Heisenberg, lattice): $\mathrm{Ob}_g = 0$ at all g because $m_k = 0$ for $k \geq 3$; every graph with an internal vertex of valence ≥ 3 vanishes.
- Class **L** (affine KM): $\mathrm{Ob}_g = 0$ by the Whitehead lemma at each genus: the obstruction class lives in $H^2(\mathfrak{g}; M_g)$ for a $\mathfrak{g}$-module M_g determined by lower-genus data, and $H^2 = 0$ for semisimple $\mathfrak{g}$.
- Class **M** (Virasoro, $\mathcal{W}_N$): $\mathrm{Ob}_g = 0$ in cohomology (proved); the chain-level lift at genus g requires the harmonic mechanism $\delta_r^{\mathrm{harm}} = c_r(\mathcal{A}) \cdot m_0^{\lfloor r/2\rfloor-1}$ from Hodge decomposition. For class **M**, the chain-level lift is conjectural at $g \geq 2$; the coderived formulation is clean.

Uniqueness: at each genus, the space of lifts modulo gauge is an affine space under $H^1(\mathrm{gr}^g_F L^{\Theta^{<g}})$. For standard families this space is one-dimensional (spanned by ∂_c or ∂_k), so the lift is unique up to reparametrisation of the central charge.

11.7. $\mathcal{W}_3$ computation in detail. A filtered quadratic PVA is generated by $u^1, \ldots, u^N$ with λ-bracket $\Pi^{ab}(\partial)$ of total generator degree ≤ 2. Classical $\mathcal{W}_3$ fits: generators T (weight 2), W (weight 3), with the composite $\Lambda =: TT : -(3/10)\partial^2 T$ controlling the nonlinear channel; the WW OPE has the $c = -22/5$ pole in $16/(22+5c)$ where Λ becomes null.

The WW lambda-bracket. In lambda-bracket notation (Vol II convention):

$$\begin{aligned}\{W_\lambda W\} &= \frac{c}{3}\lambda^5 + 2T\lambda^3 + T'\lambda^2 + \Big[\frac{16}{22+5c}\Lambda + \frac{3}{10}T''\Big]\lambda \\ &\quad + \Big[\frac{8}{22+5c}\Lambda' + \frac{1}{15}T'''\Big],\end{aligned}$$

where $\Lambda =: TT : -(3/10)T''$ and primes denote ∂-derivatives. The coefficient $\alpha(c) = 16/(22+5c)$ is fixed uniquely by quasi-primarity ($L_1\Lambda = 0$) and the Jacobi identity for the triple $\{W, W, W\}$. At $c = -22/5$: $\alpha \to \infty$, Λ becomes null, and $\mathcal{W}_3$ degenerates (the $(2,5)$ minimal model). At $c = 0$: $\alpha = 16/22 = 8/11$, a finite value corresponding to the free-field limit.

The genus-1 obstruction. The genus-1 obstruction for $\mathcal{W}_3$ decomposes into three channels:

$$\mathrm{Ob}_1(\Theta_0^{\mathcal{W}_3}) \;=\; \mathrm{Ob}_1^{TT} + \mathrm{Ob}_1^{TW} + \mathrm{Ob}_1^{WW}.$$

The TT channel is the Virasoro genus-1 obstruction (vanishes by the same mechanism as for Vir_c alone). The TW cross-channel vanishes by conformal weight parity: the fields T and W have weights 2 and 3 (different parity), forcing the contraction $\sum_i m_2(T^i, m_2(\Theta_0^{TW}, T_i))$ to zero by grading. The WW channel: the key computation. The contraction $\Delta_{\mathrm{cyc}}(\Theta_0^{WW})$ produces terms involving Λ and T''; the W-normal form (expressing everything in the basis $\{T, W, \Lambda, T'', \ldots\}$ with Λ as the independent weight-4 quasi-primary) reduces the obstruction to a single scalar proportional to ∂_c.

The triangular W-normal form vanishing theorem: in the local, translation-invariant, filtration-preserving sector, $\mathrm{Ob}_1(\Theta_0) = 0$ for classical $\mathcal{W}_3$. The relevant cohomology is $H^1((\mathrm{gr}^1 L)^{\Theta_0}, d_0^{\Theta_0}) \cong \mathbb{C} \cdot \partial_c P_c$, one-dimensional and spanned by the central-parameter direction. Every genus-1 lift is gauge equivalent to a renormalisation $c \mapsto c + \quad c_1$; the genus-1 quantum $\mathcal{W}_3$ is the unique quantisation in the direction ∂_c.

Genus-2 cross-channel obstruction. At genus 2, the $\mathcal{W}_3$ obstruction has a genuinely new feature: the TW cross-channel contributes. The seven genus-2 stable graphs produce seven graph amplitudes, and the cross-channel correction $\delta F_2^{\mathrm{cross}}(\mathcal{W}_3)$ is nonzero. The universal formula:

$$\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) \;=\; \frac{(N-2)(N+3)}{96} + \frac{(N-2)(3N^3+14N^2+22N+33)}{24c}.$$

At $N=2$: $\delta F_2^{\mathrm{grav}} = 0$ (Virasoro is single-generator, no cross-channel). At $N=3$: $\delta F_2^{\mathrm{grav}}(\mathcal{W}_3, c) = 5/96 + (c+204)/(16c)$. The scalar uniform-weight formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ fails at $g \geq 2$ for multi-weight algebras (ALL-WEIGHT, WITH CROSS-CHANNEL CORRECTION).

11.8. The Khan–Zeng sigma model. Khan and Zeng construct a 3d holomorphic-topological Poisson sigma model that realises the modular PVA quantisation programme geometrically. The construction applies to every freely generated PVA with conformal vector. The field content on $\mathbb{R}_+ \times \mathbb{C}$:

- $A_{\bar{z}} \in \Omega^{0,1}(\mathbb{C}) \otimes \mathfrak{g}$: the $(0,1)$-form field (holomorphic gauge field).
- $\phi \in \Omega^0(\mathbb{C}) \otimes \mathfrak{g}^*$: the scalar field (conjugate momentum).
- $\eta \in \Omega^1(\mathbb{R}_+) \otimes \mathfrak{g}$: the topological 1-form on the half-line (boundary condition data).

The action functional:

$$S_{\mathrm{KZ}} \;=\; \int_{\mathbb{R}_+\times\mathbb{C}} \Big(\langle \phi, \bar{\partial} A\rangle + \langle \phi, \tfrac{1}{2}\{A_\lambda A\}|_{\lambda=0}\rangle + \langle \eta, d_t\phi\rangle \Big)\, dt \wedge dz \wedge d\bar{z}.$$

Gauge invariance is the PVA Jacobi identity: the variation $\delta S_{\mathrm{KZ}} = 0$ under $\delta A = \bar{\partial}\epsilon + \{A_\lambda \epsilon\}|_{\lambda=0}$ requires exactly the Jacobi identity for $\{-_\lambda-\}$, the Leibniz rule for μ, and the sesquilinearity of the bracket. A conformal vector promotes the holomorphic-topological theory to topological: the Sugawara construction $T = \{Q, G\}$ makes $\mathbb{C}$-translations Q-exact, collapsing the holomorphic direction in cohomology.

The boundary theory on $\{0\} \times \mathbb{C}$ carries a (-1)-shifted Poisson bracket; the quantisation of this bracket is the genus-0 MC problem. The bulk theory on $\mathbb{R}_+ \times \mathbb{C}$ provides the genus-raising mechanism: loops in the bulk produce traces on the boundary, and the genus-g obstruction Ob_g is the g-loop Feynman integral in the bulk sigma model. The entire modular PVA quantisation programme is thereby geometrised.

The scope splits. Affine Kac–Moody algebras at non-critical level and principal $\mathcal{W}$-algebras are proved through the affine/DS gauge-theoretic lanes. For finite-type freely generated finite-jet PVAs with conformal vector and inner holomorphic translation, the Khan–Zeng construction supplies the classical BV action; the all-loop boundary quantization and the E_3-topological lift require the KZ analytic SDR package and a quantized homotopy $T = [Q_{\mathrm{tot}}, G]$.

11.9. Curved extension via S-tails. In the inhomogeneous quadratic setting (Gui–Li–Zeng), the correct input is a twisted pair (B, B°, S): a CDG-algebra B, a CDG-coalgebra B°, and curvature S. An S-tail expansion attaches a univalent S-decorated vertex to a corolla, followed by contraction of the new internal edge. Adjoining S-tail expansions to one-edge expansions with the same codim-2 compatibilities gives a curved modular bar object. The curved MC equation is $\mu((S+\alpha)\boxtimes(S+\alpha)) = 0$; at genus 0 it is the curved A_∞ MC equation with $m_0 = S$, at genus 1 it is $\Delta_{\mathrm{cyc}}(S+\alpha) = 0$, and at genus g it is the BV quantum master equation $\Delta(e^{(S+\alpha)/\ }\ ) = 0$.

11.10. Genus-0 Gui–Li–Zeng comparison. At tree level, the modular MC equation reduces to the classical twisting morphism equation. For a Koszul pair $(\mathcal{A}, \mathcal{A}^!)$ with $\mathcal{A}^!$ a CDG algebra with curvature S,

$$\mathrm{MC}_0(L_{\mathrm{mod}}) \;\cong\; \mathrm{Hom}_{\mathsf{dg}}(\mathcal{A}, B^\circ) \;\hookrightarrow\; \mathrm{MC}(\mathcal{A}^!\widehat{\otimes}\mathcal{B}).$$

The modular theory is the genus extension of the classical Gui–Li–Zeng picture. The CDG curvature S is the bar-dual of the vacuum OPE: $m_0 = S \in \bar{B}^0 = \mathbb{C}$ is the scalar that records the quartic pole (for Virasoro, $S = c/2$). At genus 0: the curved A_∞ relation $m_1^2(a) = [m_0, a]$ is the statement that the bar differential squares to the commutator with the curvature, not to zero. At genus g: $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ is the fiberwise statement, with $\kappa = m_0$ the curvature scalar. The modular theory organises these identities across all genera simultaneously.

12. Holographic modular Koszul duality

Sections 1–9 construct E_n-chiral algebras as algebraic-geometric objects: bar complexes on configuration spaces, R-matrices, derived chiral centres with E_2/E_3 structure, the $\mathrm{SC}^{\mathrm{ch,top}}$ datum on the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$. The physical interpretation: the derived chiral centre IS the bulk of a real 3d holomorphic-topological gauge theory. The open/closed world (Section 10) supplies the 3d geometric framework; the modular PVA quantisation programme (Section 11) supplies the obstruction theory. The holographic modular Koszul datum packages the passage from algebraic geometry to physics into a single six-component structure: boundary algebra, Koszul dual, line category, spectral R-matrix, universal MC element, and modular shadow connection. Every ingredient constructed in Sections 1–9 is a projection of this datum. The datum is the smallest package from which the full physical correspondence reconstructs.

12.1. The holographic modular Koszul datum. Every 3d HT system T is controlled by

$$\mathcal{H}(T) = (\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}),$$

where:

(i) $\mathcal{A}$ is the boundary chiral algebra on a curve X, restricted from the holomorphic boundary $\{0\} \times X \subset \mathbb{R}_+ \times X$.

(ii) $\mathcal{A}^!$ is the strict chiral Koszul dual, $\mathcal{A}^! = (\mathcal{A}^{\mathrm{i}})^\vee$ with $\mathcal{A}^{\mathrm{i}} = H^*(\bar{B}(\mathcal{A}))$. Holographically, $\mathcal{A}^!$ is a dg-shifted Yangian: the bar complex $\bar{B}(A_b)$ is an E_1 coassociative coalgebra, and $\mathcal{A}^!$ is its linear dual. The $\mathrm{SC}^{\mathrm{ch,top}}$ datum is the pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}),\ \mathcal{A})$ of chiral derived center and boundary algebra.

(iii) $\mathcal{C} \simeq \mathcal{A}^!$-mod is the braided tensor category of line operators: objects are topological defect lines, morphisms junction operators, braiding from crossing in the holomorphic plane.

(iv) $r(z) \in \mathcal{A}^! \otimes \mathcal{A}^!((z^{-1}))$ is the classical r-matrix satisfying CYBE, the binary genus-0 shadow $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$.

(v) $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ is the universal MC element from the bar-intrinsic construction $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$, always MC because $D_{\mathcal{A}}^2 = 0$.

(vi) $\nabla^{\mathrm{hol}}_{g,n} = d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}})$ is the modular shadow connection on $\overline{\mathcal{M}}_{g,n}$, flat by the MC equation projected to (g, n).

12.2. Collision filtration. The modular convolution $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ carries a collision filtration $F^0_{\mathrm{coll}} \supset F^1_{\mathrm{coll}} \supset \cdots$ indexed by minimum pairwise distance on $\overline{C}_n(\mathbb{C})$. Depth 0 is free propagation; depth 1 is first collision; depth k is k-fold nested collision. The spectral sequence has $E_1^{p,*} = H^*(\mathrm{gr}^p_{\mathrm{coll}}\, \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, d_0)$ with d_1 Arnold cancellations; it converges to $H^*(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$, and collapse is governed by shadow depth: class **G** collapses at E_2; class **M** does not collapse at any finite page.

12.3. Four proved recovery theorems. *(R1) Collision residue is twisting.* At depth 1, degree 2: $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) = \pi_{\mathcal{A}}$, the universal twisting morphism producing the dg-shifted Yangian $r_{\mathcal{A}}(z)$. The MC equation becomes $\partial\pi + \pi \star \pi = 0$.

(R2) CYBE from Arnold. The three boundary divisors of $\overline{M}_{0,4}$ correspond to channels (12|34), (13|24), (14|23). Evaluating the Arnold relation on the Casimir $\Omega \otimes \Omega \in (\mathcal{A}^!)^{\otimes 3}$ produces the three terms of the CYBE $\sum [r_{ij}, r_{ik}] = 0$.

(R3) Affine shadow / KZ comparison. For $\mathcal{A} = \widehat{\mathfrak{g}}_k$, on the affine comparison surface the genus-0 degree-n shadow one-form is the KZ connection form

$$\mathrm{Sh}_{0,n}(\Theta_{\widehat{\mathfrak{g}}_k}) \;=\; \sum_{i<j} r_{ij}\, d\log(z_i - z_j) \;=\; \frac{1}{h^\vee + k} \sum_{i<j} \frac{\Omega_{ij}\, dz_{ij}}{z_{ij}},$$

where $r_{ij} = \Omega_{ij}/(h^\vee + k)$ is the propagator-weighted Casimir. Flatness is MC at genus 0, degree n. For Virasoro, the Virasoro conformal Ward extraction plus higher collision residues yields the BPZ equations. The affine Drinfeld–Kohno theorem compares monodromy with the quantum-group representation category.

(R4) Quartic as L_∞ obstruction. $[\mathfrak{R}^{\mathrm{mod}}_4(\mathcal{A})] = [o_4(\mathcal{A})] \in H^2(\mathcal{A}^{\mathrm{sh}}_{4,0})$ is the obstruction to extending $r(z)$ from degree 2 to degree 4. Vanishes for classes **G**, **L**, **C**; nonzero for class **M** (Virasoro: $Q^{\mathrm{contact}} = 10/[c(5c+22)]$).

Recovery theorem table.

Recovery	Depth	Output	Family
(R1) Collision residue	1	twisting morphism $\pi_{\mathcal{A}}$	all
(R2) Arnold $\to$ CYBE	1	Yang–Baxter for $r(z)$	all
(R3) Shadow $\to$ KZ	0–n	KZ/BPZ connection	KM, Vir
(R4) Quartic obstruction	4	L_∞ class	class **M** only

Each recovery theorem captures a *different* projection of $\Theta_{\mathcal{A}}$: (R1) the degree-2 binary collision; (R2) the degree-3 ternary collision (Arnold relation on three points); (R3) the full degree-n shadow at genus 0 (comparing n-point collisions with KZ monodromy); (R4) the degree-4 obstruction (the first nonlinear invariant). Together they reconstruct the full genus-0 MC element from its finite-degree projections.

Critical level recovery. At the critical level $k = -h^\vee$ for affine Kac–Moody: $\kappa = 0$, all monodromy is trivial (integer eigenvalues from the Sugawara formula at $k = -h^\vee$), and the degree-2 ordered chiral homology $H^1_{\mathrm{ord},2}$ on $V \otimes V$ doubles from 4 to 8 dimensions for $\mathfrak{sl}_2$ (a finite computation conserving $\chi_{\mathrm{ord},2} = -4$). The spectral sequence $E_1^{p,q}$ has $d_1 = \lambda \cdot [\partial]$ vanishing at $\lambda = 0$ (the critical specialisation); the Feigin–Frenkel centre $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee}) \cong \mathrm{Fun}(\mathrm{Op}_{^L\mathfrak{g}}(D))$ emerges as H^0 of the FULL symmetric bar, and $H^n(\bar{B}^{\Sigma}(\widehat{\mathfrak{g}}_{-h^\vee})) \cong \Omega^n(\mathrm{Op}_{^L\mathfrak{g}}(D))$ for all $n \geq 0$ (infinite-dimensional; Frenkel–Teleman 2006).

12.4. Five conjectural targets.

(i) *Boundary-defect realisation.* $\mathcal{A}_\partial(T)$ and $\mathcal{A}^!_\partial(T)$ are the two algebras of $\mathcal{H}(T)$ for every 3d HT system, with $C \simeq \mathcal{A}^!$-mod.

(ii) *Yangian-shadow.* In full derived generality, the collision residue recovers the dg-shifted Yangian $r(z)$ at the chain level, including higher A_∞ homotopies.

(iii) *Sphere reconstruction.* The Gaiotto–Zeng (2026) commuting differentials on the non-planar sphere algebra are the scalar shadow $\mathrm{Sh}_{0,n}(\Theta_{\mathcal{A}})$; sphere amplitudes are flat sections of $\nabla^{\mathrm{hol}}_{0,n}$ on established comparison surfaces.

(iv) *Quartic resonance obstruction.* $\mathfrak{R}_4^{\mathrm{mod}}$ is the first global nonlinear invariant, measuring the failure of the large-N expansion to truncate at tree level.

(v) *Singular-fibre descent.* $\nabla^{\mathrm{hol}}|_{\partial\overline{\mathcal{M}}} \sim \nabla^{\mathrm{clutch}}$ with residues from the clutching law of $\Theta_{\mathcal{A}}$: $\Theta_{\mathcal{A}}|_{\sigma_\Gamma} = \prod_v \Theta_v \cdot \prod_e P_e$.

12.5. Coloured modular deformation object. A 3d HT theory produces three observation algebras: $\mathrm{Obs}^{\mathrm{hol}}$ (bulk), Obs^{∂} (boundary), Obs^{ℓ} (line). The coloured Swiss-cheese FM space $\overline{C}^{\log}_{\bullet,\mathrm{SC}}$ encodes the collision geometry of all three. The three-colour modular deformation object is

$$\mathfrak{g}^{\mathrm{HT,log}}_{\mathrm{mod}} \;=\; \hom^{\alpha_{\mathrm{HT}}}\big(C_\bullet(\overline{C}^{\log}_{\bullet,\mathrm{SC}}), \mathrm{End}_{\mathrm{Ch}_\infty}(\mathrm{Obs}^{\mathrm{hol}}_\infty, \mathrm{Obs}^{\partial}_\infty, \mathrm{Obs}^{\ell}_\infty)\big).$$

The three colours interact through $\mathrm{SC}^{\mathrm{ch,top}}$: bulk-to-boundary, boundary-to-line, no line-to-bulk.

The modular effective action is the graph sum

$$S^{\mathrm{mod}}_{\mathrm{HT}} \;=\; \sum_{\Gamma\in\mathsf{StGraph}_{\mathrm{SC}}} \frac{\;^{b_1(\Gamma)}}{|\mathrm{Aut}(\Gamma)|} W^{\log}_{\Gamma} O_\Gamma,$$

and the BV quantum master equation $(d_{\mathrm{BV}} + \quad \Delta_{\mathrm{odd}}) \exp(S^{\mathrm{mod}}/\quad) = 0$ is the single equation governing T. At genus 0: classical BV $\{S_0, S_0\} = 0$; genus 1: one-loop anomaly cancellation; genus g: g-loop Ward identity.

12.6. Khan–Zeng bridge. Khan and Zeng construct a 3d HT Poisson sigma model on $\mathbb{R}_+ \times \mathbb{C}$ attached to a freely generated PVA: $\mathfrak{g}$-valued $(0,1)$-form $A_{\bar z}$, $\mathfrak{g}^*$-valued scalar ϕ, $\mathfrak{g}$-valued 1-form η on $\mathbb{R}_+$, with action

$$S_{\mathrm{KZ}} \;=\; \int_{\mathbb{R}_+\times\mathbb{C}} \big(\langle\phi, \bar\partial A\rangle + \langle\phi, \tfrac{1}{2}\{A_\lambda A\}|_{\lambda=0}\rangle + \langle\eta, d_t\phi\rangle\big)\, dt\wedge dz\wedge d\bar z.$$

Gauge invariance is the PVA Jacobi identity; a Virasoro element promotes HT to topological; the boundary carries a (-1)-shifted Poisson bracket and generates the genus-0 quantisation problem. The modular extension couples the bulk theory to stable curves at $g \geq 1$ and controls the full genus tower. The pipeline:

$$\text{classical coisson} \xrightarrow{\text{KZ quantisation}} \text{genus-0 quantum Koszul} \xrightarrow{\text{modular lift}} \mathcal{H}(T).$$

The bridge is completed in Volume II on the affine lane by the one-loop-exact half-space BV theorem. For finite-type nonlinear KZ PVAs, the classical BV action is constructed, while the all-loop QME, reflected weight identity, and boundary vertex algebra require the KZ analytic SDR package with image-choice Stokes pushforward. The affine Poisson–vertex sigma model satisfies the physics bridge hypotheses, and Drinfeld–Sokolov reduction preserves the logarithmic $\mathrm{SC}^{\mathrm{ch,top}}$ shadow. The proved standard landscape enters through these two gauge-theoretic gates.

The genus expansion of the modular bar differential, $D = \sum_{g\geq 0} \quad ^g D^{(g)}$, is the Feynman transform of the modular operad; the canonical MC element $\delta_{\mathcal{A}} = D - D^{(0)}$ computes positive-genus corrections. Stable graphs, automorphism weights, genus bookkeeping: the mathematical shadow of the 't Hooft expansion. The identification $\quad \leftrightarrow 1/N$ for a specific holographic model remains the principal open bridge.

12.7. Heisenberg holographic datum. Base case: shadow class **G**, shadow depth 2, trivial line-operator category, scalar R-matrix. For $\mathcal{A} = \mathcal{H}_k$:

$$\mathcal{A} = \mathcal{H}_k, \qquad \mathcal{A}^! = \mathrm{Sym}^{\mathrm{ch}}(V^*), \qquad \mathcal{C} = \mathsf{Vect},$$
$$r(z) = k/z, \qquad R(z) = \exp(k/z), \qquad \Theta_{\mathcal{H}_k} = k\cdot\eta\otimes\Lambda.$$

The shadow connection $\nabla^{\mathrm{hol}}_{g,n} = d$ is trivial (scalar MC element). All five conjectural targets are trivially verified: free boundary, curved symmetric dual (with $\mathcal{H}_k^! \neq \mathcal{H}_{-k}$, $m_0 = -k\omega$), trivial lines, vanishing quartic, no moduli dependence.

12.8. Affine $\widehat{\mathfrak{sl}}_2$ holographic datum. First nontrivial example: class **L**, shadow depth 3, Yang's $r(z) = k\,\Omega/z$, line-operator category $Y(\mathfrak{sl}_2)$-mod, KZ shadow connection, quantum-group representation theory from genus-0 monodromy.

$$\mathcal{H}(\widehat{\mathfrak{sl}}_{2k}) = \Big(\widehat{\mathfrak{sl}}_{2k},\ \widehat{\mathfrak{sl}}_{2-k-4},\ \mathcal{O}_q^{\mathrm{sh}},\ \tfrac{k\,\Omega}{z},\ \Theta_k,\ d - \kappa_k\,\omega_g\Big),$$

with $q = e^{i\pi/(k+2)}$ and $\kappa_k = 3(k+2)/4$. At $k=-2$: $\kappa = 0$, the Feigin–Frenkel centre is the algebra of $\mathfrak{sl}_2$-opers. The KZ monodromy is compared with the $U_q(\mathfrak{sl}_2)$ R-matrix by affine Drinfeld–Kohno; on the reduced evaluation comparison surface, $\mathcal{C}^{\mathrm{red}}_{\mathrm{line}}|_{\mathrm{eval}}$ is identified with the corresponding quantum-group side, and the coloured Jones polynomial is recovered from monodromy of the bar complex (Steinberg convolution of the chiral self-intersection on $\overline{C}_k(\mathbb{C})$).

This is 3d Chern–Simons at level k. The holographic datum is the $k=0$-vanishing r-matrix plus the full KZ package.

12.9. Virasoro holographic datum. First example of shadow class **M**. The r-matrix $r_c(z) = (c/2)/z^3 + 2T/z$ has both a third-order pole (gravitational) and a first-order pole (stress-tensor exchange); the quartic contact invariant $Q^{\mathrm{contact}} = 10/[c(5c+22)] \neq 0$. The datum packages 3d gravity on $\mathbb{R}_+ \times \mathbb{C}$:

$$\begin{aligned}
\mathcal{H}(\mathrm{Vir}_c) &= (\mathrm{Vir}_c,\ \mathrm{Vir}_{26-c},\ \mathcal{C}_c,\ r_c(z),\ \Theta_c,\ \nabla^{\mathrm{hol}}_c),\\
r_c(z) &= \frac{c/2}{z^3} + \frac{2T}{z}, \qquad \kappa_c = \frac{c}{2},\\
\nabla^{\mathrm{hol}}_{1,1} &= d - \kappa_c\,\omega_1 - \frac{10}{c(5c+22)}\,\delta_4 - \cdots.
\end{aligned}$$

Koszul dual Vir_{26-c} (self-dual at $c=13$, not $c=26$); shadow connection with curvature $\kappa_c\omega_g$ at leading order. The transferred coproduct Δ_z^{Vir} is strictly primitive at all degrees (Principle **??**): ghost-number obstruction intrinsic to DS reduction kills every HPL tree correction. All gravitational dynamics resides in the collision residue and the infinite A_∞ product tower; the coproduct channel is identically zero. Gravity does not produce particles, only braids them.

The six components in full.

(i) $\mathcal{A} = \mathrm{Vir}_c$: the Virasoro algebra at central charge c, generated by T of weight 2. The OPE $T(z)T(w) \sim (c/2)/(z-w)^4 + 2T(w)/(z-w)^2 + \partial T(w)/(z-w)$ has three pole orders. The three pole orders produce: at order 4, the curvature $m_0 = c/2$; at order 2, the self-coupling $T_{(1)}T = 2T$ driving infinite depth; at order 1, the translation $T_{(0)}T = \partial T$.

(ii) $\mathcal{A}^! = \mathrm{Vir}_{26-c}$: the Koszul dual, obtained by $c \mapsto 26-c$ (Feigin–Frenkel involution). Koszul complementarity (Virasoro branch): $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = c/2 + (26-c)/2 = 13$. Self-dual at $c=13$. The critical dimension $c=26$ is where the *dual* curvature vanishes: $\kappa(\mathrm{Vir}_{26-c})|_{c=26} = 0$; the algebra itself retains $\kappa = 13$. At $c=0$: $\kappa = 0$ and $\kappa' = 13$.

(iii) $\mathcal{C}_c$: the line-operator category. For large c, this should be the category of modules over a $\mathcal{W}_\infty$-type Yangian; at specific rational c (minimal models), $\mathcal{C}_c$ is a finite semisimple category (the (p,q) fusion category). The spectral braiding is the E_1 ordered structure.

(iv) $r_c(z) = (c/2)/z^3 + 2T/z$: the classical r-matrix. The $d\log$ absorption of the OPE drops each pole order by 1: quartic → cubic, quadratic → simple. The cubic pole $c/(2z^3)$ is the gravitational sector; the simple pole $2T/z$ is the stress-tensor exchange. At $c=0$: $r_0(z) = 2T/z$ (the curvature vanishes but the self-coupling persists). The CYBE is the Arnold relation evaluated on the stress-tensor Casimir.

(v) $\Theta_c = \sum_{r \geq 2} \Theta_c^{(r)}$: the universal MC element. Its low-degree projections:

$$\begin{aligned}
\Theta_c^{(2)} &= (c/2) \cdot \eta \otimes \Lambda && \text{(curvature)},\\
\Theta_c^{(3)} &= \text{Virasoro bracket} && \text{(Lie structure of } T_{(1)}T = 2T\text{)},\\
\Theta_c^{(4)} &= \frac{10}{c(5c+22)} \cdot \delta_4 && \text{(quartic contact)},\\
\Theta_c^{(5)} &= \frac{480}{c^2(5c+22)} \cdot x^5 && \text{(quintic obstruction, nonzero)}.
\end{aligned}$$

The quintic $\Theta_c^{(5)} \neq 0$ proves infinite depth: no finite truncation captures Θ_c.

(vi) $\nabla_c^{\mathrm{hol}} = d - \mathrm{Sh}_{g,n}(\Theta_c)$: the shadow connection. At $(g, n) = (1, 1)$:

$$\nabla_{1,1}^{\mathrm{hol}} = d - \frac{c}{2}\omega_1 - \frac{10}{c(5c+22)}\delta_4 - \frac{480}{c^2(5c+22)}\delta_5 - \cdots .$$

Flatness $(\nabla^{\mathrm{hol}})^2 = 0$ is the MC equation projected to (g, n); it holds at all genera simultaneously.

Distinguished values of c.

c	κ	Q^{contact}	Physics
0	0	∞	free field; curvature vanishes
1	$1/2$	$10/27$	single free boson
13	$13/2$	$10/(13 \cdot 87)$	self-dual point; $\kappa = \kappa'$
25	$25/2$	$10/(25 \cdot 147)$	critical bosonic string
26	13	$10/(26 \cdot 152)$	dual curvature $\kappa' = 0$
$-22/5$	$-11/5$	∞	$(2, 5)$ minimal model; Λ null
∞	∞	$\sim 2/c^2$	semiclassical gravity

Gravitational primitivity. The Drinfeld coproduct Δ_z^{Vir} vanishes at all degrees: $\Delta_z^{\mathrm{Vir}}(T^{(k)}) = 0$ for all $k \geq 0$, where $T^{(k)}$ denotes the k-th derivative of T. The obstruction is ghost-number: the DS BRST complex that produces Vir from $\widehat{\mathfrak{sl}}_2$ has ghost number 1, and every HPL tree correction to Δ_z involves at least one ghost propagator insertion. The ghost propagator $h\colon F_{\mathrm{gh}} \to F_{\mathrm{gh}}[-1]$ shifts ghost number by -1; the coproduct $\Delta_z\colon \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$ has ghost number 0; the mismatch kills every correction. All gravitational dynamics resides in the A_∞ product tower $\{m_k\}_{k \geq 2}$ and the collision residue $r(z)$; the coalgebra direction is trivial.

12.10. The M2 brane example. First genuinely interacting holographic example: boundary algebra $\mathcal{W}_{1+\infty}$ (infinitely many generators), Koszul dual dg-shifted Yangian for $\mathfrak{psl}(2|2)$, line-operator category a quantum affine superalgebra. All three genus-2 shells active; the planted-forest shell receives its first nontrivial contribution from Mok's log-FM strata.

Physical setup. 3d $\mathcal{N} = 4$ ADHM gauge theory, N M2 branes probing $\mathbb{C}^2/\mathbb{Z}_k$. At large N, the boundary VOA is $\mathcal{W}_{1+\infty}[c]$ at $c \sim N$, with strong generators of weights $1, 2, 3, \ldots$ and rational OPE coefficients in N. The Yang R-matrix $R(u) = u\,I + \Psi\,P$ governs the chiral quantum group structure with $\Psi = N$. The Miura factorisation $T(u) = \prod_{i=1}^{N}(u + \Lambda_i)$ produces the elementary symmetric generators $\psi_s = e_s(\Lambda_i)$.

The Koszul dual. $\mathcal{A}^!_{M2,\infty} \simeq Y^{\mathrm{dg}}(\mathfrak{psl}(2|2))$. The classical r-matrix $k\,\Omega_{\mathfrak{psl}(2|2)}/z$ vanishes at $k = 0$ and satisfies CYBE via the Arnold relation on the $\mathfrak{psl}(2|2)$ Casimir. The Lie superalgebra $\mathfrak{psl}(2|2)$ has vanishing Killing form, so the r-matrix uses the non-degenerate supersymmetric bilinear form $\mathrm{str}(XY)$. The line-operator category is $C_{M2} \simeq \mathrm{Rep}(U_q(\widehat{\mathfrak{psl}}(2|2)))$ with $q = e^{2\pi i/(k+2)}$.

The datum.

$$\mathcal{A} = \mathcal{A}_{M2,\infty}, \qquad \mathcal{A}^! = Y^{\mathrm{dg}}(\mathfrak{psl}(2|2)),$$
$$C = \mathrm{Rep}(U_q(\widehat{\mathfrak{psl}}(2|2))), \qquad r(z) = k\,\Omega_{\mathfrak{psl}(2|2)}/z,$$
$$\Theta_{M2} = \sum_{\Gamma} \frac{W_\Gamma^{\log \overline{C}}}{|\mathrm{Aut}(\Gamma)|} \otimes \Phi_\Gamma^{M2}, \qquad \nabla^{\mathrm{hol}}_{g,n} = d - \mathrm{Sh}_{g,n}(\Theta_{M2}).$$

Shadow tower. Cubic nonzero (nonabelian current subalgebra); quartic nonzero ($16/(22+5c(N))$); quintic $\{\mathfrak{C}_{M2}, Q^{\mathrm{contact}}_{M2}\}_H \neq 0$; tower infinite. At genus 2, all three shells active; the planted-forest shell is the first contribution from Mok's log-FM strata to the holographic genus expansion. The quartic contact invariant is the first obstruction to planarity: it distinguishes the full string-theoretic amplitude from its planar limit.

Conformal anomaly and the $1/N$ expansion. The conformal anomaly of $\mathcal{W}_{1+\infty}[c]$ at $c = N$ is

$$\kappa(\mathcal{W}_{1+\infty}[N]) \;=\; N \cdot (H_\infty - 1) \;\sim\; N \cdot \log N.$$

The logarithmic growth in N is characteristic of the $\mathcal{W}_{1+\infty}$ tower: the harmonic sum $H_N - 1$ diverges logarithmically as $N \to \infty$. The modular characteristic $\kappa \sim N \log N$ controls the curvature of the genus tower at large N; the $1/N$ corrections are encoded in the quartic and higher shadow invariants.

The genus-g free energy at large N: $F_g(N) = \kappa(N) \cdot \lambda_g^{\mathrm{FP}} + O(1/N)$ (UNIFORM-WEIGHT; MULTI-WEIGHT CORRECTIONS FROM CROSS-CHANNEL). The leading $N \log N$ term is the planar gravitational contribution; the subleading $O(1/N)$ terms come from the quartic contact invariant and its descendants in the shadow tower.

The modular Koszul datum $\mathcal{H}(M2)$ packages the complete holographic correspondence into a single MC problem. The computation extends to N M5 branes (boundary VOA is a 6d $(2,0)$ chiral algebra), N D3 branes (boundary VOA is $\mathcal{N} = 4$ SYM chiral algebra), and arbitrary 3d $\mathcal{N} = 4$ gauge theories; in each case, the holographic datum $\mathcal{H}(T)$ is the complete algebraic encoding.

Summary. Sections 1–2 constructed the bar complex from $\eta = d\log(z_1 - z_2)$ and the Arnold relation. Sections 3–4 identified these as projections of $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$. Sections 5–6 extracted the characteristic invariants from $\Theta_{\mathcal{A}}$ and classified algebras by shadow depth. Section 7 ran the machine on every standard family, reading both the geometric face and the E_1 algebraic face for each family. Section 8 lifted the shadow obstruction tower into number theory. Section 9 proved scalar saturation and the Koszulness meta-theorem. Sections 10–11 transported the machine to three dimensions via the Swiss-cheese operad and the modular PVA quantisation obstruction theory. Section 12 packaged the full holographic correspondence into a six-component modular Koszul datum controlled by a single MC equation.

13. Completion and the frontier

The holographic datum of Section 12 packages a chiral algebra, its Koszul dual, the line-operator category, the spectral R-matrix, the universal MC element, and the shadow connection into a single algebraic object. The frontier consists of structural questions the packaging forces: the completed bar-cobar problem by infinite-type algebras, the analytic sewing problem by $g \geq 1$ curvature, the factorization envelope problem by the passage from Lie conformal input to modular output, and the non-principal duality problem by Drinfeld–Sokolov reduction at arbitrary nilpotent orbits.

13.1. The Maurer–Cartan frontier. STATUS: MASTER CONJECTURES MC1 THROUGH MC4 ARE PROVED; MC5 IS PARTIALLY PROVED, WITH THE ANALYTIC HS-SEWING PACKAGE ESTABLISHED AT ALL GENERA AND THE GENUSWISE BV/BRST/BAR IDENTIFICATION CONJECTURAL. MC3 holds for all simple types on the evaluation-generated core via multiplicity-free ℓ-weights (Theorem ??, Corollary ??). The residual problem DK-4/5 (extension beyond evaluation modules) is downstream.

MC1 (PBW concentration). For every standard family (Kac–Moody, Virasoro, principal $\mathcal{W}_N$, $\beta\gamma$, lattice) and every genus $g \geq 1$, the PBW spectral sequence on the genus-g bar complex has $E_\infty^{p,q}(g) = 0$ for $q \neq 0$. Proof: genus-0 Koszulness gives concentration of $E_1^{p,q}(g{=}0)$; the genus-g enrichment from the Hodge bundle is d_0^{PBW}-exact because it arises from the regular Eisenstein series (exact under Poincaré residue). Concentration is uniform in k.

MC2 (universal $\Theta_{\mathcal{A}}$). The bar-intrinsic construction $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is automatically MC because $D_{\mathcal{A}}^2 = 0$. Scalar saturation universality: for every one-parameter algebraic family $\mathcal{A}_k$ with rational OPE coefficients, $\Theta_{\mathcal{A}_k} = \kappa(\mathcal{A}_k) \cdot \eta_{\mathrm{univ}} \otimes \Lambda_{\mathrm{univ}}$. Proof via Whitehead reduction ($E_2^{p,q} = H^p(\mathfrak{g}; \mathrm{Sym}^q V) = 0$ at $p = 2$ by Whitehead's lemma for reductive $\mathfrak{g}$) and algebraic semicontinuity (the non-vanishing locus is Zariski-closed, proper, and empty at the free-field point). Ambient $D^2 = 0$ on the planted-forest complex follows from Mok's normal-crossings result.

MC3 (thick generation). On the type-A Yangian module surface, the four-package MC3 problem reduces to a single compact/completed extension packet via (a) chromatic filtration of K_0 by q-character support, (b) prefundamental Clebsch–Gordan closure $V_n \otimes L^- = \bigoplus_{j=0}^n L^-(n-2j)$ at character level, (c) Francis–Gaitsgory pro-nilpotent completion, and (d) the remaining DK compact/completion packet. The CG decomposition is extended to all simple types via the Chari–Moura multiplicity-free ℓ-weight property, replacing the minuscule hypothesis; the remaining post-CG completion packets beyond type A are the downstream problem.

MC4 (completed bar-cobar). The strong completion-tower theorem: the strong filtration axiom $\mu_r(F^{i_1}, \ldots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$ yields an automatic degree cutoff, so for fixed N only finitely many degrees contribute to F^N, making continuity and Mittag-Leffler automatic. The completion closure $\mathrm{CompCl}(\mathcal{F}_{\mathrm{ft}})$ of complete filtered objects with finite-type associated graded carries a quasi-inverse bar-cobar homotopy equivalence. Key results: coefficient stability, MC twisting closure, completed twisting representability, and the uniform PBW bridge (MC1 implies MC4 for all standard families).

MC4 splits into two sub-problems, both proved:

- *MC4*$^+$ (positive towers: $\mathcal{W}_{1+\infty}$, affine Yangians, RTT algebras): solved by weight stabilisation.
- *MC4*$^\circ$ (resonant towers: Virasoro, non-quadratic $\mathcal{W}_N$): reduced to a finite resonance problem by the resonance-filtered bar-cobar theorem. The resonance rank $\rho(\mathcal{A})$ measures the depth; every positive-energy chiral algebra has $\rho < \infty$ (Theorem ??). The Virasoro case: Vir_{26-c} is the depth-zero resonance shadow; the genuine $\mathcal{W}_\infty$-type dual requires the full resonance-filtered completion.

MC5 (genus expansion). Partially proved. The analytic sewing package is established at all genera, together with the algebraic genus tower. The full genuswise BV/BRST/bar identification remains conjectural. At genus 0, the algebraic BRST/bar comparison is proved (Theorem ??); the tree-level amplitude pairing is conditional on Corollary ??; bar-cobar inversion $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is Theorem B. At genus 1, the curved bar complex with $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_1$ supplies the one-loop package, and the complementarity splitting $Q_1(\mathcal{A}) \oplus Q_1(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_{1,n}, Z(\mathcal{A}))$ is the genus-1 modular-invariance constraint.

At $g \geq 2$ the resolution proceeds in five steps:

(i) *Inductive genus determination* (Theorem ??): $\Theta_{\mathcal{A}}^{(g)}$ is determined by $\Theta_{\mathcal{A}}^{(<g)}$ via the graph-sum recursion on $\overline{\mathcal{M}}_{g,n}$.
(ii) *2D convergence*: no UV renormalisation on curves; the Feynman integrals $W_\Gamma = \int_{\sigma_\Gamma} \omega_\Gamma$ converge absolutely on each stratum without counterterm subtraction.
(iii) *Analytic-algebraic comparison* $D^{\mathrm{an}} = D^{\mathrm{alg}}$ on the algebraic core.
(iv) *General HS-sewing* (Theorem ??): polynomial OPE growth plus subexponential sector growth implies the Hilbert–Schmidt sewing condition, yielding trace-class closed amplitudes at every genus.
(v) *Heisenberg one-particle sewing* identifies the Heisenberg sewing amplitude with a Fredholm determinant, supplying the decisive first-case verification.

At multi-weight $g \geq 2$ (all-weight), the genus expansion admits an explicit cross-channel correction: $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$, proved for $\mathcal{W}_3$ by graph-by-graph computation over all seven stable graphs at $g = 2$ and all forty-two stable graphs at $g = 3$. The scalar formula of the uniform-weight expansion *fails* for multi-weight algebras at $g \geq 2$ (resolving Open Problem ?? negatively). A universal closed form for the gravitational cross-channel correction covers principal $\mathcal{W}_N$: $\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) = (N-2)(N+3)/96 + (N-2)(3N^3+14N^2+22N+33)/(24c)$, exact for $\mathcal{W}_3$ and a lower bound for $N \geq 4$. The correct receptacle at $g \geq 1$ remains the coderived category: curvature forces coderived/contraderived passage.

Completion programme: q-windows and completion entropy. The bar-cobar quasi-isomorphism $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is proved at each finite-type stage $\mathcal{W}_N$. The passage to the inverse limit $\mathcal{W}_\infty = \varprojlim \mathcal{W}_N$ requires completed bar-cobar; the strong filtration axiom gives a quotientwise-finite differential. The completed bar complex decomposes into finite windows indexed by reduced weight $\rho(a) = \mathrm{wt}(a) - 1$.

For Virasoro at $q = 2$: the window $K_2(\mathrm{Vir})$ has basis $\{sT, sT|sT\}$, and the bar differential on the two-dimensional complex is $d(sT|sT) = s(T \cdot T)$. The product $T \cdot T$ decomposes into the quasi-primary Λ (weight 4, reduced weight $3 > 2$, projects to zero), $\partial^2 T$ (reduced weight $3 > 2$, projects to zero), and the central term $(c/2) \cdot \mathbf{1}$. Only the central term survives:

$$d(sT|sT)|_{K_2} = (c/2) \cdot s\mathbf{1}.$$

The modular characteristic $\kappa(\mathrm{Vir}_c) = c/2$ is the sole surviving coefficient of the bar differential on the smallest nontrivial window.

Window dimensions grow exponentially. Define the completion entropy $h_K(\mathcal{A}) := \log(\rho_{\mathcal{A}}^{-1})$, with $\rho_{\mathcal{A}}$ the radius of convergence of the primitive generating series from the vacuum character. The entropy measures kinematic completion complexity and forms a monotone ladder

$$h_K(\mathcal{W}_2) \approx 0.567 < h_K(\mathcal{W}_3) \approx 0.772 < h_K(\mathcal{W}_4) \approx 0.835 < \cdots < h_K(\mathcal{W}_\infty) \approx 0.872,$$

governed at the limit by the MacMahon plane-partition generating function $\chi_{\mathcal{W}_\infty}(q) = \prod_{n \geq 2}(1 - q^n)^{-(n-1)}$.

13.2. The analytic frontier. Curvature forces this direction. At $g \geq 1$ the bar complex satisfies $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g \neq 0$; the ordinary derived category is empty, and the correct receptacle is the coderived category, requiring Hilbert-space-valued structures.

The bare VOA is a dense algebraic skeleton; the object for partition functions is the sewing envelope $\mathcal{A}^{\mathrm{sew}}$, the Hausdorff completion in the topology of all-amplitude seminorms $\|a\|_{\mathrm{amp}} := \sup_{v\in\mathcal{V}}\langle v, \pi(a)v\rangle$. The HS-sewing condition:

$$\sum_{a,b,c\geq 0} q^{a+b+c}\|m^c_{a,b}\|^2_{\mathrm{HS}} < \infty$$

for $|q| < 1$ implies closed amplitudes $\mathrm{Tr}(\pi(a)\cdot q^{L_0})$ are trace-class and the sewing operation converges absolutely. An analytic Koszul pair is one for which the analytic bar-cobar adjunction restricts to a Quillen equivalence on sewing-complete categories.

Four targets (two proved, two conjectural):

(a) *Heisenberg sewing* (decisive first case, proved): sewing envelope of $\mathcal{H}_k$ is $\mathrm{Sym}\,A_2(D) \cong \text{ind-Hilb}(\mathcal{H}_k)$ (Moriwaki 2026).
(b) *Lattice sewing* (proved): sewing envelope of V_Λ is vertex-algebraic completion by lattice theta functions.
(c) *Analytic realisation criterion* (conjectural): HS-sewing on $\mathcal{A}$ plus dual HS on $\mathcal{A}^!$ should suffice.
(d) *Boundary bar duality* (conjectural): $B^{\mathrm{an}}(\mathcal{A}_\partial) \simeq B^{\mathrm{top}}(\mathrm{Fact}^{\mathrm{bulk}}_T)$.

13.3. Factorization envelopes and the modular Koszul datum. The bar construction starts from a chiral algebra, but the natural input in most geometric constructions is a Lie conformal algebra. The factorisation envelope is the functor producing a chiral algebra from Lie conformal data; the programme asks whether the entire modular Koszul package can be constructed functorially at the Lie conformal level, without passing through the vertex algebra. The target is a universal modular factorization envelope $U^{\mathrm{mod}}_X(L)$, left adjoint to extracting primitive currents.

A cyclically admissible Lie conformal algebra L (conformal grading, compatible decreasing filtration, bounded poles, invariant bilinear pairing) produces the modular Koszul datum

$$\Pi_X(L) = \big(\mathrm{Fact}_X(L),\ \bar{B}_X(L),\ \Theta_L,\ \mathcal{L}_L,\ (V^{\mathrm{br}}_L, T^{\mathrm{br}}_L),\ R^{\mathrm{mod}}_4(L)\big),$$

with each component functorial in L. The target universal envelope $U^{\mathrm{mod}}_X(L)$ satisfies the adjunction

$$\mathrm{Hom}_{\mathsf{Fact}^{\mathrm{mod}}}(U^{\mathrm{mod}}_X(L), \mathcal{F}) \cong \mathrm{Hom}_{\mathsf{LCA}^{\mathrm{cyc}}}(L, \mathrm{Prim}^{\mathrm{mod}}(\mathcal{F})),$$

and $\Theta_L = \Theta^{\mathrm{env}}_{\leq\infty}(L)$ is the universal MC class. Direct sums: κ, T^{br}, Δ, R^{mod}_4 decompose additively. BRST reduction: $\Theta_{\mathcal{W}_N} = H^0_{Q_{\mathrm{DS}}}(\Theta_{V_k(\mathfrak{g})\otimes F_{\mathrm{gh}}})$, the shadow tower of the W-algebra is the DS reduction of the full current-plus-ghost tower, not of the bare affine tower.

13.4. Non-principal W-algebra duality. Drinfeld–Sokolov reduction connects affine landscape to W-algebra landscape. For principal f, the reduction commutes with bar-cobar duality by Feigin–Frenkel, and the holographic datum descends. For non-principal f, this commutation is unproved. The theory demands the extension: non-principal $\mathcal{W}$-algebras include the physically central cases ($\mathcal{W}_k(\mathfrak{sl}_N, f_{\mathrm{sub}})$ controls $\mathcal{N} = 2$ superconformal theories) and the mathematically central ones (Feigin–Tipunin at admissible levels).

The hook-type nilpotent orbit in type A is the first non-principal corridor, conditional on DS-bar compatibility (Theorem ??). For a partition $\lambda = (n-p, 1^p)$ in $\mathfrak{sl}_N$,

$$(\mathcal{W}_k(\mathfrak{sl}_N, f_\lambda))^! \simeq \mathcal{W}_{k^\vee_\lambda}(\mathfrak{sl}_N, f_{\lambda^t}),$$

where λ^t is the transpose and $k^\vee_\lambda$ is a rational function of k. Proof: for hook-type partitions, DS commutes with bar-cobar by Fehily and Creutzig–Linshaw–Nakatsuka–Sato (2023–2025).

The transport propagation lemma: hook-type seeds combined with edge-compatibility propagate duality results along edges of the nilpotent Hasse diagram in type A. The type-A transport-to-transpose conjecture: for every partition λ of N, the duality formula holds. The general obstruction is proving that DS reduction commutes with bar-cobar at the chain level, i.e. that the natural map $\mathrm{DS}(\bar{B}(\widehat{\mathfrak{g}}_k)) \to \bar{B}(\mathrm{DS}(\widehat{\mathfrak{g}}_k))$ is a quasi-isomorphism.

13.5. The boxed machine.

$$\boxed{\text{homotopy chiral input } \mathcal{A} \;+\; \text{log-FM cutting cooperad } C^!_{\mathrm{ch}} \;+\; \text{modular convolution } L_\infty \;\Longrightarrow\; \Theta_{\mathcal{A}},\ \mathcal{T}^{\mathrm{mod}}_{\Theta_{\mathcal{A}}},\ \kappa,\ \Delta_{\mathcal{A}},\ \mathfrak{R}_4^{\mathrm{mo}}}$$

The modular convolution L_∞-algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is the single organising structure. It carries a natural modular L_∞ structure from the Feynman transform of the modular operad; $\mathrm{Conv}_{\mathrm{str}}(C^!_{\mathrm{ch}}, \mathcal{P}^{\mathrm{ch}})$ is its strict model. The universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ is proved: it is the bar-intrinsic extraction $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$, automatically MC because $D^2_{\mathcal{A}} = 0$.

The shadow obstruction tower consists of finite-order projections:

$$\begin{array}{lll}
\Theta^{\le 2}_{\mathcal{A}}: & \kappa(\mathcal{A}) & \text{(modular characteristic, scalar curvature),} \\
\Theta^{\le 3}_{\mathcal{A}}: & \mathfrak{C}(\mathcal{A}) & \text{(cubic shadow, Lie structure),} \\
\Theta^{\le 4}_{\mathcal{A}}: & \mathfrak{Q}(\mathcal{A}) & \text{(quartic resonance class, contact),} \\
\Theta^{\le r}_{\mathcal{A}}: & \mathrm{Sh}_r(\mathcal{A}) & \text{(degree-}r\text{ shadow),} \\
\Theta_{\mathcal{A}}: & \varprojlim_r \Theta^{\le r}_{\mathcal{A}} & \text{(universal MC element).}
\end{array}$$

Theorems A–D and Theorem H are proved projections at the scalar level κ. The holographic datum $\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^!, C, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}})$ packages $\Theta_{\mathcal{A}}$ and its projections into the data of a 3d HT field theory.

Principle 0.3 (The four-test interface)**.** The modular Koszul machine has a complete algebraic-geometric interface with $\overline{\mathcal{M}}_{g,n}$ consisting of four independent proved tests:

(i) $D^2_{\mathcal{A}} = 0$: the bar differential squares to zero at all genera and degrees (Theorems ??, ??).
(ii) $\mathrm{obs}_1(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_1$ unconditionally at genus 1 for all families; $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ for uniform-weight algebras at all genera via clutching/GRR; for multi-weight algebras at $g \ge 2$ (all-weight), with cross-channel correction $\delta F^{\mathrm{cross}}_g$ (Theorem ??, Theorem ??).
(iii) $Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$: complementarity assembles the Koszul pair into a Lagrangian decomposition (Theorem ??).
(iv) Sewing: genus-g amplitudes converge from lower-genus data (Theorem ??).

Each test captures a distinct aspect: (i) is purely algebraic and requires no Koszulness; (ii) identifies the tautological class but does not involve $\mathcal{A}^!$; (iii) requires the Koszul pair but is genus-by-genus; (iv) is analytic and independent of the Hodge-class identification. Any three can hold with the fourth failing. Three regions lie outside: multi-weight at $g \ge 2$ (all-weight, resolved with cross-channel correction, Open Problem ??), BV/BRST = bar at higher genus (Conjecture ??), and non-perturbative completion.

The holographic frontier. The holographic datum $\mathcal{H}(T)$ is a boundary object. Its lift to a genuine 3d bulk theory is controlled by the relative holographic deformation complex $\mathfrak{h}^{\Theta}_T := \mathrm{fib}(\mathfrak{g}^{\mathrm{SC}}_T \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})_{\alpha_T}$: lifts of $\Theta_{\mathcal{A}}$ are MC elements of the twisted fibre. The graded obstruction tower $[\mathfrak{o}_r(\beta)] \in H^2(\mathrm{gr}^F_r \mathfrak{h}^{\Theta}_T)$ turns the frontier into cohomology calculus. At depth 4, the relative quartic obstruction $\mathfrak{R}^{\mathrm{rel}}_4(T)$ is the first invariant detecting when a boundary datum cannot arise from a planar holographic realisation. The Loop–Connes principle $\beta^*_{\mathrm{der}}(\Delta_{\mathrm{open}}) = \Lambda_P$: closed genus creation is the image of the open BV operator under the derived-centre map; loops in the bulk are traces on the boundary.

Cyclic trace, ribbon structure, and $1/N$. A bare modular graph sum does not define a literal 't Hooft expansion (Lemma **??**). Cyclic trace structure on the open E_1-chiral algebra resolves this: cyclicity enables ribbon thickening (Theorem **??**), and a matrix realisation gives the exact weight $N^{\chi(\Sigma_\Gamma)}$ (Theorem **??**). The three levels of language (plain modular E_1 genus expansion, cyclic traced modular E_1 't Hooft expansion, ribbonised modular Swiss-cheese holographic 't Hooft language) are made precise in the E_1 modular Koszul chapter (Chapter **??**).

13.6. Admissible levels and logarithmic CFT. At admissible levels $k = -h^\vee + p/q$ (coprime $p \geq h^\vee$, $q \geq 1$), the Zhu algebra $A(\mathcal{V}_k(\mathfrak{g}))$ is semisimple but the vertex algebra is not simple: there is a maximal proper ideal J_k with $L_k(\mathfrak{g}) = \mathcal{V}_k(\mathfrak{g})/J_k$. The bar complex of $\mathcal{V}_k$ acquires a periodic CDG structure indexed by q-th roots of unity: the curvature $m_0 = \kappa \cdot \omega_g$ factors through $\Phi_q(\zeta)$, and the weight filtration has a finite plateau at level $N = q$. This is the algebraic door to logarithmic CFT on the degenerate admissible lane. On the non-degenerate admissible lane the simple quotient is rational; on the degenerate lane the simple quotient has a logarithmic module category. Weightwise finite-page character degeneration holds on the non-degenerate lane; identification of the resulting bar cohomology with logarithmic extensions is not proved here.

13.7. Relative Feynman transform. The modular bar coalgebra $\bar{B}_{\mathrm{mod}}(\mathcal{A})$ is an algebra over the relative Feynman transform $\mathrm{FT}_{\mathsf{Com}_{\mathrm{mod}}/\mathrm{SC}^{\mathrm{ch,top}}}$, the algebraic skeleton shared by the factorisation approach (§10.8) and the operadic approach. Factorisation on $\mathrm{Ran}(\Sigma_g) \times \mathrm{Ran}(\mathbb{R})$ over $\overline{\mathcal{M}}_g$ produces it via families of factorisable $\mathcal{D}$-modules; the operadic structure on $\mathrm{SC}^{\mathrm{ch,top}}_{\mathrm{mod}}$ produces it via formal completion.

Recognition. The tree-level part D_0 is the E_1 coassociative coalgebra structure (encoding boundary A_∞ data); the genus-raising part D_1 is the $\mathsf{Com}_{\mathrm{mod}}$-modular structure. The $\mathrm{SC}^{\mathrm{ch,top}}$ datum is the pair $\big(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}),\ \mathcal{A}\big)$ of chiral derived center and boundary algebra.

13.8. The Calabi–Yau quantum group programme (Volume III). Volume III constructs the concrete examples of E_1-chiral quantum groups from Calabi–Yau geometry. The abstract framework of Sections 1–9 receives its geometric instantiation: the CY categories produce chiral algebras via a functor $\Phi\colon \mathrm{CY}_d\text{-Cat} \to E_2\text{-ChirAlg}$, and the resulting chiral algebras carry the full modular Koszul package.

The CY-to-chiral functor. For a CY_d category $\mathcal{C}$ (a smooth proper dg-category with d-Calabi–Yau structure), the functor Φ produces an E_2-chiral algebra $\Phi(\mathcal{C})$ whose:

- central charge is $\kappa_{\mathrm{ch}}(\Phi(\mathcal{C}))$, a function of the CY structure (for local surfaces $\mathrm{Tot}(K_S \to S)$: $\kappa_{\mathrm{ch}} = \chi(S)/2$, where χ is the topological Euler characteristic);
- r-matrix is the collision residue of the CY-derived MC element, carrying the BPS state structure of the CY geometry;
- shadow tower identifies with the automorphic correction of the associated BKM superalgebra (when one exists).

The functor is proved for $d = 2$ (CY_2 = K3 and abelian surfaces) and for $d = 3$ (∞-categorical proof: $\mathrm{HH}^{-2}_{E_1} = 0$ implies the E_3-lifting is contractible).

Example: $K3 \times E$. The product of a K3 surface and an elliptic curve is a CY_3 with $\kappa_{\mathrm{ch}} = \chi(K3)/2 = 12$ (since $\chi(K3) = 24$). The BKM superalgebra is the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Phi_{10}}$ with root multiplicities encoded by the Igusa cusp form Φ_{10}, weight $10 = 2\kappa_{\mathrm{BKM}}$ with $\kappa_{\mathrm{BKM}} = 5$. The Koszul conductor: $K = 0$ (free-field branch, CY complementarity).

The shadow tower of $\Phi(K3 \times E)$ identifies with the automorphic correction: the coefficients $\mathrm{Sh}_r(\Phi(K3 \times E))$ are the Taylor coefficients of $\log \Phi_{10}$ expanded in the appropriate Siegel modular coordinates. The shadow-Siegel gap (the discrepancy between the shadow tower and the full Siegel modular form) measures the genuinely genus-2 information invisible to the shadow tower.

Example: conifold. The resolved conifold $\mathrm{Tot}(\mathcal{O}(-1)^{\oplus 2} \to \mathbb{P}^1)$ is a toric CY_3 with $\kappa_{\mathrm{ch}} = 1$ (from DT computation; the formula $\kappa_{\mathrm{ch}} = \chi(S)/2$ applied to the base $\mathbb{P}^1$ gives $\chi(\mathbb{P}^1)/2 = 1$, which is numerically correct but the derivation requires care since $\mathcal{O}(-1)^{\oplus 2} \neq K_{\mathbb{P}^1}$). The CoHA (cohomological Hall algebra) of the conifold quiver $Q = \bullet \rightrightarrows \bullet$ carries the E_1-chiral quantum group structure; integrality of the DT invariants $\Omega(\gamma) \in \mathbb{Z}$ is encoded in the CoHA product.

Example: $\mathbb{C}^3$. The simplest non-compact CY_3. The shadow tower at $\kappa_{\mathrm{ch}} = \Psi$ (the deformation parameter of $\mathcal{W}_{1+\infty}$) produces the perturbative constant-map Gromov–Witten free energies $F_g^{\mathrm{GW,const}}(\mathbb{C}^3)$. On the DT side, the MacMahon function $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ counts plane partitions. The shadow IS the full GW answer for $\mathbb{C}^3$ because there are no compact curves (no worldsheet instantons); the two towers (shadow and GW) agree exactly.

13.9. The 6d holomorphic Chern–Simons hierarchy. The primary geometric source of E_1-chiral quantum groups is 6d holomorphic Chern–Simons theory on a Calabi–Yau threefold Y. The hierarchy of theories descends by dimensional reduction and defect insertion:

Theory	Space	Output	Status
6d hCS	Y (CY_3)	BPS/CoHA algebra	source
5d CS	$\mathbb{R} \times S$ (CY_2)	affine Yangian	reduction
4d CS	$\Sigma \times C$	integrable system	proved
3d HT CS	$\mathbb{R} \times C$	boundary + bulk	proved
2d chiral	C	bar complex	proved

Each step in the hierarchy corresponds to a dimensional reduction or defect insertion, and the modular Koszul datum descends functorially through each step. The bar complex of the 2d chiral algebra on C is the terminal output; the 6d theory on Y is the geometric origin. The shadow tower of the 2d chiral algebra encodes the BPS state content of the 6d theory through the automorphic correction.

Codimension-2 defects. For a holomorphic curve $C \subset Y$ (a codimension-2 defect), the boundary algebra restricted to the defect is $\mathcal{W}_{1+\infty}$ at $\Psi = -\sigma_2(N_{C/Y})$, where σ_2 is the second Chern root of the normal bundle $N_{C/Y}$. For $C = \mathbb{P}^1 \subset \mathbb{C}^3$ with $N_{C/\mathbb{C}^3} = \mathcal{O}(-1)^{\oplus 2}$: $\sigma_2 = 1$, giving $\Psi = -1$ and $c = 1$ from the Sugawara construction. The spectral parameters of the chiral quantum group come from the two independent directions in $N_{C/Y}$.

The Drinfeld centre closes the circle. The E_n operadic circle

$$E_3(\text{bulk}) \to E_2(\text{boundary chiral}) \to E_1(\text{bar/QG}) \to E_2(\text{Drinfeld centre}) \to E_3(\text{derived centre})$$

is the holographic structure of the programme. Each arrow is: restriction to codimension-2 defect, ordered bar complex, categorified averaging (Drinfeld centre), and higher Deligne (derived centre). The circle closes for 3d HT theories with conformal vector: the Drinfeld centre $\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))) \simeq \mathrm{Rep}^{E_2}(\mathcal{W}_{1+\infty})$ identifies the E_2 Drinfeld centre of the E_1 quantum group representation category with the E_2 representation category of the boundary chiral algebra. This is the categorical level of the holographic correspondence.

13.10. The deepest open problems (2026-04-17 state). The 2026-04-16 closure wave proved the periodic-CDG compatibility for admissible Kazhdan–Lusztig, iterated Sugawara and the E_∞-topologization ladder, the chiral higher-Deligne theorem, the $\mathrm{SC}^{\mathrm{ch,top}}$ heptagon, the CY-D dimension stratification, and the Francis–Gaitsgory $(\infty, 2)$-equivalence at properad level. The 2026-04-17 Beilinson-rectified audit reduced the frontier to the following residual problems.

Beilinson-rectified residuals (2026-04-17).

(i) *$W(p)$ triplet tempering*: the Gurarie 1993 and Flohr 1996 logarithmic-CFT amplitudes exhibit unbounded Massey products despite finite-dimensional Zhu algebra. The correct tempered scope is principal, non-logarithmic, non-minimal standard landscape; tempering of $W(p)$ is open pending an Adamović–Milas character-amplitude bound.
(ii) *Chain-level class M on the original complex*: the weight-completed category is proved (Milnor/Mittag-Leffler via the explicit SDR of $h \cdot m_0^{j-1}$); the direct-sum chain-level identification is genuinely false, since $S_4(\mathrm{Vir}_c) = 10/[c(5c+22)] \neq 0$ at generic c. Whether the original complex admits chain-level E_3 through a different homotopy transfer remains open.
(iii) *Exact closed form of β_N at $N \geq 4$*: W_N tempering is unconditional via β-independent Stirling dominance. The two candidate closed forms $(N+1)(N+2)/2$ and $N^2 - N + 4$ diverge at $N = 4$ (15 vs. 16). A Fateev–Lukyanov explicit structure-constant derivation for $\mathcal{W}_4$ is the outstanding computation.
(iv) *Super-complementarity canonical Verdier pairing*: the super-Yangian complementarity sum splits by normalisation. The super-trace form $\kappa^{\mathrm{str}} + \kappa^{\mathrm{str},!} = 0$ is PROVED for $Y(\mathfrak{sl}(m|n))$ on the sub-Sugawara line $m \neq n$ via super-Sugawara rational-identity cancellation. The Berezinian form $\kappa^{\mathrm{sBer}} + \kappa^{\mathrm{sBer},!} = \max(m,n)$ is CONJECTURED at Sugawara-shifted dual level, conditional on a closed-form derivation of the $\frac{1}{2}\max(m,n)$ shift magnitude induced by the central automorphism σ_{sBer} (Nazarov 1991 and Gow 2006 supply centrality of $\mathrm{sBer}(T(u))$ and non-degeneracy of the Berezinian pairing, not the shift magnitude in closed form). Two admissible pairings (super-trace and Berezinian) coexist without programme-level canonicalisation; the Verdier pairing inscription is pending.

Structural frontier.

(v) *Closing the E_n circle* (Conjecture **??**): $\mathcal{Z}(\mathbf{U}_{\mathcal{A}}) \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$.
(vi) *Drinfeld double at E_1-chiral level*: construct $\mathcal{A} \bowtie \mathcal{A}^!$ as a Hopf object and identify the antipode with orientation reversal.
(vii) *A genuinely new E_1 invariant*: a number from E_1 theory that E_∞ cannot compute.
(viii) *Equivalence of five E_1-chiral notions*: Notions A–E potentially define different categories.
(ix) *Chiral R-matrices controlling holomorphic Wilson lines*: braiding of ordered conformal blocks.
(x) *6d hCS defect algebra in full generality*: link algebra on S^3 from CY_2 normal fiber.
(xi) *E_1 Verlinde formula*: non-commutative fusion using the R-matrix.
(xii) *BKM denominator = bar Euler product*: chain-level root multiplicities from BPS degeneracies.

Closed since 2026-04-16. CY-A at $d = 3$ (∞-categorical, $\mathrm{HH}^{-2}_{E_1} = 0$); the periodic-CDG conjecture (FM251) for admissible KL; the modular-family upgrade of Theorem A over $\overline{\mathcal{M}}_{g,n}$; topologization for general chiral with conformal vector via the iterated-Sugawara E_∞ ladder; the chiral coproduct for non-gauge-theoretic families via the unified $Q_g^{k,f,\mu}$ construction.

Homotopy-involutivity. $\mathrm{FT}^2_{\mathsf{Com}_{\mathrm{mod}}/\mathrm{SC}^{\mathrm{ch,top}}} \simeq \mathrm{id}$. Genus-0 involutivity from operadic homotopy-Koszulity; closed-colour modular involutivity from the Feynman involution (Theorem **??**); assembly via the genus-filtration spectral sequence.

Three routes unified. Route B (factorisation) is global truth; Route A (operadic) is local approximation; Route C (this section) is the categorical image. The factorisation structure projects to the relative Feynman transform by forgetting $\mathcal{D}$-module data; the operadic structure injects by formal completion.

13.11. Computational infrastructure. Every formula has been verified by an independent symbolic computation layer comprising tens of thousands of tests across several hundred modules, covering the shadow obstruction tower through degree 32, the standard landscape for all seven families, the genus-2 and genus-3 planted-forest formulas, the DS-shadow cascade (degrees 2–8 for $N = 2, \dots, 5$), the $\mathcal{W}_3$ multi-generator shadow metric, the Böcherer bridge at genus 2, the Pixton shadow bridge, the entanglement programme, the non-simply-laced bar computations (B_2, C_2, G_2, F_4), minimal model bar and fusion rules, the dressed propagator engine (scalar complementarity, R-matrix $R_1 = 1/12$), and the constrained Epstein zeta. The computation is adversarial: tests encode cross-family consistency checks that cannot be fooled by hardcoded values.

Volume III extends the engine to Calabi–Yau geometry: forty-two CY/BKM compute engines with 5700+ tests cover the $K3 \times E$ programme (shadow-Siegel gap, BKM root multiplicities, moonshine multiplier, Schottky barrier), toric CY_3 CoHA integrality, and the automorphic correction as shadow obstruction tower identification.

Volume II develops the 3d theory in three new chapters: the factorisation Swiss-cheese algebra (global truth on $\mathrm{Ran}(\Sigma_g) \times \mathrm{Ran}(\mathbb{R})$), the modular Swiss-cheese operad (local approximation), and the relative Feynman transform (algebraic skeleton). Together with the 3d gravity chapter, the affine half-space BV theorem, and eleven additional computational engines, these constitute the Volume II programme.

13.12. The single open problem. Everything proved in this monograph is a finite-order projection of the universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$. Everything open is a question about its ambient:

Characterise the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ as a geometric object.

Precisely: the modular convolution algebra is the endomorphism algebra of the Lagrangian embedding $\mathcal{L}_{\mathcal{A}} \hookrightarrow \mathcal{M}_{\mathrm{vac}}(\mathcal{A})$ in the (-2)-shifted symplectic stack of vacua. Volume I computes the Taylor jets of this embedding (the shadow obstruction tower). Volume II identifies the local composition law (the Swiss-cheese operad). What remains is the global geometry of the stack itself: the derived symplectic structure, the moduli of Lagrangians, and the higher-categorical descent that would make the construction independent of choices.

The approximately 150 open conjectures in these volumes are projections of this single question:

- the $\mathcal{D}$-module purity conjecture asks whether the Lagrangian is pure (characterisation (xii) of Koszulness);
- the non-principal W-duality programme asks how the Lagrangian transforms under DS reduction at arbitrary nilpotent orbits;
- the analytic completion programme asks when the algebraic Lagrangian extends to a convergent one (the sewing envelope);
- the factorization envelope programme asks for the universal construction of the Lagrangian from Lie conformal data alone;
- the holographic completion question asks whether the genus-0 Lagrangian extends over $\overline{\mathcal{M}}_{g,n}$;
- the Calabi–Yau quantum group programme (Volume III): Sections 1–9 introduce E_1-chiral quantum groups as abstract algebraic-geometric objects; Volume III constructs the concrete examples from CY geometry. The functor $\Phi\colon \mathrm{CY}_d\text{-Cat} \to E_2\text{-ChirAlg}$ produces chiral algebras from CY categories (proved for $d = 2$ and $d = 3$; the $d = 3$ proof uses the ∞-categorical vanishing $\mathrm{HH}^{-2}_{E_1} = 0$). The CY quantum groups (CoHA, quantum toroidal algebras, BPS algebras) are the flesh on the algebraic skeleton of Sections 1–9. The theories studied include 6d holomorphic Chern–Simons on CY_3 (the primary source), 4d and 5d HT Chern–Simons, 6d little string theory, and M5 brane theory. The shadow tower of the CY chiral algebra IS the automorphic correction of the BKM superalgebra; at $\kappa_{\mathrm{ch}} = \Psi$ the shadow produces the constant-map Gromov–Witten free energies of $\mathbb{C}^3$. The Drinfeld centre $\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))) \cong \mathrm{Rep}^{E_2}(\mathcal{W}_{1+\infty})$ closes the E_n operadic circle.

Each is a different facet of the geometry of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$, and none requires going beyond the framework already established.

The entire structure unfolds from a single datum: the logarithmic 1-form $\eta = d\log(z_1 - z_2)$. At n points it became a differential on the Fulton–MacPherson compactification; across all genera it became the total bar differential $D_{\mathcal{A}}$; in the modular convolution algebra it became the Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$. One 1-form, one relation (Arnold), one extraction (residues), one ordering (the real line): a universal invariant of chiral algebras that is algebraic, computable, and complete.

Volume I constructs E_n-chiral algebras as algebraic-geometric objects on curves and their configuration spaces: factorisation algebras, bar complexes on every curve geometry, R-matrices, derived chiral centres with E_2/E_3 structure, the $\mathrm{SC}^{\mathrm{ch,top}}$ datum on the derived centre pair. Volume II interprets these constructions physically: the derived centre IS the bulk of a 3d gauge theory; 3d quantum gravity is the climax. Volume III constructs the concrete CY quantum groups from higher-dimensional HT theories: the E_1-chiral quantum groups that Volume I introduced abstractly receive their geometric instantiation. Together: holomorphic chiral factorisation (co)homology via bar and cobar chain constructions at various geometric locations, and the different locations produce the different operads at play.

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